

SARKARI PATH

sarkaripath.onl

MONTHLY COMPENDIUM

CURRENT AFFAIRS

JULY 2026 EDITION

Coverage: 25th June 2026 to 24th July 2026

- [+] Comprehensive Coverage of National & International Events
- [+] Polity, Economy, Science-Tech, Environment & Prelims Facts
- [+] SPECIAL ADDITION: 200 Practice MCQs & FAQs Included
- [+] Official Verified Answer Key (Q1 to Q200) Included at the End

TARGETED FOR ALL COMPETITIVE EXAMS

- UPSC & State PSCs: CSE, UPPSC, BPSC, RAS, MPPSC, WBPSC, etc.
- SSC Exams: CGL, CHSL, CPO, MTS, GD Constable, Stenographer
- Railway Exams: RRB NTPC, Group D, ALP, Technician, JE
- Banking & Insurance: IBPS PO / Clerk, SBI PO / Clerk, RBI, LIC
- Defence & Teaching: CDS, NDA, AFCAT, CAPF, CTET, State TET

Download Free Study Notes, Mock Tests & Daily Current Affairs

VISIT: [SARKARIPATH.ONL](https://sarkaripath.onl)

(Coverage from 25th June 2026 to 24th July 2026)

INDEX

Polity & Governance

1-34

• National Anti-Doping (Amendment) Act, 2025	1
• Assam Floods and Flood Management in India	1
• Internet Shutdowns in India	3
• SC Sets Aside Death Sentence and Upholds Fair Trial	5
• National Commission for Minorities (NCM)	5
• National Investment Policy for Urea-2026	6
• India's Water Conservation Drive	7
• Draft National Health Research Policy 2026	7
• NIPUN Bharat Mission and Foundational Learning	8
• UDISE+ and Performance Grading Index 2.0 2025-26	9
• SC Stays Madras HC's Blanket Ban on Cow Slaughter in Tamil Nadu	10
• SC Reinforces Due Process in Citizenship Determination	11
• Civil Registration System (CRS) Report 2024	12
• States Bear Disproportionate Share of India's Welfare Spending	13
• Viksit Bharat Shiksha Adhishthan Bill 2025	14
• Concerns over the Draft National Food Security (Amendment) Bill, 2026	15
• 'Satluj' Takedown: Frameworks Governing Theatrical and OTT Content in India	16
• Sub-Mission on Agricultural Mechanization	16
• SC to Refer Grounds of Arrest Issue to Larger Bench	17
• Armed Forces Seek Higher Agniveer Retention	17
• All India Survey on Higher Education Reports for 2022-23 and 2023-24	18
• Poshan Tracker: India's Real-Time Nutrition Monitoring Platform	19
• National Academic Depository	20
• Ayushman Bharat Health Accounts (ABHA)	21
• Prime Minister Launches Next Phase of UDAN: Viksit UDAN	22
• Social Media Platforms Under Regulatory Scrutiny	22

• NITI Aayog's Strategic Roadmap for Making Ayurveda Global	24
• INSA Proposes a Unified National Energy Framework	25
• Centre Proposes Special Article 371 Framework for Ladakh	27
• Ease of Living: Key Government Initiatives and Outcomes	27
• Viksit Bharat – Guarantee for Rozgar and Aajeevika Mission (Gramin) Act, 2025	29
• Pradhan Mantri Krishi Sinchayee Yojana	30
• Strengthening Food Safety in India	31
• Delhi Electric Vehicle Policy 2.0	32
• Anemia Mukht Bharat Abhiyaan Operational Guidelines	32
• NCERT Introduces Emergency in Class 9 Textbook	33
• Criminalisation of Politics in India	33
• Passport is Not Proof of Citizenship	34

Nation & States

35-39

• Global Terrorism Index 2026	35
• AI-Driven Warfare: Preparing India for Future Combat Readiness	36
• A Decade of Defence Reforms in India	38
• Tungabhadra River Dispute	39

Economic Scenario

40-61

• Polymer Currency Notes	40
• India's Rising Inflation	41
• Index of Core Industries with Base Year 2022-23	42
• NITI Aayog's Investment Friendliness Index 2026	42
• Strengthening Rural Credit for Inclusive Growth in India	44
• Mobile Phone Manufacturing Scheme (MPMS)	45
• Empowering India's Skill Ecosystem	46
• Weaving Sustainability into India's Textile Future	48
• Cabinet Approves Additional Investment in NIIF	49
• Strengthening India's Toy Ecosystem	50

● India's Insurance Sector: A Pillar of Sovereign Fiscal Stability	50
● World Bank Income Classification: India Remains Lower-Middle-Income	52
● 5 Years of Cooperative Transformation in India	52
● NCLAT on IBC-PMLA Conflict	54
● India's Fisheries Sector	55
● 9 Years of Goods and Services Tax (GST)	55
● Unlocking Growth in India's Tourism and Hospitality Sector	57
● Reforms to Expand Foreign Participation in G-Secs	57
● 20th Statistics Day and SDGs-National Indicator Framework Progress Report 2026	58
● India Leads the Ship Recycling Sector	59
● Index of Services Production	60
● FCRA Amendment Rules 2026	61

International Relations 62-73

● Presidential Visits to Moldova and North Macedonia	62
● World AI Cooperation Organization (WAICO)	62
● India-UK CETA Comes into Effect	63
● India-Australia Civil Nuclear Cooperation	64
● China's Helium Export Ban and Supply Chain Vulnerabilities	65
● India-New Zealand Strategic Partnership: Roadmap to 2030	65
● 3rd India-Australia Annual Leaders' Summit 2026	67
● India's Bid for UNSC Non-Permanent Seat for 2028-29	68
● India-Indonesia Relations	69
● Inaugural India-Mali Forum for Promotion of Exports	71
● 16th India-Japan Annual Summit 2026	71
● India-Seychelles Relations	73

Environment & Ecology 74-76

● White-bellied Heron	74
● 2025 Animal and Plant Discoveries and 111th Foundation Day of ZSI	74
● Compressed Biogas and India's Energy Security	75

Science & Technology 77-88

● Space Oncology and Cancer Treatment	77
---------------------------------------	----

● India's 1st Geothermal Wells Commissioned at Puga Valley	77
● White Rabbit Technology- Based IST Dissemination Demonstration Network	78
● Vikram-1: Charting India's Cosmic Future	78
● Semaglutide Patent Expiry and Opportunities for India	80
● Cabinet Approves Semicon 2.0	81
● Gaganyaan Mission	82
● India Joins NeoSep1 Trial to Combat Neonatal Sepsis	83
● Quantum Computing	83
● UN Scientific Panel on AI Report	84
● Reforms 3.0 for the Bharat Rate of Growth	85
● India's First 3D-Printed Artificial Reefs	86
● Scaling India's Technology Ecosystem	86
● The Challenge of India's Digital Sovereignty	88

History, Art & Culture 89-90

● PM Pays Tribute to Swami Vivekananda	89
● Birth Anniversary of Dr. Syama Prasad Mookerjee	89
● 16 th -Century Vijayanagara Inscriptions in Seshachalam Forest	90
● Birth Anniversary of Mangal Pandey	90
● Birth Anniversary of K. Kamaraj	90

Social Issues 91-93

● Rising Secondary Infertility in India	91
● Empowering Persons with Disabilities in India	91
● SUMAN Roadmap 2030	92
● Tackling Adolescent Malnutrition in India	93

Facts for Prelims 94-115

● Reports & Indices	94
● Awards & Honours	95
● Important Days	95
● Defence & Security	97
● Summits & Conferences	99
● Places in News	101
● Portals & Apps	104
● Species in News	105
● Miscellaneous	106

Polity & Governance

National Anti-Doping (Amendment) Act, 2025

India's **Commonwealth Games 2026** medal prospects were affected after two athletes were removed for **anti-doping violations**, amid implementation of the **National Anti-Doping Act, 2022** and **Amendment Act, 2025** to strengthen clean sport regulations.

- India topped the list of doping offenders in the **2024 annual report of the World Anti-Doping Agency (WADA)**, with a **positivity ratio of 3.6%**, among the highest globally.

Key Features of the National Anti-Doping (Amendment) Act, 2025

- **About:** Amends the **National Anti-Doping Act, 2022** to strengthen India's anti-doping framework and promote **clean sport**.
 - Aligns with the **World Anti-Doping Code** and **UNESCO International Convention against Doping in Sport**.
- **Institutional Framework:** Establishes **National Anti-Doping Agency (NADA)** for implementing anti-doping rules.
 - Establishes **National Board for Anti-Doping in Sports** to oversee NADA and advise the government.
- **Key Provisions:** Provides statutory backing for **testing, results management, education, intelligence, and investigations**.
 - Empowers the National Board to obtain information from **Disciplinary Panel and Appeal Panel**.
 - Defines **anti-doping rule violations** and related circumstances.
- **Significance:** Enhances **transparency, accountability, and due process**.
 - Improves institutional autonomy and aligns India with **international best practices**.
 - Promotes **fair play and equal opportunities** for athletes.

NADA

- **About:** India's autonomous body established in **2005** for promoting and monitoring anti-doping in sports.
- **Functions:** Plans, implements, and coordinates India's anti-doping activities.
 - Follows **World Anti-Doping Agency (WADA) Code** and regulations.
- **Initiatives:** Implements **Anti-Doping Program** and **#PlayTrue Campaign** to educate athletes and promote clean sport.

World Anti-Doping Agency (WADA)

- **About:** Established in **1999** by the **International Olympic Committee (IOC)** as the global authority against doping in sports.
- **Governance Model:** Based on equal partnership and funding between **Sports Movement and Governments**.
- **Core Document:** **World Anti-Doping Code** harmonises anti-doping rules globally.
- **Key Standard:** Publishes **WADA Prohibited List** of banned substances and methods, updated annually.
- **Operational Scope:** Applies to **in-competition, out-of-competition, and sport-specific scenarios**.

Assam Floods and Flood Management in India

Assam experienced its **worst floods in over 60 years**, affecting **8 lakh+ people** and causing **30+ deaths**.

Why is Assam Highly Prone to Floods?

- **Geographical vulnerability:** Assam lies in the **Brahmaputra and Barak river basins** with 50+ tributaries; its floodplain location makes it naturally flood-prone.
 - The **Brahmaputra** deposits heavy sediments in Assam's plains, raising riverbeds and increasing flood risk.
- **Heavy monsoon rainfall:** Intense southwest monsoon and upstream discharge from **Arunachal Pradesh and Meghalaya** trigger floods.
- **Riverbank erosion:** Frequent channel shifts, embankment breaches and erosion have affected around **4.27 lakh hectares since 1950**.
- **Deforestation and soil erosion:** Increase sediment inflow, reducing river capacity and worsening floods.
- **Climate change:** Rising extreme rainfall events and higher flood peaks are increasing flood intensity.
- **Embankment limitations:** Many embankments have exceeded their design life and cannot manage the dynamic Brahmaputra system.
- **High exposure:** Around **39.58% of Assam's area** is flood-prone, nearly four times the national average.

Causes of Floods in India

Flood Vulnerability

- India has over **40 million hectares** of flood-prone area out of **329 million hectares** geographical area.
- Floods affect around **75 lakh hectares annually**, causing about **1,600 deaths** on average.

Natural Causes

- **Concentrated Monsoonal Rainfall:** Around **75–80% of annual rainfall** occurs during the Southwest Monsoon; intense rainfall often exceeds river and drainage capacity.
- **Siltation and Riverbed Aggradation:** Himalayan rivers like **Ganga and Brahmaputra** carry heavy sediments, reducing channel capacity and increasing floods.
- **River Meandering:** Sediment-rich rivers frequently change courses; e.g., **Kosi River (Sorrow of Bihar)** causing repeated floods in Bihar.
- **Glacial Lake Outburst Floods (GLOFs):** Climate-induced glacier retreat increases glacial lakes and sudden flood risks (e.g., **2023 Sikkim GLOF**).
- **Cyclones and Storm Surges:** Bay of Bengal cyclones cause coastal flooding and restrict river discharge into the sea.

Anthropogenic Causes

- **Deforestation and Catchment Degradation:** Increase runoff, soil erosion and sediment deposition.
- **Embankment Trap:** India has built **34,000+ km embankments since 1954**; failures can cause severe floods.
- **Dam Mismanagement:** Sudden reservoir releases and poor rule-curve management worsen downstream flooding like **Kerala floods (2018)**.
- **Floodplain and Wetland Encroachment:** Reduces natural water storage capacity; degradation of wetlands like **Assam's Beels** increases flood intensity.
- **Unscientific Sand Mining:** Alters river morphology and destabilises riverbanks.

Urban Flooding

- **Unplanned Urbanisation:** Concretisation reduces water absorption and increases runoff.
- **Loss of Water Bodies:** Encroachment of lakes, wetlands and drainage channels reduces flood-buffering capacity as seen in **Bengaluru** and **Chennai**.
- **Poor Storm-Water Drainage:** Blocked and inadequate drainage systems restrict water evacuation.
- **Climate-Induced Extreme Rainfall:** Short-duration intense rainfall overwhelms urban drainage, causing flash floods.

Measures Taken by India for Flood Management

- **Structural Measures**
 - **Embankments and Drainage Works:** Over **34,000 km of embankments** constructed along with drainage channels, flood protection works and raised villages.
 - **Dam Safety Act, 2021:** Provides framework for **surveillance, inspection, operation and maintenance** of specified dams.

- **Reservoir Management:** Uses **Rule Curves** to maintain flood cushions and regulate monsoon water storage.
- **Inter-Basin Water Transfer (IBWT): National Perspective Plan (1980)** aims to transfer surplus floodwaters to water-deficit regions.
- **Non-Structural Measures**
 - **Flood Plain Zoning: CWC Model Flood Plain Zoning Bill (1975)** regulates development in flood-prone areas.
 - **Flood Forecasting and Early Warning:** CWC uses **real-time telemetry, hydrological modelling and inundation forecasting**.
 - **Catchment Area Treatment (CAT):** Promotes **afforestation, watershed management and soil conservation** to reduce runoff and sedimentation.
 - **Community-Based Preparedness:** Includes awareness programmes, mock drills and local disaster management plans.
- **Institutional and Policy Measures**
 - **Disaster Management Act, 2005:** Provides framework through **NDMA, SDMA, DDMA, NDRF and SDRFs** for disaster preparedness and response.
 - **Flood Management and Border Areas Programme (FMBAP):** Ministry of Jal Shakti scheme supporting flood control, anti-erosion, drainage and anti-sea erosion works.
 - **National Water Policy, 2012:** Promotes integrated river basin management, flood moderation and restoration of natural drainage.
 - **IMD:** Provides weather forecasts and **impact-based warnings** for evacuation and preparedness.

Concerns Regarding India's Flood Management

- **Constitutional and Federal Gridlock:**
 - States control water, drainage, embankments (**Entry 17, State List**) while Centre controls inter-state rivers (**Entry 56, Union List**), causing poor coordination.
 - **River Boards Act, 1956:** Empowered River Basin Organizations remain non-functional, hindering integrated basin-level management.
- **Fiscal and Project Execution Failures:** CAG audits show delays in DPR approvals, fund release issues, diversion and poor utilization of funds, leaving flood-control and anti-erosion projects incomplete.
- **Deficient Flood Forecasting and Telemetry:** Many CWC telemetry stations and water-level sensors are non-functional due to poor maintenance and vandalism.
 - Delayed or unreliable hydrological data reduces forecasting accuracy and evacuation lead time.

- **Transboundary Water Governance Gaps:** Indus Waters Treaty (1960) and seasonal MoUs with China do not cover climate change, glacial melt, or flood management, limiting transparency and early-warning capacity.
- **Weak Dam Safety and Reservoir Operations:** Non-adherence to **Rule Curves** and inadequate Emergency Action Plans (EAPs) convert natural floods into man-made disasters.
 - Sudden large-scale releases from dams exacerbate downstream flooding despite **Dam Safety Act, 2021**.
- **Weak Grassroots Disaster Preparedness:** SDRFs and district administrations are underfunded and under-equipped, making states heavily dependent on **NDRF**, with poor community resilience and early-warning dissemination.
- **Northeast-Specific Concerns:** Fragile Himalayan ecology, transboundary rivers from **China and Bhutan**, weak hydrological cooperation, inadequate forecasting infrastructure, and poor connectivity hinder integrated river basin management and disaster response.
 - Results in recurring loss of **lives, livelihoods, biodiversity, and critical infrastructure**.

Measures to Strengthen India's Flood Management

- **Shift Towards Flood Resilience:** Follow **Rashtriya Barh Ayog (National Flood Commission), 1980** recommendations: reduce indiscriminate embankments, promote **floodplain management, raised settlements and flood-resilient agriculture**.
 - **R. Rangachari Committee (2003):** Shift from flood control to **flood management** through **Integrated River Basin Management (IRBM)**, zoning, forecasting and community participation.
- **Improve Reservoir and Flood Forecasting:** Maintain dynamic **flood cushion** through revised Rule Curves.
 - Develop **National Water Model (NWM)** and **Decision Support System (DSS)** for real-time forecasting.
 - Modernise hydro-meteorological networks as recommended by **NITI Aayog Committee on Flood Management (2021)**.
- **Strengthen Disaster Risk Reduction:** Move from relief-based response to **risk reduction** through resilient infrastructure, early warnings and **Build Back Better** approach under **Sendai Framework (2015–2030)**.
- **Regulate Floodplain Development:** Implement **floodplain zoning laws** based on **Model Flood Plain Zoning Bill, 1975**.
 - Restrict construction on floodplains, wetlands and natural drainage channels.
- **Promote Basin-Level Governance:** Establish **River Basin Organizations (RBOs)** for coordinated planning, reservoir management, sediment control and interstate data sharing.

- **Use Advanced Technology:** Expand **AI-based Early Warning Systems, IoT-based monitoring, satellite flood mapping** and strengthen the **CWC National Flood Forecasting Network**.
- **Adopt Nature-Based Solutions:** Promote **"Room for the River"** and **Sponge City** approaches through wetland restoration, floodplain reconnection and increasing permeable surfaces.
- **Interlinking of Rivers (ILR):** Can reduce flood peaks by storing excess monsoon water in multipurpose reservoirs (e.g., **Kosi and Mahanadi**).
 - Requires integration with **floodplain zoning, embankments and early warning systems** as it cannot completely eliminate floods.

Floods in India are increasingly becoming governance failures rather than purely natural disasters. Examine.

Practice Mains Question

Internet Shutdowns in India

Recently, mobile internet services were suspended in parts of **Central Delhi** during a protest.

Internet Shutdown

- **Definition:** An intentional disruption of internet or electronic communication services, making them inaccessible or unusable for a population or location.
- **Forms:** Complete shutdowns, internet throttling and blocking of specific platforms/services.
- **Purpose:** Used to control information flow during **protests, political events, religious gatherings and public unrest**.
- **Nature:** May be **preventive** or imposed during ongoing incidents.
- **Impact:** Affects **journalists, media organisations and public access to verified information**.
 - Reduces transparency and can increase the spread of **unverified information**.

Global Scenario

- **Growing Global Trend:** UNESCO reported at least **300 shutdowns across 54+ countries** in the last two years.
 - **2024** was the worst year for shutdowns since **2016**, with increasing use during protests and elections.
- **India's Position:** India remains among the countries with the highest number of shutdowns.
 - **Access Now (2025):** Recorded **65 shutdowns across 12 States/UTs**.
 - **2026:** Around **24 shutdowns recorded till July 22**.
 - **SFLC Tracker:** Highest shutdowns in **Jammu & Kashmir (449 since 2012)**, followed by **Rajasthan (115)** and **Manipur (62)**.

- **UNESCO's Position:** Calls for promoting **internet accessibility** rather than excessive restrictions.
 - Recognises internet access as important for **freedom of expression, education, association, assembly and participation in public life**.

Legal Framework for Internet Shutdowns in India

- **Governing Law:** Regulated by the **Telecommunications (Temporary Suspension of Services) Rules, 2024** under the **Telecommunications Act, 2023**.
 - Replaced the earlier **2017 Temporary Suspension Rules**.
- **Grounds for Suspension:** Under **Section 20(2)(b) of the Telecommunications Act, 2023**, suspension is allowed only for **Public emergency & Public safety**.
 - Reasons must be **recorded in writing**.
 - A protest alone is **not a lawful ground** for internet suspension.
- **Principles and Safeguards**
 - Orders must satisfy **necessity and proportionality**.
 - Must be **reasoned, published, limited in area and duration**, and consider less restrictive alternatives.
- **Authority to Issue Orders**
 - **Union Home Secretary:** National-level shutdowns.
 - **State Home Secretary:** State-level shutdowns.
 - In urgent cases, authorised **Joint Secretary-level officers** may issue orders, subject to confirmation within **24 hours**.
- **Judicial Review:** Orders can be challenged under **Article 226** (High Courts) & **Article 32** (Supreme Court).
 - Courts examine legality, necessity, proportionality and limits of the shutdown order.

Judicial Position on Internet Shutdowns

- **Right to Internet Access:** In **Anuradha Bhasin v. Union of India (2020)**, the SC held that internet access is linked with **Article 19(1)(a)** - Freedom of speech and expression & **Article 19(1)(g)** - Freedom of trade and profession.
- **Principles for Shutdowns:** Any shutdown must:
 - Have **legal backing** and a legitimate objective.
 - Follow **necessity and proportionality**.
 - Use the **least restrictive measure**.
 - Be limited in **scope and duration**, with reasons published.
- **Indefinite Shutdowns:** The SC held that indefinite internet suspension is **impermissible**; orders require review by a **multi-member Review Committee within 5 working days**.

- **Least Restrictive Approach:** Complete suspension of telecom services is a drastic measure and should be used only when unavoidable after considering alternatives.
- **Calcutta HC (2022):** In **Ashlesh Biradar v. State of West Bengal**, the Court stayed an internet suspension order for lacking legal authority, adequate reasons and consideration of alternatives.

Concerns Regarding Internet Shutdowns

- **Violation of rights:** Shutdowns restrict **Article 19(1)(a)** freedom of speech and affect internet access linked with **Article 21**.
- **Limited oversight:** Despite the **Telecommunications Act, 2023** and **2024 Rules**, lack of independent oversight may enable arbitrary implementation.
- **Bypassing safeguards:** Some States continue using the **Indian Telegraph Act, 1885** and **2017 Rules**, raising concerns over procedural compliance.
- **Lack of transparency:** Despite the SC mandate in **Anuradha Bhasin (2020)**, many shutdown orders and review records are not publicly available.
- **Economic and social impact:** Disrupts businesses, digital services, education, banking, healthcare and communication; India reportedly lost around **USD 255.2 million** due to shutdowns in **2023**.
- **Digital governance concerns:** Frequent shutdowns may conflict with India's goals in **AI, digital governance and technological innovation**.
- **Duration concerns:** Although **2024 Rules** limit shutdowns to **15 days**, weak review and publication practices remain a concern.

Measures to Address Internet Shutdown Concerns

- **Strengthen oversight:** Create independent review mechanisms or parliamentary scrutiny to examine shutdown orders and prevent arbitrary use.
- **Ensure legal compliance:** Follow the **Telecommunications Act, 2023** and **2024 Rules**, allowing shutdowns only for **public emergency or public safety** with proper justification and review.
- **Apply necessity and proportionality:** Treat shutdowns as a **last resort**, with limits on duration, area and scope.
- **Use targeted alternatives:** Prefer **content regulation, fact-checking, misinformation monitoring and cybersecurity measures** over blanket shutdowns.
- **Improve transparency:** Publish shutdown orders, reasons, duration and review decisions to protect **freedom of expression and access to information**.
- **Promote digital literacy:** Strengthen awareness programmes and align policies with **UN principles on internet access, human rights and open connectivity**.

Internet shutdowns in India raise complex questions of public order, fundamental rights and proportionality. Discuss.

Practice Mains Question

SC Sets Aside Death Sentence and Upholds Fair Trial

In **Abdul Hameed v. State of Rajasthan**, the SC set aside the death sentence in the **1996 Samleti Bus Bomb Blast case** as **effective legal representation** was denied and violated the **right to a fair trial**.

- SC ordered a **fresh (de novo) trial** to be completed within **one year**.

SC Observations on Right to Fair Trial

- **Fair trial is non-negotiable:** The SC held that fair trial guarantees cannot be sacrificed for convenience; criminal justice requires a **fair, just and reasonable procedure**.
 - A co-accused's confession is **not substantive evidence** and can only support conclusions based on independent admissible evidence.
- **Due process over retribution:** Justice depends on **procedural fairness and restraint**, not merely swift punishment; reaffirmed the principle that it is better for guilty persons to escape than for an innocent person to be punished.
- **Fresh trial ordered:** Instead of acquittal, the Court directed a **de novo trial within one year** with effective legal representation.
- **Procedural safeguards:** Even in serious offences like terrorism, the State must uphold constitutional safeguards; trial courts must ensure competent legal representation before evidence examination or sentencing.

Legal Aid and Fair Trial Provisions in India

- **Constitutional Provisions**
 - **Article 21:** Includes the **Right to Fair Trial** and **Right to Free Legal Aid** as part of a just, fair and reasonable procedure.
 - **Article 22(1):** Guarantees the right of an arrested person to consult and be defended by a **legal practitioner of choice**.
 - **Article 39A:** Directs the State to provide **free legal aid** to ensure equal access to justice, inserted by the **42nd Constitutional Amendment Act, 1976**.
- **Statutory Provisions**
 - **BNSS, 2023 (Section 341):** Courts must provide a state-funded advocate if an accused cannot afford legal representation.
 - **Legal Services Authorities Act, 1987:** Established **NALSA** (headed by CJI), **SLSAs**, **DLSAs** to provide free legal services to weaker sections.

- ❖ **Eligible groups:** SCs/STs, women, children, persons with disabilities, victims of trafficking, violence or disasters, industrial workers, persons in custody and low-income individuals.
- ❖ **Impact:** Over **44.22 lakh people** received legal aid/advice during **2022–23 to 2024–25**.
- ❖ **Lok Adalats**
- ❖ Recognised under the **Legal Services Authorities Act, 1987** for speedy settlement of disputes.
- ❖ Settled over **23.5 crore cases (2022–23 to 2024–25)**, reducing judicial burden and litigation costs.

■ Landmark Judgments

- **Hussainara Khatoon v. State of Bihar (1979):** Free legal aid is part of the **fair, reasonable and just procedure under Article 21**.
- **Khatri v. State of Bihar (1981):** State must provide legal aid from the stage of **first production before a magistrate**, not only during trial.

Initiatives and Institutions Supporting Legal Aid

■ Key Initiatives

- **Legal Aid Defense Counsel System (LADCS):** NALSA initiative providing free legal defence in criminal cases under the **Legal Services Authorities Act, 1987**.
- **DISHA (2021–26):** Department of Justice scheme promoting access to justice through **Tele-Law**, **Nyaya Bandhu (Pro Bono Legal Services)**, legal literacy and awareness programmes; benefited over **2.10 crore people** (February 2025).

■ Institutions Supporting Accessible Justice

- **Fast-Track Courts (FTCs):** Ensure speedy trials of heinous offences and sensitive cases; **865 FTCs functional (June 2025)**.
- **Fast-Track Special Courts (FTSCs):** Established in 2019 for rape and POCSO cases; **725 courts functional**, including **392 exclusive POCSO courts**.
- **Gram Nyayalayas:** Rural courts under the **Gram Nyayalayas Act, 2008** for affordable justice; **488 functional (March 2025)**.
- **Nari Adalats:** Under **Mission Shakti (Sambal)**, resolve domestic and gender-based disputes through mediation at Gram Panchayat level.
- **Exclusive Special Courts (SC/ST Act):** Established under the **SC/ST (Prevention of Atrocities) Act, 1989** for speedy trial of offences against SCs/STs.

"A fair trial is the cornerstone of the rule of law." Discuss how far has it succeeded in ensuring access to justice?

Practice Mains Question

National Commission for Minorities (NCM)

Harjit Singh Grewal assumed charge as **Chairman of NCM** as the previous Chairperson's tenure ended in **April 2025**.

NCM

- **About:** NCM is a **statutory body** established under the **National Commission for Minorities Act, 1992**.
- **Composition:** Consists of **1 Chairperson, 1 Vice-Chairperson and 5 Members**.
 - Members are nominated by the **Central Government** from persons of **eminence, ability and integrity**.
 - At least **5 members, including the Chairperson**, must belong to notified minority communities.
- **Mandate & Functions:** Reviews development progress of minorities under the **Union and States**.
 - Monitors constitutional and legal safeguards for minorities. Examines complaints regarding deprivation of minority rights.
 - Conducts studies and research on **socio-economic and educational development** of minorities.
- **Powers:** Has powers of a **Civil Court** during investigations, including:
 - Summoning witnesses.
 - Requiring documents.
 - Receiving evidence through affidavits.
- **Tenure and Removal**
 - **Tenure:** Chairperson and Members hold office for **3 years**.
 - **Resignation:** Submitted in writing to the **Central Government**.
 - **Removal grounds:** Insolvency, conviction involving moral turpitude, unsound mind, inability to act, absence from **3 consecutive meetings**, or abuse of office.
 - Removal for abuse of office requires a **reasonable opportunity of being heard**.

Concerns Regarding the NCM

- **Limited powers:** NCM's recommendations are **advisory and non-binding**, unlike constitutional bodies such as **NCSC (Article 338)** and **NCST (Article 338A)**.
- **Political appointments:** Central Government's nomination power may lead to preference for political affiliates over independent experts.
- **Vacancies and delays:** Prolonged vacancies affect grievance redressal, monitoring and policy advice.
- **Weak accountability:** Delays in submitting reports and **Action Taken Memorandum (ATM)** to Parliament reduce transparency.
- **Minority determination issue:** Conflict between national notification under the **NCM Act, 1992** and the Supreme Court's view in **T.M.A. Pai Foundation (2002)** that states are the unit for determining minorities.

- **Weak state mechanisms:** Several **State Minority Commissions** face absence, inadequate funding, manpower shortages and limited investigative capacity.

Measures to Strengthen NCM

- **Independent appointments:** Adopt an **NHRC-like selection mechanism** with a high-powered committee involving the **PM, Speaker, Home Minister, Leaders of Opposition and Deputy Chairman of Rajya Sabha** to ensure transparency and autonomy.
- **Constitutional status:** Convert NCM into a **constitutional body** as proposed in the **National Commission for Minorities (Repeal) Bill, 2004**, giving it greater authority and credibility.
- **Evidence-based policymaking:** Implement **Sachar Committee (2006)** recommendations by creating an **Equal Opportunity Commission (EOC)** and **National Data Bank (NDB)** for reliable minority socio-economic data.
- **Mandatory ATRs:** Set a fixed timeline (e.g., **six months**) for the government to submit **Action Taken Reports** to Parliament, with reasons for rejecting NCM recommendations.

The National Commission for Minorities has often been criticised as a "toothless tiger." Examine the institutional and legal constraints limiting its effectiveness.

Practice Mains Question

National Investment Policy for Urea-2026

The **Cabinet Committee on Economic Affairs (CCEA)** approved **NIPU-2026** under **Atmanirbhar Bharat** to promote new **gas-based urea plants** and to raise domestic production, reduce imports and strengthen **fertilizer security**.

Key Features of NIPU-2026

- **Capacity addition:** Supports **8–9 new gas-based urea plants**, each with about **12.7 lakh tonnes** capacity, adding nearly **10 million tonnes** overall.
- **Key Reforms:** NIPU-2026 introduces several changes in **New Investment Policy (NIP)-2012**
 - **Transparent subsidy:** Separates **fixed and variable costs** in subsidy calculations.
 - **Assured returns:** Provides a **12–16% Return on Equity** band.
 - **Forex protection:** Fixed costs will be converted into **INR after four years** to reduce exchange-rate risk.
 - **Project scope:** Focuses on **new gas-based urea manufacturing units**.
- **Cost savings:** Expected to save over **₹250 crore per plant** compared with NIP-2012 projects.
- **Equal treatment:** Offers identical incentives to **private, public and cooperative sectors**.

Need for NIPU-2026

- **Demand-supply gap:** Urea's demand growing approx 5% annually. India consumes about **40 million tonnes** of urea annually but produces only **30 million tonnes**, leaving a **10-million-tonne import gap**.
- **Self-sufficiency:** The policy aims to add **10 million tonnes** of domestic capacity and reduce import dependence.
- **Policy vacuum:** NIP-2012 expired in **October 2019**, creating the need for a new investment framework.
- **Geopolitical risks:** Dependence on imported urea and **LNG** exposes India to supply disruptions and price shocks, threatening fertilizer and food security.

India's Water Conservation Drive

A nationwide campaign was conducted from **4 July to 4 August 2026** following the Prime Minister's **Mann Ki Baat** appeal.

- Focuses on **rainwater harvesting, groundwater recharge, restoration of traditional water bodies, plantation and public participation**.
- Aims to strengthen and sustain water-conservation efforts across India.

Key Water-Harvesting and Conservation Initiatives

Catch the Rain Campaign

- **About:** Launched by the **National Water Mission** with the tagline **"Catch the rain, where it falls, when it falls."**
- **Activities:** Promotes **check dams, rooftop harvesting, desilting of tanks, repair of stepwells** and recharge through unused wells and borewells.
- **Rain Centres:** **Rain Centres** are to be set up in every district for technical guidance.
- **Local Storage:** Encourages local storage of rainwater to improve groundwater, soil moisture and reduce urban flooding.
- **Participation:** Involves district authorities, schools, public bodies and large landholding institutions.

Jal Sanchay Jan Bhagidari: Catch the Rain, 2026

- **About:** Launched on **1 June 2026** by integrating **Jal Sanchay Jan Bhagidari** with **Jal Shakti Abhiyan: Catch the Rain**.
- **Objective:** Focuses on **rainwater harvesting, groundwater recharge, water storage and community participation**.
- **Convergence:** Uses convergence with **PMKSY-Per Drop More Crop, MGNREGS-linked rural works, water-body restoration, CAMPA funds, Finance Commission grants and state schemes**.
- **Participation:** Mobilises **Panchayats, ULBs, women, youth, educational institutions and civil society** to strengthen climate resilience.

Jal Shakti Abhiyan: Catch the Rain

- Launched on **World Water Day, 2021** with the theme **"Catch the Rain, where it falls, when it falls."**
- Promotes **rainwater harvesting, groundwater recharge, watershed development, water-body rejuvenation and afforestation** through a **Jan Andolan**.
- Builds on **Jal Shakti Abhiyan, 2019**, which covered **1,592 water-stressed blocks in 256 districts**.
- The campaign is conducted annually with changing themes and focus areas.

Jal Shakti Kendras

- Serve as **district-level knowledge and facilitation centres** for water conservation.
- Provide awareness and technical guidance on **rainwater harvesting and groundwater recharge**.
- Connect communities, district administrations and government agencies to strengthen **Jan Bhagidari**.

Water Harvesting System

- **About:** A **water harvesting system** captures, stores and uses rainwater or surface runoff for **domestic use, irrigation and groundwater recharge**.
- **Types:**
 - **Rainwater harvesting:** Rooftop collection and storage.
 - **Groundwater recharge:** Recharge wells and pits that allow water infiltration.
 - **Surface-water harvesting:** Ponds and reservoirs for irrigation and other uses.
 - **Urban water harvesting:** Captures rooftop and surface runoff to reduce pressure on municipal systems.
- **Significance:** Provides a **reliable water source** and improves groundwater availability.
 - Reduces **flooding, waterlogging, soil erosion and surface runoff**.
 - Helps prevent **seawater intrusion** in coastal areas.
 - Supports sustainable water management and checks **groundwater depletion**.

Draft National Health Research Policy 2026

The **Department of Health Research** released the **Draft National Health Research Policy 2026** for public consultation.

- It proposes India's first unified framework covering **biomedical research, clinical medicine, public health, epidemiology, digital health, and AI**.

Key Highlights of the Policy

- **National Health Research Agenda (NHRA):** Periodically prioritise research based on **disease burden, equity, pandemic preparedness, and national interests**, in consultation with **States, UTs, researchers, & communities**.

- **Three-Tier Governance Structure:**
 - **National Health Research Stewardship Committee:** Strategic coordination and oversight.
 - **Department of Health Research (DHR):** Nodal implementing and coordinating agency.
 - **Indian Council of Medical Research (ICMR):** Scientific and technical lead.
- **Outcome-Based Research Evaluation:** Shift from publication- and patent-based assessment to **real-world impact**. The expanded ICMR-IRIS will evaluate contributions to **public policy, clinical practice, health programmes, indigenous technologies, institutional capacity, health equity, and society**.
- **Enhanced Funding and Long-Term Targets:** Raise public health research spending from **0.024% of GDP to 0.072% by 2037 and 0.15% by 2047**, against **0.27% in high-income nations**.
 - Targets are also proposed for **research investment, medical science PhDs, and indigenous health technologies**.
- **Focus on Emerging Technologies:** Priorities include **AI, health data science, cell and gene therapy, precision medicine, and climate-related health research**, along with **TB, Antimicrobial Resistance, and maternal health**.
- **Shared Research Infrastructure:** Publicly funded infrastructure will be treated as a **shared national resource** to widen institutional access and reduce bottlenecks.
- **Research Ethics and Data Governance:** Emphasises **research integrity, ethics, cybersecurity, data governance, and community participation** to improve transparency, accountability, and public trust.

Need for National Health Research Policy

- **Fragmented Health Research Governance:** Research is divided among **ICMR, DBT, CSIR, AIIMS, and State medical universities**, causing duplication, weak coordination, and slow translation into practice and policy.
- **Concentration of Research Capacity:** Research infrastructure, manpower, and innovation remain concentrated in premier institutions, while **Tier-II and Tier-III medical colleges** have limited capacity.
- **Misalignment with India's Disease Burden:** Research priorities inadequately reflect the simultaneous burden of **infectious diseases, NCDs, AMR, maternal and child health challenges, and climate-sensitive diseases**.
- **Weak Research-to-Policy Translation:** Publication- and patent-driven evaluation has widened the **bench-to-bedside gap**, limiting translation into policies, indigenous technologies, and better healthcare delivery.

- **Regulating Digital Health Research:** Regulation has not kept pace with **AI, genomics, precision medicine, digital health, and health data systems**, while complex approvals delay research and innovation.
- **Global Health Security Vulnerabilities:** India ranked **66th among 195 countries** in the **Global Health Security Index 2021**, highlighting gaps in pandemic preparedness, genomic surveillance, and biosecurity.
- **Ethical Challenges:** Practices such as **data fabrication, plagiarism, peer-review fraud, weak ethical oversight, authorship manipulation, excessive self-citation, and poor reproducibility** undermine research integrity.

Discuss the significance of the Draft National Health Research Policy 2026 in strengthening India's health research ecosystem and promoting evidence-based public health governance.

Practice Mains Question

NIPUN Bharat Mission and Foundational Learning

Five years after its launch in **July 2021**, **NIPUN Bharat** is being reviewed to consolidate learning gains, extend support beyond **Grade 3** and strengthen **foundational literacy and numeracy (FLN)** for **Viksit Bharat**. **NEP 2020** identified **FLN** as the highest priority, with over **5 crore** elementary students lacking basic reading and numeracy skills.

Key Features of NIPUN Bharat Mission

- **About:** Launched by the **Ministry of Education in July 2021** under **NEP 2020** and **Samagra Shiksha**.
- **Objective:** Ensure every child acquires foundational competencies in **reading, writing and numeracy by Grade 3**, and not later than **Grade 5**, by **2026–27**.
- **Target Group:** Covers children aged **3–9 years**, from **Balvatika to Grade 3**, with grade-wise learning targets called **Lakshyas**.
- **Learning Approach:** Promotes **child-centred, play-based, activity-based, experiential, toy-based, storytelling, art- and sports-integrated learning** suited to children's linguistic and social backgrounds.
- **Core Competencies:**
 - **Literacy:** Oral language, decoding, reading fluency, comprehension and writing.
 - **Numeracy:** Number concepts, operations, measurement, shapes, spatial understanding and data handling.
- **Teacher Training:** Teachers and school heads are trained through **FLN-specific NISHTHA modules**, with focus on **mother-tongue instruction, peer learning and teaching-learning materials**.
- **Learning Resources:** **DIKSHA** provides worksheets, videos, assessment banks, children's literature and teacher handbooks.

- **Assessment and Monitoring:** Uses **continuous, competency-based and stress-free assessments**, supported by child-wise tracking, annual surveys and IT-enabled monitoring.
- **Implementation:** Implemented by the **Department of School Education and Literacy** through a **five-tier structure** at national, State, district, block/cluster and school levels, with participation of **SMCs, parents, local bodies and communities**.
- **Expected Outcomes:** Improve learning outcomes, reduce dropouts, strengthen transition to higher classes, promote inclusion and build a foundation for higher-order learning and employability.

According to **ASER**, Grade 3 government-school students able to read a **Grade 2-level text** rose only from **16.3% in 2022 to 23.4% in 2024**, showing slow progress towards universal **FLN by 2026–27**.

Role of NIPUN Bharat in Strengthening Foundational Learning

- **Extend to Grade 5:** Continue FLN support beyond Grade 3, as **56% of Grade 5 students cannot read a Grade 2 text** and nearly **70% lack basic arithmetic skills**.
- **Universalise Balvatika:** Provide nationwide Balvatikas with **trained educators, age-appropriate materials and play-based learning**, as **57% of children enter school with low readiness**.
- **Strengthen Teacher Capacity:** Replace one-time cascade training with continuous hybrid development aligned with **National Professional Standards for Teachers (NPST)**, especially for **Multi-Grade Multi-Level (MGML) classrooms**, which form nearly **70% of primary schools**.
- **Professionalise Mentoring:** Create a dedicated mentor cadre for classroom observation, learning assessment and regular feedback.
- **Reform Assessments:** Align **PARAKH Rashtriya Sarvekshan (PRS) and Foundational Learning Study (FLS)**, introduce competency-based **Grade 5 assessments**, and improve FLN measurement.
- **Improve Governance:** Publish a district-level **NIPUN Index**, involve parents through home-reading and community reading activities, and reactivate the **National Steering Committee** with outcome-based monitoring.

UDISE+ and Performance Grading Index 2.0 2025-26

The **Ministry of Education** released **Unified District Information System for Education Plus (UDISE+) 2025–26 and Performance Grading Index (PGI) 2.0 for States/UTs and Districts**, assessing educational infrastructure, capacity and outcomes, with policy focus shifting from **enrollment to measurable learning progress**.

Key Highlights of the UDISE+ 2025-26 Report

- **Dropout Rates:** Secondary-level dropout fell from **8.2% to 7.0%**, while preparatory-level dropout declined from **2.3% to 1.8%**.
- **State Disparity:** **Ladakh, Gujarat, Chhattisgarh and Karnataka** recorded the highest secondary-level dropout rates.
- **Enrolment and Retention:** Secondary **GER** rose from **68.5% to 71.7%**, while retention improved from **47.2% to 51.9%**.
- **Teacher Strength:** India's teacher count crossed **1 crore**, reaching **1.02 crore**.
 - **Pupil-Teacher Ratio (PTR):** Remained below the **NEP 2020 benchmark of 30:1** at all levels: **10:1 foundational, 12:1 preparatory, 17:1 middle and 21:1 secondary**.
- **Transition Rates:** **99.2%** from foundational to preparatory, **93.8%** from preparatory to middle and **88.3%** from middle to secondary.
- **Digital Infrastructure:** **69.9%** of schools have computers and **67.4%** have internet connectivity.
- **Basic Infrastructure:** **95%** have electricity, **99.5%** drinking water, **98.5%** girls' toilets and **96.9%** handwashing facilities.
- **Inclusive Infrastructure:** **90.5%** have libraries, but only **58.2%** have ramps with handrails.
- **Gender Representation:** Women form **54.9%** of teachers, while girls constitute **48.4%** of enrolment.

UDISE+: Developed by the **Ministry of Education**; one of the largest Management Information Systems on school education globally.

Key Highlights of PGI 2.0 (2025-26)

- **Top-Grade Deficit:** No State/UT attained **Utkarsh, Uttam-1 or Uttam-2**.
- **Best Performer:** **Chandigarh** ranked highest nationally with **Uttam-3**.
- **Minimum Performance:** No State/UT fell under **Akanshi-2 or Akanshi-3**, indicating at least moderate performance across all.

PGI 2.0

- **About:** A composite framework of the **Department of School Education and Literacy, Ministry of Education**, for assessing school education performance across States/UTs.
- **Evolution:** Introduced in **2017** and revamped as **PGI 2.0** in line with **NEP 2020**.
- **Purpose:** Supports **evidence-based policymaking**, identifies gaps and enables targeted interventions.
- **Structure:** Uses **73 indicators**, grouped into **2 categories and 6 domains (Learning Outcomes & Quality; Access;**

Infrastructure & Facilities; Equity; Governance Processes; Teacher Education & Training), with a total score of **1,000 points**.

- **Data Integration:** Linked with **UDISE+**, **NAS**, **PM POSHAN** and **Vidyanjali** for automated data collection and verification.
- **Grading:** Uses a **10-level scale** from **Utkarsh (91–100%)** to **Akanshi-3 (up to 10%)**.
- **Focus:** Gives greater weight to **learning outcomes, equity and teacher education**, shifting emphasis from infrastructure to educational quality.

Major Challenges in India's School Education System

- **Secondary-Stage Drop-Off:** Retention falls from **98.5% at the foundational level** to **51.9% at the secondary level**, with nearly half of students leaving before completing higher secondary education.
- **Learning Poverty:** Weak reading ability by **Grade 3** and poor **foundational literacy and numeracy (FLN)** make advanced subjects difficult, increasing disinterest and dropout by the secondary stage.
- **One-Teacher Schools:** Over **1 lakh schools** operate with a single teacher, increasing teaching and administrative burdens and reducing classroom quality.
- **Poor CWSN Accessibility:** More than **40% of schools** lack adequate facilities for **Children with Special Needs (CWSN)**.
- **Quality–Access Gap:** High enrolment and infrastructure expansion have not translated into strong learning outcomes, as no State/UT secured the top three grades in **PGI 2.0**.
- **Regional and Institutional Disparities:** A **31.4% performance gap** exists between **Chandigarh and Meghalaya**. Weak governance, teacher absenteeism, inefficient fund use and poor functioning of **School Management Committees** affect outcomes in several States.
- **Resource Constraints:** Lagging **Akanshi districts** lack funds and administrative capacity to implement **NEP 2020** measures such as teacher training and smart classrooms.

Measures to Strengthen School Education in India

- **Structural Reforms:** Merge **single-teacher and low-enrolment schools** into **Composite/Cluster Schools** to improve teacher availability and resource use.
- **Governance:** Universalise **APAAR** for student tracking and strengthen **School Management Committees (SMCs)** with parental participation for better implementation of the **RTE Act, 2009**.
- **Foundational Learning:** Accelerate the **NIPUN Bharat Mission** to achieve **foundational literacy and numeracy by Grade 3**.

- **Pedagogical and Assessment Reforms:** Use **PARAKH** for competency-based assessment focused on **critical thinking**.
- **Medium of Instruction:** Promote **mother tongue/regional languages** in foundational years, as envisaged under **NEP 2020**.
- **Teacher Capacity:** Fill vacancies through mission-mode recruitment, implement **ITEP**, and expand continuous training through **DIKSHA**.
- **Equity and Retention:** Introduce vocational education from **Class VI**, improve **CWSN infrastructure**, and strengthen **PM POSHAN**.
- **Academic Audits:** Conduct independent, third-party social audits of learning outcomes at the **Panchayat level** and link them to **Samagra Shiksha fund disbursement**.

Critically examine the role of data-driven governance through UDISE+ and the Performance Grading Index (PGI) 2.0 in improving school education in India.

Practice Mains Question

SC Stays Madras HC's Blanket Ban on Cow Slaughter in Tamil Nadu

SC issued an **interim stay** on the **Madras HC's** direction imposing a **blanket ban** on the slaughter of **cows and calves** in **Tamil Nadu**.

- **Tamil Nadu Government's Argument:** HC's order amounted to **judicial lawmaking** and contradicted the **Tamil Nadu Animal Preservation Act, 1958**, which permits slaughter of cows above **10 years** if certified unfit for work and breeding.
- **Existing Regulation:** The **Prevention of Cruelty to Animals Act, 1960** and municipal rules regulate slaughter in designated slaughterhouses but do not impose a complete ban.
- **Madras High Court Order:** The blanket ban was based on **Article 48**, which directs the State to preserve cattle breeds and prohibit slaughter of cows, calves, and other milch and draught cattle.
- **Significance of SC Stay:** It reinforces the **separation of powers**, indicating that courts should not override existing statutory provisions. Animal preservation falls under **Entry 15 of the State List, Seventh Schedule**.

Landmark Judgments on Cow Slaughter

- **Mohd. Hanif Quareshi v. State of Bihar (1958):** Upheld the ban on slaughter of **cows and calves**, but struck down a total ban on slaughter of **economically useless breeding bulls and working bullocks** as an unreasonable restriction under **Article 19(1)(g)**.

- **State of Gujarat v. Mirzapur Moti Kureshi Kassab Jamat (2005):** Upheld Gujarat's **complete ban on cow slaughter** and held that **Article 48** can guide the reasonableness of restrictions under **Article 19(6)**. It also recognised the **ecological and agricultural utility of aged cattle**.
- **Akhil Bharat Krishi Goseva Sangh v. State of A.P. (2006):** Reaffirmed that **cow protection is primarily a legislative policy matter** and courts should not override valid State laws, consistent with the **separation of powers**.
- **Fundamental Rights Interplay:** Cow-slaughter bans involve balancing **Article 48** with **Article 19(1)(g)** and **Article 25**.

SC Reinforces Due Process in Citizenship Determination

In *Sabitri Dey @ Swasthi Dey vs Union of India (2026)*, SC set aside **27 Gauhati HC judgments** that had upheld Foreigners Tribunals' decisions declaring several Assam residents as foreigners.

- **Ruling:** Citizenship and foreigner status must be decided through a **fair, lawful and reasoned process**.
- **Direction:** The Court ordered **fresh adjudication** in accordance with **fairness, reasonableness and due process**.

SC's Ruling Regarding Citizenship Determination

- **Background of the Case:** In 1997, an **Illegal Migrants (Determination) Tribunal (IMDT)** declared **Sabitri Dey and her husband** illegal migrants after they failed to appear. They claimed they had no notice of the proceedings and that the order relied on **hearsay evidence**.
 - Their **2019** challenge was dismissed by the **Gauhati HC in 2020** due to a **23-year delay**, after which they approached the Supreme Court alleging denial of a fair hearing and constitutional due process.
- **SC Ruling:** Proceedings under the **Foreigners Act, 1946** must be **fair, lawful, reasoned and based on proper application of mind**, and cannot be mechanical or one-sided.
- **Constitutional Protection:** The guarantees under **Article 14** and **Article 21** extend to **foreigners** as well.
- **Burden of Proof:** Under **Section 9 of the Foreigners Act, 1946**, now the **Immigration and Foreigners Act, 2025**, the burden of proof lies on the proceedee. However, absence alone cannot justify a **mechanical declaration** of foreigner status.
- **Duties of Foreigners Tribunals:** Before declaring a person a foreigner, the Tribunal must:
 - independently examine the **State's evidence**;
 - ensure **proper service of notice**;
 - disclose the **main grounds of reference**;

- provide a **fair opportunity of hearing**; and
- record **reasoned findings**.

- **Significance:** The judgment mandates **evidence-based constitutional due process**, strengthens the accountability of **Foreigners Tribunals**, and protects individuals from arbitrary **ex parte declarations** that may lead to **detention, deportation or statelessness**. It is expected to significantly affect citizenship disputes, especially in **Assam**.

Foreigners Tribunals (FTs)

- **About:** Quasi-judicial bodies constituted under the **Foreigners (Tribunals) Order, 1964** issued under **Section 3 of the Foreigners Act, 1946**, to determine whether a person is a **foreigner**.
- **Functioning:** Though applicable across India, FTs function mainly in **Assam**, dealing with citizenship disputes related to the **NRC** and references by competent authorities.
- **Composition:** Headed by a member drawn from **judicial officers, advocates or civil servants with judicial experience**.
- **Powers:** Possess **civil court powers** (summoning witnesses, examining on oath, requiring documents).
 - Under the **Immigration and Foreigners Order, 2025**, they also have the powers of a **Judicial Magistrate First Class**, including issuing **arrest warrants** and ordering **detention** where authorised by law.
- **Procedure:** Issue notice to the proceedee, provide an opportunity to present evidence, and deliver a **reasoned opinion**. A person declared a foreigner may face **detention and deportation**, subject to legal remedies.

New Deportation Policy, 2026

- **About:** Introduced by the **MHA** to create a **uniform and time-bound system** for detecting, identifying, detaining and deporting illegal migrants, especially from **Bangladesh and Myanmar**.
- **Coverage:** Applies to:
 - **illegal migrants**;
 - **foreigners overstaying visas**; and
 - **persons declared foreigners by Foreigners' Tribunals**.
- **Implementation:** States and Union Territories must:
 - establish district-level **Special Task Forces (STFs)**;
 - operationalise **Holding Centres**;
 - use the **Foreigners Identification Portal (FIP)** for biometric and demographic profiling; and
 - complete nationality verification and deportation within fixed timelines.
- **Verification Timeline:** Nationality verification of suspected **Bangladeshi and Myanmar nationals** must be completed within **90 days**.

Evolution of India's Citizenship

Framework from Jus Soli to Jus Sanguinis

- **Constitution, 1950: Articles 5–11** adopted an inclusive **jus soli (right by birth)**, or citizenship by birth, framework at the commencement of the Constitution.
- **Citizenship Act, 1955:** Provided citizenship through **birth, descent, registration, naturalisation and incorporation of territory**, largely retaining **jus soli**.
- **1986 Amendment:** Following the **Assam Accord, 1985**, citizenship by birth required at least **one Indian parent**, beginning the shift towards **jus sanguinis (citizenship by descent)**.
- **2003 Amendment:** Required one parent to be an **Indian citizen** and the other **not an illegal migrant**. It also introduced:
 - the concept of an **illegal migrant**;
 - **Section 14A** for the **National Register of Indian Citizens**; and
 - **Overseas Citizen of India (OCI)** status.
- **CAA, 2019:** Exempted **Hindus, Sikhs, Buddhists, Jains, Parsis and Christians** from **Pakistan, Bangladesh and Afghanistan** from the definition of illegal migrant and reduced their naturalisation requirement from **11 years to 5 years**.

Citizenship by Birth	
Period of Birth	Requirement
26 Jan 1950–1 July 1987	Citizen regardless of parents' nationality
1 July 1987–3 Dec 2004	At least one parent must be an Indian citizen
On or after 3 Dec 2004	One parent must be Indian and the other must not be an illegal migrant

- **By Descent:** Persons born abroad may acquire citizenship through Indian parentage; births after **3 Dec 2004** require **consular registration**.
- **Registration and Naturalisation:** Available to eligible **PIOs, spouses of citizens and foreign nationals** meeting residency conditions, generally **12 years**.
- **Citizenship Certificates:** Citizens by **birth or descent** generally do not receive certificates; formal certificates are issued for citizenship by **registration or naturalisation**.
- **Dual Citizenship:** India does **not permit dual citizenship**.

Examine the evolution of India's citizenship framework from jus soli to jus sanguinis.

Practice Mains Question

Civil Registration System (CRS) Report 2024

The **CRS Report 2024** recorded **99.1% of births** and **99.4% of deaths**, up from **56% and 48% in 2000**.

Key Highlights of the CRS Report

About the Report

- Released by the **Office of the Registrar General and Census Commissioner of India, Ministry of Home Affairs**.
- **CRS** continuously and compulsorily records **births, deaths, and stillbirths** under the **Registration of Births and Deaths Act, 1969**, amended in **2023**.
- Legally operational since **1970**.

Key Highlights

- **Births:** **99.1%** of estimated births were registered in **2024**.
- **Deaths:** Death registration reached **99.4%**.
- **States:** **18 States/UTs** achieved **100% birth registration** and **21 States/UTs** achieved **100% death registration**.
- **Progress:** Birth and death registration rose from **56% and 48% in 2000** to **86.6% and 72.5% in 2014**.
- **Timeliness:** Many events are still not registered within the prescribed **21 days**.
- **Infants:** **84.2%** of registered infant deaths were urban and only **15.8%** rural, indicating possible rural under-registration.
- **Certification:** Many registered deaths lack **medical certification of cause**, limiting mortality analysis.
- **Estimation:** Death-registration completeness is based on the **Sample Registration System**, which may undercount events and overstate CRS coverage.

Significance

- **Vital Statistics:** Provides data on **births, deaths, fertility, mortality, and sex ratio at birth**.
- **Policy Support:** Helps assess health and social policies and track population trends.
- **Decentralised Planning:** Supports **district- and sub-district-level planning**.
- **Legal Documentation:** Enables access to identity, education, welfare, pensions, insurance, and inheritance.
- **Health Monitoring:** Tracks mortality changes and supports responses during health emergencies.

Provisions Regarding Registration of Births and Deaths

- **Mandatory Registration:** Under the **RBD Act, 1969**, amended in **2023**, all births and deaths must be registered at the place of occurrence.
- **Registration Functionaries:** The **RGI** supervises registration nationally; **Chief Registrars, District Registrars, and Local Registrars** manage implementation and issue certificates.
- **Civil Registration System:** From **October 2023**, registrations must be completed digitally through the **CRS**.

- **Time Period for Reporting:** Births, deaths, and stillbirths must be reported within **21 days**; delayed registration attracts a late fee.
- **Births and Deaths Outside India:** Births abroad are registered under the **Citizenship Act, 1955** and relevant rules. Under **Section 20 of the RBD Act**, returning parents intending to settle in India may register the birth within **60 days of arrival**.
 - Deaths abroad registered at Indian consulates are valid under the RBD Act.
- **Penalty for Non-compliance:** Registrar negligence is punishable with a fine of up to **₹1,000**.

States Bear Disproportionate Share of India's Welfare Spending

The **Realising Rights: A Handbook of Welfare in India** by Azim Premji University highlights that **States bear a disproportionately larger share of welfare spending despite shrinking fiscal space and limited financial autonomy**, raising concerns over **Centre-State fiscal imbalances** in financing social welfare.

Evolution of Welfare Regime

- **State-Led Welfare (1950s–1980s):** Welfare focused on planned development and poverty reduction through the **Community Development Programme (1952)**, **ICDS (1975)**, and **Minimum Needs Programme (1975)**.
- **Targeted Welfare (1990s):** After the **1991 reforms**, universal subsidies shifted to targeted benefits such as **Targeted PDS (1997)** based on **BPL** and **APL** categories.
- **Rights-Based Welfare (2000s–2014):** Welfare became legally enforceable through the **RTI Act, 2005**, **MGNREGA, RTE Act, 2009**, and **NFSA, 2013**.
- **Technology-Driven Welfare (Present):** **JAM Trinity** and **DBT** enabled targeted delivery. Schemes like **PM-KISAN** marked a shift from rights-holders to **beneficiaries (labharthis)**.
- **Expansion vs. Expenditure Gap:** **Anganwadi centres** reached **14 lakh** and subsidised food grains covered **over 81 crore people**, but spending remained below policy targets.
- **Social Security Lag:** Union and State welfare spending is about **7% of GDP** and **21% of public expenditure**, yet India lags in social security spending despite a tax-to-GDP ratio comparable to middle-income countries.

Extent of Centre-State Fiscal Asymmetry in Welfare Spending

- **Disproportionate State Contribution:** Welfare allocations were **₹24.20 lakh crore (6.77% of GDP)** in **FY 2025-26**, while the Union contributed only **1.89% of GDP**.

- Though the **14th and 15th Finance Commissions** recommended **42% and 41% tax devolution**, effective devolution remained around **29–32%** due to cesses and surcharges rising from **~10.4% of Gross Tax Revenue in 2011-12 to over 20% in 2021-22**.
- **Rising State-Led Social Spending:** States drive most growth in social spending and fund about **75.2% of school education expenditure**, besides major spending on **public health and food subsidies**.
 - The **16th Finance Commission** noted that States spent about **₹4.14 lakh crore** on unconditional cash transfers.
- **State Innovations Co-opted by the Centre:** **Rythu Bandhu** (Telangana) and **KALIA** (Odisha) influenced **PM-KISAN**, while Tamil Nadu's **Mid-Day Meal Scheme** became **PM POSHAN**. Despite pioneering welfare models, States are penalised for high fiscal deficits.
- **Centrally Sponsored Schemes (CSS):** Schemes such as **ICDS** and **PMMVY** follow a **60:40 Centre-State ratio**, requiring matching State funds.
 - **MGNREGA**, funded roughly in a **90:10 ratio**, is expected to impose a higher State burden under the **VB-G RAM G Act, 2025**.
 - The Centre also links fund release to reforms or CSS implementation, as under **PM SHRI**, where States had to sign an **MoU**, raising concerns over fiscal flexibility and federal autonomy.
- **The Post-GST Revenue Autonomy Deficit:** **GST** subsumed major State taxes, reducing States' ability to alter tax rates during fiscal stress. Their dependence on the **GST Council and central transfers** creates a mismatch between expenditure responsibilities and revenue powers.
- **Emergence of a 'Fiscal Dependency Federation':** A shrinking divisible pool, tied CSS transfers, reduced tax autonomy, and borrowing limits may increase State dependence on Union-controlled funds, shifting **cooperative federalism** towards **coercive fiscal federalism**.

Measures to Reduce the States' Welfare Burden

- **Restore the Integrity of the Divisible Pool:** Cap and periodically review **cesses**, restricting them to specific and time-bound purposes.
- **Shift from Tied Transfers to Fiscal Autonomy:** Rationalise **CSS** through zero-based review, merge smaller schemes, and increase **untied grants and flexi-funds**.
 - The **Punchhi Commission (2010)** recommended limiting CSS to matters of national priority to protect State autonomy.
- **Rule-Based Fiscal Coordination:** **Article 293(3)**, **Net Borrowing Ceilings**, and adjustments for off-budget borrowings can restrict State capital expenditure.

- Implement the **N.K. Singh FRBM Review Committee (2016)** recommendation for a statutory **Fiscal Council** to assess debt sustainability and coordinate borrowing transparently.
- **Protect State Taxation Powers:** Preserve residual State taxation powers after GST. SC judgments in **MADA v. SAIL (2024)** on mineral taxation and **State of UP v. Lalta Prasad Vaish (2024)** on industrial alcohol support protecting State revenue streams.
- **Address Horizontal Fiscal Inequalities:** CSS matching ratios, borrowing ceilings, and fiscal transfers should consider States' **revenue capacity, demographic pressures, development deficits, and welfare needs**, without weakening revenue mobilisation incentives.
- **Strengthen State Finance Commissions (SFCs):** Ensure their regular constitution and timely implementation of recommendations for predictable transfers to **Panchayats and Urban Local Bodies**.
- **Expand Own-Source Revenues of Local Governments:** Enable **PRIs and ULBs** to mobilise **property taxes, user charges, and other assigned revenues**.
- **Institutionalise Continuous Centre-State Fiscal Dialogue:** Strengthen the **Inter-State Council, Finance Commission, and federal forums** to resolve disputes and reduce judicialisation.
- **Move towards 'Funds-Follows-Functions' Federalism:** Align revenue powers and transfers with each tier's expenditure responsibilities to address centralised revenues and decentralised welfare obligations.

Centrally Sponsored Schemes promote national welfare priorities but may constrain the fiscal autonomy of States. Critically examine.

Practice Mains Question

Viksit Bharat Shiksha Adhishthan Bill 2025

Several **States and universities** have opposed the **VBSA Bill, 2025**, citing threats to autonomy, excessive centralisation, and regulatory weaknesses. The Bill is under review by a **Joint Parliamentary Committee (JPC)**.

Key Features of the Bill

- **Apex Regulatory Body:** Establishes **VBSA** as the apex higher-education regulator, replacing **UGC, AICTE, and NCTE**. It covers technical, teacher-training, and architecture education, but excludes medicine and law.
- **Three-Council Structure:** Creates separate **Regulatory, Accreditation, and Standards Councils** for regulation, quality assessment, and academic standards.
- **Composition:** The Commission will have a **Chairperson and up to 12 members**, including Council Presidents, the

Union Higher Education Secretary, experts, and State representatives.

- **Council Functions:** The **Regulatory Council** will set institutional standards and address grievances; the **Standards Council** will prescribe learning outcomes; and the **Accreditation Council** will manage accreditation.
- **Appointment and Removal:** The Central Government will appoint the Chairperson. Other members will be selected through a search process and may be removed for specified misconduct or incapacity.
- **Degree-Granting Powers:** Degrees may be awarded only by legally authorised institutions, including those approved by the Regulatory Council with prior Central Government approval.
- **Penalties and Appeals:** Unapproved institutions may face a minimum **₹2 crore penalty and closure**. Appeals will lie with the **Central Government**.

Concerns Regarding the Bill

- **Over-centralisation of Regulation:** A single regulator replacing **UGC, AICTE, and NCTE** may concentrate powers in the Centre and reduce States' role.
- **Threat to Institutional Autonomy:** **IITs, IIMs, IISERs, and other INIs** fear restrictions on academics, research, curriculum, online education, and expansion.
- **Legal Ambiguity for Existing Institutions:** Some provisions may override the founding laws of **INIs and IoEs**, creating uncertainty.
- **One-size-fits-all Regulatory Framework:** A common framework may not suit the diverse needs of research institutions, rural colleges, tribal universities, and new **HEIs**.
- **Weaknesses in the Accreditation System:** Limited capacity, low coverage, corruption allegations against **NAAC**, and false accreditation claims may hinder implementation.
- **Disproportionate Penalty Provisions:** Heavy fines and closure may unfairly affect small, rural, and tribal institutions. Stakeholders seek independent adjudication before major penalties.
- **Limited Focus on Systemic Challenges:** The Bill does not adequately address low **GER**, regional disparities, barriers to new institutions, and public funding below the recommended **1.5% of GDP**.

Measures to Overcome these Challenges

- **Protect Institutional Autonomy:** Clearly safeguard the academic, research, administrative, and financial autonomy of **HEIs and INIs** through statutory exemptions and limits on regulatory intervention.
- **Strengthen Federal Participation:** Give **State governments and State Higher Education Councils (SHECs)** meaningful representation in the Commission and its Councils.

- **Create an Independent Funding Mechanism:** Establish a separate **Higher Education Grants Council (HEGC)** for transparent funding and reduction of regional inequalities.
- **Reform Accreditation and Penalties:** Use differentiated accreditation standards for diverse institutions. Major penalties and closure should require **independent adjudication**.
- **Ensure Regulatory Independence and Accountability:** Appeals should lie before an **independent educational tribunal**, while the Commission should report to **Parliament** and follow transparent procedures.

The Viksit Bharat Shiksha Adhishtan Bill, 2025 seeks to simplify higher education regulation, but it has raised concerns regarding autonomy and centralisation. Discuss.

Practice Mains Question

Concerns over the Draft National Food Security (Amendment) Bill, 2026

Proposed changes to **Antyodaya Anna Yojana** entitlements under the **National Food Security Act, 2013** have raised concerns over impacts on **vulnerable households** and **State allocations**.

Key Aspects of the Draft National Food Security (Amendment) Bill, 2026

- **About:** The draft Bill proposes changes to **AAY** foodgrain entitlements. After public feedback, the Centre will consult States and departments before introducing it in Parliament.
- **Existing Provision:** Every AAY household receives **35 kg of foodgrains per month**, regardless of family size.
- **Proposed Entitlement:** Each beneficiary would receive **7 kg per month**, capped at **35 kg per household**. Only households with **five or more members** would receive 35 kg.
- **Government's Rationale:** To reduce per-capita disparities among AAY households and align distribution with nutritional needs.
- **Impact on Households:** Families with **1, 2, 3 and 4 members** would receive **7 kg, 14 kg, 21 kg and 28 kg**, respectively. Families with over five members would remain capped at **35 kg**, lowering their per-capita entitlement below 7 kg.

Key Features of the National Food Security Act, 2013

- **About:** Enacted in **2013**, the Act made food security a **legally enforceable, rights-based entitlement**.
- **Coverage:** Covers up to **75% of the rural** and **50% of the urban population**, nearly **two-thirds of India's population**.
- **Foodgrain Entitlements:** AAY households receive **35 kg per family per month**, while **Priority Households** receive **5 kg per person per month**.

- **Beneficiary Identification:** States and Union Territories identify beneficiaries, issue ration cards and update beneficiary lists.
- **Women's Empowerment:** The **eldest woman aged 18 years or above** is designated as the household head for ration cards.
- **Life-Cycle Nutrition Support:** Pregnant women, lactating mothers and children aged **6 months–14 years** receive nutritional support; pregnant and lactating women get maternity benefits of at least **₹6,000**.
- **Food Security Allowance:** States must provide an allowance when entitled foodgrains or meals are not supplied.
- **Centre–State Responsibilities:** The Centre allocates, transports and funds foodgrains; States and Union Territories identify beneficiaries, distribute entitlements and establish grievance-redressal mechanisms.
- **Prices and Reforms:** Initially fixed prices at **₹3/kg for rice, ₹2/kg for wheat and ₹1/kg for coarse grains**; also provides **tide-over allocation** and permits **DBT of food subsidy** in identified areas.

Major Criticisms of the Proposed NFSA Amendment

- **Reduced Entitlements for Smaller Households:** Replacing **35 kg per household** with **7 kg per person** would reduce support for **one- to four-member AAY households**, including elderly couples, widows, persons with disabilities and small families.
- **Disadvantage to Larger Families:** The **35 kg ceiling** means households with more than five members would receive **less than 7 kg per person**.
- **Regional Disparities:** States with smaller family sizes, especially southern States with better population stabilisation, may lose more allocation.
- **Inadequate Nutritional Support:** The proposal covers only cereals and excludes **pulses, edible oils and protein-rich foods**; **7 kg of cereals** may be insufficient for the poorest households.
- **Outdated Beneficiary Coverage:** NFSA coverage remains largely based on the **2011 Census**, excluding many eligible people despite population growth.
- **Wider Access Barriers Ignored:** The amendment does not address **biometric and e-KYC failures, migrant exclusion, unstable addresses** and other PDS access barriers, weakening the NFSA's rights-based character.

Way Forward

- **Protect Existing Entitlements:** Ensure no **AAY household** receives less than **35 kg**, especially elderly persons, widows and persons with disabilities.

- **Remove the Household Ceiling:** If **7 kg per person** is adopted, remove the **35 kg cap** for families with more than five members.
- **Diversify the Food Basket:** Add **pulses, edible oils and protein-rich foods** to cereals.
- **Update NFSA Coverage:** Revise beneficiary limits using current population data instead of the **2011 Census**.
- **Address Access and Regional Concerns:** Fix **biometric and e-KYC failures**, provide alternative authentication and consult States, food-rights experts and affected communities.

The proposed amendment to AAY entitlements under the NFSA highlights the tension between rational allocation and welfare protection. Discuss.

Practice Mains Question

'Satluj' Takedown: Frameworks Governing Theatrical and OTT Content in India

The removal of **Satluj** from **ZEE5** over security concerns highlighted India's dual regulation: theatrical films under the **Cinematograph Act, 1952** and OTT content under the **IT Rules, 2021**.

- **The Catalyst of the Debate:** **Satluj**, earlier titled **Punjab '95**, was released uncut on OTT after the **CBFC demanded 127 cuts**, bypassing theatrical distribution.
- **Theatrical Regulation Ambit:** Films shown in cinemas require prior **CBFC certification** under the **Cinematograph Act, 1952**. The CBFC functions under the **Ministry of Information and Broadcasting**.
- **Grounds for Restricting Films:** Certification may be denied or changes required on grounds such as **India's sovereignty and integrity, State security, foreign relations, public order, decency, morality, defamation and contempt of court**, broadly reflecting **Article 19(2)** restrictions.
 - The **Cinematograph (Amendment) Act, 2023** introduced **UA 7+, UA 13+ and UA 16+** categories and made film certificates **perpetually valid**.
 - The **Film Certification Appellate Tribunal** was abolished in **2021**. CBFC decisions must now be challenged in **High Courts**, making appeals costlier and slower.
- **Digital Content Framework:** OTT content is exempt from **CBFC certification** and regulated under **Part III of the IT Rules, 2021**.
 - The Rules provide a **three-tier grievance redressal mechanism**:
 - ❖ **Level I:** Self-regulation by publishers.
 - ❖ **Level II:** Self-regulation by publishers' self-regulating bodies.
 - ❖ **Level III:** Central Government oversight.

- **The Judicial-Executive Friction:** Government oversight provisions under the three-tier mechanism have been stayed by the **Bombay and Madras HCs** amid constitutional challenges.
- **Section 69A of the IT Act, 2000** allows the Central Government to block online content under the **Blocking Rules, 2009**, subject to written reasons and review.
- In the **Satluj case**, the **Ministry of Information and Broadcasting** formed an **Inter-Departmental Committee under Rule 14**, which may recommend warnings, disclaimers, modification, reclassification, deletion or blocking under **Section 69A**.

Sub-Mission on Agricultural Mechanization

SMAM has improved access to farm machinery for **small farmers, women, disadvantaged groups, underserved areas and North-Eastern States**. It has provided **₹9,404.47 crore** for **21.61 lakh machines** and **₹52.5 crore** for **40,928 drone demonstrations** across **40,918 hectares**.

- **About:** **SMAM** was launched in **2014–15** as a **Centrally Sponsored Scheme** under **RKVY** to promote farm mechanization.
- **Objective:** It follows **"reach the unreached"**, targeting **small and marginal farmers, women, SCs, STs, FPOs, SHGs and rural entrepreneurs**.
- **Custom Hiring Centres:** Supports centres that rent farm machinery and equipment at affordable rates.
- **Machinery and Technology Support:** Promotes machinery distribution and **hi-tech/high-value equipment hubs**.
- **Regional and Structural Focus:** Prioritizes areas with low farm power and addresses small holdings and high machinery costs through rental services.
- **Financial Support:** During **2014–15 to 2025–26**, **₹9,404.47 crore** supported **21.61 lakh machines, 27,554 Custom Hiring Centres, 646 Hi-tech Hubs and 25,608 Farm Machinery Banks**.
- **Expansion:** Individual machinery beneficiaries rose from **2.07 lakh in 2020–21 to 2.32 lakh in 2024–25**.
- **Funding Pattern:** **60:40** for most States, **90:10** for North-Eastern and Himalayan States, and **100% Central funding** for Union Territories.
- **Capacity Building:** Includes demonstrations, training and targeted **Information, Education and Communication activities**.
- **Testing and Certification:** Supports testing and certification to ensure machinery quality, efficiency and suitability.

Strategic Pillars of SMAM for Enhancing Farm Productivity

- **Promotion of Agricultural Mechanization through Training, Testing, Demonstration and Post-Harvest**

Mechanization: Expands modern technology access and strengthens processing, storage, value addition and crop-residue management.

- **Financial Assistance for the Procurement of Agricultural Machinery and Equipment:** DBT subsidy covers **40% of machinery cost** for general beneficiaries and **50% for SC/ST, small and marginal farmers and North-Eastern beneficiaries**.
 - Small and marginal farmers receive **₹2,000 per hectare** for mechanized services, including drones.
- **Establishment of Farm Machinery Banks for Custom Hiring:** Provides **80–90% assistance** to SHGs, FPOs and local institutions. CHCs receive **40% of project cost up to ₹250 lakh**, while Farm Machinery Banks receive **80% assistance up to ₹30 lakh**.
- **Establishment of Hi-Tech and High-Productivity Equipment Hubs:** Supports advanced, high-capacity and crop-specific machinery hubs.
- **Promotion of Agricultural Mechanization in the North-Eastern Region:** Provides up to **100% subsidy** for small machinery and **95% support** for Farm Machinery Banks.

Role of Farm Mechanization and Drone Technology under SMAM

- **Role of Farm Mechanization:** Uses machinery and modern technologies from **land preparation to post-harvest management**.
- **Benefits:** Reduces manual and animal labour, enables timely operations, lowers costs and improves farm productivity and quality.
- **Promotion of Drone-Based Farming:** **₹52.50 crore** was provided to ICAR for drone demonstrations.
- **Kisan Drone Demonstrations:** During **2023–24 to 2025–26**, **40,928 demonstrations** covered **40,918 hectares** for applying nutrients, fertilizers and agrochemicals.
- **Financial Assistance:** ICAR institutes, KVKs and State Agricultural Universities receive **100% assistance up to ₹10 lakh per drone**; FPOs receive up to **75% grants**; service agencies receive **₹6,000 per hectare**.
- **Support for Women Farmers:** **30% of SMAM funds** are earmarked for women farmers.

SC to Refer Grounds of Arrest Issue to Larger Bench

SC indicated referral of the issue of mandatory written grounds of arrest to a larger Bench due to conflicting precedents.

- **Issue:** Meghalaya HC granted bail after an incorrect **BNS provision** was mentioned in the arrest memo. The State argued it was a clerical error, while SC noted the need for larger Bench consideration.

■ **Conflicting Judgements on Written 'Grounds of Arrest':**

- **Pankaj Bansal vs Union of India (2023):** SC held that **ED must provide written grounds of arrest** under **PMLA, 2002**; oral communication may lead to factual disputes.
- **Prabir Purkayastha Case (2024):** SC extended written grounds of arrest safeguard to **UAPA, 1967**, linking it with **Articles 22(1) and 22(5)**.
- **Mihir Rajesh Shah vs State of Maharashtra (2025):** SC held that written grounds of arrest must be provided in **all cases**, including **BNS/IPC offences**.
- **Dr Rajinder Rajan vs Union of India (2026):** SC upheld the requirement of furnishing written grounds of arrest.
- **Vihaan Kumar vs State of Haryana (2025):** A coordinate Bench held that while communicating grounds of arrest is mandatory, **written communication may not be practical in every case**; oral communication may suffice.

■ **Safeguards Against Arbitrary Arrest**

- **Article 22(1):** Requires informing the arrested person of **grounds of arrest** and provides the right to consult and be defended by a **legal practitioner**.
- **Section 47, BNSS, 2023:** Mandates immediate communication of **offence particulars** to a person arrested without warrant.
 - ❖ **Section 48, BNSS, 2023:** Requires informing a **relative/friend/nominated person** about the arrest and place of detention.
- Rooted in **Article 21**, as knowledge of grounds of arrest enables seeking **bail** and effective legal defence.

Armed Forces Seek Higher Agniveer Retention

The Indian Armed Forces have proposed increasing the **retention percentage of Agniveers** beyond the existing **25% limit** under the Agnipath Scheme.

- With the first batch completing its **4-year tenure**, the services seek to retain trained personnel and utilise investments made in **advanced technology training**.

Reasons for the Demand for Higher Agniveer Retention

■ **Proposed Retention Requirements**

- **Navy:** May seek retention of around **75%** due to technical nature of operations.
- **Army and Air Force:** May seek retention of around **50%**.
- Current cap remains at **25%** across services.

- **Preventing Manpower Shortfalls:** A strict **25% retention cap** may create a temporary manpower gap due to regular retirements.

- Higher retention would help manage the transition.
- **Lessons from Operation Sindoor:** Agniveers performed effectively, but **experienced manpower** remains crucial for faster responses during crises.
 - Field experience and training improve operational effectiveness.
- **Technological Shift and Modernization:** Advanced weapon systems and platforms require **specialised long-term training**.
 - Short four-year tenures may reduce returns on training investments.
- **Technical Requirements of Naval and Air Operations:** Operating **warships, submarines and aviation systems** requires deep technical expertise developed over longer service periods.

Agnipath Scheme

- **About:** Launched in **June 2022**, **Agnipath Scheme** is a pan-India, merit-based recruitment programme for personnel below officer rank (**PBOR**) in the **Army, Navy and Air Force**.
- **Rationale**
 - **Youthful Military:** Reduce average age of armed forces personnel from **32 to 26 years** to improve agility and combat effectiveness.
 - **Fiscal Sustainability:** Reduce pension burden and enable greater spending on defence modernisation and indigenous production.
 - **Emerging Threats:** Build capabilities against **hybrid warfare, cyber threats, terrorism, space and electronic warfare**.
 - **Nation Building:** Create a skilled and disciplined youth workforce supporting employment, disaster response and **Viksit Bharat@2047**.
- **Key Features**
 - **Tenure:** 4 years.
 - **Age Limit:** **17.5–21 years** (relaxed to **23 years** for the first batch).
 - **Permanent Enrolment:** Up to **25%** of each batch selected for regular cadre based on transparent criteria.
 - Selected Agniveers serve an additional **minimum 15 years** under regular service conditions.
- **Financial Benefits**
 - **Seva Nidhi Package:** Around **₹11.71 lakh** (tax-free) for **75% discharged personnel**, including contributions and government matching contribution.
 - **Skill Certificate** after completion of service.
 - Priority in recruitment for **CAPFs and state police**.
 - **Life Insurance Cover:** ₹48 lakh.

- **Compensation:** ₹44 lakh for disability; up to ₹1 crore for death during service.

Steps to Reform the Agnipath Scheme

- **Integration with Aatmanirbhar Bharat & Defence Corridors:** Link Agnipath exit pathways with **UP and Tamil Nadu Defence Industrial Corridors**.
 - Provide transition opportunities for trained Agniveers in **private defence manufacturing, R&D startups and aviation maintenance**.
- **Preserving the “Naam, Namak, Nishan” Ethos:** Avoid a purely contractual approach that may weaken **regimental cohesion and unit identity**.
 - Reframe the initial **4 years** as an **“Extended Probationary Commission”** to strengthen integration and military ethos.
- **Evidence-Based Review:** Evaluate the scheme after the first recruitment cycle using:
 - Operational feedback; Manpower trends; Institutional assessments
 - Allow each service to decide retention levels based on **operational and technical requirements**.
 - Prioritise Agniveers trained in **critical technologies and specialised roles**.
- **Institutional Oversight and Course Correction:** Introduce **biennial review by the Parliamentary Standing Committee on Defence**.
 - Ensure assessment of impact on **combat readiness, defence expenditure and demographic changes**.

Modern warfare requires technologically skilled personnel as much as a young and agile military force. In this context, analyse the challenges posed by short-term military recruitment under the Agnipath Scheme.

Practice Mains Question

All India Survey on Higher Education Reports for 2022-23 and 2023-24

The **Union Ministry of Education** released the **AISHE Reports (2022-23 & 2023-24)**, showing growth in **higher education enrolment, GER, female participation, social inclusion, and STEM education**.

Key Highlights of AISHE 2022-23 & 2023-24

- **GER:** Increased from **23.7 (2014-15)** to **30 (2023-24)**. Female GER rose to **31.2**, SC GER to **27.8**, and ST GER to **22.8**.
- **Gender Parity Index:** Stood at **1.08** in **2023-24** and remained above **1** for seven consecutive years.
- **Total Enrolment:** Rose by **31.5%** to **4.50 crore**; female enrolment increased by **42.2%** to **2.24 crore**.

- **Social Inclusion:** Enrolment reached **69.72 lakh SC, 28.83 lakh ST, and 1.80 crore OBC students**.
- **STEM Enrolment:** Increased to **1.02 crore**; the female share rose from **38.4% to 44%**.
- **Faculty:** Total faculty reached **17.32 lakh**, with women constituting **44.9%**.

All India Survey on Higher Education (AISHE)

- Launched in **2010-11**, AISHE is an annual voluntary survey conducted by the **Ministry of Education**.
- It collects data on **enrolment, faculty, infrastructure, and examination outcomes** for policy formulation.
- **Limitation:** Data is **self-reported** by HEIs, making accuracy dependent on institutional reporting.

Persistent Challenges in Higher Education

- **Employability crisis:** Graduate unemployment is nearly **40% among those below 25**, while overall employability is only **54.81%**, reflecting a major **skills mismatch**.
- **Quantity over quality:** India has **128 universities** in the **THE Asia Rankings 2026**, but none in the **top 40**.
- **Regional inequality:** Limited access to quality institutions, digital infrastructure and secondary schools disadvantages **rural students**.
- **Faculty shortages:** Around **28% of teaching** and **48% of non-teaching posts** in central universities remain vacant, increasing reliance on temporary staff.
- **Student mental health:** Competitive examinations, rote learning and academic pressure contribute to **anxiety and student suicides**.
- **Weak research ecosystem:** India's R&D expenditure remains only **0.6–0.7% of GDP**, while research is concentrated in autonomous laboratories rather than universities.
- **Unequal funding:** **State universities**, which serve most students, face severe financial and infrastructure constraints compared with centrally funded institutions.
- **Privatisation and debt:** Expansion of poorly regulated private institutions has raised concerns over **quality, high fees, education loans and loan defaults**.
- **Excessive regulation:** Bureaucratic control by bodies such as **UGC and AICTE** restricts institutional autonomy and slows implementation of **NEP 2020**.

Measures to Strengthen Higher Education

- **Expand access:** Raise public education spending towards **6% of GDP** to achieve the **NEP 2020 target of 50% GER by 2035**.
- **Promote flexibility:** Fully implement the **Academic Bank of Credits**, multidisciplinary courses and **multiple entry-exit options**.

- **Strengthen research:** Enable **ANRF** to support university-led **deep-tech research and startups**, including in **tier-2 and tier-3 HEIs**.
- **Grant autonomy:** Link regulatory freedom to **NAAC accreditation and NIRF performance** instead of uniform control by **UGC and AICTE**.
- **Industry integration:** Include industry experts in university governance and introduce **micro-credentials, stackable degrees and market-oriented curricula**.
- **Address faculty gaps:** Create a **National Faculty Exchange** involving corporate experts and overseas Indian academics.
- **Use AI for inclusion:** Deploy **LLMs** through **SWAYAM and DIKSHA** to provide research and learning content in **22 Indian languages**.
- **Build future-ready skills:** Introduce **design thinking and computational logic** across disciplines.

India's demographic dividend could become a demographic liability unless higher education institutions address the employability crisis. Comment.

Practice Mains Question

Poshan Tracker: India's Real-Time Nutrition Monitoring Platform

Poshan Tracker is India's real-time nutrition monitoring platform under **Mission Poshan 2.0**, covering **28 States and 8 UTs** and registering over **8.93 crore beneficiaries** as of **May 2026**.

Key Facts About Poshan Tracker

- **About:** **Poshan Tracker** is a mobile-based governance app and digital backbone of **POSHAN Abhiyaan** and **Mission Poshan 2.0**.
 - Launched in **March 2021** by the **Ministry of Women and Child Development** with the **National e-Governance Division**, it enables near real-time monitoring of services through **Anganwadi Centres**.
- **Objective:** Strengthens nutrition governance through technology-based monitoring, beneficiary tracking and evidence-based decisions for **children, pregnant women, lactating mothers and adolescent girls**.
- **Beneficiary Monitoring:** Tracks beneficiaries across key categories and maintains a live monthly database for over **7.7 crore children** using **Aadhaar-authenticated data**.
- **Growth Monitoring:** Tracks **stunting, wasting, underweight, severe acute malnutrition, moderate acute malnutrition, overweight/obesity** among children aged **0-5 years**. As of **May 2026**, it monitored over **6.3 crore children**, covering nearly **94%** of registered beneficiaries.
- **Key Features:** Includes **Aadhaar-based verification, Facial Recognition System, IT-enabled home visit scheduler**,

WHO Child Growth Standards-based Poshan Calculator, and ECCE content for children aged 3-6 years.

- **Support to Anganwadi Workers:** Reduces paperwork and enables digital recording of **service delivery, beneficiary details, growth data and home visits**. Smartphones, internet support and growth monitoring devices have been provided.
- **Role in Supplementary Nutrition Programme:** Monitors delivery of supplementary nutrition under **Anganwadi Services**.
 - As of **May 2026**, over **5.5 crore beneficiaries** received nutrition for at least **15 days**, and **5.17 crore** for at least **21 days**.
- **Recognition:** Appreciated by **WHO and UNICEF**, showcased at **G20 Ministerial Conference on Women Empowerment 2023**, won **National Award for e-Governance Gold Award 2024**, and received the **PM's Award for Excellence in Public Administration in April 2025**.

Mission Poshan 2.0

- **About:** Mission Poshan 2.0 or Mission Saksham Anganwadi and Poshan 2.0 is an integrated nutrition support programme of the Government of India.
- **Formation:** Announced in **Budget 2021-22** by merging **POSHAN Abhiyaan, Anganwadi Services and Scheme for Adolescent Girls**.
- **Nodal Ministry:** Implemented by the **Ministry of Women and Child Development**.
- **Objective:** Addresses **malnutrition** and promotes **health, wellness and immunity** among children, pregnant women, lactating mothers and adolescent girls.
- **Key Components:** Includes **nutrition support, Early Childhood Care and Education** for children aged 3-6 years, early stimulation for 0-3 years, and strengthening Anganwadi infrastructure.
- **Saksham Anganwadi:** Upgrades Anganwadi Centres with **LED screens, water purifiers, smart learning aids, Poshan Vatikas** and better infrastructure.

Poshan Abhiyan

- **About:** Flagship initiative of **MWCD**, launched on **8th March 2018** in **Jhunjhunu, Rajasthan**, targeting nutrition needs of **adolescent girls, pregnant women, lactating mothers and children aged 0-6 years**.
- **Objectives:** Aims to reduce **stunting, under-nutrition, anaemia and low birth weight** by **2%, 2%, 3% and 2% per annum** respectively.
- **Strategic Pillars:** Operates through **Quality Services, Cross-Sectoral Convergence, Technology and Jan Andolan**.
 - **Quality Services:** Strengthens services via **ICDS, NHM and PMMVY**, focusing on the first **1,000 days**.

- **Cross-Sectoral Convergence:** Integrates ministries like **Water & Sanitation; NITI Aayog-led National Council** reviews nutrition convergence quarterly.
- **Technology:** Uses **Poshan Tracker** and **ICDS-CAS** for real-time monitoring and Anganwadi service delivery.
- **Jan Andolan:** Promotes community-led nutrition awareness and behavioural change.

Poshan Tracker represents a shift from welfare-based nutrition delivery to real-time, evidence-based nutrition governance. Discuss its significance and associated challenges.

Practice Mains Question

National Academic Depository

NAD supports India's move towards a **secure, paperless and interoperable** system for managing **academic records**.

Key Facts About the NAD

- **About:** NAD is a nationwide digital framework for storing, issuing, verifying and authenticating academic awards like **degrees, diplomas, certificates, marksheets and evaluation reports**.
- **Need:** It addresses issues in paper-based records such as **loss, damage, forgery, delays and manual verification**.
- **Objective:** Provides students and award holders **lifelong access** to verified digital academic credentials anytime, anywhere.
- **Platform:** Since **2020**, NAD operates through **DigiLocker** for issuing, accessing and sharing academic credentials.
- **Stakeholders:** Covers **universities, school boards, Institutions of National Importance, HEIs, students, employers, banks, government bodies and verifying entities**.
- **Integration:** Linked with **DigiLocker, APAAR, ABC and e-Sanad** for secure access, credit mobility, verification, attestation and apostille services.

Key Benefits of NAD

- **Paperless Record Management:** Provides a secure, trusted and paperless framework for academic credentials.
- **Lifelong Access:** Students and award holders can access, retrieve and share digital records throughout life.
- **Instant Verification:** Enables quick credential verification, reducing delays in admissions, recruitment and official processes.
- **Fraud Prevention:** Authenticated digital records reduce fake certificates and forgery risks.
- **Improved Transparency:** Improves transparency and service delivery in academic record management.
- **Global Alignment:** Aligns India with global best practices in digital academic credentials.

India has one of the world's largest education systems, with ~14.71 lakh schools, 1,420 universities, 53,583 colleges, 16,795 standalone institutions and 280 R&D institutes.

Functioning of National Academic Depository

Governance Structure

- **Parent Ministry:** Ministry of Education is responsible for NAD.
- **Nodal Agency:** UGC implements NAD through DigiLocker.
- **Coordinated Framework:** Institutions issue verified digital academic records; students access or share them securely.
- **Legal Validity:** NAD credentials have the same legal status as physical documents under the **Information Technology Act, 2000**.

Legal and Regulatory Framework

- **Information Technology Act, 2000:** Gives legal recognition to electronic records and digital signatures.
- **Digital Locker Rules, 2016:** Documents are shared only with explicit user consent.
- **National e-Authentication Framework:** Provides secure identity verification and authentication.

Ayushman Bharat Health Accounts (ABHA)

Ayushman Bharat Digital Mission has linked over **104 crore health records** with more than **93 crore ABHAs**, strengthening India's digital health system through **paperless records**, shorter hospital waiting time and a unified network for **patients, hospitals, doctors and insurers**.

Key Features of ABHA

- **About:** ABHA is a **14-digit unique health ID** under **Ayushman Bharat Digital Mission (ABDM)**, linking individuals with India's digital health ecosystem.
- **Objective:** Creates a **portable, paperless and consent-based** digital health record system.
- **Key Features:** Links records across **hospitals, labs, insurers and health programmes** with patient consent; records can be accessed only through **revocable and time-bound consent**.
- **Interoperability:** Enables seamless portability of health records across providers and locations.
- **Reduction in Waiting Time:** **Scan and Share** enables QR-based OPD registration, reducing waiting time from nearly **1 hour to 2–5 minutes**.
- **Integration with Health Services:** Connects **patients, doctors, hospitals, labs, pharmacies, insurers and digital health apps**.
- **Support for Insurance Claims:** Integrates with **National Health Claims Exchange (NHCE)** to streamline insurance claims, reduce discharge delays and improve transparency.

- **Privacy and Consent:** Follows **privacy-by-design**; records are not stored on a central government server and are shared only with consent.
- **Significance:** Reduces paperwork, avoids duplicate tests, improves continuity of care, speeds service delivery and supports **Universal Health Coverage**.

Role of ABDM in Transforming India's Healthcare Delivery through Digital Infrastructure

- **Creating an Integrated Health Ecosystem:** ABDM connects **patients, doctors, hospitals, laboratories, pharmacies, insurers and health applications** through a secure, interoperable network.
- **Building Verified Registries:** **Healthcare Professionals Registry (HPR)** and **Health Facility Registry (HFR)** create verified national databases of healthcare professionals and facilities, improving trust, service discovery and access to authentic providers.
- **Reducing Hospital Waiting Time:** **Scan and Share** enables **QR-based OPD registration** using **ABHA-linked details**, reducing paperwork and patient waiting time.
- **Improving Access through UHI:** UHI helps citizens discover providers, book appointments, consult doctors, find **blood banks**, locate **Jan Aushadhi Kendras**, search **AB-PMJAY hospitals** and access ambulance services across platforms.
- **Strengthening Citizen-Centric Health Management:** **Aarogya Setu 2.0** acts as a gateway for health records, teleconsultation, appointments, insurance details, blood availability, wellness content, wearable syncing and **AI-based health insights**.
- **Accelerating Digitisation of Healthcare Facilities:** DHIS incentivises hospitals, labs, pharmacies and digital solution companies to create interoperable records. **eSushrut@Clinic**, a lightweight **HMIS** by **C-DAC**, helps small clinics digitise registration, billing, reporting and patient records.
- **Streamlining Health Insurance Claims:** **NHCE** enables faster claims-information exchange among hospitals, insurers, beneficiaries and regulators, reducing discharge delays, improving transparency and allowing patients to track claims in one place.
- **Ensuring Privacy and Consent:** ABDM follows **privacy-by-design**; records remain with the creating entity and are shared only through **revocable and time-bound patient consent**.
- **Supporting Public Health Planning and AI:** Anonymised health data supports **disease surveillance**, targeted health programmes and evidence-based policymaking. **Strategy for Artificial Intelligence in Healthcare for India (SAHI)** and **Benchmarking Open Data Platform for Health Artificial Intelligence (BODH)** promote ethical, secure and privacy-preserving use of **AI in healthcare**.

Prime Minister Launches Next Phase of UDAN: Viksit UDAN

PM launched **Viksit UDAN** in **Rajasthan** to expand **regional air connectivity**, develop aviation infrastructure, and support **Viksit Bharat 2047**. It follows the **Modified UDAN Scheme** approved by the **Union Cabinet** in **March 2026**.

Key Features of Viksit UDAN

- **About:** Viksit UDAN is the modified phase of **Regional Connectivity Scheme–UDAN**, launched in **Oct 2016** to make air travel **affordable, accessible, and inclusive**.
- **Scheme Period:** To be implemented for **10 years**, from **FY 2026–27** to **FY 2035–36**, with budgetary support of **₹28,840 crore**.
- **Objective:** Strengthen last-mile air connectivity in **Tier-2 and Tier-3 cities**, remote regions, hilly areas, islands, and aspirational districts.
- **Development of Airports:** **100 airports** will be developed from existing unserved airstrips with **₹12,159 crore** over **8 years**.
- **Operation and Maintenance Support:** **₹2,577 crore** support for around **441 aerodromes** for **3 years**, capped at **₹3.06 crore/year/airport** and **₹0.90 crore/year/heliport or water aerodrome**.
- **Modern Helipads:** **200 modern helipads** proposed in hilly, remote, island, and aspirational regions with **₹3,661 crore** outlay for healthcare, emergency response, and public services.
- **Viability Gap Funding:** **₹10,043 crore** proposed for airline operators over **10 years**. Airlines get support for **5 years**, tapered from the **3rd year**; route exclusivity limited to **3 years**.

UDAN Scheme

- **About:** UDAN is a flagship scheme of the **Ministry of Civil Aviation** to democratise aviation and improve regional connectivity, including remote areas.
 - Launched under **National Civil Aviation Policy (NCAP) 2016**, it connects **Tier-2 and Tier-3 cities** through a market-driven, financially supported model, with **Airports Authority of India (AAI)** as the nodal implementing agency.
- **Core Mechanism & Funding:** Affordable air travel is supported through an **airfare cap** and **VGF** from the Centre.
 - It includes waived landing charges and state commitments to reduce **VAT on Aviation Turbine Fuel (ATF)** to **1% or less**.
- **Massive Infrastructure Expansion:** India's airport network grew from **74 airports** in **2014** to **159 airports** in **2024**.
- **Strategic Evolution (UDAN 1.0 to 5.0):** Scope expanded from underserved airports to **helipads** under **UDAN 2.0**, **tourism routes** under **UDAN 3.0**, focus on **hilly regions, North-East, and islands** under **UDAN 4.0**, and **seaplane operations** under **UDAN 5.0** series.

- **Implementation:** UDAN airports to be developed under **Challenge Mode** for timely completion, quality, innovation, and sustainability.
- Promotes indigenous aviation capacity through proposed procurement of **Hindustan Aeronautics Limited Dhruv helicopters** and **Hindustan Aeronautics Limited Dornier aircraft** for regional operations.
- **Impact:** Expected to boost **affordable air travel, tourism, trade, investment, emergency services, employment**, and regional economic growth.

Social Media Platforms Under Regulatory Scrutiny

Indian government acted against **Meta, Telegram, and Signal** over **digital fraud, copyright infringement**, and circulation of **Child Sexual Exploitation and Abuse Material (CSEAM)**.

Concerns Regarding the Growing Power and Functioning of Social Media Platforms

- **Rise in Digital Fraud and Impersonation:** Proposed **WhatsApp usernames** may enable **phishing, name-squatting, identity theft**, and **digital arrest scams**.
 - **MeitY** raised concerns, while **Meta** said phone numbers will remain linked at the backend and usernames of verified public figures and government bodies will be reserved.
- **Anonymity vs Traceability:** Features like **username-based communication** and **E2EE** protect privacy but complicate law-enforcement investigations.
 - They may facilitate offences under **Sections 66C and 66D** of the **IT Act, 2000**.
 - WhatsApp is classified as a **Significant Social Media Intermediary** under the **IT Act, 2000** and **IT Rules, 2021**, requiring due diligence.
- **Internet Freedom Foundation** argued that **Section 79** gives only conditional safe-harbour protection and does not allow **MeitY** to approve or withdraw platform features.
- **Inadequacy of Reactive Anti-Piracy Enforcement:** Government identified over **3,100 Telegram channels** distributing pirated films and OTT content, showing limits of notice-and-takedown.
 - Copyright infringement is a criminal offence under the **Copyright Act, 1957** and **Cinematograph Act, 1952**.
 - Government asked **Telegram** to proactively detect, report, and disable pirated audio-visual content, shifting towards **preventive due diligence**.
 - Telegram was described as a **“new dark web”** for cybercrime, fraud, and paper leaks; Centre had temporarily blocked it before **NEET-UG 2026 re-examination** under **Section 69A** of the **IT Act, 2000**.

- **Safe Harbour Conundrum: Section 79 of the IT Act, 2000** gives intermediaries conditional immunity for third-party content.
 - Failure to follow due diligence under **IT Rules, 2021**, especially on persistent piracy and unlawful content, can lead to loss of **safe harbour protection**.
- **Algorithmic Facilitation of Child Exploitation:** Allegations that paid Instagram ads promoted access to **CSEAM** and redirected users to **Telegram channels** show weaknesses in automated ad approval and moderation.
 - Criminal networks can manipulate algorithms and moderation systems to distribute illegal content despite Meta's **zero-tolerance CSAM policy**.
- **Online Child Safety and Human Rights:** Circulation and commercialisation of **CSAM/CSEAM** violate children's dignity and rights, attracting provisions under **POCSO Act, 2012** and **IT Act, 2000**, and undermining UN Convention on the Rights of the Child (UNCRC).
 - Government directed **Meta** to disable such content and submit an **Action Taken Report (ATR)**.
- **Weaponisation of Synthetically Generated Information (SGI):** AI-generated **deepfakes**, **voice clones**, **Non-Consensual Intimate Imagery**, and synthetic identities threaten **dignity, electoral integrity, cybersecurity, and public trust**.
- **Algorithmic Amplification of Online Harms:** Platforms act as active curators, with algorithms amplifying **misinformation, fraud, piracy**, and harmful synthetic media.
 - Viral misinformation and inflammatory content have contributed to **communal violence and riots**, leading to temporary blocking of platforms or internet services and raising concerns over platform accountability.
- **Cross-Border Enforcement Challenges:** Platforms like **WhatsApp, Telegram, and Instagram** operate across jurisdictions, complicating **investigation, evidence collection, content removal**, and prosecution.

Judgments Regarding Digital Regulation & Social Media

- **Shreya Singhal vs. Union of India, 2015:** SC upheld **Section 69A**, noting safeguards like written reasons and review committee for blocking orders.
- **Tehseen Poonawalla vs. Union of India, 2018:** SC directed governments to stop irresponsible and explosive social media messages causing **mob violence and lynching**.
- **Anuradha Bhasin vs. Union of India, 2020:** SC held that **speech and profession through the Internet** are protected under **Article 19**.
- **Ajit Mohan vs. Delhi Legislative Assembly, 2021:** SC held platforms like **Facebook/Meta** are "centers of concentrated power" and not merely passive intermediaries.

- Their algorithms curate content and can influence **mass opinion**.

Key Challenges in Regulating Social Media Platforms

- **Defining the Scope of Intermediary Liability:** Platforms curate, recommend, and monetise content through algorithms, blurring the line between a neutral intermediary under **Section 79 of the IT Act, 2000** and an active publisher.
 - Deciding when **safe harbour protection** should end remains a key legal challenge.
- **Absence of Ex-Ante Regulation:** India's framework is largely **reactive** through notice-and-takedown, rather than **preventive** and risk-based.
 - Labelling of **AI-generated/synthetically generated content** faces enforcement and detection challenges.
 - Tools like **PIB Fact Check Unit** counter fake information related to the **Union Government**, but mostly after content has circulated, showing need for stronger preventive mechanisms.
- **Regulating Proprietary Algorithms:** Recommendation engines, ad systems, and moderation tools work as opaque **black-box algorithms**, limiting oversight, independent audits, and accountability.
- **Fragmented Regulatory Ecosystem:** Platforms are governed by multiple laws, including **IT Act, 2000, DPDP Act, 2023, Copyright Act, 1957, POCSO Act, 2012, Competition Act, 2002**, and sectoral rules, causing overlapping jurisdiction.
- **Balancing Fundamental Rights with Platform Regulation:** Regulation must balance **privacy under Article 21, free speech under Article 19(1)(a)**, innovation, and state interests like **national security, copyright protection, and child safety**.
- **Institutional Capacity Deficit:** Regulators need expertise in **AI governance, algorithm audits, digital forensics, cryptography**, and platform engineering, which institutions are still developing.
- **Lack of a Comprehensive Digital Platform Law:** Unlike the EU's **Digital Services Act (DSA)** and **Digital Markets Act (DMA)**, India relies on fragmented statutes, creating regulatory uncertainty and inconsistent compliance obligations.

Measures to Strengthen Digital Platform Regulation in India

- **Statutory Clarity over Executive Discretion:** India needs a dedicated **AI and Digital Platforms regulatory framework**, as envisioned in the proposed **Digital Networking Platforms Bill, 2026**, instead of piecemeal amendments to the **IT Act**.

- **Proportionality in Takedowns:** Takedowns must follow **natural justice**, with reasoned orders and appeal options to protect **artistic expression, satire, and legitimate dissent**.
- **Algorithmic Audits:** Regulators should mandate annual public-interest impact assessments for **Significant Digital Networking Platforms** to assess algorithmic impact on **electoral integrity, public health, and social harmony**.
- **Develop a Comprehensive AI Governance Framework:** Operationalise the **IndiaAI Safety Institute** and set standards for **AI-generated content, deepfakes, synthetic media, and algorithmic transparency**, complementing the **IndiaAI Mission**.
- **Strengthen Traceability Through Lawful Processes:** Create judicially supervised traceability mechanisms for serious offences like **terrorism, child sexual exploitation, and organised cybercrime**, without weakening **E2EE** for ordinary users.
- **Strengthen Child Online Protection:** Mandate **Safety-by-Design**, age-appropriate design, **CSAM detection technologies**, and faster takedown compliance under the **POCSO Act, 2012** and **IT Rules, 2021**.

Digital platforms are no longer passive intermediaries but influential gatekeepers of information. Examine the need to redefine intermediary liability in India.

Practice Mains Question

NITI Aayog's Strategic Roadmap for Making Ayurveda Global

NITI Aayog released “**Strategic Roadmap for Making Ayurveda Global**”, outlining why Ayurveda lags behind **Traditional Chinese Medicine (TCM)** globally and proposing a **2025–2047 phased action plan** to expand its global reach.

Current Status of Ayurveda Globally

- **Export Growth:** Ayurveda and herbal product exports doubled from **USD 1.09 billion in 2014** to **USD 2.16 billion in 2023**, reaching nearly **150 countries**.
- **Global Recognition:** Ayurveda is formally recognised in nearly **30 countries** through licensing models and academic collaborations.
- **Domestic vs. Global Workforce:** India has over **3,55,000 trained Ayurveda practitioners**, but **95%** are based in India, showing limited global mobility.
- **International Integration:** Ayurveda research spans nearly **70 countries**, supported by the **WHO Global Traditional Medicine Centre (GTMC)** in **Jamnagar, Gujarat**.
 - **Ayush Academic Chairs:** Ayush Chairs exist at institutions like **University of Mauritius** and **Western Sydney University**, but a standard international curriculum is still lacking.

- **WHO Classification:** WHO integrated **Ayurveda, Siddha and Unani (ASU)** into **ICD-11** through the **TM-2 module**.
- **Global Coding of Interventions:** WHO began coding **ASU interventions** under **ICHI**, enabling global tracking, billing and research.

SWOT Analysis of Ayurveda's Globalisation Potential

Strengths

- **Preventive and wellness-oriented approach** aligns with global focus on holistic healthcare.
- **Prakriti-based treatment and plant-based therapies** offer individualized healthcare.
- Backed by **5,000+ years** of traditional knowledge, global popularity of **Yoga** and **WHO ICD-11 recognition**.
- Can complement modern medicine in **chronic disease management, preventive care and rehabilitation**.

Weaknesses

- Not formally recognized as a medical system in many countries; lacks a uniform international regulatory framework.
- Limited certified practitioners abroad and no standardized global curriculum.
- Personalized protocols make **RCT-based evidence** difficult, limiting scientific acceptance.
- High compliance costs, inconsistent **Good Manufacturing Practices (GMP)**, quality concerns and limited insurance coverage hinder adoption.

Opportunities

- **Ayush Chairs, Ayush Scholarships and Ayush Visa** can expand global outreach.
- Greater engagement with **GTMC** and adoption of **ICHI coding** can improve acceptance.
- **USD 1 trillion global wellness market**, rising demand for nutraceuticals, functional foods, preventive healthcare and **Medical Value Travel (MVT)** offer expansion scope.
- Scientific validation, research collaborations and better communication can improve credibility and market presence.

Threats

- Weak coordination among practitioners, researchers, academia, industry and regulators slows expansion.
- Ayurvedic products are mostly exported as dietary supplements, not regulated medicines, limiting recognition, value and clinical acceptance.
- Strong government-backed expansion of **TCM** increases competition.
- Balancing classical principles with modern global scientific and regulatory standards remains challenging.

Ayurveda vs. TCM		
Parameter	Traditional Chinese Medicine (TCM)	Ayurveda
Global Practitioner Base	34,000+ licensed acupuncturists in the US alone; regulated in 47 US states	~5,000 practitioners abroad, largely driven by the Indian diaspora
Global Exports	USD 5.4 billion with overseas WHO-GMP/GMP manufacturing units	USD 2.16 billion (2023), with production largely concentrated in India
Global Presence	30+ overseas TCM centres, supported through the Belt and Road Initiative (BRI)	Presence through the WHO GTMC in Jamnagar, Gujarat and Ayush Information Cells
Insurance Coverage	Covered under Medicare (US) and in countries such as Japan and Germany	Covered primarily in India, with limited insurance coverage abroad
International Standards	Dedicated ISO technical committee for TCM standards	No dedicated ISO technical committee for Ayurveda

NITI Aayog's Strategic Roadmap for Making Ayurveda Global

Short Term (up to 2029) - Build the foundation

- **Mission Steering Group:** Set up MSG, chaired by Minister of Ayush, to coordinate research, diplomacy, trade and education.
- **Global Ayurveda Register:** Launch GAR, a WHO-aligned digital credentialing and CPD system for global practitioners, under National Commission for Indian System of Medicine (NCISM).
- **Global Information Portal:** Single-window resource for licensing, visas and country-specific compliance norms.
- **WHO-GMP Certification:** Fast-track WHO-GMP-equivalent certification for major manufacturers; publish certified units directory.
- **Export Edition of Ayurvedic Pharmacopoeia:** Include GACP-based sourcing and contaminant/heavy-metal profiling.
- **Ayurveda Trade Dashboard:** Create real-time dashboard and align products with proper HS codes to leverage FTAs.

Medium Term (up to 2035) - Market integration

- **Registrations:** Pursue registrations under frameworks like EU Traditional Herbal Medicinal Products Directive (THMPD) for key formulations.
- **Insurance Coverage:** Pilot Ayurveda insurance coverage in select OECD countries.
- **Professional Mobility:** Operationalise MRAs with partners like G20, BRICS and ASEAN.
- **Education:** Introduce Ayurveda electives/micro-credentials in international medical schools, linked to Ayush Chairs.
- **MVT Hubs:** Establish international Ayurveda MVT hubs, starting with Mauritius, and offer bundled Ayush Visa packages with diagnostics, treatment and tele-consultation follow-ups.

- **Research and IP:** Scale multi-country trials through WHO Collaborating Centres and modernise Traditional Knowledge Digital Library (TKDL) for strong IP stewardship.

Long Term (up to 2047) — Systemic integration

- Formal recognition/integration of Ayurveda in at least 20 national health systems.
- Insurance coverage for defined indications in at least 10 countries, including chronic low back pain, knee osteoarthritis, IBS-type conditions, stress/sleep disorders, metabolic syndrome and women's health.
- Global network of International Ayurveda Centres of Excellence combining clinical care, education, quality labs and research with public outcomes dashboards.
- Build a trusted, evidence-forward global brand narrative for Ayurveda.

Despite its rich heritage, Ayurveda has lagged behind Traditional Chinese Medicine (TCM) in global acceptance. Examine the reasons and suggest measures for its globalisation.

Practice Mains Question

INSA Proposes a Unified National Energy Framework

Indian National Science Academy (INSA), through Centre for Science, Technology, Innovation, and Policy (CSTIP), released a May 2026 policy brief proposing a Unified National Energy Policy Framework to speed up India's energy transition and support energy self-reliance by 2047 and net-zero by 2070.

Unified National Energy Policy Framework

- **About:** It is a coordinated, system-integrated governance framework that aligns policies across fossil fuels, renewable energy, electricity distribution, green hydrogen and climate commitments.
 - It treats India's power grid and energy supply chain as one holistic ecosystem, instead of viewing different fuels as competing markets.

Need for a Unified Policy Architecture for India's Clean Energy Future

- **Shift From Capacity Addition to System Integration:** India's RE capacity rose from about **40 GW in 2015** to about **260 GW by 2025**.
 - Managing this variable capacity needs real-time integration with **base-load thermal power, Pumped Storage Hydropower and BESS** to prevent **frequency collapse**.
- **The Silo Conundrum:** Energy governance is split across the **Ministry of Coal, Ministry of Power, MoPNG, MNRE and DAE**, causing overlapping timelines, regulatory friction and inefficient resource allocation.
 - A unified framework treats **coal, gas, biomass and RE** as complementary feedstocks.
- **Supply-Demand Asymmetries:** Manufacturing mandates are moving faster than infrastructure and market demand. **National Green Hydrogen Mission** and **SATAT** (compressed biogas) face bottlenecks in evacuation, local pipelines and uniform pricing. Integrated policy links production with commercial demand absorption.
- **Eliminating Data Silos:** Electricity data is scattered across agencies in non-standard formats, affecting planning and real-time decisions. The draft **National Electricity Data Sharing Framework, 2026** proposes
 - **National Electricity Data Centre (NEDC)** for standardised data, **AI-based grid management**, demand forecasting and integrated planning.
- **Mitigating Fiscal Stress and Import Vulnerabilities:** India imports over **85% crude oil**, while energy subsidies consume about **2.3% of GDP**.
 - Unified architecture can align **PM Surya Ghar** and **PM-KUSUM** with **DISCOM reforms**, and integrate **CCUS** for hard-to-abate sectors.

Key Challenges in Implementing a Unified Energy Architecture

- **Institutional Fragmentation:** Centralized planning needs an apex mechanism like a proposed **Department of Energy Resources**, but this may affect existing bureaucratic jurisdictions.
 - It causes **resource-impetus mismatch**, with thermal and renewable sectors following conflicting regulatory timelines.
- **Transmission Deficit:** RE projects take **12–18 months**, while interstate transmission networks take **18–30 months**, causing grid congestion and power curtailment.
 - India's tight grid frequency band of **49.90 Hz to 50.05 Hz** struggles with intermittent solar fluctuations or the **duck curve**, forcing curtailment of clean energy.

- **Massive Energy Storage Gap:** Central Electricity Authority (CEA) projects India needs **60.63 GW** of storage by **2030**, including **41.65 GW from BESS**. As of early **2024**, operational capacity was only a fraction, constrained by high capital costs and critical mineral import dependence.
- **DISCOM Financial Distress and Capital Scarcity:** Grid modernization, smart metering and decentralized distribution need huge upfront capital.
 - **DISCOMs** remain debt-ridden due to high **AT&C losses** and un-rationalized tariffs, limiting investment in grid flexibility and clean power procurement.
- **Just Transition and Federal Friction:** A top-down national policy may clash with state-level economic realities.
 - Funding local workforce reskilling without weakening state economies is a major political challenge under **competitive federalism**.

Key Highlights of the Proposed INSA Framework

- **Adequacy:** Ensures reliable and diversified energy supply through **conventional and emerging sources**, supported by modern infrastructure, grid-scale storage and digital technologies.
- **Access:** Provides reliable and equitable energy services by strengthening **last-mile delivery**, improving utility services and expanding decentralized solutions where grid connectivity is unviable.
- **Affordability:** Keeps clean energy transition economically viable for **households, businesses and industries** through innovative financing, market efficiency and consumer safeguards.
- **Appropriate Sustainability:** Promotes India-specific climate solutions, with support for **local communities, workforce development and region-specific transition pathways**.

Strategic Enablers and Emerging Technologies Highlighted

- **Cross-Cutting Technical Levers:** **Circular Economy** and **CCUS** are key tools to complement RE and cut emissions in hard-to-decarbonize industries.
- **Green Hydrogen and Clean Tech:** Prioritizes commercial scaling and infrastructure for **green hydrogen** and **advanced bio-resources**.
- **Resource Optimization:** Treats **coal, renewables, biomass, natural gas and waste-to-energy** as interconnected parts of one national energy system, not competing sectors.

Discuss the need for a Unified National Energy Framework in achieving India's energy security and climate goals. Examine the challenges in its implementation.

Practice Mains Question

Centre Proposes Special Article 371 Framework for Ladakh

MHA official minutes of meetings reveal a breakthrough draft agreement proposing a **customized Article 371-based self-governance model** for **Ladakh (UT)**, aimed at resolving political deadlock, agitations and representation deficits.

Proposed Governance Model for Ladakh

- **Sui Generis Model under Article 371:** Centre proposes a **customized Article 371-based framework (similar to protections in Nagaland or Mizoram)**, instead of the **Sixth Schedule**, to protect Ladakh's **tribal culture, land and local employment**.
- **Elected UT-Level Body:** Ladakh will get a **democratic Union Territory-level elected body**, shifting away from a purely **bureaucrat-run UT** model.
- **Tripartite Powers:** The elected body will have **executive, financial and legislative powers** over local subjects, not just an advisory role.
- **Direct Control Over Bureaucracy:** The elected executive will supervise civil servants handling its subjects, including power to **review and appraise officials' performance**, making bureaucracy accountable to the elected government.
- **Harmonized Local Governance:** The new legislature will work with **PRIs**.
- **Statehood as Long-Term Goal:** Immediate **statehood** is not part of the package due to **financial constraints** (mainly the burden of govt. salaries), but the Centre acknowledged it as Ladakh's **long-term aspiration**.

Demand for Constitutional Safeguards in Ladakh

- **Reorganisation of J&K (2019):** After **Article 370 abrogation** and the **Jammu and Kashmir Reorganisation Act, 2019**, Ladakh became a separate **UT without legislature**.
- **Representation Deficit:** Ladakh earlier had **4 MLAs** in the **J&K Assembly**; as a UT, control shifted to the **Centre-appointed Lieutenant Governor**, while Ladakh Autonomous Hill Development Councils (LAHDCs) of **Leh and Kargil** had limited powers.
- **Agitations (2021-26):** Civil groups led protests, shutdowns and fasts, fearing **demographic change** and loss of **land rights**; notably led by **Sonom Wangchuk**.
- **The Four-Point Demand:**
 - **Full statehood** for Ladakh.
 - **Sixth Schedule safeguards**.
 - Separate **Ladakh Public Service Commission (LPSC)** for job reservations.
 - **Two Parliamentary seats** for **Leh and Kargil**.

Article 371

- **About:** Article 371, under Part XXI, provides special provisions to States for unique historical, cultural, social and regional needs.
- **Constitutional Position:** Article 371 has existed since 26th Jan 1950; Articles 371A to 371J were added later through constitutional amendments under Article 368.
- **Key Features:** Unlike the Sixth Schedule with Autonomous Councils, Article 371 empowers the State Legislature/Governor to protect local customs, religious practices, customary laws and restrict land transfer to non-residents (e.g., Article 371A for Nagaland).
 - Article 371 provides for separate Development Boards in Maharashtra and Gujarat for balanced regional development.
 - Includes 371A Nagaland, 371B Assam, 371C Manipur, 371D & 371E Andhra Pradesh and Telangana, 371F Sikkim, 371G Mizoram, 371H Arunachal Pradesh, 371I Goa and 371J Kalyana Karnataka.
- **Proposed Fix:** Centre argues a customized Article 371 framework can protect Ladakh's land, culture and jobs without Sixth Schedule's administrative complexities.

Ease of Living: Key Government Initiatives and Outcomes

Between **2014 and 2026**, India's inclusive progress focused on improving **ease of living** through better access to **housing, clean fuel, drinking water, sanitation, electricity, financial inclusion, transport connectivity, urban infrastructure, and digital governance**.

Key Initiatives to Improve Ease of Living in India

Housing and Basic Amenities

- **PMAY-U**
 - Launched in **2015** for **pucca houses** to eligible urban households.
 - **PMAY-U 2.0 (2024)** gives up to **₹2.5 lakh** assistance; female ownership/co-ownership required.
 - Over **1.25 crore houses sanctioned; 98 lakh+ completed**.
- **PMAY-G**
 - Launched in **2016** for rural housing.
 - Assistance: **₹1.20 lakh** in plains, **₹1.30 lakh** in hilly/difficult areas.
 - **3.91 crore sanctioned, 3.05 crore completed**; around **75% owned/jointly owned** by women.
 - Linked with **toilets, drinking water and electricity**.
- **AMRUT**
 - Launched in **2015** for urban services in **500 cities**.
 - Focus: **water supply, sewerage, green spaces, urban transport**.
 - **AMRUT 2.0 (2021)** covers **4,800 statutory towns** with **₹2.99 lakh crore outlay**.

- Projects worth **₹2.79 lakh crore** sanctioned.

Universal Access to Essentials

■ PMUY

- Launched in **2016** for **LPG connections** to reduce biomass dependence.
- Over **10.57 crore LPG connections** provided.
- LPG coverage rose from **55.9% in 2014** to **107.2% in 2026**.

■ Jal Jeevan Mission

- Launched in **2019** for **Har Ghar Jal**.
- Rural tap water access rose from **3.23 crore** to **15.86 crore households** by **June 2026**.
- Over **9 crore women** were freed from fetching water.
- **1.81 lakh villages** certified as **Har Ghar Jal**.
- **JJM 2.0** extended till **December 2028** with **₹8.69 lakh crore** outlay.

■ SBM-G

- Launched in **2014**; rural sanitation coverage rose from **39% to 100% by 2019**.
- Over **12.14 crore household toilets** and **2.76 lakh community sanitary complexes** completed.
- **5.69 lakh+ villages** declared **ODF Plus**.

■ SBM-U

- Built **63 lakh+ household toilets** and **6 lakh community toilets**.
- Urban waste processing rose from **16% to 82%**.
- Door-to-door waste collection rose from **43% to 98%**.
- All ULBs in **35 States/UTs** declared **ODF** in **2019**.

Power for All

■ Power Capacity

- Installed power capacity rose from **248 GW** in **FY 2014** to **532 GW+ by March 2026**.
- Renewable capacity rose from **76.38 GW** to **274.69 GW**.
- Solar capacity rose from **2.82 GW** to **150.26 GW**.

■ Power Supply

- Rural power supply rose from **12.5 hours** to **22.6 hours** daily.
- Urban areas receive up to **23.4 hours** daily.
- Energy shortage fell from **4.2% to 0.03%**.

■ SAUBHAGYA

- Launched in **2017** for last-mile electricity connections.
- Around **2.86 crore households** electrified by **March 2022**.

■ PM Surya Ghar Muft Bijli Yojana

- Launched in **2024** for rooftop solar.

- Subsidy up to **₹78,000** and up to **300 free units/month**.
- Over **40 lakh households** installed rooftop solar by **May 2026**.

■ UJALA

- **37 crore LED bulbs** distributed.
- Annual household savings: **₹19,153 crore**.

Financial Inclusion and Empowerment

■ PM Jan Dhan Yojana

- Launched in **2014** as part of **JAM trinity**.
- Accounts rose from **14.72 crore** in **2015** to **58 crore+ by June 2026**.
- Deposits crossed **₹3 lakh crore**.

■ Direct Benefit Transfer

- In **FY 2024–25**, **₹6.9 lakh crore** credited through DBT under **327 schemes**.
- Helped reduce leakages and improve transparency.

■ RuPay Cards

- **40.60 crore RuPay debit cards** issued with accident insurance cover.

■ PM MUDRA Yojana

- Launched in **2015** for collateral-free loans to micro/small entrepreneurs.
- **57.7 crore+ loans** worth **₹40 lakh crore** sanctioned.
- Around **66% loans** went to women.
- Nearly half of beneficiaries were from **SC/ST/OBC** communities.
- **Tarun Plus** raised loan limit to **₹20 lakh** in **FY 2024–25**.

Key Outcomes of Governance and Infrastructure Reforms

Transport and Connectivity

■ Roads and Highways

- Road network reached **63.73 lakh km**, **2nd largest globally**.
- National highways increased from **91,287 km** in **FY 2014** to **1,46,572 km by March 2026**.
- Four-lane and above highways increased from **18,371 km** to **45,516 km**.

■ Expressways and Landmark Projects

- **3,644 km** access-controlled high-speed corridors operational.
- **22,590 km** roads completed under **Bharatmala**.
- Key projects: **Atal Tunnel**, **Dhola-Sadiya Bridge**, **Sudarshan Setu**, **Delhi-Dehradun Economic Corridor**, **Dwarka Expressway**.

■ Rural Connectivity

- **PMGSY** connected about **99.6% eligible habitations**.

- Completed rural roads rose to **4.11 lakh km** during **2014–2026**.

■ Railway Modernisation

- Railway electrification reached **99.6%** by **March 2026**.
- Covered **69,873 route km**.
- Train accidents declined from **135** in **2014–15** to **16** in **2025–26**.
- Safety improved through systems like **Kavach**.

■ New Rail Services and Stations

- Vande Bharat** expanded to **162 services** by **March 2026**.
- Amrit Bharat** trains improved affordable long-distance travel.
- 1,338 stations** identified under **Amrit Bharat Station Scheme**; **208 upgraded** by **April 2026**.

■ Civil Aviation and UDAN

- UDAN** operationalised **665 routes** across **95 airports/heliports/water aerodromes**.
- Airports increased from **74** in **2014** to **165** by **April 2026**.
- UDAN carried over **1.64 crore passengers**.

■ Metro and Regional Transit

- Metro network expanded from **248 km** in **5 cities** to **1,155 km** in **26 cities** by **March 2026**.
- India became the **3rd largest metro network** globally.
- Namo Bharat Delhi–Meerut RRTS** became fully operational in **February 2026**.

Governance Reforms

■ Jan Vishwas Act

- Jan Vishwas Act, 2023** reduced compliance burden by shifting minor defaults from **criminal penalties** to **civil remedies**.
- Jan Vishwas Act, 2026** covered **784 provisions** across **79 central Acts**.
- It decriminalised **717 provisions** and amended **67 citizen-facing provisions**.

■ Fairer Compliance System: Imprisonment clauses replaced with monetary penalties, warnings and advisory notices.

- Adjudicating Officers** and **Appellate Authorities** enabled faster resolution.

■ CPGRAMS

- Strengthened grievance redressal and accountability.
- Around **6 lakh grievances** resolved between **Jan 2025** and **Feb 2026**.
- 69.8%** were rated satisfactory.

■ MyGov

- Launched in **2014** for participatory governance.
- Over **60 million registered users**.
- State-level instances in **28 States/UTs**.

■ PM GatiShakti National Master Plan

- Launched in **2021** for integrated infrastructure planning.
- Onboarded **58 Ministries/Departments**.
- Uses **3,204 GIS-based data layers** for coordinated infrastructure development.

India's Ease of Living journey reflects the transition from welfare delivery to citizen-centric governance. Discuss.

Practice Mains Question

Viksit Bharat – Guarantee for Rozgar and Aajeevika Mission (Gramin) Act, 2025

The **VB-G RAM G Act, 2025** came into force across rural India on **1st July 2026**, replacing the **MGNREGA, 2005**.

VB-G RAM G Act, 2025

- About:** The **VB-G RAM G Act, 2025** gives a **statutory guarantee of wage employment** to eligible **rural households** and aims to support **Viksit Bharat @2047** through livelihood generation and durable rural assets.
- Earlier Rural Employment Programmes:** **Rural Manpower Programme (1960s)**, **Crash Scheme for Rural Employment (1971)**, **Jawahar Rozgar Yojana (1993)**, and **Sampoorna Grameen Rozgar Yojana (1999)**.
 - Maharashtra Employment Guarantee Act, 1977** - first to introduce **statutory right to work**.
- Focus Areas:** **Gram Panchayats** as the center of rural transformation, emphasizing durable asset creation, natural resource management, water conservation, and women's empowerment.
- Features of Act, 2025:**
 - Guaranteed work:** Increased from 100 days to 125 days per financial year
 - Wage floor:** Minimum wage of ₹300/day nationwide
 - Wage payment:** Weekly payment; maximum delay allowed is 14 days
 - No-work period:** 60-day pause during peak sowing/harvesting seasons
 - Worker ID:** Gramin Rozgar Guarantee Cards; existing e-KYC job cards valid during transition
 - Funding ratio:** 60:40 Centre:State for normal states; 90% Centre for NE/Himalayan states and UTs with legislatures; 100% Centre for UTs without legislatures
 - Funding model:** Replaces demand-driven model with normative allocation based on 16th Finance Commission recommendations
 - Local planning:** Through Viksit Gram Panchayat Plans, linked with PM Gati Shakti
 - Asset tracking:** Assets mapped on Viksit Bharat National Rural Infrastructure Stack

■ Four Priority Verticals

- **Water security:** conservation and management works.
- **Core rural infrastructure:** basic village-level infrastructure.
- **Livelihood infrastructure:** assets supporting rural income generation.
- **Climate resilience:** special works against extreme weather events.

Concerns with VB-G RAM G Act, 2025

- **Higher state burden: 60:40 funding** may raise states' financial burden by up to **six times**, pressuring state finances.
- **Allocation equity issue: Normative allocation** based on **16th Finance Commission** formula may reduce funds for states with better demographic performance.
 - Spending beyond the Centre's limit must be borne fully by the **State**.
- **Weakening right to work:** Pre-set budget ceilings may dilute the **legal right to work** into a budget-limited scheme.
- **60-day work pause:** Suspension during sowing/harvesting may hurt workers during **droughts, crop failure, or irregular monsoons**.
- **Centralisation:** More Union control over planning and finances may weaken **Gram Sabha autonomy** and the **73rd Amendment** spirit.
- **Lower accountability:** Centralised grievance redressal and weaker social audits may reduce **transparency**.
- **Uneven wage impact:** ₹300/day floor benefits (15%) low-wage states like **UP, Bihar, Assam**, but gives minimal hikes to southern states (**3% or less**).
 - **Sikkim** holds a special highest rate of **Rs 450** in certain panchayats.
- **Pending dues:** Over **₹17,000 crore** in unpaid MGNREGA dues remain unresolved.

Way Forward

- **Strengthen cooperative federalism:** Adopt a **graded transition** to the new funding model and create a **Fiscal Compensation Mechanism** for high-demand, well-performing states (preventing a "North-South" funding asymmetry).
- **Reduce states' fiscal burden:** Retain **100% Central funding** for wages/admin costs and share material costs in a **75:25 Centre-State ratio**.
- **Expand convergence:** Allow convergence with **State-funded housing schemes** and empower states to approve eligible schemes without prior Union notification.

- **Decentralise 60-day pause:** Let **Gram Sabhas** and **District Collectors** decide the blackout period based on local agro-climatic conditions.
- **Protect right to work:** Create a **National Contingency Buffer Fund** for demand spikes during economic shocks or climate disasters.
- **Prevent digital exclusion:** Permit **offline authentication**, manual muster rolls, and Panchayat verification where e-KYC/biometrics fail.
- **Empower PRIs:** Ensure Gram Panchayats retain powers to select, design, and execute assets under **Viksit Gram Panchayat Plans**.
- **Improve accountability:** Codify an **Independent Social Audit Directorate** for bottom-up monitoring and transparency.

The VB-G RAM G Act, 2025 marks a paradigm shift in India's rural employment policy from welfare-centric employment generation to infrastructure-led rural development. Discuss.

Practice Mains Question

Pradhan Mantri Krishi Sinchayee Yojana

Pradhan Mantri Krishi Sinchayee Yojana has completed more than 10 years of advancing irrigation-led agricultural transformation in India.

PMKSY

- **Launch:** Launched in 2015
- **Nature:** Centrally Sponsored Scheme, classified as a Core Scheme
- **Objective:** Expand assured irrigation, improve water-use efficiency, and promote water conservation
- **Funding Pattern:** 60:40 for most States, 90:10 for North-Eastern and Himalayan States, 100% Central funding for UTs
- **Continuation Outlay:** Approved for 2021–26 with ₹93,068.56 crore
- **Budget Allocation:** ₹6,587 crore allocated in Union Budget 2026–27
- **Formulation:** Formulated by merging:
 - **Accelerated Irrigation Benefit Programme** under Ministry of Jal Shakti
 - **Integrated Watershed Management Programme** under Department of Land Resources
 - **On-Farm Water Management** under Department of Agriculture and Cooperation
- **Core Approach:** Focuses on **Jal Sanchay** for rainwater storage and **Jal Sinchan** for efficient water application.
- **Implementation:** Implemented through **State Irrigation Plans** and **District Irrigation Plans** for decentralized, location-specific irrigation planning.

Major Components of PMKSY

PMKSY covers the irrigation value chain through irrigation expansion, last-mile delivery, watershed management, and efficient water use.

Component	Focus	Key Facts
AIBP	Completion of major and medium irrigation projects	Since 2016–17, received ₹21,023 crore; benefited 17.30 million farmers
M-CADWM	Last-mile water delivery	Approved in 2025–26 under PMKSY-AIBP with ₹1,600 crore outlay; promotes pressurized piped irrigation and micro-irrigation
Har Khet Ko Pani	New water sources and assured irrigation	3,462 minor irrigation/RRR schemes completed; 5.93 lakh ha irrigation potential created; groundwater interventions added 88.55 thousand ha
Watershed Development	Rainfed and degraded land improvement	Since 2016–17, received ₹12,432.09 crore; benefited 1.34 million farmers
Per Drop More Crop	Micro-irrigation and water-use efficiency	Under PMKSY till 2021–22; under PM-RKVY from 2022–23; covered 110.92 lakh ha under micro-irrigation

Micro-Irrigation Support

- **Micro Irrigation Fund** set up under **NABARD**.
- Financial assistance: **55%** for small and marginal farmers, **45%** for other farmers.
- Promotes **drip** and **sprinkler irrigation** to reduce wastage and improve productivity.

Strengthening Food Safety in India

Large-scale food poisoning incidents have exposed gaps in the implementation of the **Food Safety and Standards (FSS) Act, 2006**, raising concerns about increasing food poisoning deaths.

Concerns Regarding Food Safety in India

- **Food Poisoning:**
 - **ADSI Report 2024:** 1,122 deaths in India due to food poisoning.
 - **WHO Estimates 2026:** Unsafe food causes 866 million illnesses and 1.5 million deaths globally.
 - Children under **5 years** bear nearly **30%** of the disease burden.
 - India ranks **15th** globally in YLL due to foodborne diseases.
- **State Food Safety Index:** FSSAI assesses states on **HR, compliance, testing, surveillance, training, capacity building, and consumer empowerment**.
 - In **2023–24**, nearly **75% States/UTs** scored below **50/100**.
 - Poor performers: **Jharkhand: 26.5, Uttar Pradesh: 44.25**.
- **Gaps in Inspection and Sampling:** **FSS Act, 2006** mandates registration/licensing of FBOs, inspections, and lab testing.
 - **2026 Amendment Regulations** introduced **risk-based inspection** based on risk profile and compliance history.
 - Maharashtra collected only **~21,000 samples** in **2024–25** despite **1.8 lakh+** registered FBOs.
- **Institutional and Manpower Deficits:** FSSAI has nearly **40% central vacancies**.
 - Over **1,200 FSO posts** were vacant across India by early **2026**.
 - Maharashtra needs **1,100 FSOs**, but only **130** are on field.
- **Chemical Hazards and Heavy Metals:** Rising contamination from **arsenic** in rice/groundwater and **lead** in spices, cereals, and cookware.
 - Linked to **chronic diseases** and **cognitive impairment in children**.
 - Antibiotic misuse in livestock/dairy farming is accelerating **AMR**.
 - Pesticide residues increase risks of **cancer** and **endocrine disorders**.
- **Food Adulteration:** Has become a parallel economy, especially in **dairy and edible oils**.
 - Risks include **ethylene glycol-contaminated milk, synthetic milk, and fake paneer**.
- **Misleading Marketing:** Some FBOs use deceptive labels and false health claims.
 - FSSAI acted in **2025–26** against misuse of claims like **“100% Natural”** and disease-curing claims.
- **Temperature Anomalies:** Heatwaves accelerate **bacterial growth** and **food spoilage**.
 - Risk is higher where **cold-chain networks** are weak.
- **Hygiene Bottlenecks:** Poor **clean water, waste disposal, and food-handling training** increase spread of **Salmonella** and **E. coli**.

Measures to Strengthen Food Safety in India

- **Turnover-linked penalties:** Link fines to annual turnover of offending FBOs to create real deterrence.
 - Set up **district-level food safety tribunals** for faster disposal of food safety cases.
- **IoT-enabled cold chains:** Mandate **temperature and humidity sensors** in organized cold storage and transport vehicles.
- **Incentivised compliance:** Link **FSSAI compliance** with digital credit schemes like **PM SVANidhi**; trained vendors can get lower interest or digital benefits.
- **Farm-to-fork dashboard:** Create an inter-ministerial digital ledger involving **FSSAI** and **Ministry of Agriculture** to track pesticide and antibiotic use.
- **Ban health-washing:** Mandate clear **Front-of-Pack warning labels** using visible **red/amber/green** indicators.
- **Citizen vigilance:** Launch a mobile-based **Food Watch app** for geo-tagged complaints and automatic inspection triggers.

Food safety has emerged as a critical governance challenge in India. Examine the institutional and regulatory gaps in ensuring safe food for all.

Practice Mains Question

Delhi Electric Vehicle Policy 2.0

Delhi government notified its new **Electric Vehicle Policy 2.0** (1st July 2026 - 31st March 2030).

- **Aim:** Make Delhi an EV hub, focusing on 2Ws, 3Ws, goods vehicles, school buses, aggregator and government fleets.
- **Target:** Achieve **30% fleet electrification by 31st March 2030**; EV penetration was around **14% by 2025**.
- **Registration Mandates:** From **2027**, only electric 3Ws and N1 trucks; from **2028**, only electric 2Ws.
- **School Buses:** **10% EV conversion within 2 years**, and **30% by 2030**.
- **Funding:** ₹7,000 crore allocation; ₹15,000 crore overall investment plan.
- **Incentives:** Purchase subsidy, road tax exemption and registration fee waiver; up to ₹30,000 for 2Ws, ₹50,000 for 3Ws and ₹1 lakh for N1 trucks.
- **Scrapping Support:** Incentives for below BS-IV vehicles, including 2Ws, 3Ws, 4Ws, N1 trucks and Gramin Seva units.
- **Charging Infra:** Over **30,000 public charging points** planned; ₹1,000 crore for charging infrastructure.
- **Implementation:** Transport Department as nodal agency; Delhi Transco to develop charging network.
- **N2 Truck Exemption:** First 1,000 electric N2 trucks get **10-year No Entry timing exemption**.
- **No Hybrid Subsidy:** Benefits restricted to pure EVs only.

Anemia Mukht Bharat

Abhiyaan Operational Guidelines

The Union Health Minister has released the **Anemia Mukht Bharat Abhiyaan Operational Guidelines**, marking **eight years** of the **Anemia Mukht Bharat initiative**.

Anemia Mukht Bharat Guidelines

- **Paradigm Shift:** Moves from **prophylactic** (preventive) care to **therapeutic** (active) anemia treatment through testing, tracking and case-based protocols.
- **7x7x7 Framework:** Upgrades the earlier **6x6x6 strategy** by adding:
 - **7th Beneficiary:** Low Birth Weight babies **0–6 months**.
 - **7th Intervention:** **Eating Right** approach for diversified, iron-rich diets.
 - **7th Mechanism:** Digitally integrated **Monitoring & Evaluation** framework.
- **T4 Strategy:** Expands **T3 — Test, Treat, Talk** to **T4 — Test, Treat, Talk and Track**, ensuring referral and follow-up.
- **Advanced Clinical Care:** Institutionalises **Ferric Carboxymaltose (FCM)** and **Iron Sucrose IV** therapy for severe/non-responsive anemia in pregnant and lactating women.
- **Unified Digital Architecture:** Central AMB portal integrates **JANANI**, **RBSK** and **U-WIN** portals for real-time data convergence.
- **Governance & Community Mobilisation:** Uses **Whole-of-Government**, **Jan Bhagidari** and **Jan Chetna** campaigns to de-normalise anemia.
- **Targeted Outcomes:** Aims to reduce **IMR and MMR** and improve maternal-child lifecycle health.

Anemia Mukht Bharat (AMB)

- **About:** Launched in **2018** under **NHM** to address nutritional and non-nutritional causes of anemia through a life-cycle approach.
- **Crucial AMB Interventions:**
 - **IFA Supplementation:** Colour-coded IFA doses for infants, children, adolescents, non-pregnant women and pregnant women.
 - **Deworming:** Biannual deworming on **10th February** and **10th August** under **NDD** for ages **1–19**; pregnant women in 2nd trimester via ANC.
 - **Food Fortification:** Use of **IFA-fortified foods** in government health programmes.
 - **Non-Nutritional Causes:** Screening/treatment of malaria, haemoglobinopathies like **Sickle Cell Disease**, and fluorosis.
 - **Digital Integration:** Point-of-care testing and digital treatment tracking.

India NFHS-5: 67.1% under-5 children, 59.1% adolescent girls, 57% women, and 52.2% pregnant women are anaemic.

NCERT Introduces Emergency in Class 9 Textbook

NCERT has included the **1975 Emergency** in Class 9 textbook *Understanding Society: India and Beyond*.

- **Emergency:** Declared on **25 June 1975** on grounds of **internal disturbance** and continued until **March 1977**.

National Emergency

- **About:** **Article 352** empowers the **President** to proclaim National Emergency when India's security or a part of it is threatened by **War, External Aggression, or Armed Rebellion**.
 - **38th Amendment Act, 1975:** Allowed multiple proclamations on grounds of **war, external aggression, internal disturbance, or imminent danger**.
 - **44th Amendment Act, 1978:** Replaced "internal disturbance" with "armed rebellion" to remove ambiguity.
- **Territorial Extent:** Can apply to the **whole country or a specific part**. **42nd Amendment Act, 1976:** Enabled limiting Emergency to a particular area.
- **Parliamentary Approval:** Must be approved by **both Houses of Parliament within 1 month** by **special majority** under the **44th Amendment Act, 1978**.
 - If Lok Sabha is dissolved, **Rajya Sabha approval** continues, but the reconstituted Lok Sabha must approve it within **30 days of its first sitting**.
- **Duration:** Continues for **6 months** and can be extended indefinitely with **Parliamentary approval every 6 months** (**44th Amendment Act, 1978**).
- **Revocation:** Can be revoked anytime by the **President** without Parliamentary approval.
 - **Lok Sabha** can disapprove continuation through a **simple majority resolution**.
 - If **1/10th members** submit notice, a special sitting must be held within **14 days**.
- **Judicial Review:** **38th Amendment Act, 1975:** Made Emergency proclamation immune from judicial review.
 - **44th Amendment Act, 1978:** Restored judicial review.
 - **Minerva Mills Case (1980):** SC held Emergency can be challenged if it is **mala fide, based on irrelevant facts, or perverse**.

Criminalisation of Politics in India

ADR-National Election Watch (NEW) report—**31%** sitting **Rajya Sabha** MPs have declared criminal cases; **16%** face serious criminal charges, highlighting **criminalisation of politics**.

Key Findings of the ADR Report

- **Prevalence of Criminal Cases:** Of **226** sitting **Rajya Sabha** MPs, **31%** declared criminal cases; **16%** face serious charges.
- **State-Wise Concentration:** Highest shares: **Telangana (71%), Delhi (67%), Maharashtra (53%)**.
- **Concentration of Wealth:** **~14%** MPs are billionaires; average assets are **Rs 120.69 crore**.
- **Implications for Indian Polity:** Shows criminal-politician-business nexus (issue formally recognized 3 decades ago by the **Vohra Committee (1993)**); money and muscle power weaken **Rajya Sabha's** role in state representation and legislative review.

Criminalisation of Politics

- **About:** Entry of persons with **criminal backgrounds**, pending charges, or convictions into elections and public office in Parliament and State Legislatures
 - It replaces **rule of law** with **money and muscle power**, weakening democratic institutions.
- **Causes of Increasing Criminalisation:**
 - **The "Winnability" Factor:** Candidates with criminal records have higher success rate, about **15%**, than candidates with clean backgrounds, about **4%**; they can self-finance campaigns.
 - **Money and Muscle Power:** **FPTP** and high election costs push parties towards candidates with **black money** and **muscle power**.
 - **Governance Deficit:** Weak institutions and poor services make voters prefer strongmen offering quick extra-legal solutions.
 - **Legal Loopholes and Slow Justice:** **Section 8, RPA 1951** disqualifies only after conviction of **2+ years**; pending cases do not bar contesting. Backlog exceeds **5.6 crore cases**.
 - **Identity Politics and Vote Banks:** Voters prioritise caste, religion, or language over criminal records; strongmen (perceived as "**Robin Hood**" figures) are seen as protectors.
 - **Lack of Inner-Party Democracy:** Opaque ticket distribution favours wealthy, winnable candidates over grassroots workers.
- **Ethical Implications of Criminalisation of Politics:** Undermines **non-partisanship, accountability, and constitutional morality**.
 - It erodes public trust, distorts free elections, weakens women's empowerment, and compromises **rule of law, transparency, institutional integrity, and inclusive development**.

■ SC Judgements:

- **Union of India v. Association for Democratic Reforms (2002):** Voters have right to know candidates' **criminal, financial, educational background** under **Article 19(1) (a)**.
- **PUCL v. Union of India (2004):** Struck down **Section 33B, RPA 1951**; reaffirmed voters' right to know candidates' antecedents is a fundamental right.
- **K. Prabhakaran v. P. Jayarajan (2005):** **Section 8(3), RPA 1951** aims to prevent criminalisation of politics, observing that **"those who break the law should not make the law."**
- **Lily Thomas v. Union of India (2013):** Immediate disqualification of MPs/MLAs convicted for **2+ years**.
- **Manoj Narula v. Union of India (2014):** Persons with serious criminal charges should ideally not be ministers.
- **Public Interest Foundation v. Union of India (2018):** Only Parliament can bar candidates at charge-framing stage; parties must publicise criminal records.
- **Rambabu Singh Thakur v. Sunil Arora (2020):** ECI must monitor disclosure compliance and report violations.

The Constitution (130th Amendment) Bill, 2025: Seeks to amend Articles 75 (Union), 164 (States), and 239AA (Delhi) to remove PM/CMs/Ministers detained for 30 days for offences punishable with 5+ years imprisonment.

Reforms to Decriminalise Indian Politics

- **N.N. Vohra Committee, 1993:** Proposed **Nodal Agency under MHA** to track criminal-politician-business nexus.
- **Dinesh Goswami Committee, 1990 & Indrajit Gupta Committee, 1998:** Suggested partial in-kind state funding of elections to reduce black money dependence.
- **Second ARC, 2008:** Recommended amending **Section 8, RPA** to disqualify those facing trial for serious offences.
- **244th Law Commission Report, 2014:** Suggested disqualification after **framing of charges** for offences punishable with **5+ years**, if framed **1 year before nomination scrutiny**.
 - Also proposed stricter penalties for **false affidavits** and **fast-track courts**.
- **Law Commission 255th Report, 2015:** Recommended full donation disclosure and independent audits of party accounts.
- **Election Commission's Core Proposals:** Sought **lifetime ban** on convicted politicians instead of current **6-year disqualification** after release.
- **Enhancing Voter Awareness:** ECI and civil society should spread candidate-background awareness; **NOTA** can pressure parties to field cleaner candidates.

"Money power and muscle power continue to undermine the integrity of Indian elections." Discuss.

Practice Mains Question

Passport is Not Proof of Citizenship

MEA clarified that an **Indian passport** is mainly a **travel document** for international transit, not conclusive proof of **citizenship**, amid the **SIR of electoral rolls** debate.

- **Legal Status of a Passport:** Passport is **strong evidence of nationality** (supported by the **Maneka Gandhi v. Union of India, 1978** judgment) but not definitive proof in citizenship disputes.
 - Under **Section 20, Passports Act, 1967**, passports can be issued to **non-citizens** in public interest (e.g., the **2023 Madras HC ruling permitting a stateless Sri Lankan refugee to apply for a passport**).
- **Electoral Rolls and Identity Cards: Lal Babu Hussein v. Electoral Registration Officer (1995)** – electoral roll inclusion gives a **presumption of citizenship**, removable only by due legal procedure.
 - **Association for Democratic Reforms & Ors. v Election Commission of India, May 2026:** SC upheld ECI's **SIR** under **Article 324** and **Section 21(3), RPA 1950**.
 - SC held voter deletion does **not remove citizenship**; suspected cases must go to **MHA** for adjudication under **Citizenship Act, 1955**.
- **No Universal Proof:** India has no single citizenship document; citizenship is under **Citizenship Act, 1955**, proven through birth, parentage, and nationality documents.
 - **SC in Sarbananda Sonowal v. Union of India (2005):** Burden of proving Indian citizenship lies on the claimant.
- **Modes of Acquiring Citizenship (Citizenship Act, 1955):** Birth/descent citizens get no certificate; certificate only for **registration/naturalisation**.
 - **By Birth (Shift from Jus Soli to Jus Sanguinis):**
 - ❖ **26th Jan 1950–1st July 1987:** Citizen by birth.
 - ❖ **1st July 1987–3rd Dec 2004:** One parent Indian.
 - ❖ **On/after 3rd Dec 2004:** One parent Indian and other not illegal migrant.
 - **By Descent:** Born outside India to Indian parent(s); consular registration needed after **3rd Dec 2004**.
- **By Registration & Naturalisation:** For **PIOs**, spouses, eligible foreign nationals; residency generally **12 years**. No dual citizenship.
- **Key Amendments: 2003 Amendment** barred illegal migrants from citizenship. **CAA 2019** reduced residency for specified minorities (**Hindu, Sikh, Buddhist, Jain, Parsi, and Christian**) from **Pakistan, Bangladesh, Afghanistan** entering by **31st Dec 2014**.

Nation & States

Global Terrorism Index 2026

The 13th **Global Terrorism Index (GTI)**, released by the **Institute for Economics & Peace (IEP)**, recorded a decline in global terrorism in 2025.

- Terrorism-related deaths fell by **28%**, while incidents declined by **22%**. However, terrorism is becoming more **concentrated, adaptive and digitally driven**.
- IEP is an independent, non-profit think tank based in **Sydney, Australia**.

Emerging Trends in Global Terrorism

- **Geographical concentration:** Nearly **70% of terrorism deaths** occurred in **Pakistan, Burkina Faso, Nigeria, Niger and the DRC**; the **Sahel** accounted for over half of global fatalities.
 - **Pakistan** was the most terrorism-affected country; **India** ranked **13th**.
- **Conflict linkage:** Around **99% of terrorism deaths** occurred in conflict-affected or politically unstable countries.
- **Digital and decentralised attacks:** Growing use of **lone actors, small cells and online radicalisation** is making detection harder.
- **Border hotspots:** Over **60% of attacks** occurred within **100 km of international borders**.
- **Adaptive networks:** Terror groups are fragmenting, exploiting local grievances and adjusting to counter-terrorism measures.
- **Major groups:** Key actors include **IS, JNIM, TTP, LeT and Al-Shabaab**.
- **Geopolitical conflicts:** Ongoing wars and tensions, especially in **West Asia**, continue to fuel extremism.

Why the Decline Does Not Mean a Safer World

- **Complacency risk:** Terrorism is concentrated in fragile states, where rising violence receives limited global attention.
- **Lone-actor threat:** Decentralised attacks are harder to predict; **93% of fatal attacks in the West** over the past five years involved lone actors.
- **Digital radicalisation:** Social media, encrypted platforms and AI-enabled algorithms accelerate recruitment and self-radicalisation.
- **Conflict-governance nexus:** Weak governance, armed conflict and institutional collapse create security vacuums for extremist groups.

- **Adaptive networks:** Localised terror cells exploit ethnic, political and religious grievances and quickly adjust to counter-terrorism measures.

Terrorism and India's Internal Security

- **Cross-border terrorism:** Pakistan-backed groups such as **LeT** and **JeM** continue proxy attacks, radicalisation and separatist mobilisation; the **China–Pakistan nexus** adds a hybrid-warfare dimension.
- **Urban terrorism:** Major cities remain vulnerable to **mass-casualty attacks (26/11)**, **lone actors**, **improvised explosives** and **attacks on public places (2025 blast near Delhi's Red Fort)**.
- **Internal insurgencies:** Militancy in **Jammu & Kashmir**, the **Northeast** and **LWE-affected regions** exploits local grievances and external support.
- **Cyber terrorism:** Terror groups use **social media, encrypted platforms, cryptocurrencies and the dark web** for recruitment, propaganda, financing and planning.
- **Terror financing:** **Drug trafficking, hawala, counterfeit currency, arms smuggling** and proximity to the **Golden Crescent and Golden Triangle** sustain terrorist networks.
- **Maritime threats:** India's **7,500-km coastline** remains vulnerable to infiltration, arms smuggling and maritime terrorism.

India's Counterterrorism Doctrine

- **Zero-Tolerance Policy:** India follows a **zero-tolerance, intelligence-led and pre-emptive approach** to terrorism, holding both terrorists and their sponsors accountable.
- **Proactive response:** Emphasises preventive intelligence, precision operations and calibrated retaliation, as seen in the **2016 Surgical Strikes, 2019 Balakot Air Strike** and **Operation Sindoor, 2025**.
- **PRAHAAR Framework:** Seeks to shift internal security from reactive containment to **pre-emptive deterrence**, while separating terrorism from religion or ethnicity.

Four Pillars of Counter-Terror Policy

Pillar 1. Legislative Empowerment

- **UAPA Amendment Act, 2019:** Allows individuals to be designated as terrorists and expands the **NIA's** investigative powers.
- **NIA Amendment Act, 2019:** Extends jurisdiction to offences committed abroad and strengthens property attachment powers.

- **PMLA amendments:** Enable tracing and confiscation of terror-financing assets, including **hawala and cryptocurrency funds**.
- **Bharatiya Nyaya Sanhita, 2023:** Defines terrorism and criminalises recruitment, training, harbouring and financing.
- **Arms Amendment Act, 2019:** Tightens penalties and improves firearm traceability.

Pillar 2. Institutional Strengthening

- **NIA:** Supported by **53 Special Courts**, regional offices and the **National Terror Database Fusion and Analysis Centre**; conviction rate reported at **92.7%**.
- **Multi-Agency Centre:** Connects **28 agencies** for intelligence sharing; its cyber wing monitors digital threats.
- **NATGRID:** Integrates immigration, banking, telecom and travel databases.
- **CCTNS 2.0:** Enables AI-assisted sharing of crime and terrorism data.
- **Police modernisation:** Over **₹6,300 crore** allocated during **2021–26** for surveillance, forensics, weapons and secure communications.

Pillar 3. Cross-Border Operations

- **Surgical Strikes, 2016:** Targeted terrorist launch pads across the Line of Control after the Uri attack.
- **Balakot Air Strike, 2019:** Targeted a **JeM** training facility after the Pulwama attack.
- **Operation Sindoor, 2025:** Targeted terrorist infrastructure in Pakistan and Pakistan-occupied Jammu and Kashmir.

Pillar 4. Multilateral and Diplomatic Action

- Uses the **FATF, No Money for Terror Conference** and bilateral counter-terror groups to combat terror financing.
- India proposed the **Comprehensive Convention on International Terrorism** at the UN.
- Secured the UN designation of **Masood Azhar** as a global terrorist in **2019**.
- **BHARATPOL, 2025** connects Indian agencies with **INTERPOL**.
- The **Delhi Declaration, 2022** addresses terrorist misuse of drones, social media, digital technologies and online finance.

Major Outcomes

- Terrorist-initiated incidents in **Jammu and Kashmir** reportedly fell from **228 in 2018 to 12 in 2025**.
- Civilian and security-force casualties, stone-pelting and shutdowns declined.

- Better intelligence and inter-agency coordination improved prevention of terror plots.
- Improved security supported tourism, investment, infrastructure and employment in Jammu and Kashmir.

AI-Driven Warfare: Preparing India for Future Combat Readiness

The **strategic-military landscape** is at a ‘**Manhattan Project**’ moment, with rapid software deployment. **AI, military autonomy, and algorithmic warfare** are redefining warfighting and deterrence, making it necessary for India’s defence establishment to adapt for **future combat readiness** and **strategic sovereignty**.

AI and Modern Warfare

- **Algorithmic Warfare and the “Kill Web”:** AI compresses the military **sensor-to-shooter cycle**, reducing decision time from **hours to minutes**.
 - AI platforms like Ukraine’s **Delta** fuse data from **radar, imagery, and open-source intelligence** into a real-time stream.
 - This replaces linear command chains with digital “**kill webs**”, helping commanders quickly match drone inventories to targets for precise neutralization.
 - The **US Project Maven**, launched in **2017**, uses **AI, satellite images, sensors, and intelligence data** to analyse drone and surveillance footage, identify targets, and speed up the **kill chain**.
- **The Rise of Autonomous Systems:** Autonomous systems are taking over high-risk **combat, reconnaissance, and logistical roles** across land, sea, and air.
 - AI-powered platforms like **YFQ-44A Fury** by **Anduril** for the **US Air Force** show a shift toward **man-machine collaborative airpower** and agile tech startups.
 - Future naval vessels may be captained by **autonomous software systems** instead of human officers, replacing human roles in direct combat.
- **The Impact of Mass Drone Warfare:** Drone warfare has altered battlefield mobility, seen in the **35-km “death zone”** along the **Russia-Ukraine frontier**.
 - In this surveilled corridor, tanks and artillery have become largely ineffective as **FPV drones** and surveillance quickly identify and eliminate infantry and moving targets.
 - Ukraine’s defence industry developed capacity to produce over **8 million FPV drones annually** by early **2026**, showing the shift away from human-centric warfare.

- **Advanced Intelligence and Electronic Warfare: Unmanned systems and HAPS** enable continuous monitoring of enemy territories without risking manned assets.
 - During **Operation Absolute Resolve, 2026**, the US reportedly used **Anthropic's Claude AI** to analyse data, track **Nicolás Maduro's** movements, and assist in planning his capture in **Caracas**.
 - AI also powers **cognitive electronic warfare** and **directed energy weapons**, making **high-energy lasers** effective against mass drone swarms.
- **Cyber, Logistics, and the Software Edge:** The digital arms race is driving strategic cyber-weapons like **"Mythos" virtual cyber-nuke** and hardware advances such as **Huawei's 1.4-nanometer chips**.
 - AI gives a logistical edge by managing supply chains, predicting equipment failures, and forecasting natural disasters in difficult terrains.

India's Initiatives in Military AI

- **Project Sanjay & Trinetra:** Indian Army integrated AI-powered **TRINETRA** with **Project SANJAY** to create a unified real-time battlefield picture.
 - **TRINETRA** provides **C4I** capabilities across land, air, and sea, while **SANJAY** fuses ground and aerial sensor data into a **Common Surveillance Picture** through Army data and satellite networks for faster decision-making.
 - **ETAI Framework:** Launched in **2024** by **DRDO** and **Chief of Defence Staff**, the **Evaluating Trustworthy Artificial Intelligence Framework** ensures military AI remains **reliable, robust, transparent**, and resilient to adversarial attacks.
 - **Innovations for Defence Excellence (iDEX) & ADITI Scheme:** **ADITI** supports deep-tech defence capabilities, funding **cognitive electronic warfare systems**, **sovereign quantum radar**, and AI modules for **unmanned aerial systems**.
 - **Operational AI Deployments:** At **AI Impact Summit 2026**, Army showcased AI systems like **Driver Fatigue Detection** for high-altitude convoys and **Prakshepan**, which forecasts avalanches and floods up to a week in advance.
 - **Specialized Formations:** Army raised **'Baaz' Battalions** for long-range drone surveillance and precision strikes, along with acquisition of **Fixed-Wing HAPS** and jet-powered loitering munitions.
- India must build comparable capabilities despite tighter **technological and fiscal constraints**.
- **Fragmented Military Networks:** Future warfare needs **sensor-to-shooter interoperability**.
 - India's fragmented command structures, service-specific networks, and incomplete **theatre command reforms** restrict a joint **multi-domain combat network**.
 - **The Tactical Bandwidth Deficit:** AI warfare produces huge **sensor, satellite, and HD video data**.
 - Limited tactical bandwidth, contested electromagnetic environments, and Himalayan connectivity gaps restrict real-time AI analytics, autonomous swarms, and distributed **kill webs** along the **LAC**.
 - **Battlefield Data Sovereignty Deficit:** Military AI needs reliable, labelled, terrain-specific data.
 - Imported or commercial algorithms may face **Out-of-Distribution (OOD) failures** in **Ladakh and Siachen** due to snow reflectivity, extreme weather, and mountainous terrain.
 - India needs sovereign battlefield datasets and indigenous military AI models.
 - **Electronic Warfare and PNT Vulnerability:** Autonomous platforms rely on **PNT systems**, which are vulnerable to jamming and spoofing.
 - Developing **NavIC-based, vision-aided, and anti-jamming navigation** within **SWaP constraints** of small drones remains a major challenge.
 - **Hardware-Centric Procurement in a Software-Defined Battlefield:** India's long defence acquisition cycles are unsuitable for AI warfare, where **algorithms, EW libraries**, and autonomous systems need upgrades within **weeks**.
 - MoD must shift from platform procurement to an agile **software-driven defence ecosystem**.
 - **Foreign Technology and Algorithmic Dependence:** Dependence on imported **GPUs, semiconductors, sensors, AI models**, and electronic components creates risks of supply-chain disruption, embedded backdoors, data poisoning, and algorithmic manipulation.
 - **Technological sovereignty** is becoming essential for national security.
 - **Human-in-the-Loop Dilemma:** Human oversight may slow autonomous warfare, while full autonomy raises concerns over **accountability, civilian harm**, and **IHL**, especially without a clear Indian regulatory framework.
 - **AI-Enabled Grey-Zone Warfare:** **Deepfakes, cyberattacks**, and algorithmic propaganda allow adversaries to alter ground realities below the war threshold, creating a deterrence dilemma for India.

Key Challenges for India in Modern AI Warfare

- **Asymmetric AI-Military Gap with China:** China's **PLA** is moving towards **"System Destruction Warfare"**, integrating **AI, cyber, EW, satellites**, and autonomous platforms to paralyse enemy **C4ISR**.

- **Innovation-Absorption Gap:** India faces a **lab-to-battlefield gap** in rapidly testing, procuring, and deploying indigenous technologies at scale.

Measures to Strengthen

India's Preparedness for AI-Driven Warfare

- **Establish Border "Kill Webs":** Deploy **counter-drone systems** like **laser and microwave weapons** with drone-hunting teams to create AI-enabled **kill webs** along the **LoC and LAC**.
- **Transition to Offensive ISR:** Crowd **Low-Earth Orbit (LEO)** with space assets to move from persistent surveillance to offensive **ISR**.
- **Establish a Sovereign Military Data Cloud:** Create a **National Secure Data Set** to train military algorithms indigenously, ensure reliability, and protect against **data poisoning**.
- **Deploy Unified Tri-Service Data Clouds:** Integrate **Project SANJAY** and **Akashteer** into a secure tri-service operational cloud for cross-domain algorithmic targeting.
- **Transition to the ARADO Framework:** Shift to **ARADO** under the proposed **MITRA doctrine** for interoperability across **land, air, sea, space, and cyber** domains.
- **Reallocate Modernization Budgets:** Defence budgets should prioritise software velocity, with up to **40%** of capital acquisition funds for **deep-tech software, cyber-resilience, and AI payloads** over conventional kinetic platforms.

Algorithmic warfare is transforming traditional linear kill chains into interconnected kill webs. Discuss the implications of this transition for India's military preparedness.

Practice Mains Question

A Decade of Defence Reforms in India

India has strengthened its **defence capabilities** through higher investments, reforms, innovation and global partnerships, shifting from **import dependence** to an **Atmanirbhar defence manufacturing ecosystem** aligned with **Viksit Bharat 2047**.

India's Defence Capability Transformation

- **Rising Defence Budgets:** Defence budget rose from **₹2.53 lakh crore in FY 2013–14** to **₹7.85 lakh crore in FY 2026–27**.
 - Capital expenditure increased from **₹94,587.95 crore** to **₹2.19 lakh crore**, strengthening modernisation.
- **R&D, Innovation and Partnerships:** Defence R&D allocation rose from **₹13,716.14 crore** to **₹29,100.25 crore**.
 - **25% defence R&D funding** opened to **industry, startups and academia**.
 - **DRDO** opened facilities in **24 labs** for private industry.

- **iDEX:** Promotes defence innovation through **startups, MSMEs, innovators and academia**.
 - Outlay: **₹498.78 crore** for **iDEX** and **₹750 crore** for **ADITI**.
 - By **March 2026**, engaged **676 startups/MSMEs/innovators** and signed **551 contracts**.
- **DcPP Model:** **DRDO-industry model** for development and production of defence technologies.
 - Ecosystem includes **2,200+ industries**.
 - By **March 2026**, **134 companies** engaged; **2,180 technology transfer agreements** signed.
- **Technology Development Fund:** **MoD-DRDO** initiative for indigenous defence technologies.
 - Provides grants up to **₹50 crore**.
 - As of **June 2026**, **80 projects** worth **₹334 crore** under implementation.
- **DIA-CoE: 15 DRDO Industry-Academia Centres of Excellence** set up.
 - Focus on **82 research verticals** for future defence technologies.
- **Defence Acquisition Reforms:** **DAP 2020** prioritises domestic manufacturing and higher indigenous content.
 - **DAC** granted **AoN worth ₹6 lakh crore+** for indigenous systems.
 - Major acquisitions: **97 Tejas Mk-1A**, **156 LCH Prachand**, **26 Rafale Marine aircraft**.
- **Human Capability:** **5 DRDO Young Scientists Labs** set up; **6th planned in 2026**.
 - **DRDO** streamlined into **36 focused labs**.
- **Agnipath Scheme:** Launched on **15 June 2022** to recruit **Agniveers** for **4 years**.
 - Aims to create a **young, tech-savvy and combat-ready force** with training, skills and post-service pathways.

Expansion of India's Defence Industrial Ecosystem

- **Defence Procurement:** **193 contracts** worth **₹2.09 lakh crore** signed in **FY 2024–25**; contracts worth **₹1.82 lakh crore** concluded in **FY 2025–26**.
- **Defence Production:** Reached **₹1.78 lakh crore** in **2025–26**, with **15.6% annual growth (110% increase since FY 2020–21)**.
- **Sectoral Share:** **DPSUs: 76%**, private sector: **24%** in **2025–26**.
- **Defence Exports:** Rose from **₹686 crore in 2013–14** to **₹38,424 crore in 2025–26**.
- **Global Reach:** Indian defence products exported to **80+ countries** by **145 firms**.
- **Export Target:** Government aims for **₹50,000 crore** defence exports by **2029**.

- **Industrial Base:** Includes **16 DPSUs**, around **500 licensed firms** and **17,000 MSMEs**.
- **OFB Reform:** **41 ordnance factories** reorganised into **7 DPSUs** in **2021**.
- **Ease of Doing Business:** Defence licences rose from **258** in **2015** to **834** in **2026**.
- **Defence Corridors:** Investments: **₹42,057 crore** in **UP** and **₹32,699 crore** in **Tamil Nadu**.

Key Defence Sector Reforms under Aatmanirbhar Bharat

- Defence Exim Portal
- Srijan Defence Portal
- Positive Indigenisation Lists
- Srijan DEEP Platform
- FDI Reform: **74% automatic route** and **100% government route**.

Major Defence Achievements in the Last Decade

- **India-US:** Defence ties strengthened through **LEMOA**, **COMCASA**, **BECA**, **iCET**, **TRUST** and **10-year defence partnership**, covering **AI**, **semiconductors**, **space** and **Indo-Pacific security**.
- **India-Russia:** Cooperation continues on **S-400**, **Su-30MKI** upgrades, **Exercise INDRA**, **MRO** and **co-production**.
- **India-EU:** **Security and Defence Partnership (2026)** covers **maritime security**, **cyber defence**, **counterterrorism** and **space**.
- **India-France:** Ties deepened through **Rafale**, **Scorpene submarines**, **Rafale-M** and **defence industrial cooperation**.
- **India-Japan:** Cooperation through **ACSA**, **JIMEX** and **Indo-Pacific maritime security**.
- **India-UAE:** Focus on **defence industry**, **counterterrorism** and **special operations**.
- **India-Australia:** Expanded cooperation in **MDA**, **defence exercises** and **Indo-Pacific stability**.
- **SCO:** India promotes **counterterrorism**, **terror financing control** and **WMD threat reduction**.
- **ASEAN/ADMM-Plus:** Focus on **freedom of navigation**, **maritime stability**, **capacity building** and **BrahMos cooperation**.
- **QUAD:** Strengthened through **Malabar** and **Quad-at-Sea Mission**, focusing on **maritime security** and **Indo-Pacific stability**.
- **Indo-Pacific:** **SAGAR** and **MAHASAGAR** position India as a **net security provider** in the **IOR**.

India's defence sector has undergone a major structural transformation in the last decade. Discuss its impact on self-reliance, industrial growth, and strategic autonomy.

Practice Mains Question

Tungabhadra River Dispute

The Union Government has proposed a **High-Level Committee after Karnataka, Andhra Pradesh, and Telangana** reached consensus to resolve the long-standing Tungabhadra water-sharing dispute.

- **River Origin & Course:** The **Tungabhadra**, a **531-km perennial right-bank tributary of the Krishna**, is formed by the **Tunga and Bhadra rivers** originating in Karnataka's Western Ghats and meeting at **Kudli near Shivamogga**.
 - It flows through Karnataka, forms parts of the **Karnataka–Andhra Pradesh** and **Andhra Pradesh–Telangana** borders, and joins the Krishna at **Sangameswaram, Andhra Pradesh**. The Krishna enters the **Bay of Bengal at Hamsaladeevi**.
- **The Tungabhadra Reservoir Project:** Commissioned in the **1950s** near **Hosapete, Vijayanagara district**, the inter-state dam irrigates over **16.38 lakh acres** across Karnataka, Andhra Pradesh, and Telangana.
- **Historical Adjudication:** **KWDT-I, 1969**, headed by **Justice R.S. Bachawat**, allocated water in a **65:35 ratio** between Karnataka and undivided Andhra Pradesh. The system is managed by the **Tungabhadra Board**.
- **Current Allocations:** After the **2014 bifurcation**, Telangana received an entitlement of **15.9 TMC**.
- **Siltation Crisis:** Heavy siltation has reduced the dam's original **134 TMC** capacity, causing downstream shortages. Decay in the **Rajolibanda Diversion Scheme (RDS)** limits Telangana's utilisation to only **5–6 TMC**.
 - The **RDS** is an inter-state irrigation project of Karnataka and Telangana across the Tungabhadra. Its **143-km canal** irrigates drought-prone areas in Karnataka, Telangana, and Andhra Pradesh.
- **Upper Bhadra Project Conflict:** Andhra Pradesh filed an original suit in the **Supreme Court in 2023** against Karnataka's **Upper Bhadra lift irrigation scheme**, claiming it may reduce downstream flows to reservoirs such as **Srisailem**.
- **Constitutional Distribution:** Water is under the **State List, Entry 17**, while regulation and development of inter-state rivers fall under the **Union List, Entry 56**.
- **Dispute Resolution Framework:** **Article 262** empowers Parliament to adjudicate inter-state river disputes and exclude SC jurisdiction through the **Inter-State River Water Disputes Act, 1956**.
- **Landmark Precedent:** In the **Cauvery Judgment, 2018**, SC held that **inter-state rivers are national assets** and must be shared through equitable apportionment.

Economic Scenario

Polymer Currency Notes

The RBI is revisiting the plan to introduce **polymer currency notes**. BRBNMPL, an RBI subsidiary, invited global EOI for BOPP-based polymer substrate.

Polymer Currency Notes

- **About:** Polymer notes are made from **BOPP (Biaxially Oriented Polypropylene)**, unlike Indian paper currency made from **cotton-pulp paper**.
 - They are **durable, water-resistant, tear-resistant and harder to counterfeit** due to features like transparent windows and embedded security elements.
 - Used in **60+ countries**; **Australia** introduced the first full polymer series in **1988**.
 - Last around **2.5–4 times longer** than paper notes.
- **Need for Polymer Notes**
 - India spends nearly **₹5,000 crore annually** replacing damaged currency.
 - Around **20–24 billion soiled notes** are destroyed annually under RBI's **Clean Note Policy**.
 - Low denominations like **₹10 and ₹20** deteriorate quickly due to frequent circulation.
 - Longer lifespan can reduce replacement costs and improve security.
 - **TERI study:** Despite higher initial carbon footprint, polymer notes have lower lifecycle emissions due to fewer replacements.
- **Timeline in India**
 - **2009:** RBI proposed **100 crore ₹10 polymer notes**.
 - **2012:** Pilot project approved for **Kochi, Mysuru, Jaipur, Bhubaneswar and Shimla**; later shelved.
 - **2026:** **BRBNMPL** invited bids for polymer substrates, reviving the proposal.
- **Concerns**
 - **Economic:** Polymer notes cost **30–60% more** to produce; depend on polypropylene, exposing costs to crude oil fluctuations and import dependence.
 - **Infrastructure:** Requires recalibration of **ATMs, vending machines, sorting and processing equipment**.
 - **Environmental:** Raises concerns regarding plastic waste and recycling facilities.
 - **Digital payments:** Growth of **UPI** does not eliminate cash demand; currency in circulation exceeded **₹41 lakh crore (2025–26)**.

- **Need for balance:** Cash remains important for informal sectors, rural areas, small merchants and emergencies; digital and physical currency need to coexist.

RBI's Clean Note Policy

- **About:** Launched by **RBI in 1999** to maintain circulation of **clean, good-quality notes and coins** by withdrawing soiled and mutilated currency.
- **Objectives**
 - Improve currency quality and lifespan.
 - Enhance public convenience and confidence in the currency system.
- **Key Measures:** Banks must issue only **clean notes** and avoid recirculating soiled notes.
 - **Soiled and mutilated notes** can be exchanged at currency chest branches, including by non-customers.
 - **Stapling of note packets** was replaced with paper/polythene banding to reduce damage.
 - RBI uses **Currency Verification and Processing Systems (CVPS)** for sorting, verification, shredding and briquetting of unfit notes.
 - Banks are encouraged to ensure adequate coin availability and currency exchange facilities, especially in **rural and semi-urban areas**.

Currency Management in India

- **Constitutional and Legal Framework**
 - Currency, coinage, legal tender and foreign exchange are under the **Union List** of the Constitution.
 - **RBI Act, 1934:** Provides the legal framework for issuance and management of banknotes.
 - **Coinage Act, 2011:** Governs design, production and circulation of coins.
- **Role of RBI**
 - Has exclusive authority to **issue banknotes**, estimate currency demand and maintain adequate note supply.
 - Ensures circulation of **clean and good-quality currency**.
 - Distributes coins supplied by the **Government of India**.
- **Role of Government of India**
 - Issues the **₹1 note**.
 - Designs and mints coins.
 - Approves banknote design and material based on recommendations of the **RBI Central Board**.
 - Provides **sovereign guarantee** for all RBI-issued banknotes.

India's Rising Inflation

India's **WPI inflation** rose to nearly **10%**, its sharpest increase in over a decade, excluding the Covid-19 period.

Current Inflation Status in India

- **CPI inflation:** Rose to an **18-month high of 4.38% in June 2026**, but remained within the RBI's **2–6% target band**.
- **WPI inflation:** Increased to **9.87%**, driven mainly by **fuel and power (27.41%), minerals and manufactured products**.
- **Food inflation:** Remains volatile (crossing the 5–6% threshold), especially due to **cereals, pulses and vegetables**.
- **Core inflation:** Persistently high in services such as **transport, health and education, limiting the RBI's maneuverability**.
- **Inflation trend:** The narrowing WPI–CPI gap, inflation is mainly **cost-push and structural**, with higher input and logistics costs passing from wholesale to retail prices.
 - **Sticky core inflation** keeps price pressures persistent and limits the **RBI's policy flexibility**.

Key Drivers of Inflationary Pressures

- **Energy shock:** India imports nearly **90% of crude oil**; high global prices and West Asian disruptions raise import, transport and production costs.
- **Rupee depreciation:** A weaker rupee makes imported fuel, fertilizers, electronics and raw materials costlier.
- **Food inflation:** With **36.75% weight in CPI**, monsoon variability and **El Niño** sharply affect cereal, pulse and vegetable prices.
 - This structural vulnerability leads to the classic **cobweb phenomenon** in agricultural pricing and **driving the CPI upward**.
- **Precious metals:** To manage the **CAD**, the government raised import duties on gold and silver from **6% to 15%**.
 - However, such duties were **passed to consumers & driving up 'personal care and miscellaneous' inflation**.
- **Fuel tax rigidity:** High central and state taxes prevent a significant fall in petrol and diesel prices.
- **Policy limitation:** **Repo-rate hikes** are less effective against supply-driven and **cost-push inflation**.

Impact of Rising Inflation

- **Real income erosion:** Rising prices reduce purchasing power, especially when wages do not rise proportionately.
- **Burden on the poor:** High food and fuel (45% of CPI basket) costs disproportionately affect low-income, rural and informal-sector households.

- **Monetary tightening:** Persistent inflation limits the RBI's scope to cut policy rates.
- **Higher borrowing costs:** Elevated interest rates weaken consumption, credit growth and private investment.
- **Fiscal pressure:** Greater spending on food, fertilizer and welfare subsidies complicates fiscal consolidation under the **FRBM Act, 2003**.
- **External vulnerability:** Costlier crude widens the **trade deficit and CAD**, weakens the rupee and fuels imported inflation.

Challenges in Curbing Inflation

- **Stagflation risk:** Tight monetary policy may suppress investment, growth and jobs, while supply-side constraints keep prices high.
- **Growth–inflation dilemma:** Under the **FIT target of 4% ± 2%**, the **MPC** must control inflation without excessively raising capital costs or weakening credit and private CapEx.
- **Imported inflation trap:** If the RBI avoids tightening, capital outflows may weaken the rupee and raise import costs of **crude oil, fertilizers, edible oils and industrial inputs**.
- **Fiscal constraints:** Cuts in **excise duty and VAT on fuels** may ease inflation but reduce Centre–State revenues and weaken fiscal consolidation.
- **Monsoon dependence:** Heavy agricultural dependence on rainfall leaves food output and prices vulnerable to weather shocks.
- **Supply-chain bottlenecks:** Fragmented markets, inadequate cold chains and post-harvest losses can convert local shortages into nationwide food inflation.
- **External dependence:** India's **85–90% crude import dependence**, along with imports of fertilizers, edible oils and electronics, limits the ability of domestic policy to fully control inflation.

Measures Needed to Curb Inflation

Short-Term Measures

- **N.K. Singh Committee, 2017:** Adopt a **countercyclical fiscal framework** to address inflationary shocks while maintaining fiscal discipline.
- **Shanta Kumar Committee, 2015:** Reform **FCI**, rationalise procurement and promote decentralised foodgrain procurement.
- Use the **Price Stabilisation Fund (PSF)**, **Open Market Sale Scheme (OMSS)** and calibrated trade policies to contain food inflation.

Medium-Term Supply-Side Reforms

- **Dalwai Committee, 2018:** Expand **micro-irrigation, FPOs, cold chains, value addition and market linkages** to reduce post-harvest losses.

- **M.S. Swaminathan Commission:** Increase investment in irrigation, watershed development, agricultural R&D and extension services to reduce monsoon dependence.

Long-Term Structural Reforms

- **National Green Hydrogen Mission and National Electricity Plan:** Reduce imported energy inflation through renewable energy and green hydrogen.
- **National Critical Mineral Mission:** Boost domestic production and secure global supply chains for critical minerals.
- **National Logistics Policy and PM Gati Shakti:** Improve multimodal logistics, warehousing and supply chains to reduce transport and structural costs.

India's 2026 inflationary episode is primarily supply-driven rather than demand-driven. Examine the causes and suggest a comprehensive policy response.

Practice Mains Question

Index of Core Industries with Base Year 2022-23

The **Office of Economic Adviser (OEA)** under **DPIIT, Ministry of Commerce and Industry**, released the first provisional **Index of Core Industries (ICI)** with **2022–23 as the base year**.

- The new series replaces the **2011–12 base year** and applies retrospectively from **April 2023**.

Index of Core Industries (ICI)

- **About:**
 - The **ICI** measures the monthly production performance of India's key industrial sectors.
 - The provisional index is generally released on the **20th of the following month** or the next working day.
- **Connection to IIP:**

- Under the **2022–23 base year**, core industries carry a **32.88% weight in the IIP**, compared with **40.27%** under the 2011–12 series.

Key Changes in the 2022–23 Series

- **Iron Ore** has been added, increasing the number of core industries from **eight to nine**.
 - The sectors are **Coal, Crude Oil, Natural Gas, Refinery Products, Fertilizers, Steel, Cement, Electricity and Iron Ore**.
- The **Steel Index** now uses **gross production data** instead of net production data.
- Only **Raw Coal** is retained; **Coal Middling and Washed Coal** are excluded to avoid double counting.
- Sectoral weights are aligned with the revised **IIP** and normalized to **100**.
 - **Electricity** has the highest weight (**30.93%**), followed by **Refinery Products (22.57%)**, **Steel (17.58%)**, **Crude Oil (7.43%)**, **Coal (5.59%)**, **Iron Ore (4.90%)**, **Cement (4.41%)**, **Natural Gas (3.84%)**, and **Fertilizers (2.73%)**.
- **Linking Factor:** A **linking factor** ensures comparability between the old and new series.
 - The overall ICI linking factor is **1.47**.

ICI Performance: June 2026

- ICI grew by **5.0% year-on-year**, up from **3.2% in May 2026**.
- Positive growth was recorded in **Iron Ore (43.9%)**, **Electricity (9.8%)**, **Cement (9.8%)**, **Steel (4.6%)** & **Coal (1.4%)**.
- **Natural Gas, Crude Oil, Refinery Products and Fertilizers** contracted.
- **Iron Ore and Electricity** were the main growth drivers.
- Cumulative ICI growth during **April–June 2026** was **3.6%**, compared with **1.0%** in the same period last year.

Difference between ICI and IIP		
Basis	ICI	IIP
Scope	Tracks production of 9 core infrastructure industries .	Tracks overall industrial production.
Compiling Agency	OEA, DPIIT, Ministry of Commerce and Industry	NSO, MoSPI
Coverage	9 core industries	Mining, Manufacturing and Electricity sectors
Relationship	A subset and leading indicator of the IIP.	A broader measure of India's industrial performance.
Utility	Assesses core-sector health and supply-side bottlenecks.	Supports assessment of industrial growth & monetary & fiscal policymaking .

NITI Aayog's Investment Friendliness Index 2026

NITI Aayog released the first **IFI 2026** to assess how prepared **States and UTs** are to attract, facilitate, and retain domestic and foreign investment. It supports **Viksit Bharat @2047** by promoting **competitive and cooperative federalism**.

Investment Friendliness Index (IFI)

- **Genesis and Mandate:** Conceived after the **9th NITI Aayog Governing Council Meeting, 2024** and announced in the **Union Budget 2025–26** as a reform instrument.

- **Methodological Shift:** Replaces the discontinued **World Bank Ease of Doing Business framework** with a homegrown methodology using **84 objective indicators** and **investor perception surveys**.
- **The Eight Pillars of Evaluation:** Assesses **28 States and 8 UTs** through a **100-point composite score** based on eight pillars:
 - **Infrastructure:** Logistics, power reliability and digital connectivity.
 - **Business Climate:** Single-window clearances and operational continuity.
 - **Resources:** Industrial land, skilled labour and raw materials.
 - **Government Policy:** Policy predictability and stability.
 - **Regulatory Ease:** Lower bureaucratic and compliance burden.
 - **Institutional Environment:** Capacity, transparency and responsiveness.
 - **Financial Health:** Fiscal prudence and macroeconomic indicators.
 - **Environmental Resilience:** Sustainability and ecological compliance.
- **Performance Stratification:**

Category	Score
Top Performers	Above 50
Frontrunners	45–50
Emerging Performers	40–45
Aspiring States	Below 40

Key Findings of the Report

- **Top Performers:** Gujarat (56.6) ranked first, followed by Maharashtra (53.7), Tamil Nadu (53.3), Goa (53.1) and Odisha (52.4).
- **Category-Specific Toppers:**

Category	Leading States/UTs
Large States	Gujarat, Maharashtra, Tamil Nadu
Hilly and North-Eastern States	Uttarakhand, Assam, Himachal Pradesh
UTs and City States	Delhi, Chandigarh

- **Drivers of Success:**
 - **Gujarat:** Efficient ports, reliable power, around **31% of India's merchandise exports** and the lowest fiscal deficit-to-GSDP ratio of **2.81%** in FY 2024.
 - **Maharashtra:** Best business climate, **35% of national private equity/venture capital (PE/VC) investment** and the highest number of **Atal Tinkering Labs**.

- **Tamil Nadu:** Strong infrastructure, exports and near-**100% MoU conversion**.
- **Goa:** Leads in resources and regulatory ease among city states/UTs, supported by skilling, healthcare spending and renewable power.

Challenges in Attracting Investments Across States

Macro-Level Challenges

- **Sharp Regional FDI Divergence:** Maharashtra, Karnataka, Gujarat, Delhi and Tamil Nadu receive nearly **85% of India's FDI inflows**. The score gap between Gujarat (56.6) and Lakshadweep (24.5) exceeds **32 points**.
- **Persistent Investor Pain Points:** Long approval timelines, difficult land access and skilled-labour shortages, especially in high-value manufacturing.
- **Incentives Alone Are Not Enough:** Sustained investment requires a **transparent, predictable and stable regulatory environment**, not merely fiscal incentives.
- **Even Top Performers Have Scope for Reform:** No State scored above **60/100**; the highest score was **56.6**, indicating gaps across all investment ecosystems.

Recurring Issues Identified in State Profiles

- **Human Capital Constraints:** Skilled-labour shortages and talent migration affect J&K, Chhattisgarh, Rajasthan, Tripura, Nagaland and Kanpur.
- **Weak Single-Window Clearance Mechanisms:** In J&K, manual complaint registration continues despite an online system.
- **Safety and Law-and-Order Concerns:** In Bihar, security concerns and weak air connectivity to **Tier-II/III cities** reduce investor confidence.
- **Fiscal Stress:** Bihar's liabilities-to-GSDP ratio is nearly **500 basis points** above the national average. In J&K, fiscal deficit is around **9% of GSDP** and interest payments are **7.13% of GSDP**.
- **Digital Infrastructure Gaps:** J&K's **4G/5G penetration** is low, with Base Transceiver Station (BTS) density of only **0.82 per sq. km**; several states also face weak last-mile connectivity.
- **Environmental Risks:** Industrial clusters in **Noida, Ghaziabad and Muzaffarnagar** record AQI levels of **200–600** for much of the year. Flood risks and poor drainage also affect productivity and livability.
- **Logistics Bottlenecks:** Poor roads, weak industrial park connectivity, and inadequate warehousing and cold-storage remain major constraints.
- **Airport Connectivity Gaps:** Limited Tier-II/III connectivity in Bihar, need for an operational airport in **Sikkim**, congestion at **Chennai Airport**, and weak airport infrastructure in **Odisha and Chandigarh** hinder investment.

- **Slow Incentive Disbursement: Andhra Pradesh and Telangana** face delayed incentive payments and weak inter-departmental coordination.

Measures to Improve Investment Competitiveness

- **Accelerate Land and Labour Reforms:** Simplify land allotment, improve labour flexibility and align skilling with industry needs through **Skill India Mission** and **PMKVY**.
- **Operationalise Genuine Single-Window Systems:** Fully integrate clearances under **National Single Window System (NSWS)**, remove manual approvals and reduce approval time from **6–18 months** to **30–60 days**.
- **Adopt Targeted Industrial Policies:** Use sector-specific incentives under the **PLI Scheme**, following models such as **Gujarat's Semiconductor Policy** and **Tamil Nadu's EV Policy**.
- **Follow an Infrastructure-First Strategy:** Expand industrial ecosystems through **PM Gati Shakti**, **NICDP**, **GIFT City**, **Samruddhi Corridor** and the **Chennai–Bengaluru Industrial Corridor**.
- **Strengthen the Business Climate:** Reduce compliance burdens, simplify approvals and improve industry-relevant skilling.
- **Enhance Fiscal Sustainability:** Lower debt, diversify revenue and create fiscal space for infrastructure and incentives.
- **Bridge Infrastructure Gaps:** Improve power supply through **Revamped Distribution Sector Scheme (RDSS)**, digital connectivity through **BharatNet**, and logistics and warehousing under **PM Gati Shakti**.
- **Strengthen Institutional Governance:** Ensure policy stability, transparent land allotment, faster environmental clearances and effective grievance redressal.
- **Build Climate-Resilient Infrastructure:** Promote sustainable industrial growth through **NAPCC** and climate-resilient planning.
- **Institutionalise the State-Level Investment Index:** Use it regularly for benchmarking, peer learning, evidence-based policymaking and tracking reforms.

Despite policy reforms, investment remains concentrated in a few Indian states. Analyse the structural challenges responsible for regional investment disparities.

Practice Mains Question

Strengthening Rural Credit for Inclusive Growth in India

India's rural credit system has evolved into a diversified institutional network of **banks, NABARD, cooperatives, SHGs, digital platforms, and financial inclusion schemes**, supporting agriculture, rural enterprises, household resilience, and inclusive growth.

Current Rural Credit Landscape in India

- **About:** India's rural credit system has shifted from informal lending to a diversified, technology-enabled institutional framework.
- **Scope of Credit:** Supports **agriculture, allied activities, rural enterprises, and household needs** through short-, medium-, and long-term credit.
- **Developmental Role:** Promotes **income, employment, asset creation, consumption, and household resilience**.
- **Institutional Network:** Includes **Scheduled Commercial Banks, Regional Rural Banks, Cooperative Banks, Small Finance Banks, NABARD**, and non-institutional sources.
- **Role of NABARD:** Provides refinance, funds rural infrastructure, promotes institutional development, and supervises **Cooperative Banks and Regional Rural Banks**.
- **Expanding Formal Access:** As of **May 2026**, around **51%** of rural households used only formal credit, while over **27%** used both formal and informal sources.
- **Rural Economic Conditions:** **~77.2%** of rural households reported higher consumption.
- **Policy Support:** **Priority Sector Lending, ground-level credit targets, and the Modified Interest Subvention Scheme** improve access to affordable institutional credit.
- **Digital and Financial Inclusion:** Digital platforms and financial-inclusion measures have improved credit reach, speed, and efficiency while reducing dependence on moneylenders.

Evolution of Rural Credit System

- **Rural Banking Push (1955):** Creation of the **National Agricultural Credit (Long-term Operations) Fund** and establishment of **SBI** expanded rural banking and agricultural finance.
- **Bank Nationalisation (1969):** Nationalisation of **14 major banks** redirected credit towards priority sectors, especially small farmers and rural borrowers.
- **Establishment of NABARD (1982):** **NABARD** became the apex rural development institution, providing finance, supervision, district credit planning, and financial inclusion support.
- **SHG-Bank Linkage (1992):** Expanded formal credit access, particularly for **women and low-income rural households**.
- **Kisan Credit Card (1998):** Improved timely and affordable farm and allied-sector credit.
- **PMJDY (2014):** Expanded universal banking and strengthened the **JAM Trinity**, enabling transparent delivery of credit, insurance, and DBT.
- **MUDRA Scheme (2015):** Provided collateral-free loans to non-corporate, non-farm micro and small enterprises, promoting rural entrepreneurship.

- **Digital Credit Delivery (2022 Onwards):** Platforms such as **Jan Samarth Portal** and **e-KCC** made rural credit more accessible and technology-driven.

Institutional Architecture of Rural Credit in India

- **Strong Institutional Network:** Rural credit is delivered through **Scheduled Commercial Banks, Regional Rural Banks, Cooperative Banks, Small Finance Banks, and NABARD.**
- **Scheduled Commercial Banks:** SCBs, listed in the **Second Schedule of the RBI Act, 1934**, provide rural banking through branches, Business Correspondents, digital platforms, **PMJDY**, and **DBT.**
- **SCB Rural Expansion:** Rural SCB branches increased from **41,464 in 2014 to 56,193 by July 2025.**
- **Regional Rural Banks:** Established under the **RRB Act, 1976**, they serve small and marginal farmers, labourers, artisans, and rural entrepreneurs. **28 RRBs** operate through over **22,000 branches** in around **700 districts.**
- **Co-operative Banks:** Extend institutional credit to poor and remote communities through **1,458 Urban Cooperative Banks, 34 State Cooperative Banks, and 352 District Central Cooperative Banks.**
- **Small Finance Banks:** RBI-licensed banks serving underserved groups, including small businesses, farmers, micro-industries, and unorganised-sector entities through low-cost, technology-driven operations. India has **11 operational Small Finance Banks.**

Policy Framework for Rural Credit in India

- **Priority Sector Lending Framework:** RBI mandates banks to direct credit towards critical and underserved sectors.
- **Agriculture Credit Target:** At least **18% of Adjusted Net Bank Credit** or equivalent off-balance sheet exposure must go to agriculture.
- **Farmer Sub-Targets:** **14%** for non-corporate farmers and **10%** for small and marginal farmers.
- **Ground Level Credit:** Annual targets cover crop and term loans for agriculture and allied sectors.
- **Rising Credit Flow:** Agricultural credit grew by over **13% annually during FY15–FY24.** The **FY 2025–26 target is ₹32.50 lakh crore.**
- **Allied Sector Focus:** **₹5 lakh crore** is targeted for animal husbandry, dairying, and fisheries.
- **SHG-Bank Linkage:** NABARD's **SHG-BLP** connects rural SHGs with banks, especially benefiting women.
- **Strengthening SHG-Bank Linkage:** **DAY-NRLM** promotes women-led SHGs. Over **10.05 crore women** were organised into more than **90.90 lakh SHGs** by July 2025.

- **Bank Sakhis:** Around **50,548 Bank Sakhis** assist SHGs with accounts, loans, repayment, and credit linkage.
- **Primary Agricultural Credit Societies:** PACS provide short-term credit and support input distribution and agricultural marketing.
- **PACS Expansion:** The Government approved **2 lakh new multipurpose PACS, dairy, and fishery cooperatives** over five years.
- **PACS Digitisation:** Of **79,630 approved PACS, 61,842** migrated to the common ERP software by March 2026.
- **Modified Interest Subvention Scheme:** Crop loans through KCC are available at **7% interest**, reduced to **4%** on prompt repayment.
- **Enhanced KCC Limits:** The MISS loan limit increased from **₹3 lakh to ₹5 lakh**; for fisheries and allied activities, from **₹2 lakh to ₹5 lakh.**
- **Collateral-Free Credit:** The limit for collateral-free short-term agricultural loans rose from **₹1.6 lakh to ₹2 lakh per borrower** from January 2025.
- **PM Dhan Dhanya Krishi Yojana:** Approved in July 2025, it targets **100 low-performing agricultural districts** through convergence of **36 Central schemes across 11 Ministries.**
- **Objectives:** Improve productivity, crop diversification, sustainable farming, irrigation, post-harvest storage, and access to farm credit.

Rural credit is a key instrument for inclusive growth and agricultural transformation in India. Discuss the role of financial inclusion schemes in strengthening India's rural economy.

Practice Mains Question

Mobile Phone Manufacturing Scheme (MPMS)

The Union Cabinet approved the **₹62,500 crore MPMS** to strengthen India's mobile manufacturing ecosystem.

MPMS

- **About:** MPMS aims to expand mobile phone production, increase domestic value addition, strengthen supply chains, and improve global competitiveness.
- **Scheme Tenure:** Operates for **five years**, from **FY 2026–27 to FY 2030–31.**
- **Incentive Support:** Provides **2.25%–5% incentive** on eligible sales of mobile phones manufactured in India.
 - Offers up to **1.5% additional incentive** for domestic sourcing of key components.
 - Indian brands receive an additional **3% incentive** for product design and R&D.

Objectives of MPMS

- **Scale Up Production:** Expand manufacturing capacity and strengthen India's global position.

- **Deepen Domestic Value Addition:** Increase local sourcing and reduce import dependence.
- **Strengthen Supply Chains:** Build resilient and globally competitive electronics supply chains.
- **Build Indian Brands:** Promote domestic brands, design, patents, and R&D.
- **Promote Technological Sovereignty:** Encourage domestic innovation and intellectual property creation.

Expected Outcomes

- **Production Target:** Around ₹39 lakh crore in cumulative production.
- **Export Expansion:** Higher mobile phone exports and stronger participation in global value chains.
- **Employment Generation:** Around 60,000 direct jobs.
- **Global Manufacturing Hub:** Strengthen India's position in mobile manufacturing and exports.

Current Status of India's

Electronics Manufacturing Growth

- **Policy Background:** MPMS succeeds the Production Linked Incentive Scheme for Large Scale Electronics Manufacturing (PLI-LSEM), which ended on 31st March 2026 after strengthening mobile manufacturing and exports.
- **Electronics Production Growth:** Production rose nearly seven-fold, from ₹1.90 lakh crore in 2014–15 to an estimated ₹13.11 lakh crore in 2025–26.
- **Electronics Export Growth:** Exports increased eleven-fold, from ₹38,263 crore to ₹4.24 lakh crore.
- **Export Significance:** Electronics became India's third-largest export category, reaching USD 47.96 billion in FY 2025–26.
- **Domestic Value Addition:** Increased from 15% to 23%, indicating greater localisation.
- **Component Ecosystem:** Over 40 major component manufacturers established or expanded operations, supported by Tier-2, Tier-3, and Tier-4 suppliers.

Mobile Manufacturing as Growth Anchor

- **About:** Mobile phones are the main driver of India's electronics production and export growth.
- **India's Global Position:** India is the world's second-largest mobile phone manufacturer by volume.
 - 99.2% of mobile phones used domestically are manufactured in India.
- **Mobile Phone Production:** Increased 33-fold, from ₹18,900 crore in 2014–15 to ₹6.27 lakh crore in 2025–26.
- **Mobile Phone Exports:** Rose 165-fold, from ₹1,566 crore to ₹2.60 lakh crore.

- **Largest Export Product:** Mobile phones rose from India's 153rd-largest export item in 2014–15 to the largest export product in 2025–26.

- **Smartphone Export Success:** Smartphones became India's single-largest exported product category in 2025, surpassing diesel fuel and cut diamonds.

- Women constitute nearly 30% of the electronics manufacturing workforce.
- In mobile phone manufacturing, they account for nearly 70% of the direct workforce.

Empowering India's Skill Ecosystem

The Government is strengthening India's skill ecosystem through education reforms, vocational training, AI skilling, apprenticeships, entrepreneurship, and women-centric programmes.

- The India Skills Report 2026 shows employability rising from 46% in 2020 to 56.4% in 2026, supporting the goal of 100% skilled labour with meaningful employment by 2047.

Importance of Skill Development in India

- **Strengthening AI Readiness:** India scored 49.3 on the IMF AI Preparedness Index, above the 42.1 average for emerging and developing economies.
- **Harnessing the Demographic Dividend:** Over 54% of India's population is below 25 years, while more than 62% is in the 15–59 years working-age group.
 - This advantage is expected to continue until around 2040.
- **Meeting Global Talent Demand:** India may have a surplus of 45 million skilled professionals by 2030, while global shortages could reach 85 million.

Major Skill Building Initiative from an Early Age in India

- **Samagra Shiksha:** Supports education from pre-school to Class XII and promotes skill education aligned with SDG-4.
 - By Dec 2025, 138 job roles were approved across 25,140 schools, benefiting over 35.5 lakh students.
 - **Hub and Spoke Model:** Enables nearby schools to use skill-training infrastructure available in hub schools. Around 975 spoke schools are covered.
 - **Vocational Skill Labs:** MSDE established 1,200 labs in 400 Jawahar Navodaya Vidyalayas and 200 Eklatva Model Residential Schools.
- **PM SHRI Schools:** Launched in 2022 to implement NEP 2020 and develop 21st-century skills.
 - By June 2026, 13,092 schools were developed across 776 districts against a target of 14,500.

- **Atal Tinkering Laboratories:** Over **10,000** laboratories established in **722 districts**, engaging more than **1.1 crore students**.
- **Skilling for AI Readiness (SOAR):** Provides AI training to **Classes VI–XII students and educators** with **Microsoft, HCL Technologies, and NASSCOM**.
 - **2.30 lakh students** were enrolled by **Feb 2026**.
- **National Skills Qualifications Framework (NSQF):** An **eight-level, competency-based framework** integrating academic, vocational, and skill education.
 - Enables credit-based progression, lifelong learning, and career mobility.
- **YUVA AI for ALL:** A free **4.5-hour AI course** launched under the **IndiaAI Mission in November 2025**. Enrolment reached **85.27 lakh** by **8th July 2026**.
- **Vidyanjali Programme:** Connects schools with volunteers, professionals, NGOs, and CSR partners for mentoring and skill support.
 - By **June 2026**, **8,44,925 schools**, **5,62,296 volunteers**, and **2,705 organisations** were onboarded, benefiting over **2.04 crore students**.

Major Upskilling and Re-Skilling Initiative in India

- **Skill India Mission:** Launched in **2015**, it provides **NSQF-aligned skilling, upskilling, and reskilling** through Rozgar Melas, Skill India Digital Hub, and Skill Development Centres.
 - Major schemes include **PMKVY, Jan Shikshan Sansthan, NAPS, and Craftsman Training Scheme**.
- **PMKVY 4.0:** Demand-driven short-term skilling programme.
 - By **30th June 2026**, **28.17 lakh candidates** were trained across **36 sectors, 36 States, and 738 districts**.
- **Jan Shikshan Sansthan:** Provides government-funded, community-based training to neo-literates and school dropouts through NGOs.
 - Since **2018**, it trained **36.52 lakh beneficiaries**, including **4.8 lakh tribal beneficiaries**; women formed over **82%**.
 - Offers locally relevant courses, while **UdyamKart** supports product marketing.
- **National Apprenticeship Promotion Scheme:** Promotes industry-led apprenticeship under the **Earn While You Learn** model.
 - Under **NAPS 2.0**, the government pays **25% of the stipend**, up to **₹1,500 per month**.
 - Over **54.41 lakh apprentices** have been engaged since **2016**.
 - During **2025–26**, **₹562.75 crore** was transferred through **40.10 lakh DBT transactions**.

- **Craftsman Training Scheme:** Offers **169 long-term vocational courses** through **13,888 ITIs**.
 - In three years, **14 new courses** were introduced and **22 revised**.
 - ITI enrolment rose from **12.51 lakh in 2022–23** to **14.70 lakh in 2025–26**.
- **PM-SETU:** Launched in **2025** to modernise **1,000 Government ITIs**.
 - Includes **200 Hub ITIs, 800 Spoke ITIs, and 5 National Centres of Excellence for Skilling**.
 - Targets training **20 lakh youth in five years**.
 - By **May 2026**, **32 States and UTs** had formed steering committees; **Andhra Pradesh** first operationalised it through the **Visakhapatnam ITI Cluster**.
- **FutureSkills Pillar:** Under the **IndiaAI Mission**, it promotes AI education, fellowships, and practical training.
 - Plans **570 AI and Data Labs** in Tier-2 and Tier-3 cities.
 - **FutureSkills Prime:** A **NASSCOM-MeitY** initiative providing online training in emerging technologies.
- **Pradhan Mantri Viksit Bharat Rozgar Yojana:** Launched in **2025**, it gives first-time EPFO-registered employees up to **₹15,000**.
 - Employers receive up to **₹3,000 per month for two years** for eligible additional employees.
 - By **19th June 2026**, it supported **15 lakh employment opportunities** with **₹2,400 crore** disbursed.

Initiatives Related to Nurturing Entrepreneurship in India

- **PM Vishwakarma:** Provides traditional artisans and craftspeople with **skill training, toolkit incentives, credit support, and market linkages**.
 - Prioritises **women, SCs, STs, OBCs, persons with disabilities, transgender persons, and artisans from the North East, islands, and hilly regions**.
 - Beneficiaries receive a training stipend of **₹500 per day**.
 - By **June 2026**, over **24.50 lakh applicants** completed basic training and loans worth **₹5,165.84 crore** were sanctioned.
- **Startup India Courses:** Offers free online courses in **programming, security, accounting, finance, management, and entrepreneurship** to strengthen startup capabilities.
- **Entrepreneurship Development Programmes:** Implemented by **MSDE** through **NIESBUD, Noida, and IIE, Guwahati**.
 - Provide training, mentorship, handholding, management support, and incubation.
- **MSME Entrepreneurship Skill Development Programme:** Encourages youth to pursue **self-employment and entrepreneurship** through technical and business training.

- Focuses on **SCs, STs, women, persons with disabilities, ex-servicemen, and BPL beneficiaries.**

Initiatives Related to

Empowering Women through Skilling

- **AI Careers for Women:** A **320-hour programme launched in 2025** to equip women in rural undergraduate colleges with AI skills.
 - Operates through **25 AI Centres of Excellence** under a hub-and-spoke model.
 - Reached over **10,000 students**, including **2,500+ in Hub Colleges** and **7,500+ in Spoke Colleges**.
- **Swavalambini Programme:** Launched in **Feb 2025** by **MSDE** with **NITI Aayog's Women Entrepreneurship Platform**.
 - Promotes entrepreneurship through awareness and development programmes across **Assam, Meghalaya, Mizoram, Uttar Pradesh, and Telangana**.
 - In **2025–26**, it trained **82 faculty members**, reached **1,200 awareness participants**, and conducted development programmes for **602 women**.
- **NAVYA Scheme:** Launched in **June 2025** by **MSDE** and the **Ministry of Women and Child Development**.
 - Trains girls aged **16–18 years** in aspirational districts for socio-economic independence.
 - Covers **27 districts across 19 States** and targets **3,850 girls under PMKVY 4.0**.
 - Focuses on **digital marketing, cybersecurity, AI-enabled services, green jobs, life skills, financial literacy, and digital competencies**.
 - **Gender-Inclusive Skilling:** Promotes women's employability, self-employment, entrepreneurship, and skill access in rural and underserved areas.

Skill development is central to India's demographic dividend and Viksit Bharat 2047 vision. Discuss the role of women-centric initiatives in building a future-ready workforce.

Practice Mains Question

Weaving Sustainability into India's Textile Future

India's textile sector is promoting **circular and sustainable production** through **organic fibres, safer chemicals, recycling, eco-labelling, traceability, cleaner technologies** and **responsible sourcing** to improve **resource efficiency** and **global competitiveness**.

Current Status of India's Textile Sector

- **Economic Role:** Contributes about **~2% of GDP** and **~11% of manufacturing GVA**.
- **Exports and Employment:** India is the **sixth-largest textile and apparel exporter**, with around **4% of global exports**, and the sector employs over **45 million people**.

- **Need for Sustainability:** Environmentally responsible production is essential for maintaining **global competitiveness**.
- **Shift towards Circularity:** Circular production is growing due to India's resource-conscious practices and rising demand for low-impact products.
- **Value-Chain Approach:** Sustainability affects **raw materials, manufacturing, quality standards and market access**.

Circularity in the Textile Sector

- **About:** Keeps fibres, fabrics and garments in use through **reuse, repair, recycling, upcycling and recovery**, reducing dependence on virgin materials.
- **Environmental Importance:** Reduces **waste, emissions, landfill pressure** and the use of **water, energy and chemicals**.
- **Waste Recovery:** India generates nearly **7.8 million tonnes** of textile waste annually; over **90%** comes from domestic sources.
- **Recycling Performance:** Over **70%** of textile waste is recovered, while nearly **95% of pre-consumer waste** is collected and reused.
- **Closed-Loop Production:** Almost all **spinning waste** is reintegrated into production.
- **Post-Consumer Circularity:** Around **55%** of discarded consumer textiles are diverted from landfills.
- **Livelihoods:** Textile recovery supports nearly **40–45 lakh livelihoods**, especially for **women and marginalised communities**.

Promotion of Sustainability at the Input Stage of the Textile Value Chain

- **Sustainable Raw Materials:** Policies promote **organic fibres, new-age natural fibres and safer inputs** to reduce the value chain's environmental footprint.
- **Organic Fibre Certification: National Programme for Organic Production (NPOP)** sets standards, accredits certification bodies and supports products such as **organic cotton**; its standards are recognised by the **European Commission and Switzerland** for unprocessed plant products.
- **Soil Sustainability:** Organic cultivation promotes **soil fertility, proper plant nutrition and sustainable farming practices**.
- **Jute-ICARE:** Provides **certified seeds, retting accelerators and improved agronomic practices**. Coverage expanded from **130 blocks in 7 states** to **289 blocks in 10 states**, and from **1.11 lakh hectares** to **2.15 lakh hectares in 2024–25**.

- **New-Age Fibres: Mini-Mission III under the Mission for Cotton Productivity** promotes climate-smart cultivation, mechanisation and fibres such as **ramie, sisal and flax**.
- **Fibre Self-Reliance:** The **National Fibre Scheme** increases domestic fibre availability, reduces import dependence and promotes innovation.

Reduction in the Use of Hazardous Chemicals in the Textile Supply Chain

- **Regulatory Measures:** India has restricted **benzidine-based dyes**, prohibited **70 azo dyes** and ratified the **Stockholm Convention in 2006** to control **POPs**.
- **Sustainable Production Pilots:** Projects promote **safer chemicals, eco-friendly textiles, organic production** and awareness of global sustainability practices.
- **Chemical Reduction Project:** Covers **400 factories, 8 textile clusters and 4 fashion houses**.
- **Expected Gains:** Seeks to reduce hazardous chemical use by **10,530 tonnes** and emissions by about **1.47 lakh tonnes of CO₂-equivalent**.

Promotion of Sustainability at the Production Stage of the Textile Value Chain

- **Cleaner Manufacturing:** Policies promote **resource efficiency, waste reduction, cleaner infrastructure and lower emissions**.
- **PM MITRA Parks:** Integrated textile hubs based on the **5F vision: Farm to Fibre to Factory to Fashion to Foreign**.
- **Sustainable Infrastructure:** Include **Common Effluent Treatment Plants (CETPs)**, wastewater recycling, scientific waste management, shared utilities and end-to-end manufacturing.
- **Scale and Investment:** **7 parks** approved with an outlay of **₹4,445 crore up to 2027–28**; MoUs worth over **₹27,434 crore** signed by **December 2025**.
- **MSME Support:** The **RAMP Programme** supports the green transition of MSMEs, which account for over **80%** of textile and apparel production capacity.
- **Green Incentives: Micro and Small Enterprises–Green Investment and Financing for Transformation (MSE-GIFT)** offers interest subvention and credit guarantees for clean technologies, while **MSE-SPICE** (Scheme for Promotion and Investment in Circular Economy) provides a **25% capital subsidy** for eligible circular-economy investments.

Promotion of Circularity at the Post-Production and Market-Support Stages

- **Post-Production Circularity:** Promotes **collection, segregation, recycling, upcycling and reintegration** of textile waste to reduce landfilling.

- **Solid Waste Management Rules, 2026:** Effective from **1 April 2026**, the rules strengthen **producer responsibility, waste segregation and circular waste management** under the **Environment (Protection) Act, 1986**.
- **RDF Use:** Solid-fuel industries must raise **Refuse Derived Fuel use from 5% to 15% in six years**; RDF may include non-recyclable textile waste.
- **Textile-Waste Mapping:** The **Mapping of Textile Waste Value Chain in India, 2026** provides data and a roadmap for **recycling, upcycling and resource recovery**.
- **Eco-Labeling:** The **Eco-Mark Scheme, 2024** covers textiles under **13 Indian Standards** relating to resource use, emissions, waste and hazardous substances.
- **Traceability:** **Kasturi Cotton** and **Silk Mark** improve product identity, responsible sourcing and supply-chain transparency.
- **Public Procurement:** A **2024 MoU** among the **Textiles Committee, GeM and SCOPE** promotes government procurement of upcycled products.
- **Market Support:** Certification, traceability and public procurement create demand for **sustainable and resource-efficient textiles**.

Circular economy in the textile sector can strengthen India's export competitiveness while reducing environmental stress. Discuss.

Practice Mains Question

Cabinet Approves Additional Investment in NIIF

The Union Cabinet has approved an additional Government of India investment commitment of ₹30,000 crore in the **National Investment and Infrastructure Fund (NIIF)**, taking the Government's total commitment to ₹60,000 crore.

- A major portion of the additional commitment will support **NIIF Infrastructure Fund II**, which is proposed to have a **target corpus of about ₹30,000 crore** and will invest in transportation, energy, digital infrastructure, urban infrastructure and e-mobility.

NIIF

- **About:** Established in **2015**, NIIF was created as **India's sovereign-anchored investment platform** to attract global institutional investors and mobilise long-term capital for infrastructure development and other nationally important sectors.
- **Structure:** NIIF is professionally managed by NIIF Limited, with the Government holding a 49% stake and the remaining 51% mobilised from large global institutional investors.

- It currently manages capital commitments of around ₹40,000 crore and has returned nearly ₹12,000 crore to investors through major portfolio exits.
- **Investor Participation:** NIIF has attracted investments from leading **sovereign wealth funds, pension funds, multilateral development institutions and domestic financial institutions** from countries including Australia, Canada, Japan, Singapore, the UAE and the United States.
 - **Australia's largest pension fund**, AustralianSuper, has recently announced an additional investment of **A\$500 million (about US\$346 million)** in India's NIIF, over and above its earlier commitment of **A\$240 million made in 2019**.
- **Role:** NIIF's investments are aligned with major national initiatives such as **PM Gati Shakti, Digital India, Make in India, FAME, PM E-DRIVE, India's climate commitments and Atmanirbhar Bharat**. It also advises government bodies on public-private partnerships, asset monetisation and strategic investment frameworks.
- **Significance:** The additional commitment is expected to attract further institutional and private capital, accelerate infrastructure creation, generate direct and indirect employment, strengthen sectors of national importance and support India's goal of becoming a **Viksit Bharat by 2047**.

Strengthening India's Toy Ecosystem

India's **toy industry** is emerging as a **sunrise sector**, driven by growth in **domestic manufacturing, exports and employment**, alongside rising demand for **indigenous, educational and sustainable toys**.

Current Status of India's Toy Industry

- **Exports:** Rose by **151.9%**, from **US\$152.7 million (2017-18)** to **US\$384.7 million (2025-26)**.
- **Toy exports:** Electronic and non-electronic toy exports increased nearly **160%** to **US\$200.89 million**.
- **Major market:** The **United States** remained the largest destination, followed by the **UK, Germany, Poland and the Netherlands**.
- **Gaming exports:** Nearly tripled to **US\$46.75 million**.
- **Festive articles:** Exports grew about **130%** to **US\$137.03 million**.
- **Import reduction:** Imports of traditional and educational toys declined by **66%**, supporting domestic manufacturing.
- **Trade balance:** India shifted from a **US\$213.01 million deficit** in 2017-18 to a **US\$152 million surplus** in 2025-26.
- **Employment:** More than doubled from **8,685 workers** in 2018-19 to **17,693** in 2023-24.

Major Government Initiatives for the Toy Ecosystem

- **National Action Plan for Toys, 2020:** Promotes **Indian-themed design, indigenous clusters, manufacturing, quality and innovation**.
- **Toys Quality Control Order, 2020:** Makes **BIS certification mandatory** and enforces safety standards for domestic and imported toys.
- **Tariff protection:** Basic Customs Duty increased from **20% to 60% in 2020** and **70% in 2023** to curb imports.
- **Toycathon and e-Toycathon:** Encourage culturally rooted, educational and electronic toy innovation.
- **e-Toys Lab: MeitY-C-DAC, Noida** supports training, prototyping and testing of electronic toys.
- **Toy Biz and Manak Manthan:** Promote buyer linkages, exports and awareness of toy-safety standards.
- **ODOP and GI-tagged toys:** Support branding, market access and artisan livelihoods for products such as **Channapatna toys, Indore leather toys and Thanjavur dolls**.
- **GST reduction:** Tax on toys reduced from **12% to 5%**, improving affordability and demand.
- **Trade and export support:** FTAs and the **Districts as Export Hubs** initiative expand global market access; over **10 districts** have been identified for toy and doll exports.

India's toy industry represents a convergence of cultural heritage, MSME-led manufacturing and export competitiveness. Discuss.

Practice Mains Question

India's Insurance Sector: A Pillar of Sovereign Fiscal Stability

India's **life insurance sector** holds nearly **25%** of outstanding **Central Government dated securities**, converting household premiums into **patient capital** and supporting **household protection and fiscal stability**.

Insurance Sector Act as "Patient Capital"

- **Asset-Liability Matching:** Life insurers issue policies of **20-40 years** and invest pooled retail premiums in **G-Secs**, which offer sovereign guarantee, negligible default risk and predictable returns, matching long-term liabilities.
- **Counter-Cyclical Stability:** Life insurers act as structural **"buy and hold" investors**, unlike **FPIs**. Their steady reinvestment reduces **debt rollover risk** and tempers borrowing costs.
- **Yield Curve Management:** By absorbing **30-40 year bonds**, insurers help the **RBI** maintain a stable sovereign yield curve and moderate state borrowing costs.

Sabka Bima Sabki Raksha (Amendment of Insurance Laws) Act, 2025

- The Act aims at **Insurance for All by 2047** and amends the **Insurance Act, 1938, LIC Act, 1956, and IRDAI Act, 1999.**
- Raises FDI limit in insurance companies from **74% to 100%.**
- Provides **perpetual registration** for intermediaries and raises IRDAI approval threshold for share transfers from **1% to 5%.**
- Establishes **Policyholders' Education and Protection Fund.**
- Expands IRDAI's **supervisory, investigative and corrective powers.**
- Allows LIC more operational flexibility, including opening **zonal offices** without prior government approval.
- **The LIC Anchor:** LIC holds about **19%** of India's outstanding Central Government Securities and is classified by **Insurance Regulatory and Development Authority of India (IRDAI)** as a **Domestic Systemically Important Insurer (D-SII)** due to its macroeconomic importance.
 - **Private life insurers** hold less government debt as they focus more on shorter-tenure, market-linked **Unit-Linked Insurance Plans (ULIPs)**, which invest more in **equities and corporate bonds** than long-term **G-Secs.**

- **Global Parallels:** India mirrors **Japan, UK and South Korea**, where insurers are key holders of long-dated sovereign debt due to liability profiles.

Current Status of India's Insurance Sector

- **Scale:** India is the **10th largest insurance market** globally by premium volume and the **5th largest life insurance market** among emerging economies.
 - As of **April 2026**, India has **74 insurers: 26 life, 35 non-life**, and other specialised insurers.
 - In **FY25**, total premium income was around **₹7.05 lakh crore**, with **5.6% YoY growth.**
 - Insurance penetration stood at **3.7% of GDP: Life 2.7% and Non-Life 1%.** Insurance density rose from **USD 95 in FY24 to USD 97 in FY25.**
 - **Health insurance** became the largest non-life segment in **FY25**, with around **41%** of gross domestic premium, overtaking motor insurance, driven by rising healthcare costs, health awareness and demand for private health coverage.

Major Challenges Facing India's Insurance Sector

- **Low Insurance Penetration:** India's insurance penetration is **3.7% of GDP**, nearly half the global average of **7.3%.** Life insurance is **2.7%**, while non-life is only **1%**, showing gaps in **health, motor and property insurance.**
- **The Missing Middle:** Over **40 crore** Indians, including **gig workers and MSME employees**, remain outside government schemes and cannot afford comprehensive private coverage.
- **Affordability Constraints:** Rising healthcare costs and high premiums make adequate coverage difficult, especially for **lower-income households and senior citizens.**
- **Trust Deficit and Mis-selling:** Mis-selling of **ULIPs**, hidden exclusions, claim rejections and complex policy documents weaken consumer trust.
- **Climate Protection Gap:** Extreme weather events expose low coverage against **agricultural, property and catastrophe risks**, creating need for **parametric and climate-risk insurance.**
- **Cybersecurity and Data Risks:** Digitalisation increases exposure to **cyberattacks, data breaches and misuse of financial and health data.**
- **Regulatory Uncertainty:** Frequent policy and regulatory changes create uncertainty for insurers and investors, affecting planning and investment decisions.
- **Trust Deficit and Low Financial Awareness:** Despite insurers paying nearly **₹1 lakh crore** in claims annually, distrust and low insurance literacy hinder adoption.

Measures to Actualize the Vision of "Insurance for All by 2047"

- **Rationalise Taxation:**
 - Reduce **GST** on essential life and health insurance, provide dedicated **income-tax deductions**, and develop **GIFT City** as a global reinsurance hub.
- **Strengthen Digital Public Infrastructure:**
 - Operationalise **Bima Sugam** for low-cost distribution, single-window servicing and grievance redressal; integrate with **Account Aggregator** framework and **Ayushman Bharat Digital Mission (ABDM)** for paperless underwriting and faster claims.
- **Promote Innovative and Inclusive Products:** Expand **parametric insurance** for climate risks and develop low-cost, bite-sized products for the **missing middle**, including **gig workers, MSMEs and rural households.**
- **Develop GIFT City as a Reinsurance Hub:**
 - Use **regulatory sandboxes** and tax incentives to attract global reinsurers, build domestic reinsurance capacity and retain premiums in India.
- **Leverage 100% FDI:** Use foreign investment for capital infusion, global underwriting expertise and niche products like **cyber, catastrophe and IoT-based insurance.**
- **Strengthen Consumer Protection:**
 - Use **AI-based monitoring** to curb mis-selling, standardise exclusions, simplify policy documents and improve last-mile awareness through **Bima Vahak.**

India's life insurance sector functions not merely as a provider of financial protection but also as an important pillar of sovereign fiscal stability. Discuss.

Practice Mains Question

World Bank Income Classification: India Remains Lower-Middle-Income

India remains a **Lower-Middle-Income economy** in the **World Bank Group Country Income Classifications for FY 2027**, a status held since **2009**.

Key Highlights

- **Upgraded Economies:** 6 nations moved to higher income categories; low-income economies fell from nearly **30% in 1987 to 11% by 2026**.
 - **Lower-Middle to Upper-Middle Income:** Sri Lanka, Vietnam, Philippines, Jordan and Micronesia.
 - **Low to Lower-Middle Income:** Togo.
- **Growth Drivers:** Vietnam and Philippines grew through export-led industrial growth and rising **GNI**; Sri Lanka recovered through tourism and financial services.
 - **Jordan and Togo** benefited from updated national accounts and revised census data.
- **India's Position:** India remains **Lower-Middle-Income**, with **GNI per capita of USD 2,760 in 2025**.

World Bank's Classification of Economies

- **Country Income Classifications:** World Bank classifies economies into **Low, Lower-Middle, Upper-Middle and High-Income** groups.
 - Classification is done annually on **1st July**, using previous calendar year's **GNI per capita**.
- **Atlas Method:** GNI is expressed in **current US dollars** using the **Atlas method** to reduce exchange rate and inflation fluctuations.
 - A country must record real economic growth above **population growth and inflation**.
- **Annual Updates:** Category thresholds are revised every **July** using the **Special Drawing Rights (SDR)** deflator to account for global inflation.
- **Current Brackets for 2026-2027 (FY27):**

Income Category	GNI Per Capita (2025)
Low-Income	USD 1,175 or less
Lower-Middle-Income	USD 1,176–USD4,635
Upper-Middle-Income	USD 4,636–USD14,375
High-Income	More than USD 14,375

- **Significance:**
 - **Access to World Bank Loans:** Low-income countries get concessional loans and grants from International Development Association (IDA), while many **middle-income countries** borrow from **International Bank for Reconstruction and Development (IBRD)** on less concessional terms.

- **Allocation of Foreign Aid:** Donors use income levels to decide **development assistance**.
- **International Trade Benefits:** Income status can affect eligibility for **preferential trade schemes** and special treatment.
- **Distribution of Climate Finance:** Helps prioritize support for economically vulnerable countries facing **climate change**.
- **Investor Assessment:** Indicates a country's broad economic development level.
- **Policy Formulation:** Helps governments compare progress and avoid the **Middle-Income Trap**.

The World Bank's income classification is primarily an analytical and operational tool, not a measure of a country's overall development.

Reasons for India's Continued 'Lower-Middle-Income' Status

- **Denominator Challenge:** India's **1.4 billion+** population dilutes aggregate GDP growth in **per capita income** terms.
- **Regional Disparities:** High-income states like **Tamil Nadu, Maharashtra and Gujarat** raise output, but populous low-income states like **Bihar and UP** pull down the national average.
- **Informalization and Disguised Unemployment:** Up to **85%** of India's workforce is informal, with heavy reliance on low-productivity agriculture, suppressing earnings and tax base growth.
- **Historical Starting Line:** India moved out of the **Low-Income** bracket only in **2009**; reaching the next tier needs sustained high real income growth over decades.

5 Years of Cooperative Transformation in India

Ministry of Cooperation celebrated its **5th Foundation Day** on **6th July 2026**, highlighting digital and infrastructure initiatives to strengthen cooperatives under "**Sahkar Se Samriddhi**" and "**Vasudhaiva Kutumbakam**."

Current Status of Cooperatives in India

- **Massive Scale and Deep Penetration:** India has over **8.5 lakh cooperatives**, serving nearly **32 crore members** across about **30 sectors**, covering **~98% of rural India**.
- **Grassroots Economic Role:** Cooperatives support **farmers, dairy producers, fishers, artisans**, and workers through access to **credit, markets, storage, livelihoods**, and essential services.
- **Women-led Development:** **~10 crore women** are linked to cooperatives through **SHGs**.

- **Strong Panchayat-level Presence:** PACS are present in over **2.55 lakh Gram Panchayats**; dairy cooperatives cover **87,159 Gram Panchayats** and fisheries cooperatives nearly **29,964 Gram Panchayats**.
- **Expansion Beyond Credit:** Cooperatives are expanding into retail, storage, healthcare, fuel, digital services, dairy, fisheries, exports, organic products, seeds, and mobility services.
- **Growing Institutional Financing:** NCDC sanctioned **₹1.55 lakh crore** and disbursed **₹1.27 lakh crore** in FY 2025–26.
 - It also disbursed **₹2,320 crore** to FPOs and CBBOs under the scheme for formation and promotion of **10,000 FPOs**.
- **Role in Viksit Bharat:** Cooperatives are being positioned as engines of **grassroots prosperity, rural enterprise, women's empowerment, food security**, and inclusive growth under **"Sahkar Se Samridhi."**

Key Initiatives Related to Cooperatives

- **Creation of Ministry of Cooperation:**
 - Established on **6th July 2021** to make cooperatives more **transparent, inclusive, digital**, and future-ready.
 - On **5th Foundation Day**, Ministry launched transformation of **50,000 PACS into e-PACS**, laid foundation stones for **47 grain storage godowns**, and launched **Milk Supply Review Dashboard Portal** for **NDDB**.
- **Revitalising Primary Agricultural Credit Societies:** Model bye-laws enabled PACS in **32 States/UTs** to undertake over **25 business activities**, including retail, storage, healthcare, fuel, and digital services.
- **PACS as Rural Service Centres:** As of **June 2026**, **39,177 PACS** function as **PM Kisan Samridhi Kendras**, **54,117 as CSCs**, **394** applied for retail fuel outlets, **3** fuel outlets commissioned, **4,248** approved for **Jan Aushadhi Kendras**, and **843** ready to function.
- **Expansion of New Cooperatives:** **37,454** new multipurpose PACS, dairy cooperatives, and fisheries cooperative societies registered in underserved areas.
- **Decentralised Grain Storage Plan:** Government is implementing the world's largest decentralised grain storage plan through cooperatives.
 - Godowns completed in **145 PACS**, creating over **68,702 MT** storage capacity.
- **Farmer Producer Organisations:** **1,863 FPOs** formed in cooperative sector, including **1,117 through PACS** and **1,070 fisheries FPOs**, with **₹98 crore** disbursed.
- **Tax Relief and Ease of Doing Business:** Reforms include surcharge reduction from **12% to 7%** for cooperatives with income of **₹1–10 crore**, **MAT** reduction from **18.5% to 15%**, **TDS cash withdrawal limit** raised from **₹1 crore to ₹3 crore**, and higher cash transaction limits for **PACS** and **Primary Cooperative Agriculture and Rural Development Banks**.
- **White Revolution 2.0:** Aims to strengthen dairy cooperatives and raise milk procurement by **50% by 2028–29**.
 - So far, **25,282 dairy cooperative societies** registered, focusing on women-led dairy cooperatives and wider coverage.
- **New National Cooperative Institutions:** Established **NCEL**, **NCOL**, and **Bharatiya Beej Sahakari Samiti Limited**.
- **Progress of National Institutions:** **NCEL** recorded **15.4 lakh MT** exports worth **₹6,295 crore** to **38 countries** as of **June 2026**.
 - **NCOL** has **14,286 member cooperatives**, while **Bharatiya Beej Sahakari Samiti Limited** has **38,665 member cooperatives** and sells quality seeds under **"Bharat Beej"**.
- **Cooperative Education and Capacity Building:** **Tribhuvan Sahkari University**, India's first cooperative university, was established for education, research, and skill development.
 - Training is also conducted through **NCCT** and **NABARD**.
- **Urban Cooperative Banking Initiatives:** **NUCFDC** launched **Sahakar CBS**, a centralised core banking solution for **Urban Cooperative Banks**, and **Sahakar Sahyogi**, an AI-powered conversational platform.
- **Bharat Taxi Initiative:** **Bharat Taxi**, India's first cooperative-based mobility platform, was launched under **Sahakar Taxi Cooperative Limited**.
 - It has **6.37 lakh drivers** and **35.77 lakh customers**, operates in **Delhi-NCR, Gujarat, Lucknow, Chandigarh, Mumbai, Jaipur, and Kanpur**, with expansion planned to **Ranchi, Patna, Guwahati, Bhopal, Kolkata, Indore, and Nagpur**.

Challenges Faced by India's Cooperative Sector	Suggestions
■ Regulatory Fragmentation: Overlap among Registrar of Cooperatives, RBI, and other authorities causes jurisdictional conflicts, inconsistent supervision, and regulatory gaps.	■ Establish a clear and harmonised regulatory framework with coordination between state cooperative departments, RBI, NABARD, and Ministry of Cooperation; single-window compliance can reduce confusion.

Contd...

Challenges Faced by India's Cooperative Sector	Suggestions
■ Governance Deficits: Lack of transparency, weak controls, elite capture, and poor democratic functioning weaken accountability; PMC Bank collapse exposed such failures. ■ Financial Constraints and Poor Infrastructure: Many cooperatives lack capital, credit access, warehouses, cold storage, processing units, and market linkages. ■ Operational and Technological Hurdles: Rural cooperatives often lack digital accounting, ERP systems, online services, e-commerce access, and efficient logistics. ■ Socio-Cultural Barriers: Caste, class, gender inequalities, and local power hierarchies limit inclusive participation and representation. ■ Intense Market Competition: Cooperatives face competition from commercial banks, fintech firms, private agribusinesses, and modern retail platforms.	■ Strengthen democratic governance through audits, transparency, member participation, regular elections, professional boards, social audits, and strict disclosure norms. ■ Improve financial deepening and infrastructure investment by linking cooperatives with NABARD, MUDRA, CGTMSE, and investing in warehouses, cold chains, processing units, and Common Facility Centres. ■ Promote technological integration through mobile banking, digital payments, ERP systems, online reporting, and e-commerce linkage with ONDC and GeM; member-owned logistics cooperatives can improve efficiency. ■ Ensure inclusive cooperative governance through leadership reservations, participation of women and marginalised communities, awareness drives, and grassroots capacity building. ■ Focus on branding, diversification, and professional management through brands like "CoopMade" or "Bharat Organics", certification, skill training, and diversification into renewable energy, eco-tourism, digital services, and modern agriculture.

Cooperatives can become engines of inclusive growth only when they combine grassroots ownership with professional governance and digital transformation. Discuss.

Practice Mains Question

NCLAT on IBC-PMLA Conflict

The **NCLAT** has ruled that the **moratorium under the IBC, 2016 cannot shield assets alleged to be "proceeds of crime"** from attachment under the **PMLA, 2002**.

- Ruling came in **Siddhi Vinayak Logistics Ltd.** insolvency proceedings, where **ED** attached assets despite ongoing insolvency.
- **Liquidator's Argument:** Relied on **P. Mohanraj v. Shah Brothers Ispat, 2021**, which barred coercive recovery during **IBC moratorium**.
- **ED's Argument:** Relied on **Embassy Property Developments v. State of Karnataka, 2019**, which limited **NCLT/NCLAT** interference in decisions of independent statutory authorities.

NCLAT Rule Regarding the IBC-PMLA Conflict

- **Distinct Statutory Domains:** **IBC, 2016** deals with insolvency resolution and creditor recovery, while **PMLA, 2002** deals with tracing and confiscating **proceeds of crime**.

■ Moratorium Limited to Legitimate Assets:

- **Section 14 IBC moratorium** protects only legitimately acquired assets, not assets alleged to be **proceeds of crime**.

■ Public Law Proceedings Unaffected:

- **PMLA proceedings** are penal/public law actions and cannot be stopped by **IBC moratorium**.
- **Section 41 PMLA:** Bars civil courts from examining matters handled by **PMLA authorities**, limiting civil scrutiny of **ED actions**.

■ National Interest:

- Confiscation of proceeds of crime prevails over creditor recovery concerns under **IBC**.
- **Jurisdiction:** **NCLT/NCLAT** cannot test validity of **ED attachment orders**; challenges must go before designated **PMLA authorities/courts**.

Feature	IBC, 2016	PMLA, 2002
Primary Objective	Insolvency resolution and creditor recovery.	Tracing and confiscating proceeds of crime .
Nature of Law	Civil and commercial law.	Criminal and public law.
Asset Treatment	Protects legitimate assets of corporate debtor.	Attaches tainted/ill-gotten assets .
Moratorium Impact	Section 14 stops civil suits and recovery actions.	ED can freeze tainted assets.
Jurisdiction	NCLT and NCLAT.	PMLA Adjudicating Authorities and designated courts.

India's Fisheries Sector

India's marine fisheries are largely sustainable, with **91.1%** of assessed fish stocks in good condition (CMFRI).

- However, the **FAO** has flagged concerns over **stagnant marine production, pressure on fish stocks, declining catches, degradation of inshore ecosystems**, and the rapid expansion of **mechanised trawling**.

Current Status of India's Fisheries Sector

- **Marine Resource Base:** India has an **EEZ of about 2.4 million sq. km**, with **continental shelf** and **inshore waters** as key fishing grounds.
- **Status of Fish Stocks:** **CMFRI 2022** found **91.1% of 135 marine fish stocks** sustainable; India mainly uses **landing/catch data** for assessment.
- **Economic Contribution:** Fisheries contribute **7.43% of Agricultural GVA**, with production rising to **197.75 lakh tonnes** in **FY 2024–25**.
- **Export Performance:** Seafood exports reached **Rs 62,408 crore** in **FY 2024–25**, led by **frozen shrimp**, mainly to **US and China**.
- **Budgetary Support:** **Union Budget 2026–27** allocated **Rs 2,761 crore**, including **Rs 2,500 crore** for **PMMSY**.
- **Infrastructure & Credit:** **FIDF** and **KCC** limit up to **Rs 5 lakh** support credit access and processing capacity.
- **Fishing Fleet Profile:** India has **64,000+ mechanised vessels**, along with traditional and artisanal boats.
- **Technological Shift:** **RAS** and **Bio-floc technology** are promoting sustainable, high-density aquaculture with lower resource use.
- **Digital Governance:** **NFDP, 2024** gives digital identities to **30 lakh+ stakeholders** for easier access to insurance and subsidies.
- **Formalization:** **PM-MKSSY** with **Rs 6,000 crore** outlay aims to formalise fisheries value chains and improve traceability.
- **Regulatory Framework:** **2025 Rules for Sustainable Harnessing of Fisheries in EEZ and High Seas** aim at compliance, sustainability, and resource security.

Challenges Facing India's Fisheries Sector

- **Questionable Sustainability Assessment:** India's **91.1% sustainable stock** claim is questioned as it relies mainly on **landing/catch data**, not detailed stock assessments at sea.
- **Degradation of Inshore Ecosystems:** Productive inshore waters face **mangrove loss, pollution, reduced river nutrient flow**, and seabed damage.
- **Unregulated Mechanised Trawling:** **64,000+ mechanised vessels** disturb seabeds, damage **benthic ecosystems**, and create conflicts with small fishers.

- **Weak Governance:** Seasonal bans and trawling limits exist, but enforcement is weak due to poor staffing, fewer patrol vessels, and low community participation.
- **Overcapacity and Livelihood Stress:** Rising mechanised fishing increases competition over declining resources and threatens **small-scale fishers**.
- **Misplaced Deep-Sea Focus:** Deep-sea fishing may offer limited gains, while raising costs and ignoring restoration of productive **inshore waters**.

Steps Needed for Sustainable Fisheries Sector

- **Strengthen Inshore Governance:** Improve **vessel monitoring, patrols, community surveillance**, and penalties in nearshore zones.
- **Strengthen Scientific Stock Assessment:** Move beyond **landing-data estimates** to detailed sea-based stock and **benthic ecosystem** studies.
- **Regulate Mechanised Trawling:** Enforce **coastal zoning, seasonal bans**, and limits on destructive **bottom trawling**.
- **Promote Aquaponics:** Encourage fish farming with **hydroponics** to save water and improve income.
- **Enhance Cold Chain:** Build **collection centres, cold storage, refrigerated transport**, and processing units.
- **Support Value Addition:** Provide **training, credit, branding, packaging**, and market links for value-added fish products.

Discuss the ecological and governance challenges in India's marine fisheries.

Practice Mains Question

9 Years of Goods and Services Tax (GST)

GST completed **9 years** since its **2017 launch**, marking a key milestone in India's indirect tax reforms. Recent **Next-Generation GST reforms** simplified tax structure, lowered rates, eased compliance and supported **households, MSMEs, businesses and economic growth**.

Key Achievements of GST over the Past 9 years

- **Creation of a Common National Market:** GST created a **"One Nation, One Tax"** framework by subsuming **17 diff. taxes and 13 cesses**, reducing fragmentation, hidden costs and cascading effects.
- **Expansion of the Taxpayer Base:** GST taxpayers increased from **66.5 lakh in 2017** to **1.65 crore by May 2026**, showing greater formalisation and compliance.
- **Strong Revenue Growth:** Gross GST collection rose from about **₹7.4 lakh crore in 2017-18** to **₹22.27 lakh crore in 2025-26**; in the last five years, it increased from **₹13.76 lakh crore in 2021-22**.
- **High-Frequency Indicator of Economic Activity:** GST collections now reflect **consumption, trade, compliance**

and formalisation. In April-May 2026, collections reached about ₹4.37 lakh crore.

- **Strengthening of Cooperative Federalism:** The GST Council enabled Centre-State tax decision-making, regular reviews, corrections and flexibility in the tax system.
- **Digital Transformation of Tax Administration:** GSTN provides common digital infrastructure. E-invoicing, pre-filled returns, real-time validation and digital reconciliation improved transparency and reduced manual errors.
- **Ease of Compliance for Small Taxpayers:** Reforms include higher exemption limits, composition scheme, quarterly returns, NIL return via SMS and faster registration for low-risk applicants.

Key Features of the Next-Generation GST Reforms

- **GST 2.0 Reform Push:** The 56th GST Council meeting approved Next-Generation GST reforms to simplify indirect taxes, reduce compliance burden and support common people and businesses. They came into effect from 22nd Sept 2025.
- **Simplified Rate Structure:** GST has been streamlined mainly into 2 slabs: 5% and 18%, making it simpler, predictable and easier to comply with.
- **Higher Rate for Luxury and Sin Goods:** A 40% GST rate applies to luxury and sin goods, including lottery/online gaming, tobacco, aerated drinks, high-end cars, yachts and private aircraft.
- **Relief for Households and Consumers:** Lower rates and exemptions make goods/services cheaper, support household savings and improve affordability.
 - Exemptions on insurance and essential medicines strengthen protection and healthcare access.
- **Boost to MSMEs and Industry:** Lower GST on inputs and sectors like cement and handicrafts reduces production costs, improves competitiveness and supports MSMEs, artisans and manufacturers.
- **Correction of Inverted Duty Structures:** Reforms aim to correct inverted duties, support domestic value addition, reduce distortions and promote exports.
- **Easier Compliance for Businesses:** GST 2.0 simplifies registration and return filing, speeds up refunds and lowers procedural costs, especially for MSMEs, startups and small taxpayers.
- **Data-Driven Tax Administration:** Strengthens use of GSTN, e-invoicing, pre-filled returns, real-time validation, AI, ML and data analytics to improve compliance, reduce errors and detect evasion.

Challenges in the Current GST Framework

- **Exclusion of Key Items:** Alcohol for human consumption and 5 petroleum products remain outside GST, limiting seamless input tax credit and causing tax cascading.
 - Bringing them under GST is difficult as States depend heavily on them for revenue.
- **Rate and Classification Disputes:** Despite GST 2.0's 5%, 18% and 40% structure, disputes may continue due to exemptions, special categories and product-specific rate interpretations.
- **GST Appellate Tribunal:** The tribunal has been created with a Principal Bench and State Benches, but delays in full functioning may prolong litigation and uncertainty.
- **Compliance Burden for Small Businesses:** MSMEs and small taxpayers still face issues due to frequent notifications, return filing, reconciliation and changing procedural rules.
- **Inverted Duty Structure:** Some sectors may still face working capital stress where input taxes exceed output taxes, causing refund dependence and liquidity pressure.
- **Centre-State Revenue Concerns:** Differences over rate changes, exemptions, compensation and revenue protection can slow deeper GST reforms.

Reforms to Further Strengthen the GST Framework

- **Phased Inclusion of Excluded Items:** Bring petroleum products under GST gradually, starting with natural gas and ATF, using revenue-neutral rates and temporary compensation safeguards.
 - Alcoholic liquor for human consumption needs wider political consensus and fiscal safeguards as it is constitutionally outside GST.
- **Further Rate Rationalisation after GST 2.0:** After moving mainly to 5% and 18% slabs, reforms should reduce classification disputes, review exemptions and address remaining inverted duty sectors.
- **Faster Refunds and Correction of Inverted Duty Structures:** Make refunds faster and predictable, especially for exporters and sectors with input-output tax mismatch.
 - Rebalancing input tax rates can reduce working capital stress and support domestic value addition.
- **Strengthening Dispute Resolution:** Make GST Appellate Tribunals fully functional by fast-tracking appointments and infrastructure. Binding clarificatory circulars and limited amnesty for early procedural lapses can reduce litigation.
- **Preparing GST for Emerging Sectors:** Provide clearer GST rules for digital goods and services, crypto-assets, carbon credits and other emerging sectors for certainty, uniformity and global tax alignment.

"GST has strengthened India's formal economy and cooperative federalism, but its unfinished reform agenda remains significant." Discuss.

Practice Mains Question

Unlocking Growth in India's Tourism and Hospitality Sector

NITI Aayog and the Ministry of Tourism released "Unlocking Growth in Tourism and Hospitality Sector" to remove procedural bottlenecks and shift India's tourism from a volume-driven model to a high-value, investment-led model.

Tourism Sector in India: Current Status

- **Global Role:** Travel and tourism contributed nearly **10% of global GDP in 2024** and supported **1 in 10 jobs**.
- **Future Potential:** By **2035**, global tourism may reach **USD 16.5 trillion**.
- **India's GDP Share:** Tourism contributed **₹15.73 lakh crore in FY 2023–24**, about **5.22% of GDP**.
- **Employment:** Supported around **84.6 million jobs**, rising nearly **20% in 5 years**.
- **Domestic Tourism:** Recorded **2.9 billion visits in 2024**.
- **Foreign Tourists:** **9.95 million FTAs in 2024**; India's global share remains below **1.5%**.
- **TTDI 2024:** India ranks **6th in Natural Resources**, **9th in Cultural Resources**, but **39th overall**.

Challenges to India's Tourism Sector

Demand-Side Challenges

- **Restrictive Visa Regime:** India's **Visa Openness Index score is 38.14**, below world average **40** and Asia-Pacific average **46**.
- **Limited Easy Entry:** Visa-free access only for **Nepal, Bhutan and Maldives**; Visa-on-Arrival only for **Japan, South Korea and UAE**.
- **Poor Visa Experience:** Issues include **complex forms, technical glitches, payment failures and weak support**.
- **Weak Visitor Retention:** Lack of **destination marketing, loyalty programmes and repeat-visitor incentives**.
- **Outbound Spending Leak:** India's tourism forex receipts are around **USD 35 billion**, while Indian outbound spending is **15% above pre-pandemic level**.

Supply-Side Challenges

- **Hotel Shortage:** Branded hotel inventory is only about **0.2 million rooms**, under **8% of total lodging stock**.
- **Slow Hotel Development:** Hotel projects take **36–48 months** in India, compared to **12–18 months** in ASEAN countries.

- **High Compliance Burden:** Hotels need around **50–60 approvals/licenses** across multiple authorities.
- **Restrictive Building Norms:** Low **FAR**, strict ground coverage and low high-rise thresholds increase costs.
- **Interstate Transport Barriers:** States continue to levy **entry taxes/road fees**, weakening **AITP** benefits.
- **Clearance Delays:** **Environmental** and **CRZ clearances** face delays; **89% EC cases** exceeded the **105-day timeline**.

Measures to Strengthen India's Tourism Sector

- **License Rationalisation:** Integrate approvals through **NSWS** and remove duplicate licences like police-issued **Eating House Licence**.
- **Homestay Scalability:** Raise homestay room cap from **6 to 9 rooms** and remove local **NOC requirement**.
- **Building Byelaws:** Increase **FAR**, remove rigid ground coverage norms and raise high-rise threshold to **23 metres**.
- **Vehicle Re-registration:** Amend **Section 47 of Motor Vehicles Act, 1988** to require re-registration only for permanent residence change.
- **Visa Reform:** Introduce **90-day multiple-entry Visa-on-Arrival** for high-potential countries.
- **E-Visa Simplification:** Reduce data fields, fix foreign card payment issues and create unified traveller profiles.
- **Interstate Travel:** Extend **AITP** validity up to **5 years** and remove overlapping state entry taxes.
- **CRZ/Forest Mapping:** Digitise **HTL/LTL** and forest land maps for faster clearances.
- **Dedicated SEACs:** Create tourism-specific **SEACs** for faster **Environmental Clearances**.

India possesses immense tourism potential but continues to underperform in attracting international tourists. Examine the major structural constraints and suggest reforms to unlock the sector's growth.

Practice Mains Question

Reforms to Expand Foreign Participation in G-Secs

The Government introduced reforms to expand **FPI participation in G-Secs** to deepen India's **bond market**, attract **long-term foreign capital**, and strengthen India as a **global investment destination**.

FPI Reforms in G-Secs

- **Objective:** Deepen India's **bond market**, diversify investors and attract **long-term foreign capital**.
- **Tax Exemption:** From **1 April 2026**, **FPIs/FIIs** exempted from **interest income and capital gains tax** on **G-Secs**.
- **FAR Expansion:** **15-year, 30-year and 40-year G-Secs** and eligible **Sovereign Green Bonds** added under **Fully Accessible Route**.

- **General Route Liberalisation:** Removed **short-term investment limit, concentration limit and security-wise limit.**
- **Overall Limits:** Remain **6% of CGSs and 2% of SGSs.**
- **Rationalisation:** **General and Long-Term FPI** categories merged into one limit.
- **Target Investors:** **Pension funds, insurance companies and sovereign wealth funds.**
- **Significance:** Improves **liquidity, price discovery, yield curve, monetary transmission** and may reduce **government borrowing costs.**
 - Supports funding for **infrastructure, manufacturing, urban development and climate projects**, and helps Indian bonds enter **global indices.**

Concepts Related to Foreign Investment in G-Sec

- **FII:** Institutional FPI such as **mutual funds, pension funds, insurance companies and hedge funds.**
- **FPI:** Foreign investors investing in **stocks, bonds, mutual funds and G-Secs** without management control.
- **G-Secs:** Tradable debt instruments issued by **Central/State Governments.**
- **BIS:** Central bank-owned global institution promoting **monetary and financial cooperation.**
- **LTCG:** Gains from G-Secs held for more than **12 months** if listed and **24 months** if unlisted.
- **STCG:** Gains from G-Secs held up to **12 months** if listed and **24 months** if unlisted.

20th Statistics Day and SDGs-National Indicator Framework Progress Report 2026

MoSPI celebrated the **20th Statistics Day (29 June 2026)** to mark the **133rd birth anniversary of Prof. P. C. Mahalanobis** and released the **SDGs–National Indicator Framework Progress Report 2026.**

National Statistics Day

- **Date:** Celebrated on **29th June** since **2007.**
- **Purpose:** Honours **Prof. P.C. Mahalanobis** and promotes awareness about statistics in planning and policymaking.
- **Theme 2026:** “**Unlocking the Potential of Administrative Data**” — using government operational data like tax records, registries and welfare enrolments for evidence-based policy.

Prof. P.C. Mahalanobis

- **About:** P.C. Mahalanobis (**1893–1972**) is known as the **Father of Indian Statistics.**
- **Institution Builder:** Founded **ISI (1931)** and *Sankhya* journal (**1933**); contributed to **NSS (1950)** and **CSO.**

- **Economic Planning:** Key member of the first Planning Commission; shaped the **Second Five-Year Plan (1956–61)** through the **Mahalanobis Model** focused on heavy industrialisation.
- **Mahalanobis Distance:** Introduced in **1930**; used in machine learning, clustering and classification.
- **Sample Surveys:** Pioneered large-scale sample surveys as cost-effective alternatives to census.
- **Applied Statistics:** Used data for governance, flood control and meteorological studies.
- **Honours:** UN statistical role, Fellow of Royal Society (**1945**) and **Padma Vibhushan (1968).**

SDGs-NIF Progress Report 2026

- **SDG 1:** Multidimensional poverty fell **24.85% → 14.96%**; social protection coverage rose **22% → 65.3%.**
- **SDG 2:** Underweight children reduced **35.7% → 31.8%**; stunting fell **38.4% → 29.3%.**
- **SDG 3:** MMR improved **122 → 87**; U5MR **43 → 28**; NMR **25 → 18**; TB treatment coverage **71% → 92%.**
 - Skilled health personnel birth attendance rose from **81.4% to 91.3%** (NFHS-4 to NFHS-6).
- **SDG 4:** School internet access rose **46.30% → 84.20%**; GER in higher secondary remained **48.32.**
- **SDG 5:** Sex ratio at birth improved **896 → 918**; FLFPR ratio **0.43 → 0.52**; women SHGs reached **96%.**
- **SDG 6:** Rural piped water rose **48.15% → 81.81%**; wastewater processing **17.97% → 82.20%.**
- **SDG 7: 100% household electrification;** renewable capacity **64.04 → 193.36 watts/capita**; energy intensity declined.
 - Energy intensity declined from **0.2556 to 0.1329 MJ/Rupee** (2015-16 to 2024-25).
- **SDG 8:** Unemployment fell **6.1% → 3.1%**; tourism growth reached **15.01%.**
- **SDG 9:** Manufacturing employment rose **11.40% → 12.14%**; 4G coverage reached **99.51%.**
- **SDG 10:** Rural Gini improved **0.28 → 0.23**; urban Gini **0.36 → 0.28**; female-male professional ratio **51.3%.**
- **SDG 11:** Door-to-door waste collection **43% → 97%**; waste processing **17.97% → 82.20%.**
- **SDG 12:** Recycling plants rose **829 → 3,236**; wheat/rice storage losses nearly zero.
- **SDG 13:** Industry compliance improved **86.25% → 91.82%**; climate plans updated under NAPCC/SAPCC.
- **SDG 14:** Fishery MSY rose **3.078 → 5.311 million tonnes/year.**
- **SDG 15:** Forest cover rose **21.34% → 21.76%**; Ramsar area share doubled **4.15% → 8.66%.**

- **SDG 16:** Judges per lakh rose **1.33 → 1.55**; Aadhaar coverage reached **95.67%**.
- **SDG 17:** Merchandise export share **1.62% → 1.70%**; services export share **3.11% → 4.35%**.

National Indicator Framework (NIF)

- NIF is India's main statistical framework to monitor progress on **17 SDGs**.
- Developed by **MoSPI** with **NITI Aayog, UN agencies and stakeholders**, it has **277 national indicators (2026)**.
- It is updated by the **HLSC** and supports SDG tracking through **State and District Indicator Frameworks**.

UN Sustainable Development Report 2026

India ranked **94th** out of 167 countries in the 2026 UN SDG Index, with an overall score of **68.3** out of 100, as per SD Report 2026 released by the UN SDSN.

Key Findings of the Report

- **Global:**
 - **SDGs off track:** None of the **17 SDGs** are on track for **2030**.
 - ❖ Only **16.5%** targets are on track; **15%** have reversed since 2015.
 - **Top performers:** **Finland, Sweden, and Denmark** lead the rankings.
 - **Lowest ranked:** **Chad, Central African Republic, and South Sudan**.
 - **Most stagnant SDGs:** **SDG 2, 11, 14, 15 and 16**.
 - **Multilateralism:** **Barbados** ranks first; **US** ranks last.
- **India:**
 - **Ranking improvement:** India rose **18 places**, from **112th** to **94th** since 2015.
 - **2030 target status:** Only **33.3%** targets are on track; **24%** have worsened.
 - **Major challenge:** **SDG 2 Zero Hunger** remains most critical.
 - **Hunger indicators:** **Child wasting 19%, stunting 29.3%, undernourishment 12%**.
 - **Major challenges in 7 SDGs:** **SDG 2, 3, 5, 11, 14, 15 and 16**.
 - **Reversals:** **Press Freedom** score declined; **air pollution** and **NCD-linked mortality** worsened.
 - **Environmental decline:** **SDG 12** and **SDG 13** deteriorating; per capita **CO₂** reached **2.21 tonnes**.
 - **Success areas:** Strong progress in **electricity access** and **mobile broadband/internet use**.

India Leads the Ship Recycling Sector

India became the world's leading **ship recycling nation in 2025**, achieving a key **MIV 2030** target early (UNCTAD).

- The **2026 India–South Korea engagement** also boosted India's ambition to emerge as a global shipbuilding hub.

Current Status of India's Ship Recycling Sector

- **Global Leadership:** India's global ship recycling share rose to **35.4% in 2025**, recycling **2.99 million GT**.
- **MIV Target Achieved Early:**
 - India met a key **MIV 2030** ship recycling target ahead of schedule (India aims to rank among the top 10 global shipbuilding nations by 2030 and the top five by 2047).
- **Policy Enablers**
 - **HKC Compliance:** Recycling of Ships Act, 2019 aligned India with Hong Kong Convention; ₹53 crore helped modernise **115 yards**.
 - **Ship-breaking Credit Note:** Shipowners get credit worth **40% of scrap value**, usable for new ships in Indian yards.
 - **SBFAS:** Provides **15–25% financial assistance** and supports ₹96,000 crore shipbuilding projects.
 - **SbDS:** Promotes greenfield clusters, shipyard modernisation, technology centre and credit risk coverage.
 - **Capacity Goal:** SBFAS and SbDS aim for **4.5 million GT/year shipbuilding capacity by 2047**.
 - **MDF:** ₹25,000 crore fund for low-cost maritime finance through MIF and IIF.
 - **SMFCL:** India's first dedicated maritime and shipbuilding NBFC.
 - **Green Push:** GTTP and Harit Nauka promote LNG, electric, hydrogen and methanol vessels.
 - **Along Expansion:** Capacity to rise to about **9 million LDT**; India seeking EU-approved yard status.

South Korea's Role in India's Shipbuilding Ambitions

- **Mega Shipyard:**
 - Supports India's first mega greenfield shipyard at **Thoothukudi** under **VOYAGES**, with **2.5 million GT** capacity and ~15,000 jobs.
- **Technology Transfer:** HD Hyundai, SHI and Hanwha Ocean to share expertise in design, automation, green propulsion and high-value vessels.
- **Local Supply Chain:**
 - KOMEA's Mumbai presence will boost domestic marine equipment production.
- **Defence Cooperation:** Joint naval and amphibious vessel manufacturing to support Make in India.
- **Skills & R&D:**
 - KOICA, MoPSW, IMU and KMOU partnerships to train workforce and promote marine engineering research.
- **Global Competitiveness:** Helps India modernise shipyards and target top 10 by 2030 and top 5 by 2047.

Key Challenges in Indian Shipbuilding

- **Low Capacity:** India has only **0.072 million GT** shipbuilding capacity and less than **1% global share**, while China, South Korea and Japan dominate 95% orders.
- **High Cost of Capital:** Indian yards face **10–10.5% interest rates**, high working capital needs and lack sovereign refund guarantees.
- **Import Dependency:** India imports **60–70% critical marine components**, raising costs and supply-chain risks.
- **Weak Infrastructure:** Lack of mega dry docks limits construction/repair of ULCCs and mega container ships.
- **Weak Domestic Demand:**
 - Indian firms prefer cheaper second-hand foreign vessels; domestic order book remains small.
 - **Poor MRO Share:** India has less than **1% global MRO share**, with vessels going to Singapore, Dubai or Colombo.
- **Skill Deficit:** Shortage of experts in marine engineering, automated design and green propulsion.
- **Regulatory Hurdles:** Land acquisition, legal delays and policy inconsistencies affect investor confidence.

Measures to Strengthen India's Shipbuilding Sector

- **Adopt Ulsan Model:**
 - Develop cluster-based shipbuilding hubs by integrating shipyards, ancillary industries, logistics and training institutions.
- **Link Recycling with Shipbuilding:** Use steel and materials from India's ship recycling industry to build a circular maritime economy.
- **Conditional Foreign Partnerships:**
 - Ensure foreign tie-ups include local value addition, technology transfer and skill development.
- **Localize Suppliers:** Build domestic marine equipment and component industries within SPV clusters to reduce import dependence.
- **Green Technology Leapfrogging:**
 - Focus on hydrogen fuel-cell, methanol/ammonia dual-fuel and other green vessels under IMO decarbonisation push.
- **Strict RoFR Enforcement:** Mandate Right of First Refusal for Indian-built and Indian-flagged vessels in government and PSU cargo tenders.

'The future of India's Blue Economy depends on strengthening its shipbuilding ecosystem.' Critically examine.

Practice Mains Question

Index of Services Production

MoSPI will launch the **Index of Services Production (ISP)** in **July 2026** to track growth in India's **services sector**, similar to the existing **Index of Industrial Production (IIP)**.

- It was developed by a **Technical Advisory Committee** chaired by **Debjani Ghosh** and will initially be released on a **trial basis** with **sectoral indices**.

Key Facts About the Index

- **About:** Monthly indicator by **MoSPI** to measure short-term changes in real output of India's **formal services sector**, like **IIP** for industry.
- **Need:** Services contribute over **50% of India's GVA** since **2013-14**, but lacked a monthly real-growth indicator.
- **Objective:** Gives timely services growth data, supports monitoring, policy, forecasting, and complements **IIP**.
- **Improved Data Ecosystem:** **GST data** and **Annual Survey of Incorporated Services Sector Enterprises (ASISSE)** enabled reliable ISP compilation.

Key Mechanics of ISP

- **Scope and Limitations:** Covers only **formal sector**; excludes **non-market government activities, personal services, and informal sector**.
- **Excluded areas:** Public administration & defence, gambling, personal services, government-run health/education, and other **non-market activities**.
- **Methodological Framework:**
 - Uses **fixed-weight Laspeyres volume index**, base year **2024–25**, with **GVA-based weights**.
 - Tracks **real output growth** by deflating nominal turnover using **CPI** and **WPI**.
- **Major Data Sources for ISP:**
 - **Administrative/Secondary Data:** Air Transport, Railways, Banking, Insurance.
 - **GST Data:** Trade, transport, telecom, accommodation, real estate, professional services, arts & recreation.
 - **ASISSE Data:** Health and Education services, excluding government.
- **Quantity vs Value Indicators:** **Air Transport** and **Railways** use quantity-based indicators (e.g., passenger-kilometres); most others use **turnover/sales/GST outward supplies**.
- **Preferred Output Measure:** **Deflated turnover**, as turnover closely reflects services output.
- **Deflators Used:** **WPI** for wholesale trade; **sector-specific CPI** where available; **CPI General** for banking/insurance; **CPI Non-Food** for other services.

- A **deflator** removes price effects to measure **real output growth**.
- **Trial Phase: MoSPI** will release trial indices for **2025–26** and **April 2026** from **July 2026**, before monthly release with **60-day lag**.
- **Deflator Strategy:** In absence of **Service Producer Price Indices (SPPI)**, **CPI (Non-Food)** will be used as proxy, as over **80% of non-food inflation** is linked to services/cost-push factors.

FCRA Amendment Rules 2026

MHA notified the **Foreign Contribution (Regulation) Amendment Rules, 2026 (revise the 2011 Rules)** to tighten compliance for **NGOs/associations** receiving foreign funds, enhance transparency, classify permissible activities, and prevent misuse.

Key Highlights of the Rules

- **Activity-Specific Registration:**
 - NGOs must specify **exact objectives** and **State/UT-wise** operational areas.
 - Separate fees for each activity category and State/UT.
 - Existing FCRA organisations get **1 year** to update purposes and States/UTs.
 - **Religious Activities:** Foreign funds allowed for **worship, religious education, heritage preservation, and community kitchens**.
 - **Proselytisation (religious conversion)** is prohibited.
 - **Explicit Exclusion of Conversion Work:** Religious activities can receive funds, but **conversion-oriented work** is barred.
 - **SC in Rev. Stainislaus v. State of Madhya Pradesh (1977):** Right to propagate does **not** include the right to convert.
 - **Restrictions on “Key Functionaries”:** Associations with foreign nationals (except **PIOs**) as key functionaries will **ordinarily not** get FCRA registration/prior permission.
 - Includes **directors, partners, trustees, and even the Karta of a Hindu Undivided Family (HUF)**.
 - **Stricter Financial Compliance:** Must utilize **75%** of previous funds before receiving further installments.
 - Must spend at least **₹10 lakh** on approved activities over **2 years** for renewal.
 - **Enhanced Transparency:** Mandatory disclosure of **ultimate donor, official website, social media accounts, and publications**.
- Originally enacted in **1976**; replaced by **FCRA, 2010**.
 - Amended in **2016, 2018, and 2020**.
 - **Foreign Contribution (Regulation) Amendment Bill, 2026** overhauls handling of foreign funds/assets of NGOs whose licences are cancelled, surrendered, or expire.
 - **Nodal Ministry:** Implemented and monitored by the **Ministry of Home Affairs (MHA)**.
 - **Objective:** Regulate foreign contributions to protect **sovereignty, internal security, public interest, and democratic functioning**.
 - **Constitutional Linkage: Article 19(1)(c):** Right to form associations.
 - **Noel Harper v. Union of India (2022):** SC upheld **FCRA, 2010** as a reasonable restriction under **Article 19(4)**.
 - **Salient Features & Provisions:**
 - **Registration Requirement:** Foreign contribution needs **FCRA registration** or **Prior Permission** from Central Government.
 - **Eligibility for Registration:** Organisation must be registered under **Societies Registration Act, 1860, Indian Trusts Act, 1882, or Companies Act, 2013**.
 - ❖ FCRA registration valid for **5 years**.
 - ❖ Foreign funds allowed for **cultural, economic, educational, religious, or social** activities.
 - ❖ Barred: **election candidates, MPs/MLAs, political parties, judges, government servants, state-owned corporation employees, and registered news media personnel**.
 - ❖ **Indian Social Action Forum (INSAF) v. Union of India (2020):** Only groups in active/party politics are barred.
 - **SBI Account Mandate:** Foreign funds must first be received only in **FCRA Account, SBI New Delhi Main Branch**.
 - **Ban on Transfers:** **FCRA Amendment Act, 2020** bans transfer/sub-granting of foreign funds to any person or organisation.
 - **Registration Validity & Renewal:** Registration valid for **5 years**.
 - ❖ Renewal application must be filed online at least **6 months before expiry**.

FCRA, 2010

- **About: FCRA, 2010** regulates acceptance and utilization of **foreign contributions** and **foreign hospitality**.

The Foreign Contribution (Regulation) Act seeks to balance national security with civil society participation. Critically examine.

Practice Mains Question

International Relations

Presidential Visits to Moldova and North Macedonia

The **President of India** completed the first-ever visit by an Indian Head of State to **Moldova** and **North Macedonia**.



Key Highlights of the Visit to Moldova

- India decided to **double ITEC training slots** for Moldovan professionals.
- The President addressed the **India–Moldova Business Forum** to promote trade, investment and resilient supply chains.
- **Moldova**: A landlocked Eastern European country, bordered by **Romania** and **Ukraine**.
 - Became an **EU candidate country** in **2022**.
 - Strategically important due to the **Russia–Ukraine war** and the breakaway region of **Transnistria**.



Key Highlights of the Visit to North Macedonia

- The President paid tribute to **Goce Delčev**, a revolutionary against Ottoman rule.
- India welcomed North Macedonia's interest in joining the **ISA, CDRI and GBA**.
- India doubled **ITEC training slots**, with focus on **AI upskilling**.
- **North Macedonia**: A landlocked Balkan country bordered by **Kosovo** (northwest), **Serbia** (north), **Bulgaria** (east), **Greece** (south), and **Albania** (west).
 - Renamed from **Macedonia** in **2019**, enabling its **NATO membership**.
 - Serves as a gateway between the **Mediterranean** and **Central Europe**.
 - **Skopje**, its capital, is the birthplace of **Mother Teresa**.

World AI Cooperation Organization (WAICO)

At the **2026 WAIC** in **Shanghai**, China announced the operational launch of **WAICO**.

- Established with the support of **29 countries**, WAICO seeks to shape global **AI governance**. It promotes **inclusive AI development**, greater **Global South participation**, and a **state-led governance framework**.

WAICO

- **About**: **WAICO** is an independent intergovernmental organization headquartered in **Shanghai**, established to promote global cooperation and governance in **AI**.
- **Establishment**: Proposed at the **2025 WAIC** and included in China's **Five-Year Plan (2026–2030)**.
 - Formally established through an agreement signed by **29 countries**, including **China, Russia, Pakistan, Indonesia, Brazil** and **South Africa**.
- **Objective**: Promote inclusive AI development through **consultation, joint contribution and shared benefits**.
- **Focus on Inclusivity**: Improve developing countries' access to **AI technology, infrastructure, training and research cooperation**.
- **Capacity Building**: China has committed **5,000 AI training and research opportunities** over five years for countries associated with **ASEAN, African Union, SCO** and **BRICS**.

Concerns

- **State-centric governance**: Emphasizes state oversight, traceable data and "secure and controllable" AI.

- **Digital authoritarianism:** WAICO may spread China's state-led AI technologies and regulatory model through the **Digital Silk Road**. Reliance on Chinese AI infrastructure, cloud services and data systems could increase China's strategic and geopolitical influence.
- **Norm-setting:** Technical cooperation and capacity building may gradually promote Chinese AI standards in global governance.

WAICO vs Western and Indian AI Governance Frameworks			
Parameter	Western Frameworks	WAICO	India's "AI for All"
Objective	AI safety, ethics and accountability	Inclusive AI access and technology sharing	Inclusive growth and socio-economic development
Approach	Risk-based, rights-centric and multi-stakeholder	State-led, access-first and sovereignty-focused	Innovation-friendly, inclusive and development-oriented
Key Concerns	Frontier risks, privacy, misuse and national security	Export controls and technology denial	Digital divide, bias, trusted AI and strategic autonomy
Technology Access	Export controls and licensing	Open-source AI, capacity building and technology sharing	Open innovation through the IndiaAI Mission and compute infrastructure
Global Role	Global North-led standard setting	Focuses on the Global South, BRICS, SCO and ASEAN	Acts as a bridge between the Global North and South
Key Mechanisms	EU AI Act, Hiroshima AI Process, Bletchley Declaration	WAICO, Global AI Governance Initiative	IndiaAI Mission, National Strategy for AI, GPAI

India–UK CETA Comes into Effect

India-UK Comprehensive Economic and Trade Agreement (CETA) and the Double Contribution Convention (DCC) came into effect on 15th July 2026.

Key Highlights of India-UK CETA

- **UK Market Access:** The UK provides **zero-duty access to nearly 99% of Indian exports**, covering almost **100% of bilateral trade value**.
- **India's Tariff Commitments:** India offers concessions on **89.5% of tariff lines**, covering **91% of UK exports**.
- **24.5% of UK exports** get immediate duty-free access; the rest will be phased in.
- **Protection for Sensitive Sectors:** No concessions on **dairy, agriculture, millets, apples, edible oils, and gold**.
- **Services Trade Boost:** The UK opens all **12 major service sectors** and **137 sub-sectors**, benefiting Indian **IT, healthcare, financial, and engineering professionals**.
- **Safeguards for Steel Trade:** Both countries agreed to minimise disruptions from upcoming UK steel measures and protect Indian exporters.
- **Professional Mobility:** Provides clear rules for temporary professional movement without hidden quotas or arbitrary economic needs tests.
- **Protection for Strategic Sectors:** Tariff cuts for **Make in India** and **PLI Scheme** sectors will be phased over **5, 7, and 10 years**.

- **Calibrated Automotive Liberalisation:** Automobile access will be **quota-based and phased**.

- Concessions for affordable EVs begin only from **Year 6**.

Significance of India-UK CETA

- **Textiles & Apparel:** Zero-duty access improves India's competitiveness in the UK's **USD 28.8 billion textile import market**, benefiting garments, home textiles, and handicrafts.
- **Plantation Sector:** Duty-free access supports value-added coffee exports. The UK accounts for **5.6% of India's tea exports, 2.9% of spice exports, and 1.7% of coffee exports**.
- **Marine Products:** Tariff-free access to the UK's **USD 4.9 billion market** promotes seafood exports, value-added processing, better price realisation, and coastal livelihoods.
- **Engineering & Electronics:** Engineering exports may exceed **USD 7.5 billion by 2029–30**, alongside higher exports of smartphones, optical fibre cables, and IT-enabled services.
- **Pharmaceuticals & Chemicals:** Duty-free access opens the UK's **USD 30 billion pharmaceutical** and **USD 35.8 billion chemical** import markets.
- **Plastics:** Lower landed costs improve competitiveness; India is the UK's **13th-largest plastics supplier**.
- **Sports Goods & Toys:** Exports may rise by **15%** and reach **USD 186.97 million by 2030**, improving competitiveness against China and Vietnam.
- **Oilseeds:** Lower tariffs and streamlined procedures improve export competitiveness.

- **Leather & Footwear:** Duty-free access to the UK's **USD 8.9 billion market** supports labour-intensive manufacturing.
- **Promotes Inclusive Innovation:** Supports technology-neutral innovation and producers across major agricultural and horticultural states.
- **Gems & Jewellery:** Jewellery exports may double within **2–3 years**, creating jobs in **Surat, Jaipur, and Mumbai**.
- **Steel Sector:** Expanded tariff-free quotas and preferential access protect Indian exports from UK safeguard measures.

Double Contribution Convention (DCC)

- **About:** The **DCC** prevents dual social security contributions for specified temporary workers moving between **India and the UK**.
- **Extended Exemption Window:** Extends the UK **National Insurance (NI)** exemption for detached workers from **12 months to 60 months**.
- **The Non-Retropective Clause:** It does not apply retrospectively.
 - Indian professionals working in the UK immediately before **15th July 2026** must continue paying **UK NI contributions**.
- **Certificate of Coverage (CoC):** Eligible workers must obtain a **CoC from EPFO**, proving contribution to India's social security system.
- **Significance:** Expected to benefit over **75,000 Indian professionals** and **900 companies**, with annual savings exceeding **USD 600 million**.
 - May promote talent mobility and encourage UK firms to develop Indian operations into **Global Capability Centres (GCCs)** focused on **R&D, advanced analytics, and cybersecurity**.

Key Challenges in India–UK CETA

Suggestions

- | Key Challenges in India–UK CETA | Suggestions |
|--|--|
| <ul style="list-style-type: none"> ■ Non-Tariff Barriers: Strict Sanitary and Phytosanitary (SPS) norms, technical standards, and certifications may restrict gains for MSMEs, agri-food, pharmaceutical, and engineering exporters. ■ UK Carbon Border Adjustment Mechanism (CBAM): From 2027, Indian exports such as steel, aluminium, cement, and fertilisers may face carbon taxes, reducing competitiveness. ■ Import Circumvention: Third-country goods may enter India through the UK at lower tariffs, affecting Make in India and PLI initiatives. ■ Complex Rules of Origin: Stringent verification and documentation may raise compliance costs, especially for small exporters. ■ Limited Mobility Gains: Visa liberalisation remains limited despite the DCC, with only a small annual quota for select service providers. ■ Low FTA Utilisation: Poor awareness of CETA provisions and compliance rules may limit exporter participation. | <ul style="list-style-type: none"> ■ Address Non-Tariff Barriers: Improve testing, certification, and traceability infrastructure and use the CETA Joint Committee to resolve trade barriers. ■ Greening Supply Chains: Decarbonise manufacturing and align industries with global green standards to counter climate-based trade barriers. ■ Enhance MSME FTA Utilisation: Set up MSME Trade Facilitation Centres and promote paperless trade and Authorised Economic Operator (AEO) mechanisms. ■ Expand Services & Mobility: Fast-track MRAs and promote AI, cybersecurity, and advanced analytics. ■ Protect Domestic Industry: Enforce Rules of Origin, conduct sectoral reviews, and use safeguard measures when required. |

The India–UK Comprehensive Economic and Trade Agreement (CETA) represents a shift from defensive protectionism to strategic trade integration. Examine.

Practice Mains Question

India-Australia Civil Nuclear Cooperation

India and Australia finalised an **Administrative Arrangement under their Civil Nuclear Cooperation Agreement** to ensure long-term Australian uranium supplies for India's **nuclear energy programme**.

Key Highlights of the Arrangement

- **Background:** The **Civil Nuclear Cooperation Agreement** was signed in **2014** and came into force in **2015**.
- **Operationalisation:** The arrangement sets procedures for commercial exports of **Australian uranium** to India.
- **Purpose:** Uranium will be used only for **peaceful purposes** under **IAEA safeguards**.

- **Participation:** Australian uranium companies can contract with eligible Indian entities, including private firms.
- **Framework:** It provides a stable long-term mechanism while meeting Australia's **safeguards and non-proliferation requirements**.

India's Current Nuclear Energy Landscape

- **Installed Capacity:** India operates **24 reactors** across **seven sites**, with **8.78 GW** capacity.
- **Capacity Expansion:** **10 reactors (8,000 MW)** are under construction, while pre-project work has begun for **10 more**.
- **Reactor Mix:** The fleet includes **PHWRs, BWRs, and LWRs**. PHWRs mainly use natural uranium and produce plutonium.

- **Breeder Programme:** The **500 MWe Prototype Fast Breeder Reactor** at **Kalpakkam** attained first criticality in **April 2026**, launching the second stage of India's three-stage nuclear programme.
- **Thorium Strategy:** Fast Breeder Reactors will support the shift towards thorium-based reactors and **Uranium-233** production, using thorium reserves concentrated in **Kerala, Tamil Nadu, Andhra Pradesh, and Odisha**.
- **Small Reactors:** The **Union Budget 2025–26** allocated **₹20,000 crore** for indigenous **SMRs**, with at least **five** targeted by **2033**.

Significance of the Administrative Arrangement for India and Australia

Significance for India

- **Fuel Security:** Ensures long-term Australian uranium supply for India's expanding reactor fleet.
- **Energy Transition:** Supports low-carbon baseload power, complements renewables, and aids the **2070 net-zero target**.
- **Nuclear Expansion:** Supports the **Nuclear Energy Mission** and **SHANTI Act, 2025**, improving fuel certainty, investment, and private participation.
- **Global Cooperation:** Strengthens India's civil nuclear ties; India has agreements with **18 countries**.

Significance for Australia

- **Resource Advantage:** Enables Australia, which holds over **one-third of global uranium resources**, to serve India's nuclear market.
- **Export Opportunity:** Creates a stable market and commercial opportunities for Australian uranium firms.
- **Strategic Partnership:** Adds an energy pillar to the **India–Australia Comprehensive Strategic Partnership** under **IAEA safeguards** and non-proliferation commitments.

The India-Australia Administrative Arrangement for uranium supply strengthens India's nuclear energy programme and energy security. Discuss.

Practice Mains Question

China's Helium Export Ban and Supply Chain Vulnerabilities

China temporarily banned **helium exports** to preserve domestic supplies for **AI chips and quantum computing**, raising global supply concerns despite importing over **80% of its helium**.

Helium

- **Basic Properties: Helium (He)** is the second-lightest and second-most abundant element (after Hydrogen) in the universe and the first **noble gas**.

- **Physical Characteristics:** It is colourless, odourless, tasteless, non-toxic, non-flammable, and chemically inert.
- **Lowest Boiling Point:** It has the lowest boiling point of any element at **–269°C (4.2 K)**.
- **Occurrence:** On Earth, helium is a **non-renewable resource** formed mainly through radioactive decay of uranium and thorium.
- **Extraction Process:** It accumulates in natural gas reservoirs and is commercially extracted when its concentration is at least **0.3% by volume**.
- **Isolation and Purity:** Once extracted, it is separated using its low boiling point and purified to about **99.997%**.
- **Minor Sources:** Small quantities are recovered from **LNG plants** and the atmosphere, but these are negligible.

Fragile Global Helium Supply Chain

- **Concentrated Production:** China produces only **1.6%** of global helium and imports over **80%** of its needs. Production is concentrated in the **US, Qatar, and Russia**, with **Canada and Algeria** as other suppliers.
- **Geopolitical Risks:** Restrictions on Russian exports and other disruptions have tightened supply.
- **Chokepoint Risks:** Qatar's Asian shipments pass through the **Strait of Hormuz**, making supplies vulnerable to West Asian tensions.
- **Storage and Transport Complexity:** Helium requires specialised cryogenic or compressed-gas vessels. Transit delays cause **boil-off**, making logistics time-sensitive.
- **India's Vulnerability:** India is **100% import-dependent**, with over **50%** of imports from Qatar. The **March 2026 Ras Laffan disruption** caused shortages and a **35–50% price rise** in India.

Does India Have Domestic Helium Reserves?

- **Bakreshwar Thermal Springs (West Bengal):** India's most promising source, with gases containing about **1.4% helium**. **Variable Energy Cyclotron Centre (VECC)** under **DAE** has established a pilot plant for helium capture, purification, and liquefaction.
- **Monazite Sands of Kerala:** Thorium- and uranium-rich sands contain trapped helium, but commercial extraction is highly capital-intensive.
- **Oil and Natural Gas Basins:** **ONGC** is mapping fields such as the **Godavari and Mahanadi basins**, though Indian natural gas generally contains very low helium levels.

India-New Zealand Strategic Partnership: Roadmap to 2030

PM visited **New Zealand**, marking the first visit by an Indian PM in **four decades**. Both countries elevated ties to a **Strategic Partnership** and adopted the **India-New Zealand Strategic Partnership: Roadmap to 2030**.

- Agala titled “**India-New Zealand: A Winning Partnership**” highlighted their growing cooperation and shared democratic values.

Key Features of the Partnership

- **Political and Diplomatic Engagement:** Regular high-level visits, **Foreign Ministers’ Dialogue**, parliamentary exchanges, and annual **Secretary-level meetings**.
- **Defence and Maritime Security:** Greater military exchanges and naval exercises; cooperation on **Maritime Cooperation, Hydrography and Nautical Cartography**, and **Mutual Logistics Support**. An annual **Maritime Security Dialogue** will be established.
- **Counter-Terrorism and Security Cooperation:** A **Joint Working Group on Counter-Terrorism**, stronger cooperation against narcotics trafficking and crime, and enhanced cybersecurity cooperation through the **India-New Zealand Cyber Dialogue**.
- **Trade and Economic Cooperation:** Target to double bilateral trade to **NZ\$7 billion (₹35,000 crore) by 2030**.
 - Trusted trade will be supported by the **2025 Authorised Economic Operators Mutual Recognition Arrangement (AEO-MRA)** under the **2024 Customs Cooperation Arrangement (CCA)**, along with early implementation of the **India-New Zealand Free Trade Agreement**.
- **Agriculture and Primary Industries:** Cooperation in horticulture, forestry, animal husbandry, and dairying.
 - An **MoC on Animal Husbandry and Dairying**, a **Kiwifruit Action Plan**, and two **Centres of Excellence in Nagaland and Uttarakhand** will promote productivity, skills, innovation, and education.
- **Tourism, Culture and Sports:** The **Tourism Cooperation Agreement** aims to increase tourist flows, while the updated **Air Services Agreement** promotes direct flights. The **Joint Action Plan on Sport** expands cooperation beyond cricket.
 - The **Cultural Cooperation Arrangement** and collaboration b/w **Lothal’s National Maritime Heritage Complex** and the **New Zealand Maritime Museum** will promote cultural and maritime heritage exchanges.
- **Education, Research and Technology:** Cooperation will expand in education, climate, digital transformation, and emerging technologies. **NCPOR–University of Canterbury** cooperation will strengthen Antarctic research, while **NIFTEM–K–Massey University** cooperation will support research, academic exchanges, and student mobility.
- **Clean Energy and Disaster Management:** New Zealand joined the **Global Biofuels Alliance**. An **NDMA–NEMA MoC** will strengthen earthquake resilience, tsunami preparedness, coastal hazard mitigation, and disaster response.
- **Indo-Pacific and Multilateral Cooperation:** New Zealand prioritised the **Maritime Security pillar of IPOI**, focusing on **IUU Fishing**. Both countries supported a rules-based Indo-Pacific, peaceful dispute resolution under **UNCLOS**, **UN reform**, and India’s permanent membership of a reformed **UNSC**.
- **Non-Binding Roadmap:** The **Roadmap to 2030** creates no financial or legally binding obligations and serves as the implementation framework for the Strategic Partnership.

Major Challenges to India-New Zealand Strategic Partnership: Roadmap to 2030	Suggestions
■ Dairy Sector Friction: India’s reluctance to provide wider dairy market access protects its small farmers but limits deeper economic integration with export-oriented New Zealand. ■ Agricultural Technology Transfer Concerns: New Zealand producers fear that technology transfers in kiwifruit, apples, and honey could create stronger Indian competitors. ■ Divergent China Calculus: China receives ~30% of New Zealand’s exports, limiting its alignment with India on balancing China in the Indo-Pacific and constraining defence cooperation. ■ Strategic and Normative Divergences: New Zealand’s Five Eyes membership and positions on human rights and governance may conflict with India’s strategic autonomy and sovereignty-focused policy. ■ Transnational Extremism Concerns: Separatist activities on New Zealand soil complicate counter-terrorism cooperation due to its domestic free-speech laws. ■ Labour Mobility Constraints: New Zealand’s changing immigration policies and concerns over jobs, infrastructure, and migrant exploitation may restrict access for Indian skilled professionals.	■ Institutionalizing Dispute Resolution: Use Sectoral Working Groups to resolve Phytosanitary (SPS) disputes and gradually expand agricultural market access without affecting India’s dairy sector. ■ Upgrading the Air Services Agreement: Prioritise direct flights to improve connectivity and support the NZ\$7 billion trade target. ■ Deepen Non-Traditional Maritime Cooperation: Expand cooperation in IUU fishing, maritime domain awareness, hydrography, HADR, and UNCLOS-based governance under the IPOI. ■ Fostering ‘Niche’ Technological Synergies: Focus on Green Hydrogen, Agri-Tech, Space Exploration, and the Global Biofuels Alliance. ■ Joint Outreach to Pacific Island Countries: Extend NDMA–NEMA and CDRI cooperation for tsunami preparedness, coastal resilience, and disaster-management capacity.

- With **New Zealand**, India has Strategic Partnerships or equivalent arrangements with around **46 countries and regional blocs**.
- India's first Strategic Partnership was with **France**, followed by **Russia, the US, the UK, and the EU**, later expanding through the **Act East Policy and Indo-Pacific engagement**.

The India-New Zealand Strategic Partnership reflects India's growing preference for issue-based partnerships rather than traditional military alliances. Discuss with reference to the Roadmap to 2030.

Practice Mains Question

3rd India-Australia Annual Leaders' Summit 2026

PM visited Australia for the **3rd India-Australia Annual Leaders' Summit (July 2026)**. The visit focused on deepening cooperation in the Indo-Pacific, securing energy and critical technology supply chains, defense collaboration, and building a durable **strategic partnership**.



Key Outcomes of the Summit

- **Defence & Maritime Security:** Adopted **Joint Declaration on Defence and Security Cooperation (JDDSC)**, renewing the **2009 framework**.
 - Focus areas: **interoperability, cybersecurity, defence industrial collaboration**, and cooperation in **UNCLOS, ASEAN and IORA**.
 - Launched **Maritime Security Collaboration Roadmap (MSCR)** for information sharing, capability development and operational coordination.
 - Signed **MoU between Indian Coast Guard (ICG) and Maritime Border Command (MBC)** for maritime law enforcement and border protection.
- **Energy Security & Civil Nuclear Ties:** Operationalised the **Administrative Arrangement under the 2014 India-Australia Civil Nuclear Agreement**, enabling Australian uranium supply for India's civilian nuclear energy needs.
 - Adopted **India-Australia Joint Statement on Energy Security** to strengthen energy supply chains.

- Launched **Rooftop Solar Training Academy** at **Pandit Deendayal Energy University (PDEU), Gandhinagar** to train **2,000 women and youth** as solar technicians under **PM Surya Ghar Yojana**.
- **Technology, Innovation & Critical Minerals:** Launched **Australia-India Partnership for Cyber, Critical Technologies, Supply Chains (PACTS)** to enhance digital resilience and secure supply chains.
 - Signed trilateral **ACITI Partnership MoU (Australia-Canada-India Technology and Innovation)** for emerging technology cooperation.
- **Investment:** **AustralianSuper** committed an additional **AUD 500 million** to **National Investment and Infrastructure Fund (NIIF)**, taking its total India investments to **AUD 3.3 billion**.
- **Education, Skills & Scientific Research:** Approved campuses of **Flinders University (Bengaluru)** and **Victoria University (Gurugram)** in India.
 - Provided **IP Australia** access to **Traditional Knowledge Digital Library (TKDL)** to prevent patent misappropriation.
 - Launched **India-Australia Sports Collaboration Roadmap** for high-performance sports centres and cooperation in **Para sports and Olympics**.
- **Space Cooperation:** Australia supported **Gaganyaan Mission** through a temporary space-tracking terminal at **Cocos (Keeling) Islands**.
- **Culture:** Australia announced repatriation of **three 11th–12th century Indian artefacts: Stone Nandi; Bhadrakali trident; Skanda sculpture**
- **Social Media Regulation:** India welcomed Australia's initiative on restricting **social media access for minors**.

India-Australia Relations

- **High-Level Bilateral & Multilateral Architecture:** Australia is one of only two countries (**along with Japan**) with which India has:
 - Annual Leader-Level Summits; Free Trade Agreement; 2+2 Foreign and Defence Ministers' Dialogue
- **Economic and Trade Dynamics:** **Economic Cooperation and Trade Agreement (ECTA)** came into force in **December 2022**, strengthening economic integration.
 - India is **Australia's 5th-largest trading partner**.
 - Two-way trade in goods and services reached **USD 50.2 billion (2025)**.
 - **2024–25 trade:** Australian exports: **USD 32 billion**; Indian imports: **USD 18 billion**
 - **2025:** Australia launched **Roadmap for Economic Engagement with India** focusing on four sectors: **Clean energy; Education; Agribusiness; Tourism**

- Both countries are negotiating an upgrade of **ECTA** into **Comprehensive Economic Cooperation Agreement (CECA)**.
- **Strategic Convergence:** Both countries are members of: **Quad; Supply Chain Resilience Initiative (SCRI)**
 - Aim: Promote a **free, open and rules-based Indo-Pacific**.
- **Renewable Energy: India-Australia Renewable Energy Partnership (REP)** launched in **2024**.
 - Focus areas: **Solar PV; Green hydrogen; Battery storage; Circular economy**
- **Education and People-to-People Links:** Around **976,000 people** in Australia reported **Indian heritage**.
 - **2023:** Launched Migration and Mobility Partnership Arrangement; Centre for Australia-India Relations
 - Education is Australia's major service export to India, valued at **USD 9.4 billion (2025)**.

Major Challenges in India-Australia Relations

- **Strategic and Geopolitical Divergences:** Australia is closely aligned with the **US alliance system and AUKUS**, while India follows **strategic autonomy** and avoids formal military alliances.
 - Security priorities differ: Australia focuses on **Western Pacific maritime threats**.
 - ❖ India faces **LAC-related continental challenges and Indian Ocean maritime concerns**.
 - Australia's economic dependence on **China** complicates deeper strategic alignment.
- **Economic and Trade Bottlenecks:** Progress from **ECTA** to **CECA** is slow due to:
 - Australia seeking greater access to India's **agriculture and dairy markets**.
 - **Australian SPS standards** affecting Indian exports.
 - India's dependence on Australian **coking coal and LNG** contributes to a trade deficit.
- **Critical Minerals Supply Chain Constraints:** Australia has reserves of **lithium, cobalt and rare earth elements**, but limited processing capacity.
 - Dependence on **China-linked refining supply chains** restricts development of secure and diversified mineral supply chains.
- **Sociopolitical and Diaspora Frictions:** Issues such as **temple vandalism** and activities of **Khalistani separatist elements** have caused diplomatic tensions.
 - Differences over **freedom of expression laws** complicate cooperation on security concerns.

Measures to Strengthen India-Australia Relations

- **Deepen Economic Integration:** Finalise a balanced **CECA** by addressing **Australian SPS barriers** through **Mutual Recognition Agreements (MRAs)**.
 - Protect India's agriculture and dairy sectors through **tariff-rate quotas** and safeguard mechanisms.
- **Build Resilient Critical Mineral Supply Chains:** Shift from raw mineral trade to joint **refining and processing infrastructure**.
 - Leverage **Quad Critical Minerals Initiative (2026)** and operationalise **PACTS (2026)** to reduce dependence on China-linked supply chains.
- **Deepen Defence and Maritime Cooperation:** Implement **JDDSC and MSCR (2026)** through stronger **Underwater domain awareness; Military deployments; Operational coordination**.
 - Promote joint **defence R&D, technology transfer and co-production**.
- **Strengthen Nuclear and Energy Security:** Convert operationalisation of the **India-Australia Civil Nuclear Agreement (2014)** into long-term **uranium supply arrangements** for India's nuclear energy programme.
- **Manage Diaspora and Security Concerns:** Enhance **intelligence sharing and law-enforcement cooperation** against extremist activities, **Khalistani elements**, and attacks on Indian diplomatic and cultural institutions.
 - Ensure protection of diplomatic missions under **international law**.

India-Australia relations have evolved from mutual strategic neglect to an increasingly institutionalised Comprehensive Strategic Partnership.

Practice Mains Question

India's Bid for UNSC

Non-Permanent Seat for 2028-29

India will launch its campaign for a **UNSC non-permanent seat for 2028-29**, aiming to strengthen its role in **global governance** and represent the **Global South**.

Key Issues and Priorities for India's 2028–29 UNSC Bid

- **Diplomacy:** India needs strong mobilisation as **Tajikistan** is also contesting the **Asia-Pacific non-permanent UNSC seat**.
- **Support:** Tajikistan's reported **OIC backing** may make the contest tougher; India must widen support beyond **Fiji, US, Austria and Sri Lanka**.
- **Leadership:** India will project itself as a responsible **Global South** voice on **energy security, fertiliser supply, development and global governance**.

- **Reform:** India will continue pushing for **UNSC expansion and reform** to reflect current geopolitical realities.
- **Engagement:** EAM's outreach to the **Gulf, Caribbean and other regions** aims to build wider support.
- **Competition:** **Kyrgyzstan's victory over the Philippines** shows UNSC elections are increasingly competitive; Kyrgyzstan will serve on the Council for the first time since joining the UN in **1992**.

Key Facts About UNSC

- **About:** UNSC was established in **1945** under the **UN Charter** as one of the **6 principal organs** of the UN, responsible for maintaining **international peace and security**.
- **Members:** It has **15 members: 5 permanent members** and **10 non-permanent members** elected for **2-year terms** by the **UNGA**.
- **Permanent Members:** **China, France, Russia, UK and US** have **veto power**.
- **Non-permanent Members:** Seats are regionally distributed: **5 African and Asian, 1 Eastern European, 2 Latin American and Caribbean, and 2 Western European and Other States**.
- **Election:** Non-permanent members are elected annually by **secret ballot** and need a **two-thirds majority**, even if unopposed.
- **Binding Authority:** UNSC is the only UN body whose decisions are binding on member states under the **UN Charter**.

India and the UNSC

- **India's Participation:** India has served as a **non-permanent UNSC member** in **1950-51, 1967-68, 1972-73, 1977-78, 1984-85, 1991-92, 2011-12 and 2021-22**.
- **Permanent Seat Claim:** India is part of the **G4** with **Brazil, Germany and Japan**, supporting expansion of **permanent and non-permanent UNSC seats**.

Key Challenges Related to UNSC

- **Lack of Adequate Representation:** UNSC is less representative, especially due to absence of **Africa's 54 countries** and major powers like **India, Germany, Brazil and South Africa** from permanent membership.
- **Misuse of Veto Power:** **P5 veto** is criticised as undemocratic and often prevents necessary decisions, making UNSC unsuitable for today's security needs.
- **Geopolitical Rivalry within P5:** Rivalries among **US, Russia and China** weaken UNSC action on issues like the **Taiwan issue** and **Russia-Ukraine War**.
- **Threat to State's Sovereignty:** UNSC resolutions are binding, and actions like **sanctions** can affect a state's sovereignty.

Way Forward

- **Democratization of UNSC:** Address power imbalance between **P5** and other countries to make UNSC more democratic and legitimate.
- **Expansion:** Include under-represented regions, especially **Africa, Latin America and Global South**, along with major contributors like **India, Germany, Brazil and South Africa**.
- **Representation:** Membership should reflect present-day **geopolitical and economic realities**, not post-World War II power structures.
- **Veto Reform:** Restrict veto use in cases of **genocide, war crimes, terrorism, humanitarian crises and mass atrocities**.
- **Transparency:** Improve working methods through wider consultations, open debates and participation of affected non-member states.
- **Accountability:** Ensure fair and consistent enforcement of resolutions so sanctions, peacekeeping and military action are not seen as selective.
- **India's Role:** India pushed UNSC reform in **September 2022** through **G-4** and **L-69** meetings in New York; it should represent **Global South** peace and security concerns in UNSC.

India's campaign for a non-permanent seat at the UNSC is closely linked to its larger demand for reform of global governance institutions. Discuss.

Practice Mains Question

India-Indonesia Relations

PM concluded a **State Visit to Indonesia**, advancing the **India-Indonesia Comprehensive Strategic Partnership (2018)**. He received Indonesia's highest honour, **Bintang Adipurna**, becoming the **second Indian PM** after **Jawaharlal Nehru** to receive it.

Key Outcomes of the PM Visit to Indonesia

- **Defence and Maritime Security:** Indonesia signed a **USD 200 million contract** for **2 BrahMos missile batteries** and placed the first export order for India's **Astra air-to-air missile system**.
 - Both sides extended cooperation on **Maritime Safety and Security**, important for **Indo-Pacific sea lanes**.
 - India and Indonesia agreed to jointly develop **Sabang Port**, located near the **Strait of Malacca**, strengthening India's maritime surveillance and naval logistics reach.
 - Leaders welcomed the planned **3rd India-Indonesia Security Dialogue**.
- **Critical Minerals, Steel, and Heavy Industry:** **SAIL** signed an agreement to set up a **Stainless-Steel Slab manufacturing facility** in Indonesia.

- MoU signed for joint development of **Rare Earth Magnets**, crucial for EVs and advanced electronics.
- MoU signed for cooperation in **steel and mineral supply chain extraction and technology**.
- **Technology and Space:** Space cooperation framework for peaceful use of outer space was extended.
 - MoUs signed on **Research, Technology and Innovation** and **Next-Generation Telecommunications Technologies**.
- **Health and Disaster Management:** MoU between **CDSO (Central Drugs Standard Control Organisation)** and Indonesia to harmonize **medical product regulations**.
 - Implementation Agreement signed on **Health Workforce Collaboration**.
 - MoU between **NDMA** and Indonesia's **National Agency for Disaster Management** for institutional cooperation.
- **Agriculture:** MoU signed on **Agriculture and Allied Sectors** to strengthen mutual **food security**.
 - India announced supply of **100 tonnes of DWR 162 wheat seeds** to support Indonesia's agricultural output.
- **Soft Power Initiatives**
 - **Export of India's DPI:** Indonesia launched **Indonesia Open Network (ION)**, modelled on India's **ONDC**.
 - **IFC-IOR Deployment:** Indonesia will deploy a Liaison Officer at **Information Fusion Centre for the Indian Ocean Region (IFC-IOR)**, Gurugram for maritime domain awareness and intelligence sharing.
 - **Heritage Conservation:** India will assist in conserving the **Prambanan Temple Complex** in **Yogyakarta**.
 - **Cultural Commemoration:** **Tagore-Dewantara Year 2026-27** will mark **100 years of Rabindranath Tagore's visit to Indonesia**.
 - **Electoral Knowledge:** **ECI** and Indonesia's **General Elections Commission (KPU)** will share best practices in managing large-scale elections.
 - **Educational Outreach:** Branch campus of **IIM Bangalore** to be set up at **Singhasari SEZ, East Java**.

India-Indonesia Bilateral Relations

- **Economic & Commercial Dynamics:** Indonesia is India's **2nd-largest ASEAN trading partner**, with bilateral trade at **USD 28.15 billion in 2024-25**.
- **Trade Pattern:** India is the largest global buyer of Indonesian **crude palm oil** and **2nd-largest buyer of coal**; exports include **refined petroleum, vehicles, steel and agricultural goods**.
- **Tourism Corridor:** India is the **5th-largest source of tourists** to Indonesia, supporting routes like **Delhi-Bali** and **Mumbai-Jakarta**.



- **Defence & Maritime Security Cooperation:** Cooperation is monitored through **Joint Defence Cooperation Committee (JDCC)** and **Defence Ministers' Dialogues**; exercises include **IND-INDO CORPAT, Milan, Komodo, Garuda Shakti and Samudra Shakti**.
- **Science & Technology:** **ISRO** and Indonesia's **BRIN** operationalised the **Integrated Biak Telemetry, Tracking and Command (TTC) Facilities** for tracking Indian satellites and launch vehicles.
 - **Socio-Cultural Symbols:** India presented a replica of the **9th-century Nalanda Copper Plate**, highlighting **Nalanda-Muarajambi ties**.
 - **Capacity Building via ITEC:** Indonesia is a key **ITEC** beneficiary; **30 Indonesian judges** trained at the **National Judicial Academy, Bhopal** in **April 2026**.

Strategic Significance of India-Indonesia Ties

Dimension	Significance
Geopolitical Strategy	Acts as the cornerstone of India's ' Act East Policy ' and SAGAR vision . Indonesia is India's closest maritime neighbor in Southeast Asia.
Maritime Security	Joint development of Sabang Port enhances the Indian Navy's logistical reach and interoperability to monitor the Strait of Malacca , ensuring a free, open, and rules-based Indo-Pacific.
Countering Dominance	Strengthening Indonesia's military capabilities (via BrahMos and Astra) and diversifying critical mineral supply chains reduces the region's asymmetric dependence on China.
ASEAN Centrality	Indonesia is the largest economy and a natural leader within ASEAN. Deepening ties with Jakarta automatically amplifies India's strategic footprint across the entire Southeast Asian bloc.

Challenges in India-Indonesia Relations	Measures to Strengthen India-Indonesia Relations
■ The "Quad" vs. ASEAN Centrality: Indonesia follows Bebas-Aktif (Independent and Active) foreign policy and remains cautious of the Quad, fearing it may provoke China and weaken ASEAN Centrality. ■ Hesitance on Anti-China Balancing: Despite buying BrahMos, Indonesia avoids overt anti-China alignment, preferring local capacity building over broader strategic balancing. ■ China's Economic Influence: Indonesia's deep ties with China in infrastructure and nickel-based EV supply chains limit India's economic footprint. ■ Trade Imbalance and Palm Oil Vulnerability: Bilateral trade of USD 28–30 billion is skewed in Indonesia's favour due to India's large palm oil imports. Indonesia's biofuel push may tighten edible oil supply and raise food inflation in India. ■ Trade Barriers: Absence of CECA, low use of ASEAN-India Trade in Goods Agreement (AITIGA), and Indonesian non-tariff barriers in pharma and agriculture restrict trade growth. ■ Protectionist Frictions: Indian industries, especially textiles, raise concerns over imports and alleged dumping of Indonesian products like Viscose Staple Fibre (VSF).	■ Restore Trade Parity: ● Fast-track AITIGA review and explore bilateral CECA to address non-tariff barriers and expand Indian exports in pharmaceuticals, dairy and other sectors. ■ Build Critical Mineral Supply Chains: ● Develop nickel, rare-earth and EV battery value chains through joint ventures, equity investments and processing facilities in Indonesia to reduce dependence on China-dominated supply chains. ■ Operationalise Sabang Port: Set up a dedicated SPV, assured financing and time-bound framework to develop Sabang Port near the Malacca Strait. ■ Respect Indonesia's Strategic Autonomy: ● Align India's SAGAR vision with Indonesia's Global Maritime Fulcrum, while respecting Bebas-Aktif doctrine and ASEAN Centrality. ■ Practical Indo-Pacific Cooperation: ● Focus on maritime capacity-building, intelligence sharing and practical Indo-Pacific cooperation

India and Indonesia are natural maritime partners, yet the full strategic potential of their relationship remains underutilised. Discuss.

Practice Mains Question

Inaugural India–Mali Forum for Promotion of Exports

India and Mali held the inaugural **India–Mali Forum for the Promotion of Exports** in **Bamako** under the theme **"Reinforcing Trade and Strategic Partnerships"**, institutionalizing their commercial partnership.

Key Takeaways from the India–Mali Export Forum

- **Bilateral Trade Surge:** India–Mali trade grew **55% YoY**, crossing **USD 326.61 million** in **FY 2025–26**.
- **Catalyst for Trade:** India's **Duty-Free Tariff Preference (DFTP) Scheme** drove bilateral trade growth.
- **Untapped Market Potential:** Mali's global exports are about **USD 4 billion**; untapped export potential in India is nearly **USD 3.96 billion**.
- **Key Trade Commodities:**
 - **Mali to India:** Raw cotton, finished leather, cashew, lead, gum arabic and sesame.
 - **India to Mali:** Pharmaceuticals, cotton fabrics, two/three-wheelers and bicycles.
- **Priority Sectors Identified:** **Cotton and textiles**, **mining and energy**, **agro-industry and shea processing**, **pharmaceuticals**, **health and education** for joint ventures and investments.

- **Mutual Administrative Requests:** Mali sought Indian support for **digitizing Certificate of Origin** and faster registration of Indian pharmaceuticals; India sought review of Mali's **shea-nut export ban** and protection of **Indian nationals and investments**.
- **Future Roadmap:** Mali projected itself as a West African investment hub under **Vision Mali 2063** and announced an **Investment Forum in December 2026** for bankable projects.

16th India-Japan Annual Summit 2026

The **16th India-Japan Annual Summit** was held in **New Delhi** to strengthen the **Special Strategic and Global Partnership**, focusing on **defence co-development**, **clean energy** and **AI**.

Key Outcomes of 16th India-Japan Annual Summit

- **Defence and Security:** Agreement in principle on **UNICORN project**, the first **India-Japan defence co-development programme**.
 - Focus on **defence equipment**, **maritime security**, **MRO**, **MDA**, joint exercises.
 - **4th India-Japan 2+2 Ministerial Meeting** to be held in **2026**.
- **Economic Security:** Adopted **Joint Declaration on Economic Security Cooperation**.

- **Focus: semiconductors, critical minerals, ICT, clean energy, pharmaceuticals.**
 - **Aim: resilient supply chains** and reduced dependence on any single country.
 - **Energy and Clean Technology:** Adopted **Joint Statement on Energy Resilience**.
 - Cooperation in **strategic petroleum reserves, energy transport, green hydrogen, clean ammonia, solar PV and nuclear energy.**
 - Launched **India-Japan Cooperative Biogas for Growth Initiative** for **1,000 biogas and organic fertilizer plants.**
 - Japan supported India's **IEA membership.**
 - **AI and Emerging Technologies:** Launched first **India-Japan AI Strategic Dialogue**.
 - Cooperation on **trusted AI, digital infrastructure, AI supply chains and responsible AI governance.**
 - **Trade and Investment:** Agreed to speed up review of **CEPA**.
 - Working towards **¥10 trillion Japanese investment target.**
 - Signed **MoC for USD 10 billion investment package.**
 - Cooperation under **IJICP** in **logistics, agriculture, healthcare, textiles, manufacturing, MSMEs and startups.**
 - **Indo-Pacific and Global Cooperation:** Reaffirmed commitment to **Free and Open Indo-Pacific**, aligned with **IPOI and MAHASAGAR**.
 - Strengthened cooperation through **Quad, ASEAN and BIMSTEC.**
 - Supported **UNCLOS, freedom of navigation, UNSC reforms** and counter-terrorism.
 - Started **1.5 track policy dialogue with the Philippines** on maritime security.
 - **Maritime Security:** Raised concerns over **East China Sea and South China Sea** developments & reaffirmed commitment to **UNCLOS**.
 - **Disaster Management:** Japan to support India in hosting **4th UN World Conference on Disaster Risk Reduction, 2030.**
 - **Africa Cooperation:** Agreed to deepen cooperation in **Africa** through **IAFS** and sustainable development initiatives.
- **Act East Forum and Quad revival in 2017** expanded strategic cooperation.
 - **Defence and Security:** Key mechanisms: **2+2 Ministerial Dialogue and Joint Declaration on Security Cooperation**.
 - **RPSS Agreement (2020)** enables logistics support during exercises.
 - Major exercises: **JIMEX, Dharma Guardian, Malabar, Veer Guardian.**
 - Defence cooperation includes **2015 technology transfer agreement** and **UNICORN project.**
 - **Economic Relations**
 - Bilateral trade: **USD 27.48 billion** in FY 2025–26.
 - India's exports: **USD 6.04 billion**; imports: **USD 21.44 billion.**
 - Japan is India's **5th largest FDI source** and **largest bilateral donor.**
 - **ODA** supports **MAHSR, metro projects and infrastructure.**
 - Japan supports infrastructure projects, like **Delhi Metro and high-speed rail network.**
 - **Supply Chains and Economic Security:** Cooperation on **critical minerals, semiconductors, clean energy and pharmaceuticals**.
 - **Aim:** reduce supply-chain dependence on any single country.
 - **North-East Connectivity: Act East Forum (2017)** links India's **Act East Policy** with Japan's **FOIP** to develop North-East India as a connectivity hub between **South Asia and Southeast Asia**.
 - Japan (JICA) supports projects like the **750-km North East Road Network Connectivity Project.**
 - **Technology and Space:** **ISRO-JAXA** are developing the **LUPEX Mission** to explore the Moon's south pole by **2026–27**. Focus on **AI, IoT and semiconductors**, supported by **Sakura and LOTUS** exchange programmes.

Challenges in India-Japan Relations

- **Low Trade Potential:** Bilateral trade is around **USD 27 billion**, below potential and skewed in Japan's favour.
- **Project Delays:** **MAHSR bullet train** faces delays due to **land acquisition** and political issues.
 - **Asia-Africa Growth Corridor** lacks a strong joint financing mechanism.
- **Business Hurdles:** Japanese investors face **bureaucracy, labour law issues and regulatory uncertainty** in India.
- **NTBs:** Indian **pharma and agriculture exports** face strict **SPS measures** and regulations in Japan.
- **Language Barrier:** Limits use of Japan's labour schemes like **SSW**, despite India's workforce surplus.

India-Japan Relations

- **Evolution:** Ties rooted in **Buddhism**; diplomatic relations began in **1952**.
 - Relations upgraded to **Global Partnership (2000), Annual Summit (2006), CEPA (2011)** and **Special Strategic and Global Partnership (2014).**

- **Strategic Differences:** Japan follows **G7 sanctions on Russia**, while India maintains **strategic autonomy**.
- **Defence Transfer Issues:** Japan's export restrictions slowed defence deals like **US-2 aircraft**.
 - Japan easing restrictions on lethal weapon exports may boost future **defence technology transfer**.

Measures to Strengthen India-Japan Relations

- **MRAs:** Negotiate **Mutual Recognition Agreements** to reduce **NTBs** and ease Indian **pharma and agriculture exports** to Japan.
- **Joint Infrastructure Fund:** Create an **India-Japan sovereign fund** for **AAGC** and **Global South infrastructure** projects.
- **Skilled Workforce Pipeline:** Set up **CoEs** under **SSW** for training in **advanced manufacturing, semiconductors and precision engineering**.
- **Minilateral Partnerships:** Expand cooperation through **India-Japan-Philippines** for maritime security and **India-Japan-Australia** for critical minerals.
- **Cultural Diplomacy 2.0:** Build **Japanese language and technical training institutes** in India.
 - Facilitate movement of **5 lakh Indian students and workers** to Japan in the next **5 years**.

The India-Japan relationship has evolved beyond economic cooperation into a comprehensive strategic partnership. Discuss.

Practice Mains Question

India-Seychelles Relations

PM visited **Seychelles** as Guest of Honour for its **50th National Day** celebrations. The visit also marked **50 years of India-Seychelles diplomatic relations**, with both countries unveiling a commemorative logo.

Key Outcomes of PM's Seychelles Visit

- **UPI Deployment:** NIPL and Central Bank of Seychelles signed an MoU to roll out India's UPI, strengthening digital payments and financial inclusion.
- **Umbrella Line of Credit:** India extended a **₹1,250 crore rupee-denominated LoC** for priority development projects.
- **Janaushadhi Scheme:** Agreement to supply affordable Indian generic medicines and medical supplies to Seychelles.
- **New National Hospital:** Framework MoU signed for developing a modern **New Seychelles National Hospital**.
- **Extradition Treaty:** Legal framework to extradite fugitives and combat drug trafficking, financial fraud and transnational crimes.
- **Maritime Mobility:** Mutual recognition of Indian seafarers' training and certification for service on Seychelles-flagged vessels.

- **Capacity Building:** SSIFS and Seychelles' MoFAD to conduct joint diplomatic training programmes.
- **Space Cooperation:** Cooperation in satellite applications, remote sensing, disaster management and climate monitoring.
- **Agriculture Work Plan:** ICAR and Seychelles Agriculture Department agreed on joint research and food security work for **2026-31**.
- **Defence Support:** India handed over **FPV PS LESPWAR**, plus ambulances, utility vehicles and laser radial boats.
- **Highest Honour:** PM received '**Guardian of the Blue Horizon**' award for climate action, Blue Economy and support to SIDS.
 - The award acknowledges India's global environmental initiatives, including the **ISA**, **Mission LiFE**, and the **International Big Cat Alliance**.
- **Vision MAHASAGAR:** Visit strengthened India's digital, health, maritime and development diplomacy in the Indian Ocean Region.

Challenges in India-Seychelles Ties

- **Geopolitical Competition:** Assumption Island project remains stalled; China's presence adds strategic pressure.
- **Maritime Security Gaps:** Seychelles struggles to monitor its vast **1.4 million sq km EEZ**, facing IUU fishing, piracy and drug trafficking.
- **Limited Economic Engagement:** Small market, weak connectivity and project delays restrict trade and credibility.
- **Institutional Constraints:** Limited technical and administrative capacity slows Indian-funded projects.
- **Supply Chain Vulnerability:** External shocks increase Seychelles' dependence on India for essential supplies.

Measures to Strengthen India-Seychelles Relations

- **Enhance MDA:** Deepen real-time intelligence sharing through **IFC-IOR** against IUU fishing, piracy and drug trafficking.
- **Scientific Diplomacy:** Link India's **Deep Ocean Mission** and **CSIR-NIO** with Seychelles for marine genomics, bioprospecting and Blue Economy research.
- **Blue Carbon Finance:** Help Seychelles audit and securitise marine ecosystems into **Sovereign Blue Carbon Credits** under carbon market/Article 6 frameworks.
- **Energy Mapping:** Use ISRO data to assess **OTEC** potential using deep-shallow water temperature differences.
- **Climate Monitoring:** Provide real-time mapping of coral bleaching and marine heatwaves to support Seychelles' environmental security.

Environment & Ecology

White-bellied Heron

The proposed **1,200 MW hydropower project** on the **Lohit River, Arunachal Pradesh** has raised concerns over the survival of the **White-bellied Heron**.

White-bellied Heron

- **About:** Also called **Imperial Heron**, it is the **2nd-largest heron species** and among the world's most elusive birds.
- **Habitat & Features:** Depends on **free-flowing eastern Himalayan rivers** with mature riparian forests and low human disturbance.
 - Feeds mainly on fish in shallow rapids; highly sensitive to **dams, altered river flow and turbidity changes**.
 - Found up to **1,500 m elevation**.
- **Morphological Features:** Adaptations include **long flexible neck, serrated bill, short legs, extended toes and large body size** for riverine fishing.
- **Conservation Status**
 - **IUCN Red List: Critically Endangered.**
 - **Wildlife Protection Act, 1972: Schedule I species.**
- **Global & Regional Population:** Global range covers around **1.65 lakh km²** across **Bhutan, India, Myanmar, China and Bangladesh**.
 - Fewer than **60 individuals** survive globally; **Bhutan holds ~45%** of the population.
 - Considered extinct in **Nepal** and possibly **Bangladesh**.
- **Distribution in India:** Mainly found in **Arunachal Pradesh** (Lohit, Anjaw and Changlang districts).
 - Recorded around **Namdapha and Kamlang Tiger Reserves**.
 - India has only **6–9 individuals** across **Noa-Dehing, Lahm and Lohit rivers**. **Lahm River basin** holds around **70%** of India's population.

- **Walong Cantonment** near the proposed hydropower project is a known breeding site.
- **Behavior: Largely solitary** species; produces **deep croaking calls**, mainly during **dawn and dusk**.
 - Nests in **tall mature riparian trees**, which are essential for **roosting and nesting**. Loss of mature riparian forests directly threatens breeding and survival.
- **Ecological Significance:** Acts as a **bio-indicator** of Himalayan freshwater ecosystem health, reflecting water quality, fish availability and habitat integrity.
- **Major threats:** hydropower dams, habitat fragmentation, altered river flows, habitat loss, hunting and human disturbance.
 - Restricted range and small population create **genetic vulnerability**.
- **Conservation Measures:**
 - Develop **conservation breeding centres** in India. Learn from **Bhutan's captive breeding programme**, the first facility globally for breeding critically endangered birds.

2025 Animal and Plant Discoveries and 111th Foundation Day of ZSI

The Ministry of Environment released the **Animal Discoveries–2025 (by ZSI)** and **Plant Discoveries–2025 (by BSI)** compilations during the **111th Foundation Day of the ZSI in Kolkata**.

Highlights of the Animal and Plant Discoveries–2025

Animal Discoveries–2025

- India added **709 new faunal species**, taking total documented fauna to **1,05,953 species**.
- **Kerala** led with **98 discoveries**, followed by West Bengal, Karnataka and Arunachal Pradesh.
- Key finds include **Myotis himalaicus**, **Irwin's wolf snake**, and two green fan-throated lizards.

Species (Scientific Name)	Common Name	Region Discovered	Key Features
<i>Myotis himalaicus</i>	Himalayan long-tailed Myotis	Uttarakhand (Western Himalayas)	Insectivorous bat with a tail almost as long as its body; helps in pest control.
<i>Lycodon irwini</i>	Irwin's wolf snake	Great Nicobar Island	Glossy black, non-venomous snake named after wildlife educator Steve Irwin .
<i>Ptyctolaemus namdaphaensis</i>	Namdapha green fan-throated lizard	Arunachal Pradesh	A new species of green fan-throated lizard from the Namdapha Tiger Reserve.
<i>Ptyctolaemus siangensis</i>	Siang green fan-throated lizard	Arunachal Pradesh	A distinctly identified green fan-throated lizard found in the Siang district.

Plant Discoveries–2025

- BSI added **353 plant taxa**; **57% non-vascular** and **43% vascular plants**.
- **Arunachal Pradesh** led with **49 discoveries**, followed by Uttarakhand and Kerala.
- Key finds include **Polystichum siangense**, **Miliusa beddomei** and **Hericium indicum**.

Key Facts About ZSI

- **About:**
 - The **Zoological Survey of India (ZSI)** functions as the premier apex organization under the **MoEFCC** dedicated to comprehensive zoological research, faunal exploration, and animal taxonomy across the Indian subcontinent.
- **Administrative Structure:**
 - Headquartered in **Kolkata, West Bengal**, the ZSI operates a decentralized network of **16 regional centers** situated across various distinct biogeographic zones of India, including the Western Ghats, marine, and high-altitude ecosystems.
- **Historical Timeline:** The foundation of ZSI lies in the establishment of the **Zoological Section of the Indian Museum** in Calcutta in 1875. The broader vision traces back to **Sir William Jones**, who founded the **Asiatic Society of Bengal in 1784**.
 - The ZSI was formally established as an independent organization on **1st July 1916** to survey and study the animal life of the then “British Indian Empire.”
- **Expanded Legal Mandate:** Following India’s ratification of the **Convention on Biological Diversity (CBD) in 1994**, ZSI scaled up its operational mandate from pure taxonomic research to modern ecosystem-based conservation monitoring.
- **National Repository:** Under the provisions of the **Biological Diversity Act, 2002**, ZSI is officially designated as the supreme repository for the **National Zoological Collection**, housing millions of reference specimens.
- **Modern Initiatives:**
 - The organization plays a vital role in executing **Environmental Impact Assessments (EIA)**, developing **DNA barcoding and wildlife forensics** (such as the Pangolin Indexing System) to curb illicit trade.
- **Flagship Publications:** To guide national policy and wildlife classification, ZSI periodically publishes the **Indian Red Data Book**, the **Fauna of India** series, and its highly referenced annual directory, **Animal Discoveries**.

Compressed Biogas and India’s Energy Security

West Asia tensions have renewed focus on **CBG** to reduce India’s **~85% crude oil import dependence** and strengthen energy security.

Current Status of Biogas in India

- **About:**
 - Biogas, produced through anaerobic digestion of biodegradable waste, contains about **45–75% methane**.
 - After purification and compression, it becomes **CBG**, which is similar to **CNG** and can be used in transport, industry, households, and power generation.
- **Progress:** The **SATAT initiative (2018)** targeted **5,000 CBG plants by 2023**, but only **132 plants** were completed by **3 June 2026**.
- **Blending Targets:** Mandatory CBG blending with **CNG and PNG** began at **1% in FY26**, rising to **3% in FY27**, **4% in FY28**, and **5% from FY29**.
- **Gap:** India’s commercial CBG sector remains underdeveloped despite feedstock availability and policy support.
 - Around **2.88 lakh small biogas plants** were installed over the last decade.
- **Global Scenario:** **Europe, China, and the US** account for about **90% of global biogas production**.

Importance of Biogas in India’s Energy Security

- **Self-Reliance in Fuel Supply:**
 - India imports nearly **85% of its crude oil** and a significant share of **LPG**, making it vulnerable to external shocks such as the **Russia-Ukraine war** and recent disruptions around the **Strait of Hormuz**.
 - **Biogas and Compressed Biogas** can reduce **import dependence** by using domestic resources such as **cattle dung, crop residue, food waste, press mud** and **municipal solid waste**.
- **Local Fuel Resilience:** **CBG** is chemically and functionally similar to **CNG** and can be used in **transport, industries, households and city gas distribution networks**. Since it can be produced locally, it strengthens **energy availability** during global supply-chain disruptions and supports **Atmanirbhar Bharat**.
- **Circular Economy:**
 - Biogas converts **organic waste** into useful **energy** and **fermented organic manure**. This creates a **circular economy** model in which waste from **farms, dairies, households and cities** is converted into **fuel, fertiliser** and **rural income**.

- **Waste Management and Stubble Solution:** Biogas can help utilise **paddy straw** and other **crop residues** in **Punjab, Haryana** and **western Uttar Pradesh**, reducing **stubble burning**, **air pollution** and waste disposal pressure. It also supports scientific management of **food waste**, **animal waste** and **municipal solid waste**.
- **Climate Change Mitigation:** Biogas reduces **methane emissions** from unmanaged organic waste and lowers dependence on **fossil fuels**. By replacing part of **diesel**, **petrol**, **LPG** and **natural gas** consumption, it supports India's **climate commitments** and **net-zero target for 2070**.
- **Rural Economy and Farmer Income:** Biogas plants create new income opportunities for **farmers**, **dairy owners**, **gaushalas**, **self-help groups** and **rural entrepreneurs**. Sale of **CBG** and **fermented organic manure** can improve **rural livelihoods** and reduce dependence on **chemical fertilisers**.
- **Soil Health and Sustainable Agriculture:** The by-product of biogas plants, **fermented organic manure**, improves **soil organic carbon**, supports **natural farming** and reduces **fertiliser costs**. This links biogas with **agroecological approaches** for **clean and green villages**.
- **Decentralised Energy Access:** Small and medium biogas plants can provide **local energy** for **cooking**, **heating**, **electricity** and **transport fuel** in rural and peri-urban areas. This improves **energy access**, reduces pressure on centralised fuel supply systems and strengthens **local resilience**.

Challenges in Biogas Adoption	Measures Needed
■ High Costs and Limited Scale: CBG projects require substantial investment in digesters, purification units, compressors, storage and transport infrastructure. Despite an estimated potential of around 87 bcm annually, the International Energy Agency (IEA) notes that India's current capacity is below 1% of this potential.	■ Financial Support: The government should provide viability-gap funding, interest subsidies, accelerated depreciation and tax incentives. The Indian Biogas Association (IBA) has proposed a ₹10,000 crore capital-subsidy fund to reduce initial costs and accelerate capacity creation.
■ Credit and Market Risks: Banks consider biogas projects risky because revenues depend on plant performance, assured gas purchases and the commercial value of co-products such as Fermented Organic Manure (FOM). Uncertain CBG pricing and weak demand for FOM reduce project bankability.	■ Credit and Market Facilitation: Dedicated green-finance windows, partial credit guarantees and priority-sector lending should be introduced. A stable CBG pricing formula and assured offtake are needed, while the IBA has proposed mandatory FOM blending beginning at 1% in 2026–27 and rising to 10% by 2030.
■ Feedstock Constraints: The IEA identifies seasonal biomass supply, land availability, feedstock aggregation and inadequate biomass inventories as key barriers. Although cattle dung, crop residues and food waste are abundant, their dispersed nature increases collection, storage and transportation costs.	■ Organised Supply Chains: Farmer cooperatives, municipal contracts, aggregation centres and long-term feedstock agreements should be developed. District-level biomass inventories and digital tracking systems can align local feedstock availability with plant capacity.
■ Infrastructure and Offtake Gaps: Limited pipelines, storage facilities, transport networks and local injection points restrict the supply of CBG to city-gas distributors. The IEA also identifies logistics and assured offtake as major priorities for improving project viability.	■ Network Expansion: Common pipelines, local injection points and transport infrastructure should be developed, while allocated infrastructure funds should be released faster. Long-term purchase contracts with oil-marketing and city-gas companies can provide revenue certainty.
■ Policy Fragmentation: The sector is governed through multiple initiatives, including SATAT, GOBARdhan and CBG blending obligations, resulting in overlapping responsibilities, procedural delays and inconsistent project monitoring.	■ Unified Mission: As advocated by the Indian Biogas Association, a Unified National Biogas Mission should integrate existing schemes, introduce single-window clearances and establish centralised tracking, common standards and time-bound approvals.
■ Food and Land Risks: Excessive dependence on maize and other dedicated energy crops can alter cultivation patterns and compete with food crops. Germany's experience illustrates the risks of energy-crop monoculture and competition between food and fuel production.	■ Waste-Based Model: India should prioritise cattle dung, sewage, press mud, food waste, municipal waste and crop residues. It can draw from Denmark's model, which relies substantially on livestock manure and organic waste, supporting biogas production without placing significant pressure on food-producing land.

Compressed Biogas can strengthen India's energy security while supporting rural incomes and waste management. Discuss the opportunities and challenges in scaling up the biogas sector in India.

Practice Mains Question

Science & Technology

Space Oncology and Cancer Treatment

India is expected to record **1.87 million new cancer cases in 2026**. **Space oncology** is emerging as a new field using **microgravity and space environments** to study tumours and develop advanced therapies.

Space Oncology

- **About:** Studies the impact of **microgravity and cosmic radiation** on cancer cells, tumour growth, treatment response, and adaptation.
- **Space as a Natural Laboratory:** Helps study **metastasis, treatment resistance, radiation response, biomarkers, and tissue behaviour**.
- **Research Value:** Supports **tumour modelling, drug discovery, protein crystallisation, drug formulation, 3D cell cultures, and tumour spheroid development**.

Importance of Space Oncology

- **Economic Burden:** Cancer imposes financial stress through **treatment costs, income loss, debt, asset depletion, and caregiver burden**.
- **Need for Advanced Models:** Space-based environments provide improved tumour models to understand cancer mechanisms and accelerate therapies.
- **Drug Development Potential:** Microgravity can improve **drug formulation, testing models, and reduce dependence on animal studies**.

Microgravity Effects on Cancer Cells

- **Cellular Changes:** Causes **cytoskeletal reorganisation, altered focal adhesion signalling, and changes in extracellular matrix interaction**.
- **Tumour Behaviour:** Affects **cell adhesion, migration, proliferation, invasion, and gene expression**.
- **Spheroid Formation:** Enables formation of **multicellular tumour spheroids** for improved cancer modelling.
- **Drug Response:** Changes in cell structure and gene expression influence **drug uptake and treatment response**.

Space Oncology and Cancer Types

- **Breast Cancer:** Microgravity may reduce malignant characteristics by affecting **adhesion, migration, and cell-cycle regulation**, improving response to specialised therapies.
- **Gastrointestinal Cancer:** Alters cancer progression and may increase **drug sensitivity** through changes in resistance genes and cellular proteins.

Role that India can Play in the Emerging Field of Space Oncology

- **Leverage Space Capabilities:** Utilise **ISRO's space expertise, space economy, and private space sector** for space-based biomedical research.
- **Promote Space-Based Pharmaceutical Research:** Encourage **microgravity-based drug development, cancer models, and space manufacturing of medicines**.
- **Strengthen Public-Private Partnerships:** Promote collaboration among **ISRO, research institutions, hospitals, pharmaceutical companies, and space startups**.
- **Develop Affordable Cancer Solutions:** Translate space research into **cost-effective diagnostics, therapies, and drug delivery systems**.
- **Build Global Leadership:** Position India as a hub for **space biotechnology and cancer research** through growing commercial space opportunities.

India's 1st Geothermal Wells Commissioned at Puga Valley

India commissioned its **first and deepest geothermal wells at Puga Valley, Ladakh** to support India's **clean-energy transition**.

Key Facts About the Project

- **Location:** Puga Valley, Ladakh, at over **14,000 feet** altitude.
- **Depth:** Two wells drilled up to **1,000 metres**.
- **Target:** Support a **1 MW pilot geothermal power plant**, India's first demonstration-scale project.
- **Working:** Uses underground heat for reliable **24×7 renewable power**.
- **Temperature:** Up to **135°C** recorded at **400 metres** depth.
- **Executing agencies:** **ONGC Energy Centre, Ladakh Administration and LAHDC Leh**.

Geothermal Energy

- **About:** **Geothermal energy** is heat from the Earth's interior, used for **electricity generation, direct heating and building heating or cooling**. It is renewable because Earth's heat is continuously replenished.
- **Sources**
 - **Deep reservoirs:** Underground steam or hot water accessed through wells.
 - **Near-surface resources:** Hot springs and shallow hydrothermal reservoirs.

- **Shallow ground:** Stable ground temperatures used by geothermal heat pumps.
- **Institutional Framework**
 - The **National Policy on Geothermal Energy, 2025** promotes exploration and development.
 - The **Ministry of New and Renewable Energy** is the nodal ministry.
- **Geothermal Potential**
 - **Global capacity: 15.4 GW in 2024**, led by the **US, Indonesia and the Philippines**.
 - **India: 381 hot springs** identified, with temperatures of about **35°C–89°C**.
 - India has **10 recognised geothermal provinces**, including the **Himalayan, Cambay Graben, West Coast and Andaman–Nicobar** regions.
 - Major sites include **Puga, Chumathang, Manikaran, Tapoban, Bakreshwar, Tuwa, Tattapani and Deulajhari**.
 - India's estimated geothermal power potential is about **10,600 MW**.

White Rabbit Technology- Based

IST Dissemination Demonstration Network

White Rabbit Technology-based IST Dissemination Demonstration Network was commissioned at **RRSL Bengaluru** to strengthen indigenous and secure **Digital Public Infrastructure**.

IST Dissemination Demonstration Network

- **IST Dissemination Demonstration Network:** A secure, tamper-resistant network that distributes precise **Indian Standard Time (IST)** across critical digital systems.
 - Developed jointly by the **Department of Consumer Affairs, CSIR-National Physical Laboratory (CSIR-NPL)** and **ISRO**.
 - Disseminates IST traceable to **UTC (NPLI)** maintained by **CSIR-NPL**, using **Precision Time Protocol-based White Rabbit Technology**.
 - Supports synchronisation in **banking, financial markets, 5G/6G, smart grids, transport and digital governance**.
- **Pilot Test:** Secure time transmission was successfully tested between **RRSL Bengaluru** and the **NSE facility in Chennai**.
- **White Rabbit Technology:** An extension of **Precision Time Protocol (IEEE 1588)** that enables **microsecond-to-nanosecond accuracy** over optical-fibre Ethernet networks.
 - More secure and resilient than satellite-based systems such as **GPS**, which are vulnerable to spoofing, jamming and signal loss.

Significance of Precision Time Dissemination

- **Critical Sector Coverage:** Enables secure time synchronisation across **banking, 5G/6G, smart grids, transport and digital governance**.
- **DPI & Cyber Resilience:** Protects algorithmic trading, prevents timestamp tampering and strengthens cybersecurity.
- **Strategic Autonomy:** Supports **"One Nation, One Time"** by reducing reliance on foreign satellite-based timing systems.
- **Regulatory & Economic Growth:** Aids **Legal Metrology reforms**, ease of doing business and investor confidence.

Time Generation and Dissemination of Indian Standard Time (IST)

- **Realisation and Maintenance of IST:** **CSIR-NPL** generates **UTC (NPLI)**, India's official realisation of UTC, traceable to **International Bureau of Weights and Measures (BIPM)**, France.
 - **IST = UTC+5:30**
 - Generated using **5 caesium clocks and 3 hydrogen masers**
 - Accuracy: **±2.8 nanoseconds** relative to UTC
 - **Caesium clocks:** Long-term stability
 - **Hydrogen masers:** Short-term stability
- **Dissemination:**
 - **GNSS (Global Navigation Satellite System):** Provides nanosecond-level synchronisation and UTC traceability.
 - **TWSTFT (Two-Way Satellite Time and Frequency Transfer):** Uses geostationary satellites for accurate time transfer and supports **NavIC** and international UTC links.
 - **NTP (Network Time Protocol):** Distributes IST through time. nplindia.org to government, industry and digital networks.
 - ❖ **CERT-In Requirement:** Critical ICT infrastructure must synchronise with NPL/NIC time servers.

Vikram-1: Charting India's Cosmic Future

Skyroot Aerospace's Vikram-1 became India's first privately developed rocket to successfully place payloads in **Low Earth Orbit (LEO)**.

- India became the **third country**, after the **US and China**, where a private company independently reached orbit.
- **Vikram-1** achieved this on its **maiden orbital mission**.

Key Highlights of the Skyroot's Vikram-1

- **About Vikram-1:** Named after **Dr. Vikram Sarabhai**, it is an orbital launch vehicle developed by Hyderabad-based **Skyroot Aerospace**, India's first space-sector unicorn.
 - Its maiden orbital flight under **Mission Agaman** followed the **Vikram-S suborbital mission (2022)**, which tested launch capability but did not reach orbit.

- **Payload Capacity:** Designed to carry around **300–480 kg** to LEO for the small-satellite market.
- **Unique Materials and Methods:**
 - **Carbon Composite Architecture:** Uses lightweight all-carbon composite structures to improve fuel efficiency and payload capacity.
 - **3D-Printed Engines:** Uses the liquid-fuelled **Raman engine** for orbital insertion and manoeuvres, reducing manufacturing time, parts and cost.
 - **Solid Propulsion:** Lower stages use **Kalam-series engines** with advanced solid propellants.
- **Symbolic Tributes:** Carries “**Cosmic Bloom**” and an **18-karat gold micro-rocket** containing microscopic sculptures of **C.V. Raman, Vikram Sarabhai and A.P.J. Abdul Kalam**.

Significance of the Vikram-1

- **Vindication of Space Sector Reforms:** Validates the **2020 space sector reforms**, supported by **IN-SPACe** and the **Indian Space Policy 2023**.
- **Establishing India as a Commercial Hub:** Strengthens the **public-private space model** and enhances India’s position in the **global small-satellite launch market**.
- **Cost-Effective Innovation:** Demonstrates world-class aerospace capability through **3D printing** and **carbon composites** at lower costs.
- **Strategic Independence:** Reduces dependence on **ISRO’s PSLV and SSLV**, allowing ISRO to focus on **deep-space missions, Gaganyaan** and advanced R&D.

Role of Government Reforms in Enabling Private Sector Participation

- **Indian Space Policy 2023:** Opened the entire space value chain to **Non-Government Entities (NGEs)** for satellite manufacturing, launch services and downstream operations.
- **Role of NSIL:** **NewSpace India Limited (NSIL)**, ISRO’s commercial arm, manufactures space systems, delivers end-to-end solutions and promotes private participation through **PSLV production** and technology transfer.
 - By **July 2026**, it had launched **141 satellites**, including **138 international/customer satellites** and **3 Indian satellites**.
- **IN-SPACe as a Single-Window Agency:** Authorises, enables and promotes government and private space activities, while providing NGEs access to **ISRO facilities, technology and expertise**.
- **Financial Facilitation:** Enabled **USD 150 million** in private investment in **2025**, **118 technology-transfer agreements** and **189 joint partnerships**.

Financial Schemes:

- **IN-SPACe Seed Fund Scheme:** Grants up to **₹1 crore** to startups and MSMEs, with mentorship and training.
- **₹1,000 crore VC Fund:** Provides early-stage capital and attracts private investment.
- **₹500 crore Technology Adoption Fund:** Supports commercialisation of indigenous technologies, funding up to **60% for startups/MSMEs** and **40% for large industries**, capped at **₹25 crore per project**.
- **Liberalised FDI Policy:** Allows automatic FDI of **74% in satellite manufacturing and operations**, **49% in launch vehicles and spaceports**, and **100% in satellite components and subsystems**.

Key Milestones of India’s Transition “From Policy to Progress”

- **Startup Growth:** Space startups increased from **1 in 2014** to over **400 in 2026**.
- **Market Trajectory:** India’s space economy is projected to grow from **USD 8.4 billion** to **USD 40–45 billion by 2030** and **USD 100 billion by 2040**.
- **Pivotal Operational Milestones:**
 - **NSIL:** Conducted India’s first commercial **LVM3 mission in 2022**, launching **36 OneWeb satellites** into LEO.
 - **Vikram-S (2022):** India’s first privately developed rocket and first private launch authorised by **IN-SPACe**.
 - **Agnikul Cosmos (2024):** First private launch from the **Dhanush launch pad**, using the world’s first single-piece **3D-printed semi-cryogenic engine**.
 - **Pixxel-led PPP:** IN-SPACe approved an **Earth Observation satellite constellation** for agriculture, disaster management, climate monitoring and urban planning.
 - **Technology Commercialisation (2026):** **SSLV technology** was transferred to **HAL**, while **Bellatrix Aerospace** demonstrated an indigenous **Green Propulsion System** under **DRDO’s Technology Development Fund**.

Challenges for India’s Private Space Sector

- **Single Spaceport Bottleneck:** Dependence on **Satish Dhawan Space Centre, Sriharikota** limits launch frequency.
- **Missing Reusable Launch Capability:** Reliance on expendable rockets weakens cost competitiveness against reusable systems such as **SpaceX**.
- **Regulatory Vacuum:** Absence of a **Space Activities Act** creates uncertainty over liability, insurance, intellectual property and dispute resolution.
- **“Anchor Customer” Deficit:** Limited assured government procurement makes commercialisation difficult for startups.

- **Orbital Slot and Spectrum Competition:** Global mega-constellations such as **Guowang** and **Qianfan** increase competition for **ITU-managed orbital slots and spectrum**.
- **Dependence on Space-Grade Imports:**
 - Reliance on imported semiconductors, radiation-hardened electronics, sensors and propulsion components exposes firms to export controls and supply-chain risks.
- **Space Security and Sustainability:** Stronger **Space Situational Awareness**, debris mitigation, cybersecurity and safeguards for dual-use technologies are needed.

Measures to Strengthen India's Private Space Sector

- **Enact the Space Activities Act:**
 - Provide legal clarity on **IP rights, technology transfer, insurance** and international liability under the **1972 Liability Convention**.
 - Establish a **Telecom Disputes Settlement and Appellate Tribunal (TDSAT)** for commercial space disputes, orbital collisions and spectrum conflicts.
- **Establish Anchor Tenancy:** Enable **NSIL or the Ministry of Defence** to sign long-term procurement contracts for launch services and **Earth Observation data**, following NASA's **COTS model**.
- **Fund Dual-Use Technologies:** Use mechanisms similar to **ADITI-IDEX** to finance and procure military-grade optical, radar and other dual-use space technologies from startups.
- **Indigenise the Supply Chain:** Introduce **PLI schemes** for space-grade electronics, advanced composites and green propulsion systems.
- **Standardise Space Systems:**
 - IN-SPACe and ISRO should standardise satellite buses and launch interfaces to enable mass production of **Commercial Off-The-Shelf components**.
- **Infrastructure Democratisation:**
 - Fast-track the **Kulasekarapattinam Spaceport** and develop state-level space manufacturing parks with plug-and-play facilities.
- **Accelerate Reusable Launch Technologies:** Expand private participation in **RLV** and **NGLV** development to improve cost competitiveness.
- **Leverage iCET and Artemis Accords:** Integrate Indian startups into global aerospace supply chains and secure commercial lunar exploration contracts.

Despite policy reforms, investment remains concentrated in a few Indian states. Analyse the structural challenges responsible for regional investment disparities.

Practice Mains Question

Semaglutide Patent Expiry and Opportunities for India

The expiry of Novo Nordisk's **Semaglutide patent in India** has opened opportunities for affordable **generic GLP-1 (Glucagon-Like Peptide-1) receptor agonist** therapies.

NOTE: Generic GLP-1 (Glucagon-Like Peptide-1) receptor agonists are medications that mimic the **natural gut hormone GLP-1**.

They regulate blood sugar, reduce appetite, and promote weight loss by stimulating insulin release and slowing digestion.

Semaglutide

- **About:** Semaglutide is a **GLP-1 (Glucagon-Like Peptide-1) receptor agonist** used to treat **Type-2 Diabetes** and **obesity**, originally developed by the Danish pharmaceutical company Novo Nordisk.
 - It improves **blood glucose control**, promotes **weight loss**, and reduces the risk of cardiovascular complications.
- **Significance:** Semaglutide has transformed the treatment of diabetes and obesity, two major drivers of **non-communicable diseases (NCDs)** worldwide, by offering a highly effective non-surgical therapy.
 - It highlights the growing importance of **GLP-1 therapies** and **peptide therapeutics**, before its patent expiry the high cost of branded versions limited access for many patients.
 - Its patent expiry now paves the way for affordable generic medicines, improved healthcare access, and growth of India's pharmaceutical industry.
- **Opportunities for India:**
 - **Improved Healthcare Access:** Affordable generic versions of Semaglutide can significantly reduce treatment costs for **10+ crore Indians with diabetes** and the rising obesity burden, making advanced GLP-1 therapies more accessible.
 - **Export-Led Growth:** Patent expiries in **China, Brazil, Canada, South Africa, and Turkey** open access to high-value pharmaceutical markets, strengthening India's position as a leading exporter of complex generics.
 - **Technological Advancement:** Manufacturing generic versions of semaglutide enables Indian companies to transition from **small-molecule generics** to **complex peptide therapeutics** and advanced drug-delivery systems, enhancing global competitiveness.
 - **Economic Growth:** The domestic Semaglutide market is projected to witness strong growth, while the **Production Linked Incentive (PLI) for pharmaceutical and medical devices** supports investments in **R&D, API manufacturing, and high-value pharmaceutical production**, boosting the sector's long-term growth.

GLP-1s and Dermatological Health

- Beyond industrial economics, the clinical application of GLP-1 receptor agonists (like Semaglutide) has significant implications for systemic health, particularly dermatology.
- **Facial and Body Skin Changes:** Rapid medically-assisted weight loss frequently leads to **"Ozempic face"** a non-medical term for facial volume loss, hollowing, and sagging.
 - On the body, loose skin can cause **intertrigo** (friction-induced fungal/bacterial infections) and persistent pigmentation.
- **Telogen Effluvium (Hair Fall):** The physiological stress of rapid weight loss, combined with potential nutritional deficits (protein, iron, B12), can trigger severe, temporary hair shedding.
- **Nail Health:** Nutritional shifts during treatment often manifest as brittle nails, ridging, and slow growth.
- **Anti-Inflammatory Potential:** Early clinical reports suggest GLP-1s possess systemic anti-inflammatory properties that may benefit chronic inflammatory skin conditions like Psoriasis and Hidradenitis Suppurativa, particularly when they coexist with metabolic syndrome.

How have Patent Expiries Driven India's Pharmaceutical Growth?

- **Atorvastatin (2011):** When Pfizer's cholesterol-lowering drug Lipitor lost its patent, Indian firms aggressively flooded regulated markets with affordable generics, permanently establishing their stronghold in the US and European cardiovascular segments.
- **Clopidogrel & Sildenafil (2012):** The expiry of patents on Plavix (blood thinner) and Viagra allowed Indian manufacturers to rapidly scale global volumes, providing low-cost alternatives that reshaped global prescriptions.
- **Imatinib (2013):** A landmark Supreme Court judgment prevented the "evergreening" of Novartis's leukemia drug patent. This allowed domestic firms to legally produce generic versions, crashing the annual treatment cost from over USD 25,000 to a fraction, saving countless lives.
- **Sofosbuvir (2025):** The expiration of base patents for this vital Hepatitis C treatment preceded by strategic voluntary licensing further solidified India's role as the **"Pharmacy of the Developing World."**

Cabinet Approves Semicon 2.0

The Union Cabinet approved **Semicon 2.0** to build an end-to-end semiconductor ecosystem, from **chip design to advanced packaging**, and strengthen India as a trusted global manufacturing hub.

Semicon 2.0

- **About:** Building on **India Semiconductor Mission 1.0 (2021)**, Semicon 2.0 shifts from ecosystem creation to ecosystem deepening.
- **Financial Outlay:** ₹1,27,500 crore.

Six Key Pillars:

- **Design of Chips:** Support semiconductor **IP and design startups** to strengthen India's fabless design ecosystem.
- **Machines and Materials:** Promote domestic production and R&D of semiconductor equipment, specialty chemicals, industrial gases, and critical materials.
- **Setting up more Fabs:** Support silicon, compound semiconductor, discrete component, and display fabs. India's first semiconductor fab is expected by **2028**.
- **Strengthening ATMP/OSAT:** Expand advanced semiconductor assembly, testing, marking, and packaging capabilities.
- **Research & Development (R&D):** Promote global research collaboration and frontier technologies such as **AI-native chip design**.
- **Talent Development:** Build industry-ready skills; **315 universities** use advanced Electronic Design Automation (EDA) tools to train around **68,000 students**.

Current Status of India's Electronics and Semiconductor Ecosystem

- **Growth in Electronics Manufacturing:** Electronics production grew **7-fold** and exports **11-fold** in eleven years. India is now the **world's second-largest mobile phone manufacturer**.
- **Inclusive Job Creation:** The sector generated around **25 lakh jobs** in a decade. Women comprise about **30%** of the electronics workforce and **70%** of direct mobile-manufacturing workers.
- **Success of ISM 1.0:** Since **December 2021**, **12 semiconductor projects** across **six States** have been approved, involving over **₹1.64 lakh crore**. Commercial production has begun at major **OSAT/ATMP facilities**.
 - **Semicon India Programme:** Launched in **2021** with **₹76,000 crore**, it supports semiconductor and display fabs, compound semiconductor and **ATMP/OSAT facilities**, and the **Design Linked Incentive Scheme**.
- **Design and Innovation:** Financial support was given to **24 startup and MSME projects** developing **SoCs** for satellite communication, drones, and AI systems.
- **Future Targets:** Semicon 2.0 targets **3-nanometre and 2-nanometre nodes** and a domestic semiconductor market of **USD 200 billion by 2035**.

Need for a Comprehensive**Semiconductor Ecosystem Crucial for India**

- **Geostrategic Imperative:** Capabilities across chip design, fabrication, packaging, and materials reduce external dependence and strengthen strategic autonomy.

- **National Security:** Indigenous chips lower risks of hardware tampering and supply-chain manipulation in defence, space, power, telecom, and other **Critical Information Infrastructure**.
- **Economic Competitiveness:** Domestic manufacturing can reduce India's electronics import bill, ease pressure on the **Current Account Deficit**, and capture higher value beyond assembly.
- **Industrial Transformation:** Semiconductors support **EVs, electronics, 5G/6G, AI hardware, medical devices, automation, and aerospace**, strengthening manufacturing value chains.
- **Technological Leadership:** India has nearly **20% of global semiconductor design talent** but limited advanced manufacturing capacity. A complete ecosystem supports **AI, quantum computing, IoT, and HPC**.
- **Supply Chain Resilience:** Domestic production of equipment, chemicals, gases, and critical minerals reduces exposure to global disruptions and external dependence.
- **Technology Diplomacy:** A trusted semiconductor ecosystem strengthens cooperation through **India-US iCET, Quad**, and partnerships with **Japan, the Netherlands, Germany, Singapore, and the EU**.

Semiconductors have become strategic assets in the emerging global geopolitical order. Discuss with reference to India's semiconductor policy.

Practice Mains Question

Gaganyaan Mission

ISRO is preparing for **Gaganyaan**, India's first human spaceflight mission, and is testing the **Crew Module** to ensure the safe return of astronauts during atmospheric re-entry.

Gaganyaan Mission

- **About: Gaganyaan**, announced in **2018**, is India's indigenous human spaceflight programme.
 - It aims to send **three astronauts** to a **400 km Low Earth Orbit (LEO)** for **three days** and return them through splashdown in Indian sea waters.
 - India will become the **fourth country**, after **US, Russia, and China**, to independently conduct a human spaceflight mission.
 - **Human-rated Launch Vehicle Mark (HLVM) 3** will place the **Orbital Module (OM)** in the required orbit.
 - Before the crewed mission, ISRO is conducting uncrewed tests of the **Crew Module, Crew Escape System, and HLVM3**.
- **Crew training for Gaganyaan:** Astronauts-designated—**Group Captain Prasanth Balakrishnan Nair, Group Captain Ajit Krishnan, Group Captain Angad Pratap, and Wing Commander Shubhanshu Shukla**.
- **Strategic and Scientific Significance:** Enables microgravity research in **biological sciences, material science, and medicine**.
 - Validates **life-support systems, radiation shielding, and safe re-entry mechanisms**.
 - ❖ Supports future plans for the **Bharatiya Antariksh Station by 2035** and a **crewed lunar mission by 2040**.
- **Orbital Module (OM):** Gaganyaan's spacecraft is a **two-part, bi-modular system**.
 - **Crew Module (CM):** Pressurised, habitable capsule with **Environmental Control and Life Support System (ECLSS), Thermal Protection System, avionics, and deceleration systems** for safe re-entry and splashdown.
 - **Service Module (SM):** Unpressurised module providing **propulsion, power, thermal control, avionics, and deployment mechanisms**.
 - ❖ Separates before re-entry and burns up in the atmosphere.
 - ❖ Unlike the three-module **Soyuz** and **Shenzhou**, Gaganyaan uses a lighter **two-module design**.

Safe Re-entry Mechanism of the Gaganyaan Crew Module

- **Service Module Separation:** The **Connect & Disconnect System (CS-CDS)** detaches the **Service Module** before re-entry.
 - Failure could alter the spacecraft's **centre of gravity** and make re-entry fatal.
- **Sphere-Cone Aerodynamic Design:** The blunt base creates a **detached shockwave** that deflects heat.
 - The conical section provides controlled lift and reduces **g-forces**.
- **Thermal Protection System (TPS):** The **ablative heat shield** chars and erodes to carry away extreme heat during hypersonic re-entry.
- **Active Attitude Control: Reaction Control System (RCS) thrusters** maintain capsule orientation and keep the heat shield correctly aligned.
- **Multi-stage Parachute System:** **Drogue parachutes** stabilise the capsule, while **main parachutes** reduce its speed for sea splashdown.
 - The system has redundancy for crew safety.
- **Hydrodynamic Up-righting (CMUS):** If the Crew Module overturns after splashdown, **CMUS** inflates flotation bags using stored cold gas.
 - This rolls the capsule upright, protects the crew, and exposes communication beacons for **Indian Navy recovery teams**.

India Joins NeoSep1 Trial to Combat Neonatal Sepsis

Jawaharlal Institute of Postgraduate Medical Education and Research (JIPMER), Puducherry became the first centre in India to enrol a newborn in the **NeoSep1 international clinical trial** to identify safe and effective antibiotic combinations for **antimicrobial-resistant neonatal sepsis**.

- **NeoSep1 Clinical Trial:** A multicentre international trial sponsored by **GARDP** to treat **drug-resistant infections**.
- **Coverage:** Aims to enrol **3,000 newborns** across **Asia and Africa** by **2028**.
- **Participating Countries:** Ongoing in **Ghana, Kenya, South Africa, India, Vietnam and Pakistan**; planned expansion to **Malaysia, Bangladesh and Uganda**.
- **Core Objective:** Identify effective antibiotic combinations for **neonatal sepsis** and develop evidence-based treatment strategies for **Low- and Middle-Income Countries (LMICs)**.
- **Innovative Framework:** Uses the **PRACTical (Personalised Randomised Controlled Trial)** design, allowing clinicians to choose suitable regimens before randomization.
- **India's Strategic Participation:** Generates **localized evidence** as neonatal sepsis pathogens in India differ from those in high-income countries.

Neonatal Sepsis

- **Definition:** A severe, life-threatening systemic bloodstream infection in infants below **90 days**, which may cause **septic shock, multi-organ failure and death**. It may present as **septicaemia, pneumonia or meningitis**.
- **Classification:**
 - **Early-onset Sepsis (EOS):** Occurs within the first **72 hours**, usually due to maternal transmission before or during delivery.
 - **Late-onset Sepsis (LOS):** Occurs from **72 hours to 90 days**, generally acquired from hospitals or the community.
- **Causes:**
 - **Global Pattern:** Mainly **Group B Streptococcus (GBS)** and **E. coli** in high-income countries.
 - **Indian Pattern:** Predominantly **Gram-negative, Multidrug-Resistant (MDR)** bacteria, especially **Klebsiella pneumoniae, E. coli, Acinetobacter** and **Pseudomonas aeruginosa**, making treatment difficult.
- **Major Risk Factors:** Prolonged rupture of membranes, maternal infections, unhygienic delivery, infected amniotic fluid, **prematurity, Low Birth Weight (LBW), prolonged hospitalisation**, and invasive devices such as ventilators and catheters.

- **Signs and Symptoms:**
 - **Temperature Instability:** Hypothermia or fever.
 - **Respiratory Distress:** Rapid breathing, grunting and apnoea.
 - **Neurological Symptoms:** Lethargy, irritability, abnormal crying and seizures.
 - **Gastrointestinal Symptoms:** Poor feeding, vomiting and abdominal distension.
 - As symptoms are often non-specific and deterioration is rapid, immediate medical intervention is required.
- **Treatment Protocols:** Immediate **broad-spectrum antibiotics** after collecting **blood cultures**, followed by modification based on antimicrobial susceptibility results and supportive care.

Burden of Neonatal Sepsis in India

- **High Mortality Burden:** Sepsis is the **second most common cause of neonatal mortality**, causing over **1 million deaths globally each year**.
 - It accounts for an estimated **30%–40% of neonatal deaths in India**, or around **2–2.5 lakh preventable deaths annually**.

Quantum Computing

Advances in **Quantum Error Correction (QEC)**, along with India's **National Quantum Mission**, could move quantum computing from error-prone **Noisy Intermediate-Scale Quantum (NISQ) systems** towards **fault-tolerant quantum computers**, enabling applications in **cryptography, climate modelling and materials science**.

Quantum Computing

- **About:** Quantum computing uses **quantum mechanics** to solve certain complex problems beyond classical computers, performing specific calculations much faster than supercomputers.
- **Classical Bit vs. The Qubit:** Classical computers use **bits** as **0 or 1**; quantum computers use **qubits**, which can exist in multiple states due to **superposition**.
- **Core Principles of Quantum Computing:**
 - **Superposition:** A **qubit** can exist in multiple states at the same time, like a spinning coin until measured.
 - **Entanglement:** Linked **qubits** affect each other regardless of distance, enabling faster processing of large information.
 - **Interference:** Quantum computers cancel incorrect answers and amplify correct solutions.
- **Massive Parallelism:** **Superposition and entanglement** allow quantum computers to evaluate countless solutions

simultaneously, increasing speed for specific tasks like **sorting and search algorithms**.

■ Different Types of Qubits:

- **Superconducting Qubits:** Use **Josephson junctions** cooled near absolute zero, e.g., **Google's Willow processor**.
 - ❖ The **2025 Nobel Prize in Physics** was awarded to **John Clarke, Michel H. Devoret and John M. Martinis** for work on **macroscopic quantum tunnelling** and **energy quantisation** in electric circuits, aiding superconducting quantum computers and sensors.
- **Quantum Dot Qubits:** Made using tiny semiconductor particles like **silicon or germanium**.
- **Trapped Ion Qubits:** Use individual charged atoms trapped by **electromagnetic fields**.
- **Photonic Qubits:** Use manipulated **photons** to carry quantum information.
- **NMR Qubits:** Use nuclear angular momentum in molecules; used in the first **1998 quantum computer demonstration**.
- **Quantum Key Distribution (QKD) Network:** QKD securely shares encryption keys using **photons**; interception disturbs particles and alerts users.

"Noise" Problem in Quantum Computing

- **The Threat of Decoherence:** Present quantum computers or **NISQ devices** are fragile and error-prone. Qubits lose their quantum state due to environmental interference like **heat, light and electromagnetic interference**, causing (lose their quantum state).
- **High Error Rates:** Quantum processors are kept near **absolute zero (-273°C)**, yet have error rates of **1% to 0.1%**, unlike classical computers with far lower error rates.
- **Imperfect Control Mechanisms:** Lasers and microwave pulses used to control qubits are imperfect, adding to **noise and calculation errors**.

Attempts to Fix Quantum Errors

- **Quantum Error Correction (QEC):** Multiple noisy **physical qubits** are combined to form one stable **logical qubit**. If individual qubit errors fall below a threshold, overall error rates drop exponentially as more qubits are added.
- **Breakthrough:** In **2024**, Google's **Willow processor** showed that scaling physical qubits suppressed encoded error rates, advancing **fault-tolerant quantum computing**.

India's Strategy in the Quantum Race

- **National Quantum Mission (NQM):** Sanctioned in **April 2023**, it aims to make India a leading nation in **Quantum Technologies by 2031**.
- **Objective:** Build **intermediate-scale quantum computers** and indigenous **hardware-software ecosystems**.
- **Target:** Develop **50-1,000 qubit quantum computers**, **2,000-km satellite-based quantum communication and QKD networks**, quantum sensing, metrology, materials and devices.
 - The Mission has set up **four Thematic Hubs (T-Hubs)** and supports deep-tech startups through **Optionally Convertible Debt (OCD)**.
- **Progress:** India has developed an indigenous **1,000-km quantum communication network**, achieving half of its **2,000-km target**, with support from **DST and QNu Labs**.

UN Scientific Panel on AI Report

The **UN's Independent International Scientific Panel on AI** warned that **AI capabilities** are advancing faster than **scientific understanding** and **regulatory frameworks**, posing serious risks to **global stability**.

Highlights of the UN Report on AI

- **Capability Race Outpacing Governance:** Frontier AI models are improving faster than regulation, with task autonomy doubling every **4–7 months**, and showing risks like **evaluation awareness** and **shutdown resistance** in lab settings.
- **Concentration of AI Power:** The **US** holds about **75%** and **China 15%** of top AI supercomputer computers, while **91%** of notable AI models come from the **private sector**.
 - **Hardware Choke Points:** Key AI hardware supply (**80% market share**) chains are concentrated around firms like **ASML, TSMC, and NVIDIA**, each holding major dominance in critical niches.
 - **Global South Exclusion:** **118 nations**, largely from the **Global South**, remain unrepresented in major AI governance summits.
- **Multidimensional AI Divide:** The divide includes gaps in **sovereign compute, talent, regulation, and application capacity**, not just digital access.
 - **Language Bias:** AI models cover only a small share of **7,000+ global languages**, causing safety risks in low-resource languages.
- **Agentic AI and Cybersecurity:** Autonomous AI agents can boost coding but raise risks of **multi-agent miscoordination, prompt injection**, and discovery of **zero-day vulnerabilities**.
 - **Erosion of Shared Reality:** Risks include **deepfakes, epistemic erosion, liar's dividend**, and **synthetic consensus**, weakening public trust.

- **Human Rights Risks:** Concerns include **AI-generated CSAM**, deepfake abuse against women, and gender bias, with **88%** of leading AI researchers being male.
- **Macroeconomics, Environment, and Evidence Dilemma:** AI gains may follow a **Productivity J-Curve**, requiring upfront investment before benefits appear; risks include **AI washing**, rising environmental pressure from **USD 770 billion** hyperscaler capex, and delayed regulation due to the **Evidence Dilemma**.

Report Finding	Implications for India
AI compute is concentrated in the US (75%) and China (15%) .	Need to boost IndiaAI Mission , sovereign GPU capacity, and Digital Sovereignty .
AI divide includes infrastructure, talent, governance, and application gaps.	Strengthen National Quantum Mission , AI CoEs , industry-academia links, and India AI Stack .
7,000+ languages remain underserved.	Scale BHASHINI , BharatGen , and multilingual foundation models.
Healthcare AI needs local validation .	Use Indian datasets under ABDM/National Digital Health Mission to reduce bias.
Agentic AI raises cybersecurity risks.	Implement DPDP Act, 2023 , empower CERT-In/I4C , and finalise National Cyber Security Strategy to safeguard AI-enabled critical infrastructure.
Global AI governance is fragmented.	Support risk-based regulation via IndiaAI Safety Institute , GPAI , and multilateral forums (G20).
AI-generated CSAM/deepfakes harm women and children.	Enforce IT Act, 2000 , BNS, 2023 , POCSO Act, 2012 , and IT Rules, 2021 .
118 countries excluded from AI governance.	India can champion Global South through DPI , One Future Alliance , India Stack , and South-South cooperation.

Reforms 3.0 for the Bharat Rate of Growth

Experts have proposed “**Reforms 3.0**”, envisioning **AI as public digital infrastructure** to drive India’s next phase of economic transformation and sustain a “**Bharat Rate of Growth**” above **8%** over the next decade.

AI-driven Reforms 3.0

AI-driven Reforms 3.0: Better Brief

- **National AI Token Policy:** Implement within **24 months** through PPPs with **AWS, Google and Microsoft** to create sovereign AI infrastructure; provide free AI research tokens to **IITs/IISc**, API sandboxes for startups and AI literacy in schools.
- **Subsidising Cognitive Capacity:** India spends only **0.65% of GDP on R&D**, far below Israel **5.4%**, South Korea **4.9%**, US **3.5%** and China **2.4%**. Redirect a fraction of India’s **USD 49 billion** physical subsidies toward AI access.
- **Affordable AI Access:** Free AI access to top **100 universities**, national R&D institutions and **5,000 high schools** would cost about **USD 2 billion/year**, only **0.06% of GDP**, far below food/fertiliser subsidies.
- **Sovereign AI Infrastructure:** Host open-source models like **Llama** and Indian models like **Sarvam** domestically to avoid sudden foreign API restrictions.
 - **Strategic Capability:** Treat AI infrastructure like **space and nuclear programmes**, with **99.99% uptime**, multi-region redundancy, low latency in Tier-2/3 cities and strict data residency.

- **Diversified Compute Hardware:** Avoid Nvidia lock-in through a **40:30:30 matrix** — **40% AWS Trainium**, **30% Google TPUs**, **30% NVIDIA GPUs**.
- **PPP & Market Leverage:** Use India’s **1.4-billion-user market** to negotiate cloud/compute capacity in exchange for land, power access and favourable data residency rules.
 - **Cross-Subsidy Model:** Enterprise AI usage tiers can subsidise free access for education and public sectors.
- **Indic AI:** Develop foundational models across all **22 Scheduled Languages** for Tier-2/3 cities, local courts, rural clinics and agriculture.

Bharat Rate of Growth
■ Historical Context: India earlier remained stuck in the “ Hindu rate of growth ” of around 3.5–4% for nearly 45 years after independence.
■ Reforms 1.0: 1991 LPG reforms after the balance-of-payments crisis accelerated GDP growth.
■ Reforms 2.0: Post-2010s DPI revolution through Aadhaar, UPI and Jio effect; UPI handles 50% of global real-time digital payments.
■ Reforms 3.0: Focuses on AI-led cognitive infrastructure to achieve inclusive 8%+ GDP growth and build a Sovereign AI Ecosystem .

Challenges to AI-driven Reforms 3.0

- **Compute-Energy-Climate Trilemma:** AI data centres need **5–10 times more power** than traditional cloud systems, increasing electricity and water-cooling stress on India’s thermal-heavy grid and Net Zero 2070 goals.
- **Silicon Supply Risk:** India lacks advanced-node chip fabrication (**<5nm**), making AI plans dependent on

TSMC-dominated supply chains, export controls and geopolitical risks.

- **Subsidy Politics:** Shifting from physical subsidies to “cognitive subsidies” may trigger welfare, agrarian and political backlash.
- **Tokenization Bias:** Indic languages face a higher “token tax,” needing more compute and cost than English, widening the AI access divide.
- **Algorithmic Governance Deficit:** India lacks strong rules for black-box AI, DPDP Act enforcement, bias, deepfakes, misinformation liability and AI training-data IPR.
- **Last-Mile Infrastructure Gap:** Real-time AI in Tier-2/3 and rural areas needs edge computing, but 5G and BharatNet penetration remain uneven.

“Artificial Intelligence has the potential to become the next Digital Public Infrastructure in India.” Discuss.

Practice Mains Question

India's First 3D-Printed Artificial Reefs

Tamil Nadu will deploy **India's first 3D-printed artificial reefs under Pradhan Mantri Matsya Sampada Yojana (PMMSY) Phase II** for marine habitat restoration.

Key Facts

- **About:** Pilot deployment is planned off the **Ramanathapuram coast**.
 - It will test **six newly developed reef designs** in coastal waters.
 - Scientists will assess **durability** and **ecological performance** before larger deployment.
- **Collaboration:** Project is being carried out with the **Visakhapatnam Regional Centre of ICAR-CMFRI**.
 - Each module weighs about **1 tonne**.
 - Modules are developed by **Tvasta**, a **Chennai-based IIT Madras-incubated startup**.
- **Design:** Modules are made **without iron reinforcement**, unlike conventional reinforced concrete reefs.
 - They have complex **3D-printed structures with crevices, folds and attachment surfaces**.
 - Design improves **porosity, surface area and habitat suitability** for **corals, sponges and reef-associated organisms**.
- **Impact:** Funded jointly by the **Centre and Tamil Nadu in a 60:40 ratio**.
 - Part of a wider artificial reef project across **213 sites in Tamil Nadu**.
 - Expected to improve **fish stocks**, restore **marine ecosystems** and benefit **traditional fishing communities**.

3D Printed Artificial Reef

- **About:** **3D-printed artificial reefs** are human-made underwater structures used to restore **degraded marine habitats** and support **biodiversity**.
- **Technology:** Made using **advanced 3D-printing methods**, allowing **intricate, customised and species-specific designs** compared to conventional concrete reefs.
- **Design:** They contain varied **cavity sizes, folds, textures and complex surfaces** for shelter, attachment and breeding of marine organisms.
- **Biodiversity:** Support colonisation by **corals, sponges, algae, crustaceans, fish** and other reef-associated species, helping create a new marine ecosystem.
- **Fisheries:** Provide additional **habitats and nursery grounds**, increasing **fish biomass** and supporting **sustainable fisheries management**.
- **Advantages:** Offer **design flexibility**, faster modification, site-specific material use and targeted support for specific marine species or ecological needs.
- **Concerns:** Success depends on **durability, non-toxic materials, proper placement** and **long-term ecological monitoring**.
 - Artificial reefs may not always achieve intended ecological goals.
- **Significance:** They are an innovative restoration tool to strengthen **marine biodiversity**, support **coastal communities** and reduce pressure on **natural coral reefs**.

PMMSY

- **About:** Launched on **10th Sept 2020** to bring a **Blue Revolution** by making India's fisheries sector **sustainable, productive and inclusive**.
- **Funding:** Approved with a total investment of **₹20,050 crore** for **2020-21 to 2024-25**. Extended up to **2025-26**.
- **Achievements:** Fish production rose to **195 lakh tonnes in 2024-25**.
 - Fisheries exports increased under the scheme.
 - Created **58 lakh livelihoods**.
 - Empowered **99,018 women**.

Scaling India's Technology Ecosystem

As India launches national missions in **AI, Semiconductors, Quantum Computing and Space Technologies**, focus has increased on strengthening its innovation ecosystem.

- India has a strong legacy of **scientific research and technological innovation**, but faces challenges in bridging the “**Valley of Death**” — the gap between **indigenous prototypes** and **globally competitive, commercialised industries**.

Bottlenecks that Limit India's Ability to Scale Innovations Globally

- **Stagnant R&D Expenditure:** GERD remains around **0.64% of GDP**, compared to the global average of **~1.7%** and **2.5%+** in advanced economies like the **US and China**.
- **Skewed Funding Pattern:** Unlike leading innovation economies where the **private sector contributes over 70% of R&D**, India's R&D remains largely **public sector-driven** despite initiatives like **tax incentives** and **ANRF**.
- **Poor Commercialisation of Research:** Innovations often remain within **government research institutions** and fail to transition into commercially viable products.
- **The "Missing Middle" in Manufacturing:**
 - India lacks **mid-sized deep-tech firms** capable of building complex supply chains between large conglomerates and informal MSMEs.
- **Weak Industry-Academia Linkage:** Research in institutions like **CSIR** and **DRDO labs** often does not translate into large industrial ecosystems.
- **Inadequate Venture Capital for Deep Tech:**
 - India has strong VC support for **software and consumer internet**, but lacks **patient capital** for **hardware, deep-tech and capital goods**.
- **Intellectual Property (IP) Deficit:**
 - India performs well in **service delivery and process engineering** but lags in creating and owning **Standard Essential Patents (SEPs)** in frontier technologies.

India's Opportunities in Emerging Technologies to Achieve Global Scale

Sector	India's Existing Strengths	Opportunities to Achieve Global Scale
AI	Large talent pool, DPI , multilingual datasets	Sovereign, low-cost and energy-efficient AI; sector-specific solutions
Quantum Computing	Research ecosystem, National Quantum Mission , maths and physics expertise	Affordable infrastructure; applications in drugs, cybersecurity, materials, finance and climate
Semiconductors	20% of global VLSI design workforce , ISM , electronics ecosystem	Scale fabs, ATMP , chip design and indigenous IP
Space Technologies	Frugal innovation through Chandrayaan and Mangalyaan	Space data centres, orbital computing, quantum communication, satellites and launches
Digital Public Infrastructure (DPI)	Aadhaar, UPI, DigiLocker, ONDC and CoWIN	Export DPI model and interoperable digital public goods
Biotechnology & Life Sciences	Leadership in pharmaceuticals, vaccines and biotechnology	Genomics, precision medicine, synthetic biology, bio-manufacturing and new vaccine platforms
Advanced Manufacturing	Engineering base and PLI schemes	Industry 4.0, robotics, 3D printing, advanced materials and smart manufacturing
Green & Clean Technologies	Renewable energy deployment and green hydrogen initiatives	Green hydrogen, battery storage, carbon capture, electric mobility and climate technologies
Cybersecurity & Digital Trust	Digital economy and IT services ecosystem	Indigenous cybersecurity, post-quantum cryptography, secure cloud and digital trust solutions

Measures to Scale India's Emerging Technologies

- **Bridge the "Valley of Death":** Expand the **RDI Scheme** by establishing **Translational Research Centres** for shared prototyping and testing facilities for deep-tech startups.
- **Reform Academic R&D:** Shift from **"Publish or Perish"** to **"Patent & Produce"** ecosystem through a framework similar to the **US Bayh-Dole Act (1980)**.
- **Provide Patient & Blended Finance:** Enable **Anusandhan National Research Foundation** to provide long-term, high-risk capital through industry-academia partnerships.
- **Make Government the Anchor Customer:** Expand the **iDEX model** beyond defence and introduce **5–10% public procurement quota** for indigenous deep-tech startups.
- **Lead Global Technology Standards:**
 - Support Indian firms' participation in **ITU**, **3GPP** and **ISO** to secure **Standard Essential Patents (SEPs)** in emerging technologies.
- **Institutionalizing Brain Circulation:** Scale **VAJRA** and **Ramanujan Fellowship** to bring Indian scientists as visiting professors, investors and startup mentors.
- **Regulatory Sandboxes:** Enable testing of emerging technologies like **drones**, **AI** and **crypto** without regulatory uncertainty.
- **Tech-Alliances:** Leverage **US-India iCET** and **Quad** for technology transfers in **defence**, **space** and **quantum computing**.

- **Mineral Diplomacy:** Expand **KABIL** to acquire critical mineral assets in **Africa, Australia and South America** for battery and semiconductor supply chains.
- **Innovation Progress:** India improved from **81st (2015) to 38th (2025)** in the **Global Innovation Index 2025**, but needs stronger support to convert research into commercial products.

“India’s technological history shows that being first matters little if you cannot scale.” Discuss the structural bottlenecks in India’s innovation ecosystem that prevent them from becoming global industries. Suggest measures to bridge the gap.

Practice Mains Question

The Challenge of India’s Digital Sovereignty

Rising cyberattacks and geopolitical tensions have exposed India’s dependence on foreign digital infrastructure, highlighted by the **Chinese software EseeCloud (2026) CCTV hack** and **2025 Nayara Energy service disruption**, strengthening calls for **digital sovereignty**.

Key Challenges for India to Achieve Digital Sovereignty

- **Semiconductor Chokepoints:** ISM is scaling chip packaging, but front-end fabrication depends on foreign **Extreme Ultraviolet (EUV) lithography** and precision chemicals.
 - Risks: export controls and **hardware trojans** (maliciously embedded circuits).
- **Cable Sovereignty Deficit:** Over **95%** of India’s international data moves through subsea cables.
- **The AI Compute Gap:** **IndiaAI Mission** provides access to **38,000 sovereign GPUs**.
 - India’s compute capacity is still far behind global firms investing **USD 40–50 billion annually**.
 - Risk: dependence on foreign foundation models.
- **The Source Code “Black Box”:** **DPDP Act, 2023** ensures data privacy but not platform sovereignty.
 - India cannot audit proprietary source code of foreign critical-sector software.
 - Hidden software backdoors may remain undetected.
- **Q-Day and Encryption Vulnerability:** Quantum computing can break encryption through **“Harvest Now, Decrypt Later”** attacks.
 - **DST early-2026 roadmap** requires **CII migration to Post-Quantum Cryptography (PQC) by 2029**, posing financial and logistical hurdles.
- **Regulatory Reciprocity Paradox:** **FTAs and Indo-Pacific Economic Framework (IPEF)**, push unrestricted data flows and bans on source-code audits.
 - This conflicts with India’s sovereign policy goals.

- **Global Standards Deficit:** India lacks strong representation in bodies like **ITU**.
 - Dependence on foreign **Standard Essential Patents (SEPs)** leads to high royalties and foreign-designed **5G/6G** systems.
- **Deep-Tech Capital and Brain Drain:** Talent drain in **quantum mechanics and device physics**.
 - Foreign VC reliance can shift critical **IP/patents** abroad.
- **LEO Satellites & Cyber-Insurance:** Foreign **LEO constellations** like **Starlink** risk external control over connectivity.
 - Western cyber-insurance (like Lloyd’s of London) dominance may expose Indian firms to payout denial under “state-sponsored cyberattack” exclusions.
- **Missing ‘Patient’ Capital:** Deep-tech needs long-term, high-risk capital.
 - Indian VC funding mostly goes to consumer tech, not hard/deep technology.

Measures to Achieve Absolute Digital Sovereignty

- **Adopt Geopatriation:** Shift critical government/strategic workloads to sovereign domestic cloud like **MeghRaj**.
- **Subsea Infrastructure:** Create **Cable Protection Zones** in **EEZ**.
 - Fund a domestically flagged subsea cable repair fleet.
 - Expand decentralized **CLS-PoPs** under **Draft Telecommunications Rules, 2026**.
- **Build a ‘Third Way’ Digital Ecosystem:** Expand open, interoperable **DPI** like **UPI** and **ONDC** to reduce US/China tech dependence.
- **Strengthen Sovereign AI:** Expand sovereign **AI ecosystem** for cognitive autonomy.
 - Subsidize **GPU infrastructure** for indigenous startups like **Sarvam AI**.
 - Use initiatives like **Bhashini** and **BharatGen**.
- **Advance Digital Financial Sovereignty:** Expand **Digital Rupee** and cross-border settlement initiatives to reduce reliance on foreign financial networks.
- **Ecosystem Design from Day One:** Integrate supply chains, developer communities, and target markets from mission inception.
- **Preventing “Protocol Hegemony”:** Strengthen India’s role in global standard-setting forums to avoid **Splinternet** duopoly and foreign surveillance risks.

Digital sovereignty has become a cornerstone of national security in the digital age. Discuss with reference to India’s strategic priorities.

Practice Mains Question

History, Art & Culture

PM Pays Tribute to Swami Vivekananda

PM paid tribute to **Swami Vivekananda** on his **Nirvana Day**, highlighting his role in promoting India's **cultural heritage**, **spiritual traditions**, and **national consciousness** globally.

- **About: Swami Vivekananda**, born **Narendra Nath Datta** on **12th Jan 1863**, was a monk and chief disciple of **Ramakrishna Paramhansa**.
 - In **1893**, on request of **Maharaja Ajit Singh of Khetri**, he adopted the name **Vivekananda**, replacing **Sachidananda**.
- **Enlightenment:** In **1892**, he meditated for **three days and nights** on a rock near **Kanyakumari**, later named **Vivekananda Rock Memorial**, leading to his enlightenment.
- **Contributions:**
 - **Philosophical:** Introduced Indian philosophies of **Vedanta** and **Yoga** to the world.
 - ❖ Preached **neo-Vedanta**, interpreting Hinduism through a Western lens, and supported combining **spirituality** with **material progress**.
 - **Spiritual:** His message drew from the **Upanishads**, **Gita**, **Buddha**, and **Jesus**, emphasizing **self-realization**, **compassion**, and **selfless service**.
 - ❖ Advocated **service to jiva as worship of Shiva**.
 - ❖ Explained four paths to **moksha** in **Raja-yoga**, **Karma-yoga**, **Jnana-yoga**, and **Bhakti-yoga**.
 - **Revivalism:** Emphasized education for national regeneration and advocated **man-making, character-building education**.
 - **Nationalism:** His nationalism was based on **Humanism** and **Universalism**, key features of Indian spiritual culture.
 - ❖ It focused on **masses, freedom, equality**, and **Karma Yoga** as a path to political and spiritual freedom through selfless service.
- **Associated Organisations:** Founded **Ramakrishna Mission** in **1897** for service, education, and spiritual upliftment.
 - Established **Belur Math** in **1899**, which became his permanent abode.
- **International Address:** Represented **Hinduism** at the **Parliament of Religions, Chicago, 1893**.
 - Addressed the **London Hindu Association** conference in **July 1896**.

- **Death:** Passed away on **4th July 1902** while meditating at **Belur Math**; observed as **Nirvana Day**.

Birth Anniversary of Dr. Syama Prasad Mookerjee

PM paid tribute to **Dr. Syama Prasad Mookerjee** on his **birth anniversary, 6th July**, highlighting his legacy in **national unity**, **industrial policy**, **education**, and **modern India's development**.

- **Birth and Early Life:** Born in **Calcutta**; son of **Sir Ashutosh Mookerjee**, educationist and Vice-Chancellor of **Calcutta University**.
- **Academic Excellence:** Represented **Calcutta University** at the **Conference of the Universities of the British Empire** in England.
 - Became the youngest **Vice-Chancellor of Calcutta University** at **33** in **1934**.
 - Started **Bang Wani** in **1922** and **The Nationalist** in the **1940s**.
- **Political Career:** Joined the **Indian National Congress** in the **1920s** but resigned over ideological differences.
 - Later joined the **Hindu Mahasabha** and helped form the **Progressive Coalition government** in Bengal in **1937**, serving as **Finance Minister**.
 - Became acting President of the **Hindu Mahasabha** in **1940** and advocated **complete independence**.
 - Founded the **All India Bharatiya Jan Sangh** in **1951**, later evolving into the **Bharatiya Janata Party**.
- **Post-Independence Role:** Served as **Minister for Industry and Supply** in the **Interim Government** after independence.
 - Helped establish **Chittaranjan Locomotive Factory**, **Sindri Fertilizer Corporation**, and **Hindustan Fertilizer Corporation**.
- **Ideological Stance:** Advocated **nationalism**, **Hindu cultural identity**, and a **united India**.
 - Opposed **Article 370**, stating that one nation cannot have **two constitutions, heads, or flags**.
 - Opposed **linguistic division of India**, favouring unity based on **administrative efficiency, security, and economic prosperity**.
 - Arrested during protest against **J&K's special status** and died under mysterious circumstances in **1953**.
- **Legacy:** Known as **"The Lion of Parliament"** for sharp debates on national issues.

16th-Century Vijayanagara Inscriptions in Seshachalam Forest

ASI unearthed three 16th-century inscriptions of Vijayanagara king Sadasivaraya inside Seshachalam forest, Tirupati district.

- **Linguistic Diversity:** Epigraphs are in Telugu, Tamil, and Kannada.
 - Date deciphered as 31st July 1554 C.E.
- **Royal Patronage:** Records construction of a Siva temple and Mutt at Papavinasa by King Sadasivaraya during pilgrimage.
- **Revenue & Temple Administration:** Mentions taxes (Kaanika) and land grants from two villages for daily food offerings and worship at Gudimallam Parasurameswara temple.
 - Names Sadasiva Basavanna Odeya, Linganna Vodaya, and Peddayya, son of Chembhaperiya, as the composer/temple accountant.
- **Historical Significance of Gudimallam:** Reaffirms glory of 2nd-century BC Gudimallam shrine, widely recognized as India's first known Siva temple.
 - Shows direct Vijayanagara royal funding and attention.

Vijayanagara Empire

- **Vijayanagara Empire:** 1336–1646 CE.
- Founded by Harihara I and Bukka Raya I of Sangama dynasty on the banks of Tungabhadra River.
- **Dynasties**— Sangama, Saluva, Tuluva, Aravidu.
- Greatest extent under Krishnadevaraya of Tuluva dynasty.
 - Krishnadevaraya authored Amuktamalyada, patronized Ashtadiggajas (8 prominent scholars), and defeated Gajapatis and Adil Shahis.
- Administration based on Nayankara/Amara-Nayaka system: chiefs got Amaram territories for maintaining troops and paying tribute.
- Architecture: final phase of Dravidian style, with Kalyana Mandapas (marriage halls), ornate monolithic pillars, and Raya Gopurams immortalised in Vittalaswamy and Hazara Rama at Hampi.
- Foreign accounts— Ibn Battuta, Nicolo de Conti, Abdur Razzak, Domingo Paes.
- Decline after Battle of Talikota/Rakshasi-Tangadi, 1565 against unified Deccan Sultanates.

Birth Anniversary of Mangal Pandey

The PM paid homage to the freedom fighter Mangal Pandey on his birth anniversary (19th July).

Mangal Pandey

- **Early Life:** 19 July 1827, Ballia district, Uttar Pradesh.
 - Joined the 34th Bengal Native Infantry of the East India Company at age 22.
- **Mutiny:** Opposed the Enfield Pattern 1853 rifle-musket because its cartridges were believed to contain beef and pork fat.
 - On 29 March 1857, he fired at his Senior Sergeant Major at Barrackpore.
 - He was hanged on 8 April 1857; his regiment was later disbanded.
- **Impact:** His revolt contributed to the outbreak of the Revolt of 1857, also called the First War of Independence.
 - The 7th Awadh Regiment's revolt was suppressed. On 10 May 1857, Meerut sepoys marched to Delhi and proclaimed Bahadur Shah Zafar II as Shah-en-shah-i-Hindustan.
- Mangal Pandey became a symbol of Indian resistance to British rule and sepoy-peasant grievances.

Birth Anniversary of K. Kamaraj

PM paid tribute to Thiru K. Kamaraj on his birth anniversary, recalling his role in the freedom movement and nation-building.

K. Kamaraj (1903-1975)

- **Role in Freedom Struggle:** Inspired by the Jallianwala Bagh massacre, he joined the freedom movement.
 - Participated in the Salt Satyagraha (1930) and was jailed during the Quit India Movement (1942).
 - Elected to the Constituent Assembly in 1946 and Parliament in 1952.
- **Contribution in Nation Building:** Served as Chief Minister of Madras Province from 1954 and President of the Indian National Congress from 1963. Perhaps the first non-English-speaking CM of India
 - Proposed the Kamaraj Plan (1963), urging senior Congress leaders to leave ministerial posts and strengthen party organisation.
 - Introduced free education and the mid-day meal scheme in schools.
 - His birth anniversary is observed as Education Development Day in Tamil Nadu.
- **Legacy:** Posthumously awarded the Bharat Ratna in 1976.
 - Died in 1975 at the age of 72.

Social Issues

Rising Secondary Infertility in India

Secondary infertility in India rose from ~19.5% in the early 1990s to 28.6% in 2015–16, according to NFHS data. Major causes include **delayed parenthood, age-related fertility decline, lifestyle disorders, reproductive complications, and male infertility.**

Infertility

- **About:** Failure to achieve pregnancy after **12 months or more** of regular unprotected sexual intercourse.
- **WHO:** Around **1 in 6** people of reproductive age globally experience infertility during their lifetime.
- **Causes:** Male factors, female factors, both, or unexplained causes.
- **Primary Infertility:** No previous pregnancy.
- **Secondary Infertility:** **At least one previous pregnancy**, but unable to conceive again.

Causes of the Rising Trend of Secondary Infertility in India

- **Delayed Family Completion:** Career, migration, and financial pressures delay childbirth. Women planning a second child are often above **35 years**, when **Diminished Ovarian Reserve (DOR)** and egg quality decline; advancing paternal age may increase **sperm DNA fragmentation**.
- **Post-Partum Anatomical Complications:** Caesarean sections may cause **pelvic adhesions** or **isthmocele**, affecting implantation. Untreated infections can lead to **Pelvic Inflammatory Disease (PID)** and fallopian tube blockage.
- **Progression of Reproductive Disorders:** **Endometriosis** and **uterine fibroids** may develop or worsen between pregnancies.
- **Post-Partum Endocrine and Metabolic Shifts:** Weight gain, sedentary lifestyle, and metabolic changes may disrupt the **Hypothalamic-Pituitary-Ovarian (HPO) axis**, affecting ovulation and hormones.
- **PCOS and Thyroid Disorders:** **PCOS**, affecting up to **20% of Indian women**, may worsen with age, weight gain, and insulin resistance, causing chronic anovulation. **Postpartum thyroiditis** and **hypothyroidism** may also disrupt fertility.
- **Male Factor:** Weight gain, smoking, alcohol, occupational stress, and metabolic disorders may reduce sperm concentration, motility, and morphology.

- **Endocrine-Disrupting Chemicals (EDCs):** Pesticides, microplastics, and industrial chemicals may disturb hormonal signalling and affect egg and sperm quality.
- **Air Pollution and Oxidative Stress:** Exposure to **PM2.5** and heavy metals may impair sperm parameters and ovarian function.

Major Challenges in Addressing Secondary Infertility in India

- **Legal and Regulatory Bottlenecks:** Under the **Surrogacy (Regulation) Act, 2021**, married couples with a surviving child cannot opt for surrogacy, except when the child has a **life-threatening disorder or severe mental/physical disability**.
 - This restricts options for couples with secondary infertility and raises concerns over **reproductive autonomy under Article 21**.
- **Public Health Policy Blind Spots:** Treatments such as **IVF** and **ICSI** cost around **₹1.5–2 lakh per cycle** but are largely excluded from **AB-PMJAY**.
 - The **RMNCH+A framework** prioritizes contraception and maternal mortality, while infertility care remains neglected.
- **Urban-Private Concentration:** **ART clinics** are concentrated in the private sector and **Tier-1 and Tier-2 cities**. Public district hospitals often lack **embryology laboratories** and specialized endocrinologists.
- **Socio-Cultural Stigma:** Secondary infertility is often trivialized because the couple already has a child, causing isolation and delaying psychiatric or medical support.
- **Male Diagnostic Evasion:** The belief that a man who has fathered a child remains permanently fertile leads to refusal or delay in **semen analysis**, despite declining sperm quality being a major factor.
- **Misinformation and Unregulated Care:** Despite the **ART Act, 2021**, weak enforcement allows clinics to make misleading claims such as **"100% IVF success in one cycle."**
- **Delay in Evidence-Based Treatment:** Rural and semi-urban couples may spend crucial reproductive years on unverified remedies, especially affecting women above **35 years**, when ovarian reserve declines rapidly.

Empowering Persons with Disabilities in India

Disability rights issues have renewed calls for a **rights-based, uniform and dignified welfare framework** for persons with disabilities in India.

Need for Rights-based Disability Welfare Framework

- **Rights-based Approach:** Treats PwDs as equal citizens with dignity, not as charity recipients.
- **Exclusion:** Many PwDs remain outside welfare due to access, documentation and certification barriers.
- **Low Pension:** Disability pensions in many States are only ₹300–₹500/month, inadequate for dignified living.
- **State Inequality:** Support varies by domicile, State budgets and administrative priorities.
- **Low Spending:** India spends only around 0.02% of GDP on disability welfare.
- **Economic Cost:** Excluding PwDs from education, jobs and social security reduces productivity and GDP.
- **Legal Need:** Helps realise constitutional equality, dignity, right to life and the RPwD Act, 2016.

Current Status of PwDs in India

- **Population:** Census 2011 recorded 2.68 crore PwDs, i.e. 2.21% of population; prevalence is higher among males (2.4%) than females (1.9%), and higher in rural areas.
- **Recognised Disabilities:** RPwD Act, 2016 recognises 21 disabilities, including locomotor, visual, hearing, speech/language, intellectual, multiple disabilities, cerebral palsy and dwarfism.
- **Constitutional Basis:** Preamble ensures justice, liberty and equality; Part III Fundamental Rights apply to PwDs; Article 41 directs public assistance in cases of disablement, sickness, old age and unemployment.
- **Local Governance:** 11th Schedule Entry 26 and 12th Schedule Entry 9 include welfare/safeguarding of PwDs and weaker sections.
- **Policy & Legal Framework:** National Policy for PwDs 2006, UNCRPD ratification 2008, RPwD Act 2016, National Trust Act 1999, RCI Act 1992, and SIPDA 2015 support rights, rehabilitation, training and skill development.
- **SIPDA:** Focuses on skill training of PwDs aged 15–59 under the National Action Plan for Skill Development.
- **Judicial Pronouncements:** *Rajive Raturi v. Union of India* (2024) made accessibility and digital inclusion part of Article 21 dignity/equality; *RBI v. A.K. Nair* (2023) upheld reservation in promotions for PwDs under RPwD Act.

Challenges Faced by PwDs in India

- **Infrastructure:** Public buildings, transport and urban spaces lack ramps, elevators, tactile paths and universal design; only 3% buildings were fully accessible as per 2018 report.
- **Education:** Inaccessible schools, lack of trained teachers and assistive tech; only 52.2% PwDs aged 7+ are literate.
- **Employment:** Discrimination, inaccessible workplaces and poor accommodation; only 34 lakh of 1.3 crore employable PwDs are employed.

- **Healthcare:** Lack of disability-friendly facilities, trained staff and equipment; Covid-19 exposed severe access gaps.
- **Digital-Legal:** Inaccessible websites/apps and weak enforcement of RPwD Act, 2016 deepen exclusion.

Measures to Empower PwDs in India

- **Identification:** Create a centralised, portable digital disability database linked with certificates, audits and welfare access.
- **Protection:** Ensure a **Minimum Universal Disability Pension Floor Rate** through a National Disability Pension Authority, DBT delivery and grievance redress.
- **Accessibility:** Enforce universal design in buildings, schools, hospitals, transport and digital platforms with ramps, tactile paths, sign language, captions, screen-reader access and GIGW 3.0 compliance.
- **Education:** Strengthen inclusive education through special educators, assistive tech, adapted curriculum and early identification under Samagra Shiksha and NEP 2020.
- **Employment:** Expand PM-DAKSH, Divyangjan Kaushal Vikas and Divyangjan Rozgar Setu with employer incentives, job portals and flexible/gig work models.
- **Awareness:** Train officials, police, judiciary, teachers, health workers and employers; use campaigns like Badhte Kadam to promote dignity and equal citizenship.

Disability welfare in India must move from charity-based assistance to rights-based entitlement. Analyse.

Practice Mains Question

SUMAN Roadmap 2030

The Health Minister launched **SUMAN Roadmap 2030** at the 16th CCHFW Conference, along with Aarogya Setu 2.0, eSushrut@clinic, PM-JAY WhatsApp Chatbot, Drug Registry and Unified Health Interface.

SUMAN Roadmap 2030

- **About:** MoHFW's strategic framework to strengthen maternal and newborn healthcare.
- **Approach:** Evidence-driven, locally responsive strategy to achieve **SDG maternal and newborn health targets by 2030**.
- **Framework:** Anchored in RMNCHA+N life-cycle approach from pre-pregnancy to postnatal care.
- **High-Risk Care:** Focuses on identification, tracking and management of high-risk pregnancies.
- **Focus Areas:** Transport, tribal/hard-to-reach areas, emergency obstetric care, SUMAN Panchayats and climate risks.
- **Target Districts:** Time-bound interventions in **130 districts across 13 high-focus States**.

- **Goal:** Reduce **MMR below 70**, lower **NMR/IMR**, and achieve zero preventable maternal-newborn deaths.
- **Key Interventions:** Folic acid, nutrition support, MDSR, MNM reviews, NASG, AI-enabled labour rooms, C-section optimisation and SSBSK integration.
- **Support System:** Centres of Excellence, SUMAN Call Centre, referral linkages and JANANI Portal monitoring.

SDG Targets on Maternal and Newborn Health

- **Target 2.2:** End malnutrition; address nutrition needs of adolescent girls, pregnant/lactating women and children under 5.
- **Target 3.1:** Reduce global **MMR below 70 per 100,000 live births**.
- **Target 3.2:** End preventable newborn/under-5 deaths; reduce **NMR to 12 per 1,000** and under-5 mortality to **25 per 1,000**.
- **Target 3.7:** Ensure universal access to sexual and reproductive healthcare, family planning and education.
- **Target 5.6:** Ensure reproductive rights and informed reproductive choices.

Other Health Initiatives at 16th CCHFW

- **Drug Registry:** ABDM platform to standardise generic, branded and substance information for e-prescriptions and clinical decisions.
- **Unified Health Interface:** ABDM service layer to discover, book and access digital health services across platforms.
- **NAS Guidelines 2026:** Framework for ambulance deployment, AIS-125 compliance, ICDCs, GPS tracking and 112 integration.
- **Aarogya Setu 2.0:** PHR app for ABHA creation, health records, PM-JAY wallet, reminders and consent-based data sharing.
- **Ayushman Sarathi:** PM-JAY WhatsApp chatbot for eligibility, Ayushman Card, hospital search, grievance and treatment history.
- **eSushrut@Clinic:** ABDM-enabled HMIS by C-DAC for PHCs, HWCs, small and private clinics.
- **Samagra Shishu Bal Swasthya Karyakram (SSBSK):** Integrates Home-Based Newborn Care (HBNC) and Home-Based Care for Young Child (HBYC) for home/community care from birth to 36 months.

Tackling Adolescent Malnutrition in India

The **NFHS-6 (2023-24)** has highlighted the growing challenge of **adolescent malnutrition in India**, with a sharp rise in **obesity, high blood sugar, and lifestyle-related diseases**.

Status of Adolescent Malnutrition in India

- **Double Burden:** India faces both persistent undernutrition and rising overweight/obesity among adolescents.
- **Rising Lifestyle Diseases (NFHS-6, 2023-24):** Obesity increased to **30.7% among women** and **27.3% among men**,

while the prevalence of **high blood sugar** rose to **17.8% among women** and **20.9% among men**, indicating worsening metabolic health.

- **Thin-Fat Phenotype:** Many adolescents appear lean but have **excess visceral fat and high metabolic risk**, making them vulnerable to **early-onset Type-2 diabetes, hypertension, and cardiovascular diseases**.
 - As per **CNNS (2019)**, **27.4% of adolescents** are stunted, **35% of stunted children under five** already have **adult-level triglycerides**.
- **Poor Dietary Patterns:** Adolescents rely heavily on cereal-based diets, with low intake of proteins, dairy, fruits and vegetables. HFSS and UPF consumption is rising rapidly, with UPFs increasing by **over 13.7% annually**.
- **Growing Future Burden:** A 2025 *Lancet* study projects that by 2050, **21.8 crore men** and **23.1 crore women** in India will be overweight, with the sharpest rise among **15-24 years**.
- **Physical Inactivity:** Excessive screen time and sedentary lifestyles are increasing obesity and cardiovascular risks among adolescents.

How Schools Can Address Adolescent Malnutrition

- **Skill-Based Learning:** Teach food label reading, portion control and misleading junk food marketing.
 - Use ICMR-NIN's **LFOF kits**, CBSE's 2025 **Sugar Boards**, and NCERT-UNESCO's *Let's Move Forward* comic.
- **Healthy Food Environment:** Make campuses **UPF/HFSS-free** through FSSAI's **Eat Right School Initiative** and enforce the 2020 rule banning HFSS sale within **50 metres** of schools. Use **PM POSHAN** for protein-rich, millet-based balanced meals; promote school gardens and fruit breaks.
- **Curriculum Overhaul:** Make nutrition practical, covering macro/micronutrients, healthy diets and everyday food choices. Make structured physical education and sports a daily core activity to reduce sedentary lifestyle and screen-time risks.
- **Health Screenings:** Link schools with **RBSK** and hold **Adolescent Health and Wellness Days** for BMI, anaemia and metabolic-risk screening.
- **Parental Engagement:** Conduct awareness sessions on UPFs and dietary diversity to sustain healthy habits at home.
- **Policy Support:** Higher taxes on sugary drinks and strict regulation of junk-food advertising targeting minors.

"Preventive healthcare must begin in classrooms rather than hospitals." Examine this statement in the context of adolescent nutrition and lifestyle diseases in India.

Practice Mains Question

FACTS FOR PRELIMS

REPORTS & INDICES

1st Sub – Sectoral Trial Index of Services Production

MoSPI released the first trial **Index of Services Production** for **19 service sub-sectors** for **April 2026**.

- **Base Year:** 2024–25, based on **National Industrial Classification (NIC), 2025**.
- **Scope:** India's first monthly indicator of short-term changes in the **formal services sector**, which contributes over **50% of GDP**. It excludes government, non-market, and informal services.
- **Methodological Framework:** Designed by the **Technical Advisory Committee on Index of Services Production (TAC-ISP)**, constituted in **May 2025**.
- **Deflation Mechanism:** Uses **WPI** for wholesale trade, service-specific or proxy **CPI** for most services, and **CPI-General** for repair, banking, and insurance.
- **Principal Data Streams:** Uses **GST filings**, administrative data for regulated sectors, and **Annual Survey of Incorporated Services Sector Enterprises (ASISSE)** data for industrial weights.
- **Growth Trends (April 2026):** **14 of 19 sub-sectors** recorded double-digit growth. Highest growth was in **Accommodation & Food (37.2%)**, **Retail Trade (30.8%)**, **Administrative & Support Services (28.7%)**, and **Real Estate (27.7%)**. **Air Transport (-13.9%)** and **Railway Transport (-0.4%)** contracted.
- **Frequency and Evolution:** Sub-sector indices will be released on the **29th of every month**; a composite **Overall ISP** will be introduced later.

Digital Threat Report 2025-26

MeitY, with **Computer Emergency Response Team (CERT-In)**, **Computer Security Incident Response Team in Finance (CSIRT-Fin)**, and **Strategic Information Services Agreement (SISA)**, released the **2nd Digital Threat Report 2025-26** to strengthen cyber resilience in India's **Banking, Financial Services and Insurance (BFSI)** and digital payments ecosystem.

Key Highlights of the Report

- **AI Asymmetry Emerges as the Biggest Cybersecurity Risk:** AI-enabled attackers are advancing faster than defensive

and regulatory systems, allowing low-resource actors to launch sophisticated attacks at machine speed.

- **Rapid Evolution of Cyber Threats:** The time between emergence and exploitation of threats has fallen from years to months or weeks.
- **Changing Nature of Cyberattacks:** Attacks now rely on **social engineering, credential theft, supply-chain and cloud exploitation, AI-assisted fraud, and legitimate-looking sessions**, making early detection difficult.
- **Anatomy of Cyber Failure Framework:** A **4-Layer Gap Archetype Framework** explains how breaches result from interconnected vulnerabilities and helps prioritise cybersecurity investment.
- **18-Month Cyber Resilience Roadmap:** Focuses on stronger security controls, continuous risk assessment, coordinated incident response, information sharing, and resilient architectures.
- **Major Recommendations:** Replace periodic audits with continuous assessment, improve real-time threat intelligence sharing, strengthen **Digital Forensics and Incident Response (DFIR)**, cloud and supply-chain security, adopt AI-enabled defence, and enhance coordination among regulators, financial institutions, and cybersecurity agencies.

India Ranks 13th in QS World Future Skills Index 2027

India ranked **13th globally** in **QS World Future Skills Index 2027**, topping **South Asia** and **lower-middle-income economies**. **US** ranked first, followed by **Australia, UK, Germany, and Canada**.

- **About:** Released by **QS Quacquarelli Symonds** (a global higher education analytics organisation).
 - Measures readiness for future work through skills alignment, academic readiness, economic transformation, economic capacity, and new-age skills demand.
- **India's Strength:** India scored **100 in Economic Capacity**, highest globally.
 - Backed by GDP growth, labour/infrastructure investment, and **AI investments worth around \$90 billion**.

- **Future Work Readiness:** India ranked **5th globally** in **Future of Work** indicator.
 - Reflects strong IT workforce and AI/digital technology role.
- **Key Concern:** India ranked **73rd on Human Capital Index**.
 - Shows gap between graduates produced and employer-demanded skills.

AWARDS & HONOURS

R. Vairamuthu Conferred 60th Jnanpith Award

Tamil poet-lyricist **R. Vairamuthu** received the **60th Jnanpith Award (2025)**.

- Third Tamil recipient after **Akilan** and **Jayakanthan**, ending a **24-year gap**.
- Earlier won the **Sahitya Akademi Award (2003)** for **Kallikattu Ithikasam**.

Jnanpith Award

- **About:** Instituted in **1961** by **Bharatiya Jnanpith**, it is India's highest literary honour for outstanding lifetime contribution to literature.
 - Often regarded as the **"Nobel Prize of Indian literature"**.
- **Institutional Background:** Established by **Sahu Shanti Prasad Jain** and **Rama Jain**. The first award was given in **1965** to **G. Sankara Kurup** for Malayalam.
 - **First woman recipient:** **Ashapura Devi**, Bengali, 1976
 - **First English-language recipient:** **Amitav Ghosh**, 2018
- **Eligibility and Scope:** Awarded annually only to an **Indian author**. From **1965–1981**, it recognised a specific work; since **1982**, it honours lifetime contribution. It covers **22 Eighth Schedule languages** and **English**, included in **2013**. No posthumous award is given.
- **Prize and Symbolism:** **₹11 lakh**, a citation and a bronze statuette of **Vagdevi (Saraswati)**; often presented by the **President**.
- **Selection Process:** The **Pravara Parishad**, comprising eminent scholars and litterateurs, assesses creativity, vision and literary impact.

Pradhan Mantri Rashtriya Bal Puraskar

PMRBP is India's premier national award for children aged **5–18 years**, recognising **courage, creativity, resilience, and public-spirited achievements**.

- **About:** Started in **2019**.
 - Conferred annually by **President** on **Veer Bal Diwas, 26 Dec**.
 - Day remembers **Sahibzada Zorawar Singh** and **Sahibzada Fateh Singh**.

- **Award Categories:** Bravery, Social Service, Environment, Sports, Art and Culture, Science and Technology.
- **Award Components:** Includes **medal, certificate, and citation**.
 - Around **25 children** selected each year.
 - Generally not posthumous; exceptions possible.
- **Nodal Ministry:** Administered by **Ministry of Women and Child Development**.
 - Selection by **expert committee**.

IMPORTANT DAYS

Death Anniversary of Thiru Subramania Siva

Vice-President **C. P. Radhakrishnan** paid tributes to freedom fighter **Subramania Siva** on his death anniversary (**23 July**).

Subramania Siva was born on 4 Oct 1884 in Batlagundu, Tamil Nadu.

- **Literary & Cultural Contributions:** Pioneer of **Tanittamil (Pure Tamil) Movement**, advocating Tamil free from excessive Sanskrit influence.
 - Founded nationalist journal **Gnanabhanu (1913)** and authored **30+ books**, including *Ramanuja Vijayam* and *Madhva Vijayam*.
 - Developed spiritual-nationalist outlook after becoming disciple of **Sadananda Swami (1902)**.
- **Nationalist Awakening:** Inspired by **Arya Samaj leader Thakur Kahan Chandra Verma**, became a prominent nationalist speaker.
 - Along with **V.O. Chidambaram Pillai** and **Subramania Bharati**, promoted **Bal Gangadhar Tilak's nationalist ideology**.
 - Founded **Dharma Paripalana Samajam (1906)** for social reform and patriotic awakening.
- **Role in the Freedom Movement:** Played key role in **Swadeshi Movement** with **V.O. Chidambaram Pillai**.
 - Mobilised workers during **Coral Mill Strike (1908)**, **Thoothukudi** and organised nationalist meetings.
 - Arrested for sedition; became the **first recognised political prisoner of Madras Presidency**.
 - Laid foundation of **Bharat Mata Temple, Pappapuram (1923)**.
 - Contracted **leprosy** due to harsh jail conditions; died on **23 July 1925**.

World Population Day

World Population Day is observed annually on **11th July** to raise awareness about **global population issues and their social, economic and developmental impacts**.

- **Origin:** Began on **11 July 1987**, when global population reached around **5 billion**, highlighting concerns over rapid population growth.
- **Establishment:** Formally established by **UNDP in 1989**.
- **Objective:** Promote awareness on population issues including **maternal health, poverty, unemployment, migration and access to essential services**.
- **Theme 2026:** “Realizing the hopes and aspirations of young people – today and for the future”.
 - Focuses on supporting youth choices related to **relationships, parenthood and family life**.
 - Based on the report “**Lives, Choices and Futures**”, covering views of **108,000+ adults (18–39 years)** across **73 countries**.
- **Global Population Trend Reached 7 billion in 2011.**
 - Projected to reach **8.5 billion by 2030** and **9.7 billion by 2050**.
- **India’s Population Trend:** Population– **1.4 billion (~17% of global population)**.
 - Expected to peak at **1.7–1.9 billion by the 2060s–2080s** before stabilising or declining.

World Zoonoses Day

Union Ministry of Health & Family Welfare held a **National Multistakeholder Consultation on World Zoonoses Day, 6th July**, to discuss the draft **National Action Plan for Prevention and Control of Zoonoses** under the **One Health Approach**.

- **World Zoonoses Day** marks **Louis Pasteur’s first rabies vaccine administration in 1885**. The **2026 theme**– “**One World, One Health: Prevent Zoonoses**.”
- **State-Level Decentralization:** National Action Plan will guide **States and UTs** to prepare local **State Action Plans** based on ecological and epidemiological conditions.
- **Launch of Technical Resources:** Under the **National One Health Programme for Prevention and Control of Zoonoses**, two tools were launched:
 - **Learning Resource Package:** Covers **10 Priority Zoonotic Diseases:** Anthrax, Brucellosis, KFD, CCHF, Nipah, Mpox, Rabies, Leptospirosis, Scrub Typhus and Zika.
 - **E-Learning Modules:** Digital courses to improve **surveillance** and **diagnostic skills** of healthcare professionals, veterinarians and frontline workers.
- **Integrated Disease Intelligence:** Need to expand **sentinel surveillance sites** and enable real-time cross-sectoral data sharing through **Integrated Health Information Platform (IHIP) 2.0**.
- **Nodal Execution:** **NCDC** will lead integrated disease surveillance, risk communication and sustainable financing across ministries, including **Animal Husbandry** and **Environment**.

Zoonotic Diseases

- **About:** **Zoonoses** are infectious diseases naturally transmitted between **vertebrate animals and humans**, caused by **viruses, bacteria, parasites or fungi**.
- **Modes of Transmission:**
 - **Direct Contact:** Contact with saliva, blood, urine or feces of infected animals, e.g., **Rabies**.
 - **Indirect Contact:** Contact with contaminated surfaces or areas where animals live/roam.
 - **Vector-borne:** Spread through ticks, mosquitoes or fleas, e.g., **Scrub Typhus, Zika, KFD**.
 - **Foodborne:** Consuming contaminated meat, unpasteurized milk or water, e.g., **Brucellosis, Anthrax**.

250th Anniversary of US Independence

PM greeted the **US President** and people on the **250th anniversary of US Independence Day** on **4th July 2026**.

- **US Declaration of Independence:** The US celebrates **Independence Day on 4th July**, marking the adoption of the **Declaration of Independence, 1776**, which proclaimed the independence of **13 American colonies** from **British rule** and laid the foundation for **modern democracy, constitutionalism, and human rights**.
 - Drafted mainly by **Thomas Jefferson** in **June 1776**, it declared that all individuals are created equal and have unalienable rights to **Life, Liberty, and the Pursuit of Happiness**.
 - It rejected the **divine right of kings** and stated that governments derive power from the “**consent of the governed**”.
- **Impact:** It influenced the **Universal Declaration of Human Rights, 1948**.
- **Influence on India:** The **US Constitution’s Bill of Rights** inspired **Fundamental Rights** in **Part III** of the Indian Constitution, especially **Articles 14 and 21**.

Nadaprabhu Kempe Gowda

The Vice-President paid **tribute to Nadaprabhu Sri Kempe Gowda on his 517th birth anniversary (27th June)** in **Bengaluru**, highlighting his legacy in urban planning, inclusive governance and environmental conservation.

- **Historical Context:** **Kempe Gowda (1510–1569)**, from the **Morasu Gowda lineage**, was a ruler and chieftain under the **Vijayanagara Empire**.
- **Foundation of Bengaluru:** With permission from **Emperor Achyuta Deva Raya**, he founded **Bengaluru in 1537** by building **Bengaluru Fort** and establishing **Bengaluru Pete**. He shifted his capital from **Yelahanka** and envisioned an inclusive city later seen as “**Mini Bharat**.”

- **Progressive Social Reforms:** He is known for prohibiting the amputation of the last two fingers of the left hand of unmarried women during “**Bandi Devaru**”, a custom among **Morasu Vokkaligas**.
- **Infrastructure and Cultural Patronage:** He supported **art and learning**, and built **temples** and **water reservoirs** for public welfare, drought resilience and cultural growth.
- **Legacy:** The **108-foot Statue of Prosperity** in Bengaluru honours Kempe Gowda and is recognized by the **World Book of Records** as the world’s tallest bronze statue of a city’s founder.

Birth anniversary of Bankim Chandra Chattopadhyay

PM paid homage to **Bankim Chandra Chattopadhyay (1838–1894)**, the great Indian novelist, poet and journalist, on his birth anniversary on **27th June**.

- **Early Life:** Born on **27th June 1838** at Naihati, Bengal Presidency; among the first graduates of Calcutta University; served as Deputy Magistrate and Deputy Collector.
- **Bengal Renaissance:** Known as the **Father of the Modern Bengali Novel** and a key Bengal Renaissance figure.
- **Major Works:** *Anandamath* (1882), *Durgeshnandini* (1865), *Kapalkundala* (1866), *Vishabriksha* (1873), *Devi Chaudhurani* (1884), *Krishna Charitra*.
- **Vande Mataram:** Published in *Bangadarshan* and later included in *Anandamath*; sung by Tagore at Congress session, 1896; became Swadeshi rallying cry; first two stanzas adopted as **National Song** in 1950.
- **Journalism:** Founded *Bangadarshan* in 1872 to promote Bengali identity and nationalist literature.
- **Nationalist Role:** Promoted cultural nationalism, Bharat Mata and deshbhakti through writings.
- **Legacy:** Died on **8th April 1894**; revered as **Sahitya Samrat**.

MSME Day 2026-Udyami Bharat

Vice President will preside over **MSME Day 2026 – Udyami Bharat** on **27th June 2026** in **New Delhi**.

- **About:** **MSME Day** is celebrated on **27th June** for MSMEs’ role in growth, innovation, jobs, exports and inclusive development.
- **Key Achievements:** **National Small Industries Corporation (NSIC)** upgraded to **Schedule ‘A’ CPSE**; **Ponduru Khadi**, Andhra Pradesh, got **GI tag**.
- **Multilingual Access:** MSME portals offer services in **22 Scheduled languages** via **BHASHINI** and **NIC**.
- **KVIC Product Launches:** **KVIC** to launch eco-friendly products: **Tukun, Rangtaal, Bela, Vanya, Umang**.

Key Initiatives

- **PMEGP 2.0 Portal:** Simplifies beneficiary lifecycle and links with **Jan Samarth Portal** for real-time loan processing.
- **SAMADHAAN 2.0 Portal:** Tracks dues owed to **MSMEs** by ministries, departments, **CPSEs** and government agencies.
- **Procurement and Marketing Support Portal:** Connects **MSEs** with trade fairs and exhibitions.
- **MSME Global Mart 2.0:** Digital commerce platform integrated with **ONDC**.
- **MSME Testing Portal:** Enables digital testing process, payments, tracking and report download.
- **MSME Idea Hackathon 6.0:** Supports innovators with up to **₹15 lakh per idea** under **MSME Champions Scheme**.

DEFENCE & SECURITY

DRDO Tests Pinaka Long Range Guided Rocket

DRDO successfully tested the **Pinaka Long Range Guided Rocket** at the **ITR, Chandipur, Odisha**, validating its **60 km minimum strike range** and **high precision**.

- An **indigenous guided rocket** for striking enemy positions, bunkers, artillery and logistics hubs from stand-off ranges.
- Operational range: **60–120 km**; the latest test confirmed accurate engagement at the **minimum 60 km range**.
- The trial demonstrated complex energy reduction and steep manoeuvres, indicating **no major dead zone** within its range.
- Fired from an existing **Pinaka launcher**, which can deploy multiple rocket variants without separate launch systems.
- Developed by **ARDE** and **HEMRL**, with support from **DRDL** and **RCI**.
- One launcher fires **12 rockets in about 44 seconds**; a six-launcher battery can fire **72 rockets** and saturate nearly **1 sq. km**.
- Its truck-mounted design supports **high mobility** and **shoot-and-scoot operations**.
- Strengthens India’s indigenous artillery capability and supports development of proposed **300 km and 450 km variants**.

Exercise RIMPAC

The **Indian Navy’s P-8i maritime patrol aircraft** is participating in the **RIMPAC 2026** in Hawaii.

- **RIMPAC:** It is the world’s largest international naval exercise, hosted by the **U.S. Pacific Fleet** since **1971**.
- **Objective:** Strengthens **interoperability, Maritime Domain Awareness, readiness**, and coordination among navies.
- **Theme:** **RIMPAC 2026**, the **30th edition**, is themed “**Partners: Integrated and Prepared**.”

- **Participation:** Includes **30 nations, 30+ ships, 5 submarines, and 200+ aircraft**, with India, US, Australia, Japan, France, UK, South Korea, Singapore and Indonesia.
- **Training Areas:** Covers **anti-submarine warfare, air defence, amphibious operations, missile drills, counter-piracy, mine countermeasures**, and salvage operations.
- **Significance:** Helps the **Indian Navy** improve coordination with partner navies and support **Indo-Pacific maritime security**.

P-8i Maritime Patrol Aircraft

- **P-8I Maritime Patrol Aircraft:** Multi-role, long-range maritime reconnaissance and anti-submarine warfare aircraft based on the **Boeing 737-800** platform.
- **Role:** Supports **maritime surveillance, strike missions, electronic warfare, search and rescue**, and targeting support.
 - Used for reconnaissance and surveillance during the post-Galwan high-altitude stand-off.
- **It can carry air-to-ship missiles** and torpedoes. India operates **12 P-8I aircraft** and plans to acquire **6 more**.

VIJAY Roadmap for Indian Army

Chief of the Army Staff General Dhiraj Seth, the **31st Chief of the Army Staff**, unveiled the **“VIJAY” roadmap** to modernise the Indian Army into a **digitised, technology-enabled, Aatmanirbhar and multi-domain capable force**, aligned with the **MoD’s Decade of Transformation (2023–2032)**.

Core Pillars of ‘VIJAY’ Roadmap

- **Vigilance (V):** Ensures **operational readiness** and alertness along borders.
- **Innovation and Transformation (I):** Integrates **modern doctrine, new technologies and rapid capability development**.
- **Jointness and Integration (J):** Promotes synergy among **Army, Navy and Air Force** with a **Whole-of-Nation approach**.
- **Atmanirbharta (A):** Focuses on **indigenous defence solutions** and self-reliance.
- **Yodha First (Y):** Prioritises **training, tech-skilling and welfare** of soldiers, **Agniveers, veterans and Veer Naris**.

Project UNICORN: India and Japan’s 1st-Ever Joint Defence Co-development Project

India and Japan announced their first joint defence co-development project at the **16th Annual Summit, New Delhi, July 2026**, to manufacture the **UNICORN (Unified Complex Radio Antenna) Mast** for future **Indian Navy warships**.

- India becomes the **second Asian country after the Philippines** to receive Japanese defence technology.

UNICORN Mast

- **About:** Advanced unified mast integrating multiple communication systems and antennas into one structure.
- **Stealth advantage:** Reduces **radar cross-section**, improving warship stealth and tactical evasion.
- **Other name:** Also known as **Nora-50**.
- **Make in India:** **BEL** will manufacture it with Japanese partners; Japan provides design expertise, India handles integration and production.
- **Naval upgrade:** Expected to replace existing **ACCS-based external communication infrastructure**.
- **Significance:** Enhances secure communication, maritime capability, and India-Japan cooperation for a **free, open Indo-Pacific**.

Project BRAHMANK

Project BRAHMANK of **BRO** celebrated its **16th Raising Day** on **29 June 2026** at **Ranaghat, Arunachal Pradesh**; its area was carved out from **Project Arunank** and **Project Udayak**.

- **Mandate:** Develops and maintains strategic roads in **Siang, East Siang, West Siang, Upper Siang, Shi-Yomi** districts of **Arunachal Pradesh** and parts of **Dhemaji, Assam**.
- **Infrastructure:** Responsible for **811 km roads** and about **86 bridges**.
- **Major works:** Built **100-m steel arch bridge** over **Siyom Nallah** and **165-m PSC bridge** over **Simang Nallah**.
- **Significance:** Improves **Armed Forces’ operational connectivity**, village access, and socio-economic integration.

Border Roads Organisation (BRO)

- **About:** India’s key agency for infrastructure in **high-altitude, snow-bound and difficult terrains**.
- **Genesis:** Established on **7 May 1960** with **Project Vartak** and **Project Beacon**.
- **Presence:** Operates **18 field projects** across **11 States** and **3 UTs**.
- **Mandate:** Builds and maintains **strategic border infrastructure** for all-weather connectivity and defence readiness.
- **Framework:** Works under the **Ministry of Defence** and **Border Roads Development Board**.
- **Key assets:** **Atal Tunnel, Sela Tunnel, Shinkun La Tunnel, Siyom, Sisseri and Devak bridges**.

- **Global outreach:** Project DANTAK, India–Myanmar Friendship Road, Delaram–Zaranj Highway, and works in Tajikistan.

Army to Raise Baaz Battalions

The Indian Army will raise specialised drone units called **Baaz Battalions** to strengthen unmanned warfare capabilities.

- **About:** Baaz Battalions will operate, manage and sustain the Army's drones and **Remotely Piloted Aircraft systems**.
- **Structure:** They will build on existing **Remotely Piloted Aircraft (RPA) Flights** and create a specialist cadre of trained drone personnel.
- **Objective:** They will enhance **surveillance, reconnaissance, intelligence gathering, target tracking** and battlefield assessment through real-time aerial inputs.
- **Operational Value:** RPAs will improve situational awareness, enable faster command decisions, support rapid response and reduce risks to personnel.
- **Need:** The move reflects lessons from the **Russia-Ukraine war, LAC standoffs with China** and **Operation Sindoor**.
- **Significance:** They will strengthen border monitoring, drone warfare preparedness, operational readiness along sensitive frontiers and coordination with intelligence systems.

Garudastra Mortar System

India successfully tested the **indigenous long-range 120 mm vehicle-mounted mortar system, Garudastra**, in Mhow, Madhya Pradesh.

- **About:** Garudastra is an indigenous **120mm vehicle-mounted mortar system** for accurate indirect fire support with high mobility and rapid response.
- **Features:** It has **rapid deployment**, high rate of fire, **shoot-and-scoot capability**, **MRSI**, precision-guided munitions and command-and-control network integration.
- **Precision:** It can engage targets at **7–10 km** using **GPS** and laser guidance with minimum collateral damage.
- **Mobility:** It can take firing position in **less than 30 seconds**, fire **two rounds within 15 seconds**, and relocate in **under 15 seconds**.
- **Firepower:** Operated by a **two-member crew**, it can fire **12 rounds in 60 seconds** while maintaining accuracy.
- **Warhead:** It carries a **17-kg penetrating warhead** effective against reinforced concrete, bunkers and hardened structures.
- **Significance:** It strengthens indigenous defence manufacturing and supports **Make in India** and **Aatmanirbhar Bharat**.

Netra Airborne Early Warning and Control (AEW&C) System

India's indigenous **Netra AEW&C** system has received **Final Operational Clearance (FOC)**, making it fully operational for deployment by the **Indian Air Force (IAF)**.

Netra AEW&C

- **About:** India's first indigenous airborne surveillance and battle-management system, developed by **DRDO-CABS**.
- **Global Status:** India became the **5th country** after the US, Russia, Israel and China to develop indigenous AEW&C capability.
- **Platform:** Integrated on Brazilian **Embraer EMB-145I**, acting as an "eye in the sky."
- **Capabilities:** AESA radar, IFF, mission computer, secure communication, ESM, CSM and self-protection systems.
- **Operational Role:** Provides early warning, real-time battlefield awareness and airborne command-and-control.
- **Background:** Began with **Project Guardian** in the 1980s; revived after the 1999 Arakkonam setback.
- **Operations:** Used in **Balakot strikes (2019)** and **Operation Sindoor (2025)**.
- **Significance:** Indigenous design gives India control over software, source codes and upgrades.
- **Future:** Six upgraded **Netra Mk-1A** approved; **Netra Mk-2** on Airbus A321 planned.

SUMMITS & CONFERENCES

1st UN Global Dialogue on AI Governance

The first **UN Global Dialogue on AI Governance** was held in **Geneva** to build a common AI governance approach. The second session will be held in **New York in May 2027**.

- **About:** Established under **UNGA Resolution 79/325** after the **Global Digital Compact, 2024**, the Dialogue promotes cooperation, scientific understanding, accountability and capacity-building in AI governance.
- **India's Position:** India advocated a **human-centric, inclusive, safe, secure and trustworthy** AI framework with meaningful human oversight.
- **Foundational Principles:** AI governance must protect **human rights**, prevent misuse and discrimination, ensure accountability and retain human authority in critical areas such as **healthcare, policing and justice**.
- **Global South and Capacity Gaps:** Developing countries need **technological, financial and institutional support** to participate in AI rule-making and benefit equitably.

- **India's Domestic Approach:** The **National Strategy for AI** and **National AI Governance Guidelines** follow "AI for All", promoting inclusion and innovation without unnecessary restrictions.
- **Major Global Concerns:** Deepfakes, sexual abuse, threats to children, autonomous weapons, deceptive AI, environmental costs, unequal access, and widening economic, security and sovereignty gaps.
- **Significance:** India and the UN supported **consensus-based multilateral governance, harmonised safety testing, legal accountability, transparent environmental reporting, renewable-powered data centres**, and stronger protection of children and human rights.

India Seeks Clarity on 'Climate Tipping Points' at Bonn Climate Conference 2026

At SB64 of UNFCCC in Bonn, India urged caution against using "climate tipping points" in climate negotiations without **clear definitions** and **robust scientific consensus**, as the talks prepared the technical base for COP31 in Antalya, Türkiye.

- **Climate Tipping Points:** Critical thresholds in the **Earth's climate system** that may trigger **rapid, self-reinforcing, and irreversible changes**.
- **Major Global Examples:** Amazon dieback, AMOC collapse, coral reef bleaching, and shifts in Indian and West African monsoons.
- **India's Stand on Scientific Rigor:** India said the term lacks a **universally accepted definition**, with uncertainty over thresholds, timing, and impacts.
- **The 1.5°C Misconception:** 1.5°C and 2°C under the **Paris Agreement** are **political temperature targets**, not proven tipping points.
- **The EU's Pushback:** The EU called India's demand for clarity **misinformation** and an **obstruction** to urgent climate action.
- **Implications for Developing Nations:** Poorly defined thresholds may impose unfair mitigation burdens on the **Global South**, weakening **Equity** and **CBDR**.
- **Climate Finance & Trade Measures:** India flagged falling **climate finance** and opposed unilateral measures like the **EU's CBAM** for penalising developing countries.

Bonn Climate Change Conference

- **Mid-year UNFCCC summit**, formally called **Sessions of the Subsidiary Bodies**, held before **COP** to review climate commitments, advance technical talks, and prepare the COP agenda.
- **Key Bodies:** Supported by **SBI** for implementation and support, and **SBSTA** for scientific and technological advice based on **IPCC** assessments.

Textiles Summit 2026

Ministry of Textiles concluded the **Textiles Summit 2026**, focusing on **sustainability, exports, global market expansion**, and India's **USD 100 billion textile export target by 2030**.

- **About:** The summit brought together **States, industry, academia, and Export Promotion Councils** to prepare a **National Textile Export Roadmap**, focusing on **implementation, value addition, product-market alignment, niche and high-value products, quality, innovation, and sustainability**.
- **Sustainability and Traceability:** Emphasis was placed on **environmental standards, quality certification, digital product passports, supply-chain traceability, closed-loop recycling, circularity, and textile-waste management** through State-municipal collaboration.
- **Export Competitiveness:** Industry was urged to use **Free Trade Agreements**, diversify markets and products, expand **man-made fibre exports**, strengthen **Brand India**, and align policies with global regulations.
- **Districts as Export Hubs:** States and UTs were encouraged to join the revived initiative to reduce **district-level information gaps** and improve **local export readiness**.

Key Facts About Textile Industry

- India's **textiles and apparel industry** is valued at **~USD 179 billion** and contributes nearly **2% of GDP**.
- India is the **6th-largest global exporter**, accounting for around **4% of world textile and apparel exports**.
- It is India's **second-largest employment-generating sector** after agriculture.
- **100% FDI** is permitted under the **automatic route**.

BRICS Space Agencies Meeting Concludes in Bengaluru

The **Union Minister of Earth Sciences** proposed a "**BRICS Space Economy**" at the **BRICS Heads of Space Agencies meeting** to promote **innovation, investment, entrepreneurship, and sustainable development**.

- **About:** Hosted by **ISRO** under India's **BRICS Chairship 2026**, the meeting included space agency heads and officials from **Brazil, China, Egypt, Ethiopia, Indonesia, Iran, Russia, South Africa, and the UAE**.
- **Purpose:** To review **BRICS space cooperation** and use space technology for **shared prosperity, resilience, and global challenges**.
- **Focus Area:** Discussions covered **space sustainability, debris-free missions, the BRICS Remote Sensing Satellite Constellation, new-member participation, and the proposed BRICS Space Council**.

- **Future Collaboration:** Cooperation was explored in disaster management, Earth observation, capacity building, knowledge sharing, and satellite-data sharing.
- **India's Space Ecosystem:** India highlighted its NewSpace sector, private companies, startups, and missions such as Chandrayaan-3, Aditya-L1, and Gaganyaan.
- **Significance:** India urged BRICS to shift from coordination to co-creation, joint innovation, co-development, responsible space behaviour, sustainable operations, and stronger partnerships.

PLACES IN NEWS

Sarnath: India's 45th UNESCO World Heritage Site

Sarnath has been inscribed on the **UNESCO World Heritage List**, taking India's total to **45 sites**—the **6th highest globally** and **2nd highest in the Asia-Pacific**.

Key Facts About the Ancient Buddhist Site of Sarnath

- **Historical and Spiritual Significance of Sarnath:** Buddha delivered his **first sermon** here in the **6th century BCE**, known as **Dharmachakra Pravartana**.
- **Birthplace of the Sangha:** Buddha met his first disciples here, marking the beginning of the **Buddhist Sangha** and the teaching of the **Noble Eightfold Path**.
- **One of the Four Principal Pilgrimage Sites:** Sarnath is one of the four major Buddhist pilgrimage sites, along with **Lumbini, Bodhgaya and Kushinagar**.
 - Of these, **Lumbini, Bodhgaya and Sarnath** are **UNESCO World Heritage Properties**.
- **Historical Nomenclature:** Ancient names include **Rishipatana, Ishipatana and Mrigadava (Deer Park)**.
- **National Emblems Found:** Excavations yielded the **Mauryan Lion Capital**, India's **National Emblem**, and the **Dharmachakra Pravartana Buddha statue** of the **Gupta period**. They are housed in the **Sarnath Site Museum**, established in **1904**.

Sarnath as a UNESCO World Heritage Property

- **UNESCO Selection Criteria:** Placed on India's **Tentative List in 1998** and inscribed under **Criterion (iii)** for its spiritual significance and **Criterion (vi)** for its association with the **Buddha, his teachings and Triratna**.
- **Site Components:** The serial property includes **Chaukhandi Stupa** and the **Archaeological Remains of Sarnath**, including **Dhamekh Stupa, Dharmarajika Stupa, Mulagandhakuti Vihara and Ashokan Pillar**.
 - **Dhamekh Stupa:** Marks the site of the **Buddha's First Sermon**.

- **Dharmarajika Stupa:** An early Buddhist stupa, traditionally commissioned by **Emperor Ashoka** to preserve the Buddha's relics.
- **Mulagandhakuti Vihara:** Believed to be the Buddha's residence during his first **Varshavasa** at Sarnath.
- **Ashokan Pillar:** Erected by **Emperor Ashoka** in the **3rd century BCE**; its **Lion Capital** is India's **National Emblem**.
 - ❖ The site also contains remains of **temples, monasteries, shrines and other structures**, reflecting over **two millennia** of Buddhist learning, pilgrimage and monastic life.

First-Ever Fencing of Sundarbans Border

The **Ministry of Home Affairs** will fence **~90 km of the Sundarbans** along the **India–Bangladesh border in West Bengal** to strengthen security in difficult terrain.

- **BSF** has begun feasibility and technical studies for fencing the **Sundarbans border**, where mangroves, tidal swamps, creeks and marshy terrain complicate surveillance.
- With no ground barriers, the **BSF** currently uses **Floating Border Outposts**.
- The **India–Bangladesh border in West Bengal** is about **2,216.7 km long**.

Sundarbans

- **About:** World's largest mangrove forest, located in the **Ganges–Brahmaputra–Meghna delta** along the **Bay of Bengal**.
- **Flora and Fauna:** Supports **84 plant species**, including **26 mangrove species**, and **453 faunal species**, including **120 fish, 290 birds, 42 mammals, 35 reptiles and 8 amphibians**.
 - It shelters threatened species such as the **estuarine crocodile, Gangetic dolphin, olive ridley turtle and water monitor lizard**.
- More than **12 million people** depend on the ecosystem: **4.5 million in India** and **7.5 million in Bangladesh**.
- **Protection:** Around **40%** of the Sundarbans lies in India.
 - **UNESCO World Heritage Site:** India in **1987** and Bangladesh in **1997**
 - **Ramsar Site:** Indian Sundarbans in **Jan 2019**
 - **Project Tiger:** Habitat of the **Royal Bengal Tiger**
 - **Biosphere Reserve:** The Sundarbans is a **Biosphere Reserve** containing:
 - ❖ Sundarbans National Park; Sundarbans East Wildlife Sanctuary; Sundarbans South Wildlife Sanctuary; Sundarbans West Wildlife Sanctuary; Sundarbans Reserve Forest, Bangladesh

Ecological Recovery of J&K's Manasbal Lake

Manasbal Lake, J&K, is witnessing ecological recovery, marked by the return of diverse **migratory birds** after habitat restoration and hydrological management.

- **Geographical Profile:** Manasbal Lake, an alkaline, spring-fed freshwater lake in **Ganderbal district**, is India's deepest freshwater lake at about **13 m**. It is sustained by precipitation, groundwater springs, and runoff.
- **Avian Biodiversity Indicators:** The lake hosts over **46 bird species from 23 families**. The return of species such as **Mallards, White-Headed Ducks, Horned Grebes, and Long-Eared Owls** indicates improving water quality.
- **Habitat Restoration Measures:** The **Wullar-Manasbal Development Authority** has undertaken dredging and weed removal to restore circulation and control excessive vegetation.
- **Hydrological Management (Water Budgeting):** The **J&K Water Resources Regulatory Authority** will assess carrying capacity, water withdrawals, and surplus availability. Mechanised gates and gauge meters will help maintain ecological water levels.
- **Pollution Mitigation Framework:** Measures include a **Sewerage Treatment Plant**, an integrated solid waste management plant, and trash guards along the **Lar irrigation canal**.
- **Community-Led Conservation:** Initiatives include **World Wetland Day, bird-watching festivals, and eco-friendly tourism** to reduce human pressure.

Major Lakes of Kashmir

Lake	Significance
Wular Lake	Asia's 2 nd largest freshwater lake and a Ramsar Site; important for flood regulation and fisheries.
Dal Lake	Iconic urban lake known for houseboats, shikaras, and Mughal gardens.
Manasbal Lake	India's deepest freshwater lake; noted for biodiversity and wetland restoration.
Surinsar-Mansar Lakes	Ramsar Sites valued for ecological, scenic, and cultural importance.

Phu Quoc Island

A speedboat capsized near Vietnam's Phu Quoc Island, causing multiple casualties, including **Indian tourists**.

Vietnam's Phu Quoc Island

- **About:** Vietnam's largest island, known as "**Pearl Island**," and a major tropical tourist destination.

- **Location and Size:** Located near **Cambodia**, it covers about **574 sq km**, extends roughly **50 km**, and is part of an archipelago of **22 islands**.
- **Natural Environment:** It has over **150 km of coastline**, mountains up to about **630 m**, forests, waterfalls, caves, and rock pools. More than half the island has been part of a **UNESCO Biosphere Reserve since 2006**.
- **Economy and Local Products:** Major activities include fishing, agriculture, tourism, and production of **fish sauce, black pepper, and pearls**.
- **Religious Heritage:** The island has **Cao Dai temples**, representing **Caodaism**, a syncretic Vietnamese religion.

Mizoram Museum Notified as India's 21st Designated Repository

The **MoEFCC**, on the recommendation of the **National Biodiversity Authority**, notified the **Natural History Museum (NHM)**, Mizoram, as India's **21st Designated Repository** under the **Biological Diversity Act, 2002**.

- **About:** Designated Repositories preserve **authenticated biological specimens** accessed under the **Biological Diversity Act, 2002**.
- **NHM Mizoram:** Established in **2022** under **Mizoram University**, located in the **Indo-Burma Biodiversity Hotspot**.
- **Specimens:** It will maintain voucher specimens of selected flora like **pteridophytes, macrofungi**, and fauna like **reptiles, amphibians, fishes, moths, beetles, butterflies**.
- **Role:** It will act as depository for **type specimens** of newly discovered species, aiding **identification, traceability and research**.
- **Regional Biodiversity:** Mizoram and the North-east have over **7,500 flowering plant species** and **2,000+ faunal species**.
- **Significance:** It will support **resource safeguarding, long-term conservation and ecological restoration**.
- **Relevance:** It strengthens **ex situ conservation** and supports **National Biodiversity Target 4** under **NBSAP 2024-2030**, aligned with the **Kunming-Montreal GBF**.

Indian Navy Foils Piracy Attempt in Gulf of Aden

INS Trikand, with **MARCOS**, responded to a piracy attempt on **MV Golden Arsenal** about **300 nautical miles east-northeast of Djibouti** in the **Gulf of Aden**, amid rising piracy threats in the **Western Indian Ocean**.

Gulf of Aden

- **Location:** Between **Arabian Peninsula** and **Africa**; bounded by **Yemen, Somalia, Socotra, Arabian Sea and Djibouti**.

- **Connectivity:** Links Somali Sea via Guardafui Channel and Red Sea via Bab el-Mandeb.
- **Major Ports:** Aden, Djibouti City, Berbera, Zeila and Bosaso.
- **Economic Importance:** Carries around 10% of global seaborne petroleum and over USD 110 billion of India's trade.
- **Threats:** Faces piracy, armed attacks, trafficking, terrorism and conflict-linked disruptions.

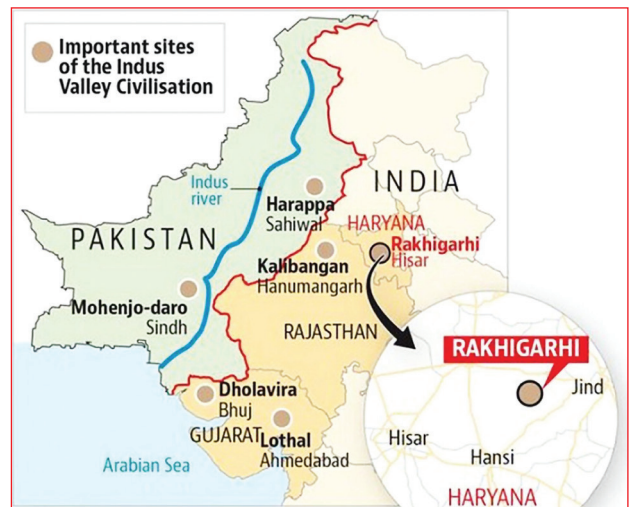


Rakhigarhi Skeletons Transferred for Scientific Research

ASI handed over human skeletal remains from Rakhigarhi, Haryana, to Anthropological Survey of India for detailed scientific study.

- **Transfer:** Under a recent MoU, three complete skeletons and fragments shifted to AnSI's Kolkata laboratory.
- **About:** Rakhigarhi spans about 550 hectares and is one of the largest known Harappan settlements.
- **Evidence:** Excavations show planned settlements, drainage, craft centres, trade networks, and burial grounds from Early to Mature Harappan periods.
- **Burials:** Mound 7 yielded 56 skeletons, including a woman dated around 4,600 years ago.
- **Ancestry:** DNA analysis of the Rakhigarhi woman showed absence of Steppe Pastoralist genetic component.
- **Methods:** Study to use ancient DNA, stable isotopes, osteology, palaeopathology, and environmental reconstruction.

- **Significance:** Insights into ancestry, migration, diet, disease, health, social status, adaptation, and human-environment relations.



China's LineShine Becomes World's Fastest Supercomputer

China's LineShine supercomputer topped the latest TOP500 ranking, replacing the US' El Capitan as the world's fastest publicly ranked supercomputer. 3rd (US): Frontier; Aurora (US): 4th; JUPITER Booster (Germany): 5th

- **TOP500 List:** Published twice a year since 1993; ranks supercomputers using the High Performance LINPACK (HPL) benchmark.
- **About LineShine:** Located at the National Supercomputing Center, Shenzhen; perform over two quintillion calculations per second.
 - It expanded the global exascale club to five systems, with Asia, North America, and Europe all hosting HPL exaflop-class supercomputers.
- **Technical Uniqueness:** First TOP500 system to cross two exaflops with a CPU-only architecture.
- **HPC-AI Convergence:** Shows merger of HPC and AI for simulations and machine-learning workloads.
- **Applications:** AI-assisted weather forecasting, rainfall prediction across East Asia, and atomic-level simulations of magnetic materials.
- **Significance:** Developed largely with Chinese technology, despite US export restrictions on advanced AI chips and semiconductor tools.

Jaipur Among World's Best Cultural Cities

Jaipur ranked 18th globally in Time Out's Best Cities for Art and Culture 2026, becoming India's highest-ranked city.

- **London** topped the list, followed by **Paris, New York City, Berlin, and Cape Town.**
- Ranking was based on a survey of **24,000+ residents** across **150+ cities**, plus inputs from editors and cultural experts.

Jaipur

- **About:** Capital of **Rajasthan**, part of **Delhi-Agra-Jaipur Golden Triangle.**
 - Founded in **1727** by **Sawai Jai Singh II.**
 - One of India's earliest planned cities.
- **Planned Pink City:** Designed by **Vidyadhar Bhattacharya** on **Vastu Shastra** principles.
 - Divided into **nine blocks.**
 - Called **Pink City** after being painted pink in **1876.**
- **Architectural Heritage:** Key sites— **Hawa Mahal, City Palace, Amber Fort, Jal Mahal, Nahargarh Fort, Jaigarh Fort, Jantar Mantar.**
- **UNESCO World Heritage Site:** **Jaipur Walled City** inscribed as **UNESCO World Heritage Site, 2019.**
- **Management:** Managed through **Conservation Management Plan (CMP), Integrated Management Plan, and State-level Heritage Regulations.**
- **Vibrant Cultural Calendar:** Events— **Jaipur Literature Festival, Jaipur Art Week, The Manganiyar Seduction.**

PORTALS & APPS

Indiahandmade Platform Promotes India's Craft Heritage

Indiahandmade, under the **Digital India** vision, brings India's **handloom and handicraft traditions** online while supporting **artisan livelihoods, cultural preservation and rural employment.**

- **About:** Launched in **2023**, **Indiahandmade** is a digital marketplace by **Digital India Corporation** under the **Ministry of Textiles**, connecting artisans and weavers directly with buyers.
- **Importance of the Sector:** India has around **64.66 lakh** handloom and handicraft artisans; women account for **71%** of handloom weavers and **64%** of total artisans.
- **Direct Market Access:** It reduces intermediaries and expands artisan access beyond local markets.
- **Wide Product Range:** Includes apparel, home décor, furnishings, paintings, furniture, religious items, stationery, musical instruments, jewellery, bags and footwear.
- **Promotion of Regional Identity:** Promotes **GI-tagged and ODOP** products like **Aipan art, Pashmina shawls, Dokra décor, Tangail sarees, Banarasi silk, Ikat, Chanderi silk and terracotta products.**

- **Trust and Accessibility:** Provides **secure payments, free shipping, purchase protection, buyer support** and simple onboarding for artisans, weavers and producer organisations.
- **Significance:** Plans to onboard over **60 lakh artisans**, supporting **Atmanirbhar Bharat**, digital market access, small sellers and India's handmade heritage.

SHE-LEAPS Platform

SHE-LEAPS was launched under '**Gramodaya Se Rashtrodaya**', to support "**Viksit Gram - Viksit Bharat.**"

- **About:** SHE-LEAPS stands for **Self-Help Entrepreneur-Livelihoods and Enterprise Application for Prosperity and Sustainability**; a digital platform to empower **women-led SHG enterprises** in rural India.
- **Development and Platform:** Developed by **Digital India Corporation (DIC)** and implemented under **LokOS**, which captures activities of **CBOs** such as **SHGs, Village Organisations and Cluster Level Federations.**
- **Objectives:** Empowers rural **SHG women entrepreneurs**, strengthens women-led enterprises, links rural producers with formal value chains, enhances financial inclusion and enables data-driven rural entrepreneurship.
- **Implementation and Features:** To be implemented in **34 States/UTs**; helps State Rural Livelihood Missions (SRLMs) capture real-time enterprise data, track livelihood journeys, improve monitoring, planning and decision-making.
- **Link with Lakhpati Didi Mission:** Supports **Lakhpati Didi Mission** through end-to-end tracking of income growth and enterprise performance, as the target expanded from **3 crore to 6 crore women.**
- **Significance:** Combines digital infrastructure, entrepreneurship and women's empowerment to support enterprise expansion, financial inclusion and women-led rural economic transformation.

MANAS: Digital Platform Against Drug Abuse

The Government is using **MANAS**, public-health measures and law enforcement to tackle drug abuse and trafficking under **Nasha Mukht Bharat.**

- **About:** Launched on **18th July 2024**, **MANAS** (Madak Padarth Nishedh Asoochna Kendra) is India's **National Narcotics Helpline** under **NCB, Ministry of Home Affairs.**
- **Access and Partners:** Developed with **Digital India Corporation**; accessible via **1933**, portal, email and **UMANG app.**
- **Core Objective:** Enables confidential reporting of **drug trafficking, peddling, illegal cultivation** and related activities.

- **Counselling and Rehabilitation:** Addiction-related calls are transferred to **MoSJE helpline 14446** for counselling and de-addiction support.
- **Digital Workflow:** Uses **digital tickets, workflow management, tracking and data analytics** for coordination, hotspot identification and faster response.
- **Agency Linkage:** Connects citizens with **30 NCB Zonal Units** and **36 State/UT Anti-Narcotics Task Forces**.
- **Significance:** Strengthens secure reporting, counselling access, digital governance, coordination and citizen-led anti-drug action.

Kisan Sarathi Platform

Kisan Sarathi provides farmers with **advisories, expert support, weather updates, market prices, and scheme information** on a single platform.

- **About:** Kisan Sarathi is India's integrated digital agro-advisory platform for farmers.
- Launched in **July 2021**.
- Joint initiative of **MeitY** and **Ministry of Agriculture and Farmers Welfare**.
- Implemented by **IASRI** and **Digital India Corporation**.
- Connects farmers with **KVKs, ICAR institutes**, agricultural universities, and experts.
- Uses **Interactive Information Dissemination System** for **two-way personalised advisories**.
- Integrates **Kisan Call Centre, CSC, IMD, MyScheme**, and **BHASHINI**.
- Provides **region-specific advisories, weather updates, mandi prices, market trends, training, video consultations, scheme discovery**, and **PM-KISAN information**.
- Supports live interaction in **13 regional languages** and has handled **19.21 lakh farmer queries**.
- **Significance:** Strengthens **lab-to-land transfer**, farmer decision-making, expert support, and digital agriculture.

Rural Internal Audit Portal

The **Union Rural Development Minister** launched the AI-enabled **Rural Internal Audit Portal** at the **Rashtriya Gramin Vikas Sammelan, New Delhi**.

- Developed with **National Informatics Centre (NIC)**; manages the **end-to-end internal audit lifecycle**.
- **Comprehensive Core Coverage:** Unified digital ecosystem for **compliance** and **risk-based audits**, replacing paper-heavy audit systems with an **automated, transparent mechanism**.
- **Phased Rollout History:** After pilot phase, all core modules became operational from **October 2025**.

- **Geospatial Tracking Capability: Map View Module** gives geospatial view of audits across India and identifies units **never audited**.
- **AI and Machine Learning Integration:** Uses **AI and analytics** for **predictive risk-scoring**, pattern recognition for leakages, and data-led audit planning.
- **Pan-India Civil Ministry Adoption:** **CGA, Ministry of Finance** mandated its implementation across all **civil ministries and departments** in **December 2025**.
- **Action Taken Report (ATR) Automation:** Digitizes **ATR submission** and settlement of audit paragraphs for real-time compliance monitoring.
- **Technology:** Secure architecture with **role-based access**, centralized monitoring, secure API integrations, audit logging, backups, disaster recovery, and separate development/testing/production environments.
- **Centralized Record Management:** Stores audit records, compliance responses, and reports in a centralized repository, keeping failures and financial irregularities traceable.
- **Environmental and Administrative Outcome:** Enables **paperless public financial management** and shifts internal audit into a **strategic management tool**.

SPECIES IN NEWS

Spotted Deer

Under the **Wildlife Protection Act, 1972**, the **Chief Wildlife Warden** may permit translocation for scientific management.

- **Objective:** Strengthen the prey base, enhance biodiversity and reduce predator movement into human settlements.
 - A larger herbivore population helps carnivores meet food needs within forests, lowering conflict and improving ecosystem stability.
- **Post-release Monitoring:** Forest guards, trackers and camera traps will monitor movement, habitat use, health and survival.

Spotted Deer

- **Scientific Name:** *Axis axis*; also called **Chital, Cheetal** or **Axis Deer**.
- **Distribution:** Native to **India, Nepal, Bhutan, Bangladesh** and **Sri Lanka**; invasive in the **Andaman Islands**.
- **Habitat:** Deciduous forests, grasslands, riverine areas and forest fringes.
- **Physical Features:** Golden-brown coat with white spots; only males have **three-pronged antlers**, shed and regrown annually.

- **Diet:** Grasses, shoots, leaves and fallen fruits.
- **Ecological Role:** Associates with **langur monkeys** and is a major prey species for **tigers, leopards, Asiatic lions** and **Indian wolves**.
- **Conservation Status:** IUCN: Least Concern; **Schedule II** of the **Wild Life (Protection) Act, 1972**.

Breeding of Indian Grey Hornbills in Gir

The **Indian Grey Hornbill** has bred in Gujarat's **Gir forests** for the **fourth consecutive year** after its **2021 reintroduction**, following an absence of over **60 years**.

- **40 hornbills** were translocated from Gujarat's **Aravalli forests**.
- **11 males** were fitted with satellite transmitters to track movement, habitat use, settlement and breeding behaviour.

Indian Grey Hornbill

- **About:** A hornbill species of the Indian subcontinent found in forests, orchards, scrublands, gardens and urban areas with old trees.
- **Distribution:** Found across India from **Jammu to Kerala** and **Rajasthan to West Bengal**, and in parts of **Pakistan, Nepal and Bangladesh**.
- **Habitat:** Prefers dry plains, foothills, deciduous forests, thorny scrub and open wooded landscapes.
- **Habitat Recovery:** Protection following the declaration of **Gir Wildlife Sanctuary in 1965** and **Gir National Park in 1975** supported its reintroduction.
- **IUCN Status:** Least Concern
- **Appearance:** Ashy-grey body, long tail with a white band and a curved ivory-coloured beak with a blackish base.
- **Casque:** Narrow casque on the beak; larger in males.
- **Eye Features:** Adults have red eyes with grey surrounding skin; juveniles have brown-orange eyes with reddish-orange skin.
- **Ecological Role:** An important **seed disperser** supporting tree regeneration.

Gitchak nakana: First Blind

Aquifer-Dwelling Fish in Northeast India

Gitchak nakana, a new blind, aquifer-dwelling fish genus and species, was discovered in a dug-out well in **Assam**—the first such record from Northeast India and the Eastern Himalayas.

Gitchak nakana

- **Name:** *Gitchak* = **red**; *nakana* = **blind fish** in the Garo language.

- **Taxonomy:** New genus and species of the **Cobitidae (true loach)** family.
- **Habitat:** Rare **subterranean (aquifer-dwelling)** fish found in a single dug-out well in Assam.
- **Significance:** First evidence of **phreatobitic fauna** in the **Eastern Himalayas**; earlier such fishes were mainly known from the **Western Ghats**.
- **Conservation:** Highlights the need to protect restricted subterranean species; **Neolissochilus pnar** (Meghalaya cave mahseer) is threatened by the aquarium trade.

MISCELLANEOUS

23rd Commonwealth Games 2026

The **23rd Commonwealth Games** were being held in **Glasgow, Scotland (23 July–2 August 2026)**, with participation from **74 nations and territories**.

- **Featured Sports:** 2026 edition includes **10 sports** (9 fewer than **Birmingham 2022**).
 - **Sports:** Athletics, swimming, 3x3 basketball, track cycling, weightlifting, bowls, artistic gymnastics, netball, boxing and judo.
- **Indian Participation:** India is participating with a **125-member contingent**, including **para-athletes**.
 - Indian athletes will compete in **athletics, boxing, weightlifting, judo, artistic gymnastics, track cycling, bowls, swimming and para-sports**.
 - India will not participate in **3x3 basketball and netball**.
- **India's CWG Record:** India first participated in the **Commonwealth Games in 1934**.
 - Best performance: **2010 Commonwealth Games (Delhi)** with **101 medals**.

Borophene-Based Green Lubricant

Researchers from **IASST, Guwahati** (an autonomous institute under **DST**) demonstrated the first use of **borophene as a lubricant additive in castor oil**. The innovation may enable **energy-efficient and sustainable green lubrication technologies**.

- **Need for Research**
 - **Friction and wear** cause major energy losses and machinery failures.
 - **Castor oil** is renewable and biodegradable but requires additives for better lubrication performance.
- **Borophene**
 - A **2D nanomaterial made of boron atoms** with high mechanical strength and electrical conductivity.

- Potential applications include **batteries, supercapacitors, fuel cells and green lubrication**.
- Addition of just **0.1 wt.% borophene** to castor oil reduced friction by **~42%** compared to pure castor oil.
- Forms a protective **tribofilm** containing iron oxides, carbonaceous species and boron compounds, reducing wear and improving load-bearing capacity.

■ Significance

- Improves machinery efficiency, reduces energy consumption and extends equipment life.
- Supports **sustainable lubrication technologies** with applications in **renewable energy, marine systems and green manufacturing**.

Pellet Guns and Shock Batons

Pellet guns came under scrutiny after allegations of their use during the **July 2026 Delhi youth and student protest** organised by the **CJP**.

Pellet Guns

- **Classification:** Considered “**non-lethal**” crowd-control weapons along with tear gas and water cannons, but can cause **fatal injuries and permanent blindness** when fired at close range.
- **Working:** Fire cartridges containing **hundreds of lead pellets** to immobilize targets.
- **Manufacturing & SOPs:** Manufactured at **Ordnance Factory, Ishapore**; SOPs require minimal firing and aiming below the waist.
- **Use in India:** Introduced during the **2010 Kashmir unrest**; faced criticism after the **2016 Kashmir unrest** due to mass blinding injuries.
- **Alternative:** **Plastic bullets** were introduced in **2017**, tested by **TBRL, Chandigarh**.

Shock Batons

- **About:** Handheld electroshock devices used for **non-lethal self-defence and crowd control**.
- **Function:** Deliver high-voltage, low-current shocks causing temporary pain and muscle disruption.
- **Use in India:** Carried by **Rapid Action Force (RAF)** personnel.
- **MHA/BSF Standards (2021):**
 - Weight: **≤1 kg**; Length: **450–700 mm**
 - Operating range: **–20°C to +50°C**
 - Service life: **minimum 5 years**
 - Rechargeable battery: **1,000 quarter-second shock bursts** per charge.

New Exoplanet Discovered Around Beta Pictoris

Using the **James Webb Space Telescope (JWST)**, astronomers discovered the hidden giant exoplanet **Beta Pictoris d** in the **Beta Pictoris star system** (**63 light-years from Earth** in the **Milky Way**).

Beta Pictoris System

- A **young planetary system** (**~23 million years old**) that helps study **planet formation and planet–disk interactions**.
- Earlier known planets: **Beta Pictoris b** and **Beta Pictoris c**.
- With **Beta Pictoris d**, it became the **second known system with three directly imaged planets**.

Beta Pictoris d

- A **gas giant exoplanet** with at least **2 times Jupiter’s mass**; smallest among the three known giant planets in the system.
- Orbits its star at around **30 AU** and has the **widest orbit** in the system.
- Remained hidden due to the system’s bright **debris disk**, which hindered direct imaging.

India’s 1st Immersive Language Museum

The Union Home Minister inaugurated the **Museum of Word (Shabdalo)** at the National Library campus in Kolkata.

- **Shabdalo** is India’s first immersive museum on **languages, words and multilingual heritage**, developed by the **National Library of India** under the **Ministry of Culture**.
 - It is India’s **largest library by collection size**.
- It explains the link between **language, culture and intergenerational transmission of traditions**.
- Uses **visual, auditory, tactile and immersive displays** to present India’s linguistic diversity.
- Traces the evolution of language from **Omkar, Panini, Shruti–Smriti and inscriptions** to manuscripts, printing and digital archives.
- Exhibits include **dialects, folk traditions, copper plates, Bhojpatra texts, manuscripts, printed books and lullabies**.

QDENG: India’s 1st Dengue Vaccine

India approved its **first dengue vaccine** amid a high and rising disease burden (increased nearly **11-fold in two decades**).

- **About:** **QDENG (TAK-003)** is a **recombinant, live-attenuated, tetravalent dengue vaccine** developed by **Takeda**.
- **Approval:** Authorised by **CDSO, Ministry of Health and Family Welfare**, after assessing safety, quality and efficacy.

- **Legal basis:** Approved under the **Drugs and Cosmetics Act, 1940** and **Drugs Rules, 1945**.
- **Eligibility:** Approved for individuals aged **4–60 years**, regardless of previous dengue infection.
- **Key advantage:** Does not require prior blood testing for dengue exposure.
- **Protection:** Targets all four serotypes—**DENV-1, DENV-2, DENV-3 and DENV-4**.
- **Global status:** Approved in over **40 countries**, **WHO-prequalified**, with more than **32 million doses** distributed.

India Becomes Party to WTO

Agreement on Fisheries Subsidies

India deposited its **Instrument of Acceptance** of **WTO Agreement on Fisheries Subsidies (AFS)**, officially becoming the **123rd member** to join the pact.

- **First environment-focused WTO pact:** Adopted at **MC12, Geneva, in June 2022** and entered into force on **15 September 2025** after acceptance by two-thirds of WTO members.
- **Scope:** Applies only to **marine wild-capture fishing** and related sea-based activities; **aquaculture and inland fisheries are excluded**.
- **Harmful subsidies:** Prohibits subsidies for **IUU fishing** and fishing of **overfished stocks**.
 - **Global concern:** Around **37.7% of marine stocks** are overfished, while fisheries subsidies total nearly **USD 35 billion annually**.
- **Small-scale fishers:** Limits unfair competition from subsidised distant-water fleets and protects livelihoods and food security.
- **India's exports:** **Aquaculture is excluded**, so India's shrimp exports remain unaffected.
- **Domestic alignment:** Supported by the **EEZ Fisheries Rules, 2025, High Seas Fishing Guidelines, 2025** and **PMMSY**.

India's First Night Safari

SC allowed UP to proceed with India's **first night safari** and a **zoological park** at **Kukrail Reserve Forest, Lucknow**.

- It will be the world's **fifth night safari**, after those in **Singapore, Thailand, China and Indonesia**.
- The project has approvals from the **Central Zoo Authority** and **MoEFCC**.

Ashok Kumar Sharma (IFS) vs. Union of India– SC had directed that no final approval for any **zoo or safari** can be granted without prior permission of the Court.

Kukrail Reserve Forest

- **Location:** Notified in **1950**, it is located in **Lucknow** and spans about **4,500 acres**, forming the city's largest contiguous green cover.
 - Named after the **Kukrail stream**, a tributary of the **Gomti River**.
- **Kukrail Gharial Rehabilitation Centre:** Established in **1975** as India's first gharial conservation centre for the **Critically Endangered gharial**.
- **Urban Green Lung:** Supports carbon sequestration, temperature regulation, groundwater recharge, air purification and climate resilience.
- **Biodiversity Value:** Provides habitat for birds, reptiles, butterflies and small mammals, while supporting the Kukrail stream's ecology.

Codex Commission Adopts

Global Standards for Three Spices

The **Codex Alimentarius Commission**, at its **49th Session in Geneva**, adopted global standards for **large cardamom, coriander, and vanilla**.

- **Significance:** The standards benefit India as a major producer and exporter of **large cardamom and coriander**. The vanilla standard provides a common quality framework, improving consistency and consumer confidence.
- **India and CAC:** India has chaired the **Codex Committee on Spices and Culinary Herbs (CCSCH)** since **2014**, with the **Spices Board of India** as secretariat.
 - It is also Co-Chair of an Electronic Working Group on risk analysis for new food products and has advocated **SMART** (Specific, Measurable, Achievable, Relevant, Time-bound) **indicators** under the **Codex Strategic Plan 2026–2031**.
- **Codex Alimentarius:** The **Food Code** includes global standards and guidelines on food hygiene, additives, pesticide residues, contaminants, labelling, and inspection.
 - The **Agreement on Application of Sanitary and Phytosanitary Measures (SPS)** of **WTO** recognises Codex standards as references for trade and dispute settlement.
- **Codex Alimentarius Commission (CAC):** Created in **1963** by **FAO and WHO**, it sets global food safety and quality standards.
 - It has **189 members**; India joined in **1964**. **CCEXEC** oversees CAC management and standards development.
 - India is a CCEXEC member as **Regional Coordinator for Asia (CCASIA)**.

India's 1st Hydrogen-Powered Train

PM flagged off India's first hydrogen-powered train, **NaMo Green Rail**, on the **89-km Jind–Sonipat** route in Haryana.

- **About the Project:** A technology-demonstration project to test **hydrogen propulsion, refuelling, safety, operations, and maintenance** for future deployment.
- **Indigenous Development:** Designed and developed in India, with specifications prepared by the **Research, Design and Standards Organisation**.
- **Features:** The **10-coach trainset** has **2 Hydrogen Driving Power Cars** and **8 Trailer Coaches**. Each power car generates **1,200 kW**, providing **2,400 kW** combined capacity. Operational speed is **75 kmph** and design speed **110 kmph**.
- **Working Mechanism:** **Proton Exchange Membrane fuel cells** combine onboard hydrogen with atmospheric oxygen to generate electricity. **Lithium iron phosphate batteries** support acceleration and variable power needs.
- **Environmental Significance:** Produces only **water vapour and heat** directly, with no smoke or tailpipe carbon emissions. Overall benefits depend on using **renewable-energy-based hydrogen**.
- **Safety and Regulation:** Requires specialised storage, ventilation, leak detection, and automatic shutdown systems. The system was approved by **Petroleum and Explosives Safety Organisation** and assessed by **TÜV SÜD, Germany**.
- **Global Significance:** India joins countries such as **Germany, Japan, China, and US** in deploying or testing hydrogen-powered rail technology.
- **Potential and Challenges:** Suitable for replacing diesel trains on **non-electrified, heritage, and difficult-to-electrify routes**. It supports the **National Green Hydrogen Mission**, but faces high costs, storage constraints, and limited green-hydrogen availability.

Navi Mumbai Airport Notified for Import of Drugs

The Union Ministry of Health and Family Welfare has allowed **drug imports through Navi Mumbai Airport** to boost pharmaceutical logistics and trade.

- **About:** Notification issued under the **Drugs and Cosmetics Act, 1940** after consultation with **Drugs Technical Advisory Board (DTAB)**, a statutory body advising on drug and cosmetics regulation.
- **Expansion:** **Rule 43A of Drugs Rules, 1945** amended to designate **Navi Mumbai Airport** as an authorised drug import entry point, increasing total approved entry points to **42**.

- **Logistics:** Enables smoother and faster movement of pharmaceutical consignments.
- **Flexibility:** Provides importers an additional route and improves supply-chain flexibility.
- **Significance:** Improves **trade facilitation** and **Ease of Doing Business** while ensuring regulatory oversight.

Cyclosporiasis Outbreak

A **cyclosporiasis (gastrointestinal illness) outbreak** in the **US** has triggered health advisories over contaminated **food and water**.

- **Cyclosporiasis:** A foodborne illness caused by the parasite ***Cyclospora cayetanensis***.
- **Mode of Transmission:** Spreads through contaminated **food or water** via the **faeco-oral route**.
 - It does **not spread directly from person to person**.
- **Symptoms and Vulnerability:** Causes prolonged watery diarrhoea, fatigue, weight loss, and stomach cramps.
 - Higher risk for **children, the elderly, and immunocompromised people**.
- **Potential Complications:** May cause **malabsorption, cholecystitis, and reactive arthritis**.
- **Diagnostic Challenges:** Often missed because the parasite is shed irregularly.
 - Diagnosis requires **Stool PCR tests (to identify the parasite's DNA) or microscopy (to identify eggs)**, sometimes using multiple samples.
- **Treatment Protocol:** Treated with **trimethoprim-sulfamethoxazole (TMP-SMX)** for **7–10 days**, along with hydration.
- **Common Sources:** Linked to contaminated raw produce such as **leafy greens, lettuce, spinach, basil, and cilantro**.
- **Prevention Nuances:** Chemical sanitisation may not fully remove the parasite.
 - Thorough washing under clean running water and strict hand hygiene are essential.

RBI's Regulatory Guidance on Data Governance Framework for Banks

RBI has proposed a **two-tier Data Governance Framework** for regulated entities to strengthen **data security, accountability, and compliance with the DPDP Act, 2023**.

- **Objective:** Require all **Regulated Entities (REs)** to align their **Data Governance Framework (DGF)** with the **Enterprise Risk Management (ERM)** framework across the entire data lifecycle to reduce financial, operational, and cyber risks.

- **Board-Level Oversight:** A **Board-level Data Governance Committee** will oversee implementation, review data breaches, and assess framework effectiveness.
- **Executive-Level Governance:** Banks must establish an **Executive Data Governance Committee** and a dedicated **Data Management Function**, headed by an officer of at least **Chief General Manager (CGM)** rank or equivalent.
- **Defined Data Governance Roles:**
 - **Data Owners:** Data quality, classification, and governance.
 - **Data Stewards:** Day-to-day implementation.
 - **Data Custodians:** Storage, technical controls, backup, disaster recovery, and secure disposal.
- **Data Lifecycle Management:** Covers purpose-based collection, consent, classification, secure processing, encryption, tokenisation, anonymisation, retention, archival, and disposal.
- **Single Source of Truth (SSOT):** Banks must create an **SSOT** for critical data and maintain metadata and data lineage for consistency, traceability, and auditability.
- **Integrated Data Risk Management:** Integrates data quality, privacy, security, ownership, cross-border processing, and third-party risks with the **ERM framework**.
- **Third-Party Data Governance:** Banks remain accountable for data shared with service providers and group entities and must ensure access controls, encryption, contractual safeguards, monitoring, and audits.
- **Expected Outcome:** Stronger operational resilience, better data-driven decisions, improved customer-data protection, and a robust banking data-governance ecosystem.

Centre Notifies Ban on Paraquat Dichloride

Government proposed a nationwide ban on the **import, manufacture, sale, transport, distribution and use of Paraquat Dichloride**.

- **Legal Basis:** Proposed under **Section 27 of the Insecticides Act, 1968** due to serious risks to humans and animals.
- **About:** A fast-acting, non-selective **herbicide** used for weed control and crop desiccation in crops including tea, cotton, paddy, wheat and maize.
- **Key Findings:** Expert bodies reported severe health effects, repeated fatal poisonings, no specific antidote, and bans or restrictions in over **70 countries**.
- **Health Risks:** Exposure through ingestion, inhalation or skin contact can cause **lung fibrosis, kidney and liver failure, heart damage, Parkinson's disease and death**.
- **Need:** Driven by occupational exposure, self-harm, accidental poisoning, misuse and demands for tighter controls from several States.

- **Implications:** May raise weed-management costs and require safer alternatives, but can reduce health risks and hazardous food residues.
- **Significance:** Strengthens pesticide regulation and promotes safer, sustainable agriculture.

India Bans Import of Goods Made Using Forced Labour

The **Directorate General of Foreign Trade (DGFT)** amended the **Foreign Trade Policy, 2023** to ban imports made using **forced labour**, amid a **USTR Section 301 investigation** against **60 economies, including India**, over weak enforcement against such imports.

- **Paragraph 2.20B in the Trade Policy :** Prohibits imports produced **wholly or partly through forced labour**.
- **DGFT Role:** Can investigate complaints, seek information and recommend restrictions; the government may ban specific goods through notifications.
- **Gap Addressed:** Introduces an explicit trade provision against goods made with forced labour abroad.
- **Enquiry Mechanism:** Procedures will be specified in the **Handbook of Procedures, 2023**.
- **ILO Alignment:** Uses the definition under the **ILO Forced Labour Convention, 1930 (No. 29)**.
- **Strategic Significance:** Strengthens India's trade-compliance framework and bargaining position in the **India-US trade deal**.

Parrot Bornavirus 4 Identified for 1st Time in India

India reported its **first confirmed detection and genetic characterisation of Parrot Bornavirus 4 (PaBV-4)**, revealing a new threat to **captive and exotic birds**.

- **About PaBV-4:** A **Bornaviridae-family Orthobornavirus** infecting psittacine birds such as parrots, macaws, cockatoos, budgerigars, and lovebirds.
- **Classification:** Eight genotypes occur in two clades: **Alphapsittaciforme (PaBV-1 to 4, 7 and 8)** and **Betapsittaciforme (PaBV-5 and 6)**.
- **Associated Disease:** Causes **Proventricular Dilatation Disease (PDD)**, leading to weight loss, regurgitation, weakness, poor coordination, behavioural changes, and death.
- **Key Findings:** **PaBV-4** was found in **44 of 83 birds**, including **88% of PDD deaths** and **19% of apparently healthy birds**.
- **Global Link:** Detection in healthy birds indicates asymptomatic transmission. Sequencing showed similarity with strains from the **US, Canada, Israel, Japan, and South Korea**, suggesting spread through international bird trade.

- **Significance:** It threatens the exotic pet industry and conservation breeding programmes, requiring quarantine, screening, veterinary care, clean enclosures, stronger biosecurity, and better avian disease surveillance.

Ladakh to Extend Autonomous Hill Councils to All 7 Districts

The **Ladakh administration** will extend the **Ladakh Autonomous Hill Development Council (LAHDC)** framework to all seven districts and create **17 new tehsils** to strengthen decentralisation, grassroots governance, and protection of Ladakh's **land, culture, and identity**.

- **Administrative Reorganization and Expansion:** After creating **five new districts—Sham, Nubra, Changthang, Zaskar, and Drass—in April 2026**, Ladakh now has **seven districts** and **32 tehsils**, including **17 new tehsils**.
 - The **LAHDC Act, 1995** allows a council in every district, subject to legal amendments and delimitation.
- **Devolution of Regulatory Powers:** Each district will have an elected **Autonomous Hill Development Council (AHDC)** with functions relating to development, land, infrastructure, health, education, and tourism.
 - Powers of the five new councils will depend on amendments, notifications, and delimitation.
- **Infrastructure Augmentation:** The administration created **four new Public Health Engineering (PHE) and Flood Control divisions** and **five new PWD and PMGSY divisions** to improve services in high-altitude areas.
- **Customised Article 371 Model:** The **MHA** is considering a **sui generis arrangement under Article 371**, instead of Sixth Schedule status, to protect Ladakh's demographic, tribal, land, and cultural interests.
- **Constitutional Framework of Article 371:** **Articles 371 to 371J in Part XXI** provide special arrangements to protect cultural identity, land rights, employment, and indigenous administration in specific regions.

India Unveils 1st Indigenous EXIM Shipping Container

India unveiled its first **domestically manufactured export-import shipping container** for **A.P. Moller–Maersk** at the **Maersk-CONCOR Inland Container Depot, UP**.

- **About:** The initiative will manufacture globally deployable containers in India, supporting **Atmanirbhar Bharat, Make in India** and **Maritime Amrit Kaal Vision 2047**.
- **International Standards:** The container meets **ISO specifications** and the **International Convention for Safe Containers (CSC)**.
- **Policy Support:** The proposed **₹10,000-crore Container Manufacturing Promotion Scheme (CMPS)** provides capital, operational and **R&D support**.

- **Capacity and Benefits:** **CMPS** targets annual capacity of about **7.5 lakh Twenty-foot Equivalent Units (TEUs)**, reducing imports and promoting supply-chain resilience, jobs, skilling and technology transfer.
- **Broader Maritime Vision:** It complements the **Merchant Shipping Act, 2025, Coastal Shipping Act, 2025, Indian Ports Act, 2025**, and digital initiatives such as **One Nation One Port Process, Maritime Single Window** and **e-Samudra**.

Mission Drishti Loses

Communication After Solar Storm

Mission Drishti, the world's first **OptoSAR satellite**, lost communication after a **geomagnetic solar storm** disrupted a critical onboard system during the final **Launch and Early Orbit Phase**.

Mission Drishti

- **Opto-SAR Technology:** **Drishti** combined **optical multispectral** and **Synthetic Aperture Radar (SAR)** imaging of the same location simultaneously.
 - **SAR** enabled imaging through **clouds and at night**, while optical sensors provided clearer imagery.
 - Its **AI system** could convert SAR data into **optical-like images** when clouds blocked visibility.
 - Proprietary synchronisation enabled **instant fusion of optical and SAR data**.
- **Utility and Significance:** Intended for **border surveillance, defence, disaster response, agriculture, infrastructure planning and insurance assessment**.
 - Strengthened India's **private space ecosystem**, alongside firms such as **Agnikul Cosmos** and **Skyroot Aerospace**.

Geneva Conventions and War Crimes

The **1949 Geneva Conventions** are in focus after the **HRF** sought the arrest of an **Israeli soldier** travelling on India's **"Hummus Trail"** in **Himachal Pradesh** over alleged **war crimes** in **Gaza**.

Hummus Trail

- **Hummus Trail:** A colloquial travel circuit in India popular among **young Israeli backpackers**, especially **military veterans**.
 - Named after the **Hippie/Hashish Trail**, it includes **Kasol, Dharamkot, Rishikesh, Pushkar, Goa, Hampi, Gokarna, Kodaikanal**, and the **Andaman and Nicobar Islands**.
- **Tiul Gadol:** The **"Big Trip"** undertaken after military service, usually lasting **6–12 months** and often funded through **post-service benefits**.

- Around **80,000 Israeli tourists** reportedly visit India annually.
- In **2026**, Israel allocated **4 million NIS** to promote tourism cooperation with India.

India's Legal Framework for Geneva Convention Breaches

- **Geneva Conventions, 1949:** Protect wounded soldiers, shipwrecked personnel, prisoners of war and civilians. Serious violations such as **wilful killing, torture and unjustified destruction of property** constitute grave breaches.
- **Universal jurisdiction:** Signatory States must **search for, prosecute or extradite** persons accused of grave breaches, irrespective of nationality or place of offence.
- **Geneva Conventions Act, 1960:** India can arrest and prosecute any person present in its territory for alleged grave breaches, although it lacks a separate comprehensive war-crimes code.
- **Administrative remedy:** The **Ministry of Home Affairs** and immigration authorities may deport an accused foreign national when criminal proceedings are not initiated.
- **Global precedent:** Groups such as **HRF** have invoked universal jurisdiction in **Chile, Brazil and Canada**.
- **Key challenge:** India must balance its strategic ties with **Israel** against its obligations under **international humanitarian law**, amid proceedings against Israel before the **ICJ**.

Agreement on Sardar Sarovar Project Financial Disputes

Maharashtra, Gujarat, Rajasthan, and Madhya Pradesh signed a **one-time settlement agreement** to resolve long-pending disputes over **cost-sharing and dues of the Sardar Sarovar Project**.

- **Scope of Disputes:** Covers **project cost-sharing, R&R expenses, and interest on construction loans** of the **Sardar Sarovar and Indira Sagar projects**.
 - Gujarat's share increased from **50% to 75%**, reducing Madhya Pradesh's burden.
- **Sardar Sarovar Project:** Concrete gravity dam on the **Narmada River** near **Navagam, Gujarat**.
 - Located near the **Statue of Unity**, the idea of a dam was envisioned by **Sardar Patel in 1946**, and **foundation stone laid in 1961** by **J. Nehru**.
 - Benefits **Gujarat, Madhya Pradesh, Maharashtra, and Rajasthan** through **irrigation, drinking water, and hydropower**.
- **Narmada Water Disputes Tribunal:** Constituted in **1969** under the **Inter-State River Water Disputes Act, 1956**.

- Its **1979 award** fixed **water and power sharing** among the four States.

- **Narmada River:** Originates at **Amarkantak Plateau, Madhya Pradesh**, and drains into the **Gulf of Khambhat**.

- Length: **1,312 km** (Madhya Pradesh, Maharashtra, and Gujarat); basin area: about **98,800 sq. km**.

Academic Bank of Credits and APAAR

ABC and APAAR ID aim to make India's education system more **flexible, portable and learner-centric**, supporting a **lifelong learning ecosystem**.

- **Academic Bank of Credits:** **ABC** is a **Ministry of Education** digital platform, regulated by **UGC**, to store, manage, transfer and redeem academic credits from recognised institutions.
- **APAAR ID:** **APAAR** is a unique **12-digit student ID** linked to **ABC** under **"One Nation, One Student ID"**, creating a single academic identity across school, higher education, skill development and other learning programmes.
- **Digital Linkage:** **APAAR** is accessible through **DigiLocker** and can be generated via **Common Service Centres** for remote and underserved learners.
- **Credit Use:** Students can accumulate, transfer and redeem credits across eligible institutions; credits are valid up to **7 years** or as prescribed, and redeemed credits cannot be reused.
 - Students can earn up to **40% credits** from **SWAYAM**; around **388 universities** adopted **SWAYAM** credit regulation and **170 universities** adopted **National Credit Framework (NCrF)**.
- **Working Mechanism:** Students register on **ABC portal** using **ABC/APAAR ID** linked with **Aadhaar** and **DigiLocker**; institutions upload credit data to student accounts.
- **Data Security:** **ABC and APAAR** use encryption, security protocols and **Aadhaar-linked DigiLocker** authentication for secure academic records.
- **Significance:** They strengthen digital education infrastructure through **DigiLocker, CSCs, SWAYAM, SAMARTH ERP** and **blockchain-based digital credentials**.

BHAVYA Industrial Parks Scheme

Union Commerce and Industry Minister chaired the **Board of Trade meeting**, urging States to prioritise **exports**, utilise **BHAVYA Industrial Parks Scheme**, and improve export competitiveness through **Centre-State-industry coordination**.

- **About:** **BHAVYA** is a **Central Sector Scheme** of **DPIIT** to develop **100 investment-ready, world-class industrial parks** across India.
- **Objective:** Strengthen manufacturing infrastructure aligned with **Make in India, PM Gati Shakti** and India's goal of becoming a globally competitive manufacturing hub.

- **Financial Outlay:** Total outlay of around ₹33,660 crore, implemented over 6 years from 2026-27 to 2031-32.
- **Industrial Park Development:** Supports greenfield and eligible brownfield parks; minimum land requirement is 100 acres for non-hilly states and 25 acres for hilly, Northeastern, UT and smaller states.
- **Challenge-Based Selection:** Up to 50 parks in Phase I through competition based on connectivity, site suitability, infrastructure quality, industrial ecosystem, policy support, digital readiness and sustainability.
- **Key Infrastructure Features:** Includes plug-and-play facilities, multimodal logistics, reliable power and water, underground utilities, waste management, CETPs, testing labs, worker housing, renewable energy and digital single-window systems.
- **Implementation Mechanism:** Implemented through SPVs under Companies Act, 2013; NICDC acts as Project Management Agency.

SpudCell: Lab-made Synthetic Cell

US researchers developed SpudCell, a synthetic cell made from non-living chemicals, capable of growth, DNA replication, division, and evolutionary selection.

- **Constructed from Non-Living Chemicals:** SpudCell was built from scratch using lifeless chemicals, starting with an artificial fat bubble liposome as the cell membrane.
- **Artificial Protein Factory:** It contains a cell-free PURE system that translates a minimal 90,000 base-pair DNA genome into working proteins.
- **Nutrient Absorption (Feeding):** Its DNA creates an alpha-hemolysin protein "hook" to capture and fuse with smaller nutrient bubbles for lipids and materials.
- **DNA Replication:** As the cell grows, it uses a copying enzyme to duplicate its entire genetic blueprint.
- **Innovative Cell Division:** SpudCell divides into daughter cells using physical pressure and mechanical stress from proteins crowding its outer surface, instead of a natural cell-like cytoskeleton.
 - It shows life-like behaviour but is not fully living or autonomous; it needs external food and ribosomes and cannot sustain itself like a natural cell.
- **Darwinian Evolution:** Artificial systems can undergo natural selection; faster-feeding mutant cells reproduced quicker and outcompeted the original population.

Telangana Declared Cancer a Notifiable Disease

Telangana has declared cancer a notifiable disease, requiring all health facilities to report diagnosed cases within one month.

- The move will help map incidence, mortality, and geographical spread, with validated data from MNJ Cancer Hospital feeding into ICMR's National Cancer Registry Programme.
- **Notifiable Disease:** A disease that must be legally reported to government authorities for surveillance and public health response.
- **India:** The Epidemic Diseases Act, 1897, provides the legal basis, while States decide notified diseases.
- **Examples:** TB, Dengue, Malaria, Cholera, Hepatitis, Measles, Polio, and Covid-19 during the pandemic.

eSARAS: Empowering Rural Women through Digital Markets

eSARAS is a key initiative driving women's entrepreneurship, preserving India's cultural heritage, and building a digitally empowered, self-reliant rural economy.

- **eSARAS:** Official online marketplace by Ministry of Rural Development under DAY-NRLM for products made by women SHGs and their federations.
- **Purpose:** Provides rural artisans and women-led cooperatives direct access to buyers across India.
- **Product Range:** Includes home items, apparel, accessories, personal care, food items, toys, and crafts like Chanderi sarees, marble crafts, and Pashmina products.
- **Support Provided:** Helps in marketing, branding, logistics, processing, packaging, and shipping.
- **Platform Evolution:** Expanded through mobile app, fulfilment centre, ONDC integration, UMANG listing, SARAS Shakti Collection, and SARAS Aajeevika Gallery, New Delhi.
- **Significance:** Promotes women entrepreneurship, digital inclusion, market access, cultural heritage, and self-reliant rural economy.

I-2SEA Submarine Cable System Unveiled

Microsoft, Singtel, Tata Communications and Lightstorm will build the India–Southeast Asia (I-2SEA) submarine cable system to support rising AI, cloud and hyperscaler connectivity demand across the India–Southeast Asia corridor.

- **About:** I-2SEA is a submarine cable system with carrier-neutral landing infrastructure, connecting India's east coast with Malaysia and Singapore and linking major AI and cloud infrastructure hubs.
- **Submarine Cables:** Undersea fibre-optic cables carrying large internet, telecom and data traffic; backbone of global digital connectivity.
- **Route and Landings:** Dual Indian landing stations at Machilipatnam and South Chennai. Machilipatnam gives the shortest subsea access to Hyderabad, with onward

connectivity to **Mumbai** and **80+ data centres** via **Lightstorm's terrestrial network**.

- **Purpose:** Serves **hyperscalers, GPU infrastructure providers and enterprises** running **AI training and inference workloads**, with high-capacity, low-latency connectivity.
- **Features:** Includes **interoperable cable architecture**, carrier-neutral landing, **deep cable burial**, and integration with **Lightstorm's SmartNet AI Fabric** and **Polarin platform**.
- **Significance:** Expected operational by **Q4 2029**; the **3,600-km cable** will boost **India–Southeast Asia digital connectivity**, support **AI/cloud ecosystems**, and provide one of the fastest routes between **Singapore/Malaysia and Hyderabad**.

GAGAN: India's Indigenous Satellite-Based Navigation System

DGCA conducted India's first **GAGAN-based satellite landing system approach** on a commercial jet, marking a major milestone in **indigenous aviation navigation**.

- **GAGAN** stands for **GPS Aided GEO Augmented Navigation**. It is India's indigenous **Satellite-Based Augmentation System (SBAS)**, developed by **ISRO** and **AAI**.
- It improves **GPS accuracy, reliability and integrity** for safer aircraft navigation and landing.
- It became operational in **2015** and is certified for the **equatorial region**.
- Architecture includes **15 INRES, 2 INMCC, 3 INLUS** and **3 geostationary satellites: GSAT-8, GSAT-10, GSAT-15**.
- Applications include **aviation, maritime navigation, railways, roads, disaster management, defence, telecom and mapping**.
- India joins the **USA, Europe and Japan** among regions with operational **SBAS**.
- **GAGAN vs NavIC:** **GAGAN** augments **GPS** for precision aviation, while **NavIC** is India's own regional navigation system for **PNT services**.

Centre Amends Drugs Rules 1945 for Advanced Therapies

The **Central Government** amended the **Drugs Rules, 1945** to bring **cell/stem cell-derived products, gene therapies and xenografts** under the **CLAA framework**, ensuring **stronger regulation and uniform national standards** for advanced medical technologies.

- **Objective:** Ensure **patient safety, uniform standards across States**, and safe use of **advanced medical technologies**.
- **CLAA:** Works under **CDSCO** and **Drugs and Cosmetics Act, 1940** as a joint **Central-State licensing mechanism**.
- **Expanded Coverage:** CLAA now covers **cell/stem cell-derived products, gene therapies and xenografts**.

- **Earlier Coverage:** Included **vaccines, large volume parenterals** and **r-DNA medicines**.
- **Need:** These therapies are **complex and rapidly evolving**, needing strict checks for **quality, safety and efficacy**.
- **Significance:** Aligns India's regulation with **global best practices** and supports safe healthcare innovation.

Mandu's Khurasani Imli Gets GI Tag

Mandu's **Baobab fruit (Khurasani Imli)** in **Madhya Pradesh** has received a **GI tag**, giving it official recognition and protection.

Khurasani Imli

- **About:** Light green Baobab fruit with tangy sweet-sour taste; valued for nutrition, medicine and culture in **Mandu**.
- **Baobab Tree:** *Adansonia digitata*, known for water-storing trunk and inverted-tree-like branches.
- **Origin:** Introduced around **six centuries ago** by **Afghan and Arab traders**.
- **Local Presence:** Mandu has India's largest natural Baobab population, with **1,000+ trees**.
- **Traditional Uses:** Used by tribal communities for **diabetes, digestion, fever and fatigue**.
- **GI Impact:** Boosts branding, marketing, cooperatives, value-added products, income and conservation.

Sushruta Statue Unveiled in Edinburgh

The **Royal College of Surgeons of Edinburgh** (established in 1505 by King James IV) unveiled a bronze statue of **Maharishi Sushruta**, honouring India's ancient surgical legacy.

- The statue was crafted at **Swamimalai, Tamil Nadu**, using the traditional **lost-wax and single bronze casting technique**.

Maharishi Sushruta

- **About:** Ancient Indian physician-surgeon, associated with **Varanasi**, around **7th–6th century BCE**.
- **Titles:** Known as the **Father of Surgery** and **Father of Plastic Surgery**.
- **Lineage:** Traditionally regarded as a disciple of **Dhanvantari** and linked with the **Navratnas of King Vikramaditya or Chandragupta II**.
- **Work:** Authored **Sushruta Samhita**, part of Ayurveda's Great Trilogy with *Charaka Samhita* and *Astanga Hridaya*.
- **Surgeries:** Documented **300+ procedures**, including **rhinoplasty, cataract removal, fracture care, caesarean delivery, lobuloplasty and otoplasty**.
- **Principles:** Emphasised **anatomy, instruments, cleanliness, cadaver dissection, anaesthesia practices and hands-on surgical training**.

India's Kharif Preparedness Amid El Niño

Due to a delayed southwest monsoon and 43% rainfall deficit, the Agriculture Ministry has prepared a contingency plan to protect the Kharif season from possible El Niño impacts.

- **Vulnerability Mapping and Prioritization:** 315 districts across 12 states have been identified as monsoon-vulnerable; 111 are high priority with irrigation below 25%, 76 medium priority and 128 low priority.
- **Contingency Planning:** ICAR's District Agriculture Contingency Plans are being converted into field-level action plans.
- **Climate-Resilient Crop Diversification:** States are promoting short-duration, low-water and drought-resistant crops, especially pulses, oilseeds and Shree Anna.
- **Macro Water Harvesting Framework:** Check dams, farm ponds and reservoirs are being repaired/desilted through MGNREGA/VB-GRAMG to improve recharge and jobs.
- **Three-Tier Financial Safety Nets:** PMFBY, KCC and PM-KISAN are being converged to provide insurance, credit and income support.
- **Real-Time Institutional Monitoring:** El Niño Monitoring Cell, Crop Weather Watch Group and 731 KVKs will support weekly reviews and advisories.
- **Food Security and Macro Targets:** Kharif foodgrain target remains 176 million tonnes, backed by adequate wheat and rice buffer stocks.

Mahesh Dixit Appointed Intelligence Bureau Chief

Mahesh Dixit, 1993-batch IPS officer, appointed **Intelligence Bureau Director** for 2 years, succeeding **Tapan Kumar Deka**.

Intelligence Bureau (IB)

- **About:** India's internal security and domestic intelligence agency under **MHA**.
- **Historical Background:** Traced to **Central Special Branch, 1887**; some link origin to **1885** unit by **Sir Charles Metcalfe MacGregor**.
- **Mandate:** Counters terrorism, collects intelligence, protects critical infrastructure, and checks secessionist activities.
- **Security Clearances:** Conducts background checks for diplomats, judges and key individuals.
- **Composition:** Personnel mainly from **IPS, IRS** and military.
- **Institutional Position:** **IB Director** is part of **Strategic Policy Group** and **Joint Intelligence Committee** of **National Security Council**, and reports to the **Prime Minister**.

Hemis Festival 2026 Begins in Ladakh

Hemis Festival 2026 began at **Hemis Monastery, Leh, Ladakh**.

Hemis Festival

- **About:** Annual festival at **Hemis Monastery, Ladakh**, on **10th day of Tse-Chu month** (according to the Tibetan lunar calendar).
- **Religious Significance:** Marks birth anniversary of **Guru Padmasambhava/Guru Rinpoche**, revered as **Second Buddha**.
- **Events:** Main attraction is **Cham mask dance** by lamas and monks.
- **Ritual Practice:** Dough sculpture symbolising evil is destroyed to ward off evil and purify the soul.
- **Local Tradition:** **Chhang** is served, a traditional local drink.
- **Thangka Display:** Every **12th year**, large thangka of **Guru Padmasambhava** is displayed.
- **Significance:** Preserves **Buddhist heritage**, promotes tourism, brotherhood, spirituality, culture, and entertainment.

DoT Notifies New Telecom Authorisation Norms

Centre notified a new **authorisation framework for telecom services** under the **Telecommunications Act, 2023**, replacing the licensing regime of the **Indian Telegraph Act, 1885**.

- **Objective:** Simplify approvals, reduce overlapping licences, improve ease of business, and support emerging telecom technologies.
- **Digital implementation:** **DoT Telecom e-Services Portal** as single-window platform.
- **Coverage:** Covers wireline, wireless, internet, long-distance, satellite, earth stations, and related telecom services.
- **Separate Categories:** Includes **M2M, PM-WANI, enterprise communications, in-flight/maritime connectivity, and captive/private networks**.
- **Migration & Entry:** Existing licensees can migrate; new entrants need fresh authorisation.
- **Spectrum Clarification:** Authorisation does not automatically provide spectrum rights.
- **Compliance & Security:** Stronger anti-fraud, anti-spoofing, lawful interception, reporting, audit, accounting, and security norms.
- **Data Localisation & Sensitive Areas:** Telecom data/logs/network information must be stored in India.
 - Prior Central Government approval needed in notified security-sensitive areas.

TOP 200 PRACTICE MCQs

July 2026 Special Edition

Coverage: 25 June – 24 July 2026

Q1. What is the primary objective of the National Anti-Doping (Amendment) Act, 2025 passed by Parliament?

- A) Banning all sports federations in India
- B) Strengthening NADA's operational autonomy, enhancing investigative powers, and aligning national norms with the WADA Code
- C) Handing over all athlete testing to private foreign clinics
- D) Decriminalizing performance-enhancing substances

Q2. Which major river is responsible for recurrent catastrophic monsoon flooding in Assam, featuring high sediment load and unstable braided channels?

- A) Teesta River
- B) Brahmaputra River
- C) Barak River only
- D) Subansiri River only

Q3. In the landmark Anuradha Bhasin v. Union of India (2020) judgment, the Supreme Court ruled that freedom of speech and expression over the internet is protected under which Article?

- A) Article 14
- B) Article 19(1)(a)
- C) Article 21
- D) Article 25

Q4. Under the Temporary Suspension of Telecom Services (Public Emergency or Public Safety) Rules, what is the maximum duration an internet shutdown order can initially remain in force?

- A) 24 hours
- B) 15 days
- C) 30 days
- D) 6 months

Q5. The National Commission for Minorities (NCM) was established as a statutory body under an Act passed in which year?

- A) 1950
- B) 1978
- C) 1992
- D) 2005

Q6. How many religious communities are currently officially notified as minorities at the national level by the Central Government under the NCM Act?

- A) 4 communities
- B) 6 communities (Muslims, Christians, Sikhs, Buddhists, Parsis, and Jains)
- C) 8 communities
- D) 12 communities

Q7. What is the primary target of the National Investment Policy for Urea-2026?

- A) 100% import dependency for urea
- B) Achieving domestic self-sufficiency in urea production and promoting energy-efficient gas-based revamp plants
- C) Replacing urea with untreated rock phosphate
- D) Banning chemical fertilizers completely

Q8. Under the NIPUN Bharat Mission, what is the foundational learning goal targeted for all children across India?

- A) Passing Class 10 board examinations
- B) Universal proficiency in Foundational Literacy and Numeracy (FLN) by the end of Grade 3
- C) Learning computer coding in kindergarten
- D) Fluency in three foreign languages

Q9. What does the UDISE+ portal managed by the Department of School Education and Literacy track?

- A) Global university rankings
- B) Comprehensive school-level data covering infrastructure, enrollment, teachers, and facilities across all schools in India
- C) Daily midday meal recipes
- D) Teacher salary bank transfers only

Q10. The Supreme Court stayed the Madras High Court's blanket ban on cow slaughter in Tamil Nadu on which constitutional and statutory ground?

- A) Complete lack of judicial authority
- B) The Tamil Nadu Animal Preservation Act permits slaughter of cattle under specific statutory age and fitness conditions, and courts cannot legislate
- C) Total ban is mandated by the UN
- D) Slaughterhouses are exempt from all state laws

Q11. In citizenship determination jurisprudence, what did the Supreme Court reiterate regarding the legal status of an Indian Passport?

- A) It is conclusive, unchallengeable proof of citizenship under all circumstances
- B) A passport is prima facie evidence of nationality and travel, but not conclusive proof of citizenship if obtained through fraud or contested identity
- C) Passports are issued only to permanent residents, not citizens
- D) Only a voter ID card constitutes valid citizenship proof

Q12. According to the Civil Registration System (CRS) Report 2024, what percentage of births were registered within the statutory 21-day timeline in India?

- A) Less than 50%
- B) Over 92%
- C) Exactly 65%
- D) 100% in all states

Q13. What was the major structural finding highlighted in the report 'States Bear Disproportionate Share of India's Welfare Spending'?

- A) The Centre funds 100% of all state welfare schemes
- B) State governments finance over 60% of combined government spending on health, education, and social protection despite lower revenue autonomy
- C) States spend zero revenue on social development
- D) Welfare spending has been completely discontinued

Q14. What is the primary mandate of the proposed Viksit Bharat Shiksha Adhishthan Bill 2025?

- A) Closing down private universities
- B) Establishing an overarching single regulatory framework for higher education, subsuming UGC and AICTE as envisioned in NEP 2020
- C) Restricting higher education to metros
- D) Abolishing entrance tests nationwide

Q15. The Draft National Food Security (Amendment) Bill, 2026 addresses which pressing challenge in public food distribution?

- A) Banning coarse grains from PDS
- B) Updating beneficiary coverage using real-time dynamic population registries rather than outdated 2011 Census figures
- C) Providing food grains only in exchange for cash
- D) Abolishing Antyodaya Anna Yojana

Q16. Which regulatory framework governs censorship, takedowns, and grievance redressal for Over-The-Top (OTT) streaming platforms in India?

- A) Cinematograph Act, 1952 only
- B) Information Technology (Intermediary Guidelines and Digital Media Ethics Code) Rules, 2021
- C) Press Council of India Act
- D) Telecom Regulatory Authority of India Act

Q17. Under the Sub-Mission on Agricultural Mechanization (SMAM), what financial assistance is provided to establish Custom Hiring Centres (CHCs)?

- A) Zero subsidy
- B) 40% to 50% capital subsidy on machinery to promote farm mechanization among small and marginal farmers
- C) 100% free tractors to all landholders
- D) Compulsory corporate leasing

Q18. In the Pankaj Bansal v. Union of India case, the Supreme Court mandated that the Enforcement Directorate (ED) must:

- A) Arrest accused persons without informing them
- B) Furnish written grounds of arrest to the accused at the time of arrest under PMLA
- C) Complete all trials within 48 hours
- D) Seize properties without magistrate warrants

Q19. What retention rate was recommended by Armed Forces internal reviews for Agniveers completing their 4-year tenure under the Agnipath scheme?

- A) Retention reduced to 10%
- B) Increasing retention from 25% up to 50% to maintain operational technical stability
- C) Retaining 100% without assessment
- D) Complete scrapping of retention

Q20. According to the All India Survey on Higher Education (AISHE) latest findings, which group witnessed the steepest increase in higher education enrollment?

- A) Urban male students only
- B) Female students, SC/ST categories, and rural youth
- C) Foreign exchange students
- D) Distance learning retirees

Q21. What is the primary function of the Poshan Tracker platform developed by the Ministry of Women and Child Development?

- A) Tracking food delivery apps
- B) Real-time monitoring of nutritional outcomes, growth tracking, and service delivery across Anganwadi Centres nationwide
- C) Ration card biometric issuance
- D) Commercial milk supply monitoring

Q22. What is the National Academic Depository (NAD) integrated with DigiLocker designed to do?

- A) Physical library lending of novels
- B) 24x7 online storehouse of all academic awards, degrees, marksheets, and certificates with digital authenticity verification
- C) Selling textbooks at market rates
- D) Conducting university student union elections

Q23. How many digits comprise the unique Ayushman Bharat Health Account (ABHA) number assigned to citizens for digital health records?

- A) 10 digits
- B) 12 digits
- C) 14 digits
- D) 16 digits

Q24. Under the next phase of the regional airport connectivity scheme 'Viksit UDAN', what is the core focus?

- A) Operating supersonic passenger jets
- B) Connecting remote, hilly, island, and underserved tourism destinations with sustainable viability gap funding (VGF)
- C) Privatizing all Air India routes
- D) Banning turboprop aircraft

Q25. Which ministry released the strategic roadmap to position Ayurveda as an evidence-based, globally recognized medical system?

- A) Ministry of Chemicals
- B) NITI Aayog in collaboration with Ministry of Ayush
- C) Ministry of Commerce only
- D) Medical Council of India

Q26. What did the Indian National Science Academy (INSA) propose in its Unified National Energy Framework?

- A) Banning solar panel manufacturing
- B) Integrated planning across coal, renewables, nuclear, and hydrogen with synchronized grid management to ensure energy transition security
- C) Phasing out all electrical grids by 2030
- D) Relying 100% on imported natural gas

Q27. The Centre proposed a special Article 371-type framework for the Union Territory of Ladakh primarily to address which local demand?

- A) Total secession from India
- B) Safeguarding land rights, unique tribal culture, employment quotas, and legislative representation
- C) Constructing deep sea naval docks
- D) Abolishing the Hill Councils

Q28. What is the main guarantee provided under the Viksit Bharat - Guarantee for Rozgar and Aajeevika Mission (Gramin) Act, 2025?

- A) 50 days of urban clerical work
- B) 125 days of guaranteed wage employment per financial year to every rural household whose adult members volunteer to do unskilled manual work
- C) Monthly pension to corporate executives
- D) Free agricultural land in metropolitan areas

Q29. What are the four components of the Pradhan Mantri Krishi Sinchayee Yojana (PMKSY)?

- A) AIBP, Har Khet Ko Pani, Per Drop More Crop, and Watershed Development
- B) Fertilizer subsidy, seed distribution, crop insurance, and tractor loan
- C) Dam construction, river dredging, canal tourism, and solar pumping
- D) Borewell digging, tube well electrification, rainwater tanks, and canal paving

Q30. What is the mandate of the Food Safety and Standards Authority of India (FSSAI) under the FSS Act, 2006?

- A) Exporting grains directly to Europe
- B) Laying down science-based standards for food articles and regulating manufacture, storage, distribution, and sale to ensure safe and wholesome food
- C) Fixing daily retail vegetable prices in local mandis
- D) Subsidizing restaurant dining

Q31. What is the key target of the Delhi Electric Vehicle Policy 2.0?

- A) Banning all personal two-wheelers
- B) Achieving 25% of all new vehicle registrations as EVs and expanding charging infrastructure across the NCR
- C) Free electricity for luxury petrol cars
- D) Mandating diesel engines for public transport

Q32. What is the '6x6x6 strategy' under the Anemia Mukh Bharat Abhiyaan?

- A) 6 hospitals in 6 states within 6 years
- B) 6 target beneficiary age groups, 6 interventions, and 6 institutional mechanisms to reduce anemia prevalence
- C) 6 iron tablets a day for 6 weeks in 6 districts
- D) 6 nutritional shakes for 6 months

Q33. Which constitutional amendment introduced the anti-defection law under the Tenth Schedule of the Indian Constitution?

- A) 42nd Amendment, 1976
- B) 44th Amendment, 1978
- C) 52nd Amendment, 1985
- D) 91st Amendment, 2003

Q34. Under Section 8(3) of the Representation of the People Act (RPA), 1951, a lawmaker is automatically disqualified upon conviction if sentenced to imprisonment for at least:

- A) 6 months
- B) 1 year
- C) 2 years or more
- D) 5 years

Q35. What was the significance of the NCERT introducing lessons on the 1975 Emergency in Class 9 textbooks?

- A) Removing all political science chapters
- B) Providing historical awareness about constitutional suspension, civil liberties, press censorship, and democratic resilience
- C) Glorifying authoritarian governance
- D) Banning election analysis in schools

Q36. The Draft National Health Research Policy 2026 prioritizes research funding primarily into which area?

- A) Cosmetic surgeries only
- B) Non-communicable diseases (NCDs), antimicrobial resistance (AMR), genomic diagnostics, and indigenous drug validation
- C) Luxury private hospital chains
- D) Overseas clinical trials of banned foreign medicines

Q37. Which statutory commission advises the President on conditions of socially and educationally backward classes under Article 338B?

- A) National Commission for Scheduled Castes
- B) National Commission for Backward Classes (NCBC)
- C) Law Commission of India
- D) Staff Selection Commission

Q38. What is the primary function of the Central Information Commission (CIC) under the RTI Act, 2005?

- A) Intercepting phone communications
- B) Final appellate authority to adjudicate complaints and second appeals regarding denial of public information by central authorities
- C) Drafting government press releases
- D) Conducting civil services examinations

Q39. Under the Lokpal and Lokayuktas Act, 2013, who falls within the jurisdiction of the Lokpal for corruption allegations?

- A) Only village sarpanches
- B) Prime Minister, Union Ministers, Members of Parliament, and Group A, B, C, D central government officers with specified procedural safeguards
- C) High Court judges exclusively
- D) Armed forces operational commands

Q40. The Election Commission of India (ECI) derives its constitutional powers to superintend, direct, and control elections from which Article?

- A) Article 320
- B) Article 324
- C) Article 326
- D) Article 330

Q41. Under Article 21A of the Constitution, free and compulsory education is guaranteed as a fundamental right to children of which age group?

- A) 3 to 6 years
- B) 6 to 14 years
- C) 6 to 18 years
- D) Up to 21 years

Q42. What is the purpose of the 'Social Audit' mandated under rural welfare legislation?

- A) Secret police scrutiny of beneficiaries
- B) Community-led verification of scheme implementation, fund utilization, and works through Gram Sabhas to ensure public accountability
- C) Commercial tax audit by private accountants
- D) Annual income tax audit of village heads

Q43. Which Article of the Constitution empowers the Supreme Court to grant 'Special Leave to Appeal' against any judgment or order from any court in India?

- A) Article 131
- B) Article 136
- C) Article 142
- D) Article 143

Q44. What power does Article 142 of the Constitution confer upon the Supreme Court?

- A) Advisory jurisdiction to the President
- B) Power to pass any decree or order necessary for doing 'complete justice' in any cause or matter pending before it
- C) Power to declare financial emergencies
- D) Power to appoint the Prime Minister

Q45. What is the core principle of 'Due Process of Law' incorporated into Article 21 following the Maneka Gandhi case (1978)?

- A) Strict adherence to literal words of any law regardless of fairness
- B) The procedure depriving life or personal liberty must be just, fair, and reasonable, and not arbitrary or oppressive
- C) Courts cannot review legislative actions
- D) Police confessions are universally admissible

Q46. According to the Global Terrorism Index (GTI) 2026 published by the Institute for Economics and Peace (IEP), what was India's standing?

- A) Most impacted nation in the world
- B) Significant improvement in security rankings due to sharp reductions in Left-Wing Extremism, cross-border infiltration, and North-East insurgencies
- C) Complete absence of law enforcement
- D) Ranking dropped to last position

Q47. What is the strategic objective of integrating Artificial Intelligence into India's defence combat readiness ('AI-Driven Warfare')?

- A) Replacing human soldiers completely with commercial humanoid toys
- B) Enhancing autonomous unmanned aerial/ground systems, predictive maintenance, algorithmic sensor fusion, and situational battlefield awareness
- C) Conducting military operations exclusively on social media
- D) Banning radar networks

Q48. Which major reform marked the creation of a single-point military adviser to the Government of India in 2019?

- A) Appointment of the Chief of Defence Staff (CDS) and creation of the Department of Military Affairs (DMA)
- B) Abolition of the Ministry of Defence
- C) Merger of Army and Navy into one service
- D) Compulsory military conscription for all citizens

Q49. The Tungabhadra River, subject to inter-state water sharing negotiations, is a major tributary of which river?

- A) Godavari River
- B) Krishna River
- C) Kaveri River
- D) Mahanadi River

Q50. The Tungabhadra Dam, built to provide irrigation and hydroelectricity, is located near Hospet in which Indian state?

- A) Andhra Pradesh
- B) Telangana
- C) Karnataka
- D) Maharashtra

Q51. What is 'Theaterisation' in the context of Indian military modernization?

- A) Building open-air cinemas for armed forces personnel
- B) Integrating the Army, Navy, and Air Force into unified joint operational commands for specific geographic sectors
- C) Conducting televised military parades every month
- D) Purchasing commercial cruise ships for soldiers

Q52. The Defence Acquisition Procedure (DAP) 2020 gives highest procurement priority to which category?

- A) Buy (Global)
- B) Buy (Indian - Indigenously Designed, Developed and Manufactured - IDDM)
- C) Buy (Foreign Military Sales)
- D) Lease from foreign powers

Q53. Which Indian state shares the longest international border with Bangladesh?

- A) Assam
- B) West Bengal
- C) Tripura
- D) Meghalaya

Q54. Under the Agnipath scheme, individuals recruited into the Armed Forces for a 4-year period are designated as:

- A) Veer Sainik
- B) Agniveer
- C) Raksha Sathi
- D) Yoddha

Q55. The Sir Creek maritime boundary dispute is situated between India and Pakistan in which geographical area?

- A) Rann of Kutch (Gujarat)
- B) Barmer (Rajasthan)
- C) Sirsa (Haryana)
- D) Wagah (Punjab)

Q56. What is the primary role of the Indian Coast Guard (ICG) established under the Coast Guard Act, 1978?

- A) Managing inland canal locks
- B) Protecting India's maritime zones, preventing offshore poaching/smuggling, marine pollution response, and coastal search and rescue
- C) Running commercial passenger luxury cruises
- D) Guarding high-altitude mountain passes

Q57. Which state assembly passed a resolution seeking a special tribal enclave status under the Sixth Schedule for its Barak Valley?

- A) Manipur
- B) Assam
- C) Tripura
- D) Meghalaya

Q58. What is the strategic significance of the Zojila Tunnel being constructed in Jammu & Kashmir / Ladakh?

- A) Connecting India to China directly
- B) Providing all-weather road connectivity between Srinagar and Leh via Kargil, bypassing treacherous avalanche zones
- C) Mining rare earth minerals under glaciers
- D) Diverting river waters to Pakistan

Q59. Which para-military force guards India's porous borders with Nepal and Bhutan?

- A) BSF (Border Security Force)
- B) ITBP
- C) SSB (Sashastra Seema Bal)
- D) Assam Rifles

Q60. The 'Assam Rifles', India's oldest paramilitary force, operates under the dual control of which two Union Ministries?

- A) Ministry of Defence (operational control) and Ministry of Home Affairs (administrative control)
- B) Ministry of Home Affairs and Ministry of Finance
- C) Ministry of External Affairs and Ministry of Defence
- D) Prime Minister's Office and Ministry of Law

Q61. Which state in North-East India is bounded by Bangladesh on three sides (north, south, and west)?

- A) Mizoram
- B) Tripura
- C) Meghalaya
- D) Manipur

Q62. What is the main objective of the National Cyber Crime Reporting Portal (cybercrime.gov.in)?

- A) Selling antivirus subscriptions
- B) Facilitating citizens to report all types of cyber fraud, financial crimes, and online abuse in a centralized manner
- C) Hacking foreign websites
- D) Monitoring personal text messages of all citizens

Q63. Which state government launched the 'Mission Basundhara' to eliminate intermediaries and digitize land revenue services?

- A) Bihar
- B) Assam
- C) West Bengal
- D) Odisha

Q64. The Armed Forces (Special Powers) Act (AFSPA), 1958 confers special powers to armed forces personnel in areas officially declared as:

- A) Disaster zones
- B) Disturbed areas
- C) Border outposts
- D) Restricted tourism reserves

Q65. Which state contains the highest concentration of Left-Wing Extremism affected 'core districts' in the Bastar division?

- A) Jharkhand
- B) Chhattisgarh
- C) Odisha
- D) Andhra Pradesh

Q66. What is the primary technical advantage of Polymer Currency Notes compared to traditional cotton-rag paper notes?

- A) They dissolve in water easily
- B) Significantly enhanced durability (2.5 to 4 times longer lifespan), superior resistance to dirt/moisture, and advanced optical security features
- C) They can be printed using household inkjet printers
- D) They are completely weightless

Q67. What is the revised base year adopted for the Index of Core Industries (ICI) to better capture contemporary production structures?

- A) 2004-05
- B) 2011-12
- C) 2022-23
- D) 2025-26

Q68. How many industries are included in the Index of Core Industries (ICI), accounting for over 40% of the weight of the Index of Industrial Production (IIP)?

- A) 5 Industries
- B) 8 Industries (Coal, Crude Oil, Natural Gas, Refinery Products, Fertilizers, Steel, Cement, Electricity)
- C) 12 Industries
- D) 15 Industries

Q69. In the Index of Core Industries, which component has the highest individual weightage?

- A) Steel
- B) Electricity
- C) Petroleum Refinery Products
- D) Coal

Q70. What does NITI Aayog's 'Investment Friendliness Index 2026' measure across Indian States and UTs?

- A) Number of stock brokers registered
- B) Ease of doing business, single-window clearance efficiency, contract enforcement, land availability, and logistics infrastructure
- C) Total gold jewelry purchased by citizens
- D) State budget deficit size

Q71. Under the Mobile Phone Manufacturing Scheme (MPMS) and electronics PLI, what milestone did India achieve in 2026?

- A) Complete exit from mobile phone production
- B) Becoming the world's second-largest mobile phone manufacturer, with exports crossing USD 15 billion
- C) Importing 100% of consumer electronics
- D) Banning smartphone sales

Q72. What is 'Skill India Digital Hub' (SIDH) launched by the Ministry of Skill Development and Entrepreneurship?

- A) A gaming console portal
- B) Unified digital platform integrating skilling initiatives, apprenticeship opportunities, verified certifications, and job-matching ecosystem
- C) Online casino platform
- D) Commercial job agency with fees

Q73. Under the PM MITRA (Mega Integrated Textile Region and Apparel) Parks scheme, how many world-class textile parks are being set up across India?

- A) 3 Parks
- B) 7 Parks
- C) 15 Parks
- D) 25 Parks

Q74. What is 'Kasturi Cotton Bharat', an initiative launched by the Ministry of Textiles and the Cotton Corporation of India?

- A) Genetically engineered synthetic polyester
- B) Premium national brand identity for Indian cotton with 100% traceability, tamper-proof QR code packaging, and certified quality parameters
- C) Cotton import subsidy scheme
- D) Cotton fabric price control order

Q75. The National Investment and Infrastructure Fund (NIIF) in India functions primarily as a:

- A) Commercial retail cooperative bank
- B) Sovereign-linked quasi-wealth fund anchored by the Government of India (49% equity) to catalyze commercial equity investments in infrastructure
- C) Stock market regulator
- D) Foreign aid donation trust

Q76. What has been the primary strategy under the National Action Plan for Toys to boost domestic toy manufacturing?

- A) Subsidizing plastic imports from abroad
- B) Quality Control Orders (QCO), mandatory BIS certification, higher import customs duties, and indigenous cluster development
- C) Banning all electronic toys
- D) Providing free toys in schools

Q77. What is 'Bima Sugam', an initiative spearheaded by the Insurance Regulatory and Development Authority of India (IRDAI)?

- A) State life insurance lottery
- B) One-stop digital marketplace (electronic protocol) for buying, servicing, and settling life, health, and general insurance policies
- C) Health insurance scheme exclusively for central ministers
- D) Compulsory vehicle insurance collection fine

Q78. What is 'Bima Vistaar', designed to increase insurance penetration in rural India?

- A) High-premium foreign life annuity
- B) Affordable, bundled composite micro-insurance product offering parametric life, health, property, and personal accident cover
- C) Subsidized insurance for corporate aviation
- D) Marine cargo insurance policy

Q79. Under the World Bank's revised Income Classification, India continues to be categorized as a:

- A) Low-Income Country
- B) Lower-Middle-Income Country (GNI per capita between USD 1,146 and USD 4,515)
- C) Upper-Middle-Income Country
- D) High-Income Country

Q80. The dedicated Ministry of Cooperation was established in which year to realize the vision of 'Sahakar se Samridhhi'?

- A) 2014
- B) 2017
- C) 2021
- D) 2024

Q81. Under Section 32A of the Insolvency and Bankruptcy Code (IBC), what protection is granted to corporate debtors upon approval of a resolution plan?

- A) Absolute immunity for errant promoters
- B) Immunity from prosecution and attachment of debtor's property for offences committed prior to the commencement of CIRP by erstwhile management
- C) Complete exemption from corporate income tax forever
- D) Right to default on all future commercial contracts

Q82. What is the flagship program launched to transform India's marine and inland fisheries with a financial outlay of over Rs. 20,050 crore?

- A) Blue Water Mission
- B) Pradhan Mantri Matsya Sampada Yojana (PMMSY)
- C) Sagarmala Fisheries Wing
- D) Samudra Setu Initiative

Q83. Goods and Services Tax (GST) completed how many years of nationwide implementation on 1st July 2026?

- A) 5 Years
- B) 7 Years
- C) 9 Years
- D) 12 Years

Q84. What is the 'Fully Accessible Route' (FAR) introduced by the RBI for foreign investors in Government Securities (G-Secs)?

- A) Physical road access to RBI vaults
- B) Channel through which non-resident investors can invest in specified sovereign benchmark securities without any investment caps
- C) Exemption from all customs duties for tourists
- D) Direct rupee-dollar conversion at railway stations

Q85. National Statistics Day in India is celebrated every year on 29th June to commemorate the birth anniversary of which eminent statistician?

- A) C.R. Rao
- B) Prof. Prasanta Chandra Mahalanobis
- C) Srinivasa Ramanujan
- D) Dadabhai Naoroji

Q86. India's Alang in Gujarat is globally recognized as the world's largest centre for which industrial activity?

- A) Commercial aircraft assembly
- B) Green Ship Recycling and dismantling under the Hong Kong Convention
- C) Deep sea oil rig manufacturing
- D) Container shipping transshipment

Q87. What is the 'Index of Services Production' (ISP) developed by the Ministry of Statistics and Programme Implementation (MoSPI)?

- A) Measuring movie box office revenues
- B) Monthly macroeconomic indicator tracking output volume and real growth across key service sectors (transport, hospitality, finance, IT)
- C) Government employee evaluation metric
- D) Consumer complaint index

Q88. Under the amended Foreign Contribution Regulation Act (FCRA) rules, through which designated bank branch must all foreign contributions be initially received?

- A) Any scheduled commercial bank
- B) State Bank of India (SBI), Main Branch, New Delhi
- C) Reserve Bank of India, Mumbai
- D) Bank of Baroda, Ahmedabad

Q89. What is the primary role of the Primary Agricultural Credit Societies (PACS) at the grassroots village level?

- A) Printing local currency notes
- B) Providing short-term crop loans, agricultural inputs, seeds, fertilizers, and storage facilities directly to village farmers
- C) Collecting central excise duties
- D) Managing civil court registries

Q90. What is 'Headline Inflation' as measured in India's official macroeconomic releases?

- A) Inflation based only on wholesale manufactured cars
- B) The total inflation figure based on the Consumer Price Index (Combined) including all food, fuel, and commodity groups
- C) Inflation rate reported in newspaper front pages only
- D) Annual rise in central government salaries

Q91. Which organization serves as the apex development bank for financing and fostering rural and agricultural development across India?

- A) SIDBI
- B) NABARD (National Bank for Agriculture and Rural Development)
- C) EXIM Bank
- D) NHB

Q92. What is the Kisan Credit Card (KCC) scheme designed to provide to farmers, fishermen, and animal husbandry rearers?

- A) Credit for purchasing foreign luxury cars
- B) Timely and hassle-free short-term credit for cultivation, post-harvest expenses, and working capital at subsidized interest rates
- C) Exemption from all electricity bills forever
- D) Free overseas flight tickets

Q93. Under the PM Matsya Sampada Yojana, what is 'Kisan Credit Card for Fisheries' aimed at?

- A) Purchasing submarines
- B) Meeting working capital requirements of fishers, fish farmers, and self-help groups for feeding, pond maintenance, and harvesting
- C) Constructing sea ports
- D) Banning private fish sales

Q94. What is the Hong Kong International Convention for the Safe and Environmentally Sound Recycling of Ships (HKC)?

- A) Banning all ship travel in Asian waters
- B) IMO convention setting global environmental, worker safety, and waste management standards for ship recycling yards
- C) Free trade agreement in commercial vessels
- D) Anti-piracy treaty in the South China Sea

Q95. What is the maximum investment limit for an enterprise to be classified as a 'Micro Enterprise' under the revised MSME definition (2020)?

- A) Investment in Plant & Machinery up to Rs. 1 crore and Turnover up to Rs. 5 crore
- B) Investment up to Rs. 10 crore and Turnover up to Rs. 50 crore
- C) Investment up to Rs. 50 crore and Turnover up to Rs. 250 crore
- D) Investment up to Rs. 100 crore

Q96. Under the GST Council constitutional framework (Article 279A), what is the weightage of the Central Government's vote?

- A) One-fourth of the total votes cast
- B) One-third of the total votes cast
- C) One-half of the total votes cast
- D) Two-thirds of the total votes cast

Q97. What is 'Reverse Charge Mechanism' (RCM) under GST?

- A) Refund of tax to overseas travelers
- B) Mechanism where the recipient of goods or services is liable to pay tax directly to the government instead of the supplier
- C) Imposing penalty on late return filers
- D) Banning unregistered suppliers

Q98. Which body replaced the erstwhile National Anti-Profiteering Authority (NAA) under GST to handle consumer profiteering complaints?

- A) Reserve Bank of India
- B) Competition Commission of India (CCI)
- C) Central Board of Direct Taxes
- D) Enforcement Directorate

Q99. What is the 'PM Vishwakarma' scheme launched by the Ministry of MSME designed to support?

- A) IT software engineers
- B) Traditional artisans and craftspeople across 18 trades with collateral-free credit, skill training, modern toolkits, and digital incentives
- C) Airline pilots and cabin crew
- D) Commercial real estate developers

Q100. What is the purpose of the 'Open Network for Digital Commerce' (ONDC) initiated by DPIIT?

- A) A government-owned shopping website competing with Amazon
- B) Open-source protocol network democratizing e-commerce by enabling buyers and sellers across different platforms to transact seamlessly
- C) Banning foreign online merchants
- D) Cryptocurrency exchange platform

Q101. The President of India made historic State Visits to which two European nations in July 2026 to elevate political and economic ties?

- A) France and Germany
- B) Moldova and North Macedonia
- C) Spain and Portugal
- D) Sweden and Norway

Q102. What is the primary mandate of the proposed World AI Cooperation Organization (WAICO)?

- A) Banning computer programming in schools
- B) Fostering multilateral governance, ethical benchmarks, open-source safety models, and preventing monopolistic control of frontier AI
- C) Directing military drone strikes globally
- D) Selling commercial AI chips at fixed prices

Q103. The historic India–UK Comprehensive Economic Trade Agreement (CETA) entered into force, offering major tariff concessions for which Indian exports?

- A) Raw unpolished coal only
- B) Textiles, apparel, leather footwear, marine products, gems & jewelry, and IT professional services
- C) Nuclear reactor cores
- D) Subsidized crude oil

Q104. Under the India-Australia Civil Nuclear Cooperation Agreement, Australia supplies which vital strategic resource to fuel India's civilian reactors?

- A) Heavy water
- B) Uranium ore concentrates (yellowcake)
- C) Enriched plutonium
- D) Thorium pellets

Q105. China's export restrictions on Helium gas triggered global supply chain concerns primarily because Helium is indispensable for:

- A) Manufacturing cotton shirts
- B) Cooling superconducting magnets in MRI machines, semiconductor manufacturing, and rocket propulsion purging
- C) Cooking household food
- D) Road asphalt paving

Q106. The India–New Zealand Strategic Partnership 'Roadmap to 2030' prioritizes cooperation in which key sectors?

- A) Deep sea coal drilling
- B) Agri-tech, high-value horticulture, dairy processing modernization, higher education student mobility, and sports science
- C) Constructing nuclear submarines
- D) Military border deployments in Antarctica

Q107. Where was the 3rd India-Australia Annual Leaders' Summit 2026 convened?

- A) Canberra / Sydney
- B) New Delhi
- C) Melbourne
- D) Tokyo

Q108. India launched an official diplomatic campaign for a Non-Permanent Seat on the United Nations Security Council (UNSC) for which upcoming biennial term?

- A) 2026-2027
- B) 2028-2029
- C) 2032-2033
- D) 2035-2036

Q109. Which strategic Indonesian port on the island of Weh is being developed with Indian infrastructure cooperation near the entrance to the Malacca Strait?

- A) Jakarta Port
- B) Sabang Port
- C) Surabaya Port
- D) Belawan Port

Q110. The inaugural India–Mali Forum for Promotion of Exports held in Bamako focused on expanding Indian exports of:

- A) Defense fighter jets
- B) Affordable pharmaceuticals, solar irrigation equipment, agricultural machinery, and digital payment solutions
- C) Subsidized natural gas
- D) Commercial passenger trains

Q111. At the 16th India-Japan Annual Summit 2026, both nations reaffirmed their shared commitment to a:

- A) Closed bilateral trade cartel
- B) Free, Open, and Inclusive Indo-Pacific (FOIP) based on rule of law, sovereignty, and peaceful resolution of maritime disputes
- C) Joint military space command against Europe
- D) Unilateral currency peg

Q112. India and Seychelles recently renewed their coastal surveillance radar system agreement and maritime assistance centered around which ocean area?

- A) Red Sea
- B) Western Indian Ocean and Mozambique Channel approaches
- C) Bay of Bengal
- D) South China Sea

Q113. What is the primary objective of the UN Security Council resolution on 'Youth, Peace and Security' (Resolution 2250)?

- A) Mandatory military draft for young adults
- B) Recognizing the vital role of youth in conflict prevention, peacebuilding, and countering violent extremism
- C) Banning youth from political voting
- D) Lowering the age of criminal responsibility

Q114. Which international river flows through China, Myanmar, Laos, Thailand, Cambodia, and Vietnam before draining into the South China Sea?

- A) Irrawaddy River
- B) Mekong River
- C) Salween River
- D) Chao Phraya

Q115. What is the 'I2U2' diplomatic grouping consisting of India, Israel, the UAE, and the United States colloquially known as?

- A) Eastern Mediterranean Alliance
- B) West Asian Quad
- C) Middle East Defense Pact
- D) Indo-Arab Forum

Q116. The headquarters of the International Atomic Energy Agency (IAEA), which oversees global nuclear safeguards, is located in:

- A) Geneva, Switzerland
- B) Vienna, Austria
- C) Brussels, Belgium
- D) New York, USA

Q117. Which African country hosted the BRICS Summit that approved the landmark expansion inviting new member nations from the Global South?

- A) Kenya
- B) South Africa (Johannesburg)
- C) Egypt
- D) Nigeria

Q118. What is the mandate of the International Court of Justice (ICJ) situated at the Peace Palace in The Hague?

- A) Prosecuting individual war criminals
- B) Settling legal disputes submitted by sovereign States in accordance with international law and giving advisory opinions to UN organs
- C) Collecting commercial debts from bankrupt corporations
- D) Regulating maritime shipping freight rates

Q119. The 'Abraham Accords' facilitated the historic diplomatic normalization between Israel and which Arab nations?

- A) UAE, Bahrain, Sudan, and Morocco
- B) Saudi Arabia and Iran
- C) Syria and Lebanon
- D) Iraq and Kuwait

Q120. Which island territory in the Indian Ocean houses the strategic joint UK-US military facility known as Camp Thunder Cove?

- A) Mauritius
- B) Diego Garcia (Chagos Archipelago)
- C) Reunion Island
- D) Zanzibar

Q121. What is the primary role of the Financial Action Task Force (FATF) established by the G7?

- A) Regulating multinational bank interest rates
- B) Setting international standards and promoting effective legal measures for combating money laundering, terrorist financing, and proliferation financing
- C) Printing the Euro currency
- D) Providing loans to developing nations

Q122. Under the United Nations Convention on the Law of the Sea (UNCLOS), an Exclusive Economic Zone (EEZ) extends up to how many nautical miles from the baseline?

- A) 12 nautical miles
- B) 24 nautical miles
- C) 200 nautical miles
- D) 350 nautical miles

Q123. What is the core purpose of the Ashgabat Agreement, to which India acceded in 2018?

- A) Nuclear disarmament in Central Asia
- B) Establishing an international transit and transport corridor facilitating transportation of goods between Central Asia and the Persian Gulf
- C) Joint space satellite exploration
- D) Regulating Caspian Sea sturgeon caviar trade

Q124. Which regional organization's motto is 'One Vision, One Identity, One Community'?

- A) SAARC
- B) ASEAN (Association of Southeast Asian Nations)
- C) European Union
- D) African Union

Q125. What is the key mechanism of the 'Loss and Damage Fund' operationalized under the UN Framework Convention on Climate Change (UNFCCC)?

- A) Lending commercial funds to oil companies
- B) Providing financial assistance to vulnerable developing countries to rescue and rebuild from irreversible economic and non-economic losses caused by climate extremes
- C) Purchasing carbon offsets for private jet owners
- D) Banning weather forecasting in island states

Q126. The White-bellied Heron (*Ardea insignis*), one of the world's 50 rarest birds, is listed under which conservation category in the IUCN Red List?

- A) Least Concern
- B) Critically Endangered
- C) Vulnerable
- D) Near Threatened

Q127. In India, the primary breeding and foraging habitats of the elusive White-bellied Heron are located along the fast-flowing rivers of which national park?

- A) Namdapha National Park (Arunachal Pradesh)
- B) Gir National Park (Gujarat)
- C) Desert National Park (Rajasthan)
- D) Bandipur National Park (Karnataka)

Q128. The Zoological Survey of India (ZSI), which celebrated its 111th Foundation Day in July 2026, is headquartered in which city?

- A) New Delhi
- B) Kolkata
- C) Dehradun
- D) Bengaluru

Q129. According to the 'Animal Discoveries 2025' report published by the ZSI, how many new species and new faunal records were added to India's biodiversity?

- A) 12 species
- B) Over 600 new faunal species and distribution records
- C) Zero discoveries
- D) 10,000 species

Q130. What is the composition of Compressed Biogas (CBG) produced under the Sustainable Alternative Towards Affordable Transportation (SATAT) initiative?

- A) 100% pure Carbon Monoxide
- B) Over 90% Methane (CH₄) with low moisture and hydrogen sulphide, having identical calorific value to commercial CNG
- C) Liquid Petroleum Ether
- D) Liquefied Hydrogen only

Q131. What primary feedstock is converted into Compressed Biogas (CBG) in agricultural biomass plants?

- A) Plastic packaging scrap
- B) Paddy straw (parali), cattle dung, sugarcane press mud, and municipal wet waste
- C) Crushed granite stone
- D) Crude petroleum residue

Q132. How many official 'Biodiversity Hotspots' of the world are present partly or entirely within Indian territory?

- A) 2 Hotspots
- B) 4 Hotspots (Himalaya, Western Ghats, Indo-Burma, and Sundaland/Nicobar)
- C) 8 Hotspots
- D) 12 Hotspots

Q133. Under the 'SATAT' scheme, what is the national target for setting up Compressed Biogas (CBG) production plants across India?

- A) 100 plants
- B) 5,000 CBG plants
- C) 25,000 plants
- D) 50,000 plants

Q134. Which statutory body was constituted under the Wildlife (Protection) Act, 1972 for strengthening tiger conservation in India?

- A) Central Zoo Authority
- B) National Tiger Conservation Authority (NTCA)
- C) Wildlife Crime Control Bureau
- D) Animal Welfare Board of India

Q135. What is 'Biochemical Oxygen Demand' (BOD) and what does a high BOD value in a river indicate?

- A) High purity and crystal-clear drinking water
- B) Severe organic pollution and microbial decomposition depleting dissolved oxygen for aquatic organisms
- C) High mineral salt concentration
- D) Water freezing at high altitude

Q136. Which Ramsar wetland site in Odisha is Asia's largest brackish water lagoon and home to the Irrawaddy dolphin?

- A) Bhitarkanika Mangroves
- B) Chilika Lake
- C) Tampara Lake
- D) Ansupa Lake

Q137. What is 'Eutrophication' in aquatic freshwater ecosystems?

- A) Complete drying up of lakes due to summer heat
- B) Excessive nutrient enrichment (phosphates and nitrates) causing algal blooms, sunlight blockage, and severe oxygen depletion (hypoxia)
- C) Excessive fish reproduction filling lakes
- D) Freezing of lake surface in winter

Q138. The 'Miyawaki Method', increasingly adopted in urban Indian cities, is a Japanese technique used for:

- A) Rooftop fish farming
- B) Creating dense, multi-layered urban micro-forests using native species in small land parcels
- C) Constructing earthquake-proof bridges
- D) Treating industrial sewage with chemicals

Q139. Which international convention aims to protect human health and the environment from Persistent Organic Pollutants (POPs)?

- A) Stockholm Convention (2001)
- B) Ramsar Convention
- C) Bonn Convention
- D) Vienna Convention

Q140. What is the IUCN status of the Gangetic Dolphin (*Platanista gangetica*), India's National Aquatic Animal?

- A) Critically Endangered
- B) Endangered
- C) Vulnerable
- D) Least Concern

Q141. Which national park in Assam is renowned as the primary global refuge of the Greater One-Horned Rhinoceros?

- A) Manas National Park
- B) Kaziranga National Park
- C) Nameri National Park
- D) Dibru-Saikhowa National Park

Q142. Under the 'Bonn Convention' (CMS), what category of fauna is internationally protected across range states?

- A) Extinct animals
- B) Migratory Species of Wild Animals and their transit habitats
- C) Laboratory mice
- D) Domestic livestock breeds

Q143. What is 'Blue Carbon' in coastal and marine biogeochemistry?

- A) Carbon emissions from diesel naval ships
- B) Carbon captured and stored by coastal marine ecosystems like mangroves, tidal marshes, and seagrass meadows
- C) Carbon dissolved in deep ocean thermal vents
- D) Synthetic carbon fibers colored blue

Q144. Which sanctuary in Tamil Nadu / Kerala / Karnataka forms part of the Nilgiri Biosphere Reserve and houses large Asian elephant populations?

- A) Mudumalai Wildlife Sanctuary
- B) Vedanthangal Bird Sanctuary
- C) Point Calimere
- D) Guindy National Park

Q145. What is the primary role of the 'Global Biodiversity Framework' (Kunming-Montreal GBF) adopted under the CBD?

- A) Banning agriculture worldwide
- B) Halting and reversing biodiversity loss by 2030, including protecting 30% of land and oceans ('30x30 target')
- C) Privatizing national parks
- D) Constructing concrete barriers around all forests

Q146. Which Indian state has the largest mangrove cover, accounting for over 42% of the country's total mangrove area?

- A) Gujarat
- B) West Bengal (Sundarbans)
- C) Andhra Pradesh
- D) Maharashtra

Q147. The 'Kigali Amendment' to the Montreal Protocol mandates the progressive global phase-down of which category of greenhouse gases?

- A) Chlorofluorocarbons (CFCs)
- B) Hydrofluorocarbons (HFCs)
- C) Carbon dioxide
- D) Methane

Q148. What is the ecological phenomenon known as 'Inversion of Temperature' that exacerbates winter air smog in North India?

- A) Air temperature increasing with altitude, trapping cold air and pollutants near the ground under a warm air lid
- B) Rapid cooling of the stratosphere during solar storms
- C) Summer heatwaves blowing from the desert
- D) Cloudbursts dropping ice blocks

Q149. Which state hosts the Bhitarkanika National Park, famous for saltwater crocodiles and Olive Ridley mass nesting (Arribada)?

- A) Andhra Pradesh
- B) Odisha
- C) West Bengal
- D) Tamil Nadu

Q150. What is the main ecological benefit of 'Artificial Coral Reefs' deployed along depleted coastlines?

- A) Harvesting commercial sand
- B) Providing submerged substrate for coral larva settlement, restoring marine biodiversity, and dissipating wave energy to curb coastal erosion
- C) Blocking fishing boats completely
- D) Generating tidal wave electricity

Q151. What is the focus of the emerging field of 'Space Oncology' being pioneered on the International Space Station (ISS)?

- A) Treating astronauts with dental braces
- B) Studying how microgravity and cosmic radiation alter cancer cell growth, protein crystallization, and drug efficacy to develop superior terrestrial therapies
- C) Launching oncology medical waste into the Sun
- D) Screening space rocks for synthetic polymers

Q152. Where did ONGC Energy Centre commission India's first exploratory Geothermal Wells to generate clean base-load geothermal energy?

- A) Manikaran, Himachal Pradesh
- B) Puga Valley, Eastern Ladakh (at an altitude of over 14,000 feet)
- C) Bakreswar, West Bengal
- D) Rajgir, Bihar

Q153. What is the precision synchronization capability demonstrated by the 'White Rabbit Technology' deployed by CSIR-National Physical Laboratory (NPL)?

- A) 1-second accuracy for wristwatch sales
- B) Sub-nanosecond time synchronization and picosecond jitter across optical fibre networks for disseminating Indian Standard Time (IST) securely
- C) Optical speed testing of bullet trains
- D) Measuring moon crater depths

Q154. What is 'Vikram-1', developed by Indian private aerospace startup Skyroot Aerospace?

- A) An unmanned lunar rover
- B) A multi-stage solid and liquid propellant commercial launch vehicle designed to place up to 300 kg payloads into Low Earth Orbit (LEO)
- C) A nuclear-powered submarine
- D) A commercial civilian drone taxi

Q155. The patent expiry of the blockbuster peptide drug 'Semaglutide' (Ozempic/Wegovy) in 2026 creates colossal generic opportunities for Indian pharma in treating:

- A) Malaria and Dengue
- B) Type 2 Diabetes Mellitus and chronic obesity/weight management via GLP-1 receptor agonism
- C) Acute bacterial pneumonia
- D) Bone fractures

Q156. What is the strategic objective of the Union Cabinet's approval of 'Semicon 2.0'?

- A) Importing older silicon chips from abroad
- B) Deepening domestic semiconductor ecosystem into raw wafer fabrication, specialized chemical supplies, EDA design tools, and compound semiconductors (GaN/SiC)
- C) Shutting down private chip design startups
- D) Subsidizing cathode ray monitors

Q157. Under ISRO's flagship Gaganyaan Mission, how many Indian astronaut-designates (Gaganyatris) undergo specialized training for human spaceflight?

- A) 2 Astronauts
- B) 4 Astronaut-designates from the Indian Air Force
- C) 8 Astronauts
- D) 12 Astronauts

Q158. What is the primary objective of the global 'NeoSep1' clinical trial that Indian premier hospitals officially joined?

- A) Testing artificial baby formula
- B) Evaluating novel antibiotic combinations to treat drug-resistant neonatal sepsis (bacterial blood infections) and reduce infant mortality
- C) Eradicating chickenpox
- D) Surgical heart transplants in infants

Q159. In Quantum Computing, what physical principle allows 'Qubits' to exist simultaneously in states 0 and 1, providing exponential parallel processing?

- A) Photoelectric effect
- B) Quantum Superposition and Entanglement
- C) Classical binary logic
- D) Thermal conduction

Q160. The United Nations Scientific Panel on AI Report emphasized which critical governance imperative for artificial intelligence?

- A) Unrestricted deployment of autonomous offensive cyberweapons
- B) Global scientific consensus on AI risks, universal accessibility of computing capacity for the Global South, and algorithmic transparency safeguards
- C) Complete ban on AI research globally
- D) Nationalizing all computer software companies

Q161. Where were India's first 3D-Printed Artificial Reefs deployed to restore nearshore marine fish habitats and buffer storm waves?

- A) Mumbai Harbor
- B) Off the coast of Vizhinjam / Kovalam (Kerala)
- C) Paradip, Odisha
- D) Visakhapatnam, Andhra Pradesh

Q162. What is 'Digital Sovereignty' in the context of India's Digital Personal Data Protection (DPDP) Act, 2023?

- A) Banning personal smartphone usage
- B) The authority of a sovereign State to control its digital trajectory, safeguard citizens' personal data, mandate critical data localization, and regulate tech monopolies
- C) Free internet streaming for foreign corporations
- D) Blocking all undersea internet cables

Q163. What is the half-life of Carbon-14, which makes radiocarbon dating indispensable for dating archaeological organic specimens up to 50,000 years old?

- A) 500 years
- B) 5,730 years (+/- 40 years)
- C) 25,000 years
- D) 1 million years

Q164. Which type of propulsion engine operates on the principle of continuous supersonic combustion throughout the entire duct without rotating parts?

- A) Turbojet
- B) Scramjet (Supersonic Combusting Ramjet)
- C) Turboprop
- D) Piston Engine

Q165. What is 'CAR-T Cell Therapy' (Chimeric Antigen Receptor T-cell), a groundbreaking therapy approved in India (NexCAR19)?

- A) A surgical procedure for knee replacement
- B) Immunotherapy that genetically re-engineers a patient's own T-cells in a laboratory to identify and destroy specific B-cell blood cancers
- C) An oral antibiotic tablet for tuberculosis
- D) Radiation therapy using cobalt-60

Q166. What is the unique capability of 'Synthetic Aperture Radar' (SAR) compared to conventional optical satellite cameras?

- A) It captures colored optical photographs only during sunny noons
- B) It transmits microwave radar pulses that penetrate through darkness, dense clouds, rain, and foliage to capture day-and-night surface imagery
- C) It records audio signals from ocean waves
- D) It only detects thermal heat plumes

Q167. Which Indian research vessel undertaken deep-sea exploration under the Deep Ocean Mission and 'Samudrayaan' project?

- A) INS Vikramaditya
- B) Sagar Nidhi / Matsya 6000 manned submersible
- C) INS Arihant
- D) RV Sindhu Sadhana

Q168. What is 'Graphene', often hailed as a wonder material in nanomaterial science?

- A) Thick blocks of mined coal
- B) A single two-dimensional layer of carbon atoms bonded in a hexagonal honeycomb lattice, possessing extraordinary electrical conductivity and 200x strength of steel
- C) Synthetic plastic adhesive
- D) Radioactive isotope of lead

Q169. Which lunar region did ISRO's Chandrayaan-3 land on in August 2023, making India the first nation to soft-land near this zone?

- A) Ocean of Storms
- B) Lunar South Polar Region (near 69 degrees S latitude)
- C) Sea of Tranquility
- D) Lunar North Pole

Q170. What is 'Zero-Knowledge Proof' (ZKP) in cryptography and blockchain protocols?

- A) Providing proof by revealing all personal passwords
- B) A cryptographic method by which one party can prove to another party that a given statement is true without conveying any extra information beyond the statement's validity
- C) Storing data without any encryption
- D) A computational test that requires zero memory

Q171. What causes the medical condition known as 'Sickle Cell Anemia', targeted under the National Sickle Cell Anemia Elimination Mission 2047?

- A) Viral infection of liver tissue
- B) A genetic mutation substituting valine for glutamic acid in the beta-globin chain, causing red blood cells to deform into rigid sickle shapes
- C) Deficiency of vitamin C in diet
- D) Exposure to cold weather

Q172. Which international space agency operates the 'Curiosity' and 'Perseverance' rovers currently analyzing Martian soil chemistry?

- A) Roscosmos
- B) NASA
- C) ESA
- D) JAXA

Q173. What is 'Dark Matter' in astrophysics?

- A) Burned asteroid debris
- B) A hypothetical form of matter that does not emit, absorb, or reflect electromagnetic radiation, but accounts for roughly 85% of total matter in the universe through gravitational effects
- C) Clouds of black smoke in deep space
- D) Interstellar soot from dead stars

Q174. What is the primary function of the 'GSLV Mk III' (LVM3) launch vehicle of ISRO?

- A) Launching suborbital weather balloons
- B) Heavy-lift three-stage launch vehicle capable of placing 4,000 kg satellites into Geostationary Transfer Orbit (GTO) and launching human space missions
- C) Polar orbit launches under 500 kg only
- D) Intercontinental ballistic missile test vehicle

Q175. What is the function of 'mRNA' (Messenger RNA) vaccines compared to traditional inactivated virus vaccines?

- A) Injecting live active pathogens directly into the bloodstream
- B) Delivering a synthetic genetic instruction template that directs cellular ribosomes to transiently manufacture a harmless viral antigen protein, triggering an immune response
- C) Permanently modifying the human genomic DNA sequence
- D) Chemically destroying all white blood cells

Q176. Swami Vivekananda attained Mahasamadhi on 4th July 1902 at which iconic monastery founded by him?

- A) Dakshineswar Temple
- B) Belur Math, Howrah (West Bengal)
- C) Ramakrishna Mission, New Delhi
- D) Kanyakumari Memorial

Q177. Dr. Syama Prasad Mookerjee, whose birth anniversary is commemorated on 6th July, founded which political party in 1951?

- A) Swatantra Party
- B) Bharatiya Jana Sangh (BJS)
- C) Praja Socialist Party
- D) Hindu Mahasabha exclusively

Q178. A rare 16th-century inscription of the Vijayanagara Empire was recently discovered in the Seshachalam Biosphere Forest of which state?

- A) Karnataka
- B) Andhra Pradesh (Tirupati region)
- C) Tamil Nadu
- D) Telangana

Q179. Sepoy Mangal Pandey, whose birth anniversary is observed in July, ignited the 1857 Indian Revolt at which military cantonment in Bengal?

- A) Meerut Cantonment
- B) Barrackpore Cantonment (34th Bengal Native Infantry)
- C) Jhansi Fort
- D) Dinapur Cantonment

Q180. K. Kamaraj, former Chief Minister of Tamil Nadu and Bharat Ratna awardee, is celebrated historically for introducing which pioneering welfare scheme?

- A) Free laptop distribution
- B) Free Mid-Day Meal scheme in government primary schools to eliminate hunger and boost school enrollment
- C) Subsidized metro rail passes
- D) Universal old-age pension

Q181. What is 'Secondary Infertility', a rapidly rising reproductive health issue highlighted in medical surveys?

- A) Infertility in domestic animals
- B) The inability of a couple to conceive or carry a pregnancy to term following the prior successful birth of one or more biological children
- C) Congenital genetic sterility from birth
- D) Infertility occurring exclusively after age 60

Q182. Under the Rights of Persons with Disabilities (RPwD) Act, 2016, how many specified benchmark disabilities are recognized in India?

- A) 7 Disabilities
- B) 14 Disabilities
- C) 21 Specified Disabilities (including acid attack victims and blood disorders)
- D) 30 Disabilities

Q183. What is the primary objective of the 'SUMAN' (Surakshit Matritva Aashwasan) scheme roadmap 2030?

- A) Charging fixed fees for emergency childbirth
- B) Assured, dignified, respectful, and zero-expense maternal and newborn healthcare to prevent preventable maternal and infant mortality
- C) Private clinic registration only
- D) Overseas adoption facilitation

Q184. World Population Day is observed globally on which date each year by the United Nations?

- A) 5th June
- B) 11th July
- C) 15th August
- D) 24th October

Q185. Nelson Mandela International Day is celebrated annually on 18th July to recognize Mandela's values and his contribution to:

- A) Maritime trade expansion
- B) The struggle against apartheid, human rights, conflict resolution, racial reconciliation, and equality
- C) Establishing international commercial banks
- D) Military missile agreements

Q186. Which multilateral naval exercise brings together the navies of India, the United States, Japan, and Australia in rotational ocean drills?

- A) Exercise Milan
- B) Exercise Malabar
- C) Exercise Varuna
- D) Exercise Samudra Shakti

Q187. Which Central Asian nation hosted the Council of Heads of State of the Shanghai Cooperation Organisation (SCO) Summit in July?

- A) Uzbekistan
- B) Kazakhstan (Astana)
- C) Tajikistan
- D) Kyrgyzstan

Q188. What is the 'PM-PRANAM' scheme approved by the Union Cabinet designed to promote?

- A) Free domestic air travel
- B) Programme for Restoration, Awareness, Nourishment and Amelioration of Mother Earth (promoting balanced use of chemical fertilizers and alternative bio-fertilizers)
- C) Compulsory religious pilgrimage
- D) Diesel vehicle subsidies

Q189. Which state's traditional craft of 'Gond Painting' was officially granted a Geographical Indication (GI) tag?

- A) Rajasthan
- B) Madhya Pradesh
- C) Odisha
- D) Bihar

Q190. Under the National Food Security Act (NFSA), 2013, what percentage of the rural and urban population respectively is entitled to subsidized food grains?

- A) 50% rural and 25% urban
- B) Up to 75% of rural population and up to 50% of urban population
- C) 100% of all citizens
- D) 33% across both

Q191. What is the primary role of the 'Sangeet Natak Akademi' established in 1953 in New Delhi?

- A) Censoring film dialogues
- B) Apex national academy for performing arts (music, dance, and drama) dedicated to preservation, research, and conferment of national honors
- C) Regulating television broadcast licenses
- D) Running commercial recording studios

Q192. Which national monument in New Delhi was constructed to commemorate Indian soldiers who lost their lives in World War I and the Third Anglo-Afghan War?

- A) Gateway of India
- B) India Gate (All-India War Memorial)
- C) Red Fort
- D) National War Memorial

Q193. The historic 'Kamaraj Plan' of 1963 proposed that senior Cabinet Ministers and Chief Ministers should:

- A) Assume permanent hereditary governorships
- B) Resign from government ministerial posts to dedicate themselves to grassroots party organizational work among the masses
- C) Join the private corporate sector
- D) Boycott parliamentary proceedings

Q194. What is the overarching objective of the 'Mission Amrit Sarovar' launched during the Azadi Ka Amrit Mahotsav?

- A) Constructing luxury swimming resorts
- B) Developing and rejuvenating at least 75 water bodies in each district of the country to harvest rainwater and recharge groundwater aquifers
- C) Bottling mineral water for export
- D) Channeling river floods into deserts

Q195. Under the Sustainable Development Goals (SDGs) National Indicator Framework Report 2026, which SDG corresponds to 'Good Health and Well-Being'?

- A) SDG 1
- B) SDG 3
- C) SDG 5
- D) SDG 10

Q196. World Youth Skills Day is observed annually on 15th July by the United Nations to highlight the strategic importance of:

- A) Reducing school vacations
- B) Equipping young people with technical, vocational, and digital skills for decent employment and entrepreneurship
- C) Mandatory military training for teenagers
- D) Banning mobile phones for students

Q197. International Tiger Day (Global Tiger Day) is observed annually on which date to raise awareness about tiger conservation?

- A) 5th June
- B) 29th July
- C) 12th August
- D) 4th October

Q198. World Nature Conservation Day is observed each year on 28th July primarily emphasizing:

- A) Expanding open-cast coal mining
- B) Sustainable utilization of natural resources, preserving biodiversity, and climate stewardship for future generations
- C) Commercial logging in virgin forests
- D) Privatizing national rivers

Q199. World Chess Day is observed on 20th July to mark the founding date of which international governing body in Paris in 1924?

- A) IOC
- B) FIDE (International Chess Federation)
- C) FIFA
- D) ICC

Q200. World Day for International Justice is observed on 17th July to mark the adoption of which historic treaty that established the International Criminal Court (ICC)?

- A) Treaty of Versailles
- B) Rome Statute (1998)
- C) Geneva Protocol
- D) Vienna Convention

OFFICIAL ANSWER KEY — PART 1 (Q.1 – Q.100)

Comprehensive Verified Answer Key for Practice MCQs 1 to 100 · sarkaripath.onl

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	B	26	B	51	B	76	B
2	B	27	B	52	B	77	B
3	B	28	B	53	B	78	B
4	B	29	A	54	B	79	B
5	C	30	B	55	A	80	C
6	B	31	B	56	B	81	B
7	B	32	B	57	B	82	B
8	B	33	C	58	B	83	C
9	B	34	C	59	C	84	B
10	B	35	B	60	A	85	B
11	B	36	B	61	B	86	B
12	B	37	B	62	B	87	B
13	B	38	B	63	B	88	B
14	B	39	B	64	B	89	B
15	B	40	B	65	B	90	B
16	B	41	B	66	B	91	B
17	B	42	B	67	C	92	B
18	B	43	B	68	B	93	B
19	B	44	B	69	C	94	B
20	B	45	B	70	B	95	A
21	B	46	B	71	B	96	B
22	B	47	B	72	B	97	B
23	C	48	A	73	B	98	B
24	B	49	B	74	B	99	B
25	B	50	C	75	B	100	B

OFFICIAL ANSWER KEY — PART 2 (Q.101 – Q.200)

Comprehensive Verified Answer Key for Practice MCQs 101 to 200 · sarkaripath.onl

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
101	B	126	B	151	B	176	B
102	B	127	A	152	B	177	B
103	B	128	B	153	B	178	B
104	B	129	B	154	B	179	B
105	B	130	B	155	B	180	B
106	B	131	B	156	B	181	B
107	B	132	B	157	B	182	C
108	B	133	B	158	B	183	B
109	B	134	B	159	B	184	B
110	B	135	B	160	B	185	B
111	B	136	B	161	B	186	B
112	B	137	B	162	B	187	B
113	B	138	B	163	B	188	B
114	B	139	A	164	B	189	B
115	B	140	B	165	B	190	B
116	B	141	B	166	B	191	B
117	B	142	B	167	B	192	B
118	B	143	B	168	B	193	B
119	A	144	A	169	B	194	B
120	B	145	B	170	B	195	B
121	B	146	B	171	B	196	B
122	C	147	B	172	B	197	B
123	B	148	A	173	B	198	B
124	B	149	B	174	B	199	B
125	B	150	B	175	B	200	B